

THE UNITED REPUBLIC OF TANZANIA



MINISTRY OF HEALTH

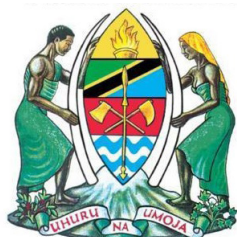
NATIONAL AIDS, STI AND HEPATITIS CONTROL
PROGRAMME



NATIONAL INTEGRATED GUIDELINES FOR MANAGEMENT OF HIV, STI AND VIRAL HEPATITIS

December, 2024

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MANAGEMENT OF HIV, STI AND VIRAL
HEPATITIS**

December, 2024

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Ministry of Health

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LIST OF ABBREVIATIONS

3TC	Lamivudine
AA	Adherence Assistant
AFB	Acid-Fast Bacilli
AIDS	Acquired Immune Deficiency Syndrome
ALAT	Alanine Aminotransferase
ANC	Antenatal Care
ART	Antiretroviral Therapy
ARV	Antiretroviral
ATV	Atazanavir
AZT	Zidovudine
BBP	Blood Borne Pathogen
BCG	Bacille Calmette-Guerin
BMI	Body Mass Index
BP	Blood Pressure
CART	Combination Antiretroviral Therapy
CBHS	Community Based HIV and AIDS Services
CBO	Community Based Organization
CHMT	Council Health Management Team
CHTC	Couples HIV Testing and Counselling
CHW	Community Health Worker
CMV	Cytomegalovirus
CNS	Central Nervous System
CoC	Continuum of Care
CPT	Cotrimoxazole Preventive Therapy
CrAg LFA	Cryptococcal Antigen Lateral Flow Assay
CrAg	Cryptococcal Antigen
CSF	Cerebrospinal Fluid
CTC	Care and Treatment Clinic
CTU	Care and Treatment Unit
CTX	Cotrimoxazole
DACC	District AIDS Control Coordinator
DBS	Dried Blood Spots
DMO	District Medical Officer
DOTS	Directly Observed Therapy, Short course
DRV	Darunavir
DTG	Dolutegravir
ECG	Electrocardiogram
EFV	Efavirenz
EIA	Enzyme Immunoassays

EID	Early Infant Diagnosis
EPI	Expanded Programme of Immunization
EPTB	Extra-pulmonary Tuberculosis
ESR	Erythrocytes Sedimentation Rate
ETR	Etravirine
FBO	Faith Based Organization
FBP	Full Blood Picture
FDC	Fixed Dose Combination
FEFO	First to Expire, First Out
FP	Family Planning
FTC	Emtricitabine
GART	Genotypic Antiretroviral Resistance Testing
GoT	Government of Tanzania
HBA	Home Birth Attendant
HBC	Home Based Care
HBCT	Home Based HIV Counselling and Testing
HCP	Healthcare Provider
HF	Health Facility
HIV RNA	Human Immunodeficiency Virus Ribonucleic Acid
HIV	Human Immunodeficiency Virus
HIVST	Human Immunodeficiency Virus Self Testing
HLD	High-Level Disinfectants
HSV	Herpes Simplex Virus
HTC	HIV Testing and Counselling
HVL	HIV Viral Load
IEC	Information Education and Communication
ILS	Integrated Logistic System
IMCI	Integrated Management of Childhood Illnesses
INH	Isoniazid
INSTI	Integrase Strand Transfer Inhibitors
IPD	In-Patient Department
IPT	Isoniazid Preventive Therapy
IRIS	Immune Reconstitution Inflammatory Syndrome
KS	Kaposi Sarcoma
KVP	Key and Vulnerable Population
LAM	Lipoarabinomannan
LFT	Liver Function Test
LIP	Lymphocytic Interstitial Pneumonitis
LPV	Lopinavir
LRTI	Lower Respiratory Tract Infection
M&E	Monitoring and Evaluation

MAC	Mycobacterium Avium Complex
MAT	Medically assisted Therapy
MCH	Maternal and Child Health
MDR	Multi-Drug Resistant
MoH	Ministry of Health
MTCT	Mother to Child Transmission
MUAC	Mid-Upper Arm Circumference
NASHCoP	National AIDS, STI and Hepatitis Control Programme
NGO	Non-Governmental Organization
NNRTI	Non-Nucleoside Reverse Transcriptase Inhibitors
NRTI	Nucleoside/Nucleotide Reverse Transcriptase Inhibitors
NSAID	Non-Steroidal Anti Inflammatory Drug
NTLP	National TB and Leprosy Programme
NVP	Nevirapine
OI	Opportunistic Infection
OPD	Out-Patient Department
ORS	Oral Rehydration Salts
OST	Opioid Substitution Therapy
PCR	Polymerase Chain Reaction
PEP	Post Exposure Prophylaxis
PHDP	Positive Health, Dignity and Prevention
PI	Protease Inhibitors
PITC	Provider Initiated Testing and Counselling
PJP	Pneumocystis Jiroveci Pneumonia
PLHIV	People Living with HIV
PMS	Patient Monitoring System
PMTCT	Prevention of Mother to Child Transmission
PPE	Pruritic Papular Eruption/Personal Protective Equipment
PrEP	Pre-Exposure Prophylaxis
PWID	People Who Inject Drugs
QI	Quality Improvement
RAL	Raltegravir
RAMs	Resistance Associated Mutations
RCH	Reproductive and Child Health
RFT	Renal Function Test
RHMT	Regional Health Management Team
RoC	Recipient of Care
RTV	Ritonavir
RUTF	Ready to Use Therapeutic Food

SDM	Service Delivery Models
SOPs	Standard Operating Procedures
SP	Sulfadoxine Pyrimethamine
STGs	Standard Treatment Guidelines
STI	Sexually Transmitted Infection
TAF	Tenofovir Alafenamide Fumerate
TB	Tuberculosis
TDF	Tenofovir Disoproxil Fumarate
TFDA	Tanzania Food and Drug Authority
THP	Traditional Health Practitioners
TLC	Total Lymphocyte Count
VCT	Voluntary Counselling and Testing
VIA	Visual Inspection with Acetic Acid
VL	Viral Load
VMMC	Voluntary Medical Male Circumcision
VZV	Varicella Zoster Virus
WBC	White Blood Cells
WHO	World Health Organization

FOREWORD

This 8th Edition National Integrated Guidelines for the Management of HIV, Viral Hepatitis, and STIs, incorporates recommendations from scientific research, policy developments, and the 2021 WHO Consolidated Guidelines on the Use of Antiretroviral Drugs for Treating and Preventing HIV Infection. These guidelines are structured along the continuum of testing, prevention, care, treatment, and support for HIV infection, Hepatitis, and STIs. It embraces a broad range of current technological innovations, interventions, and evidence-based practices, serving as a comprehensive guide for healthcare providers and professionals involved in the response to the HIV epidemic and related co-morbidities.

Tanzania is implementing prevention, care, treatment, and support interventions to achieve the global target of ending the HIV epidemic by 2030 and attaining the Triple Elimination of HIV, Hepatitis, and STI transmission from mother to child. These Guidelines prioritize interventions aimed at achieving the results and outcomes outlined in the National Health Sector Strategic Plan (HSSP), the National Multi-Sectoral Framework, and the Health Sector HIV and AIDS Strategic Plan V (2022–2026).

According to Tanzania HIV Impact Survey of 2022 – 2023 (THIS 2022 – 2023) data, Tanzania has made significant progress in its HIV response, having achieved 83 – 98 – 94 of the 95–95–95 global targets. These accomplishments highlight the collaborative efforts of various partners in supporting and delivering a robust HIV response in the country. However, recommendations based on recent developments, advancements, and experiences in addressing HIV and AIDS have necessitated the revision and updating of the 7th Edition of the National Guidelines for the Management of HIV and AIDS. This revision aims to fast-track the achievement of global and national sustainability agendas for HIV epidemic control and the Triple Elimination of HIV, Hepatitis, and STIs.

Feedback on the evolving landscape of HIV, Hepatitis, and STIs, informed by new evidence, scientific advancements, and user experiences with these Guidelines, is highly encouraged. This feedback will be instrumental in future revisions to ensure the Guidelines remain aligned with scientific and technological developments, ultimately improving the quality of health services provided.



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PERMANENT SECRETARY

ACKNOWLEDGMENT

The National Integrated Guidelines for the Management of HIV, Hepatitis, and STIs, 8th Edition, is the result of the revision of the 7th Edition. This process involved multiple consultative meetings and expert reviews. The Ministry of Health (MoH) expresses its sincere gratitude and appreciation for the invaluable technical and financial support provided by our esteemed stakeholders throughout the revision process.

The Ministry commends the collaboration of organizations within the UN Family, particularly WHO and UNAIDS, the United States Government and its agencies, Non-Governmental Organizations, Non-State Actors, academic institutions, and representatives from the community. Their tireless efforts and contributions were instrumental in updating these guidelines.

Lastly, although it is impossible to mention everyone individually, the Ministry extends its heartfelt thanks to all who participated in the revision of these Guidelines. Special recognition goes to the National AIDS Control Program for its coordination role, undertaken with great dedication and passion. The Ministry further appreciates the leadership and guidance of Dr. Anath Rwebembera, the Program Manager; the contributions of Prof. Samwel Kalluvya, Chairman of the National HIV Clinical Subcommittee; and the exceptional technical support from Dr. Christine Musanyu of WHO Tanzania Country Office, Dr. Boniface Silvan, Dr. Siraji Shabani, and Dr. Alphonse Keleman.

Your unwavering commitment and dedication were indispensable in ensuring the timely completion of this critical task.



Dr. Grace E Magembe
CHIEF MEDICAL OFFICER

EXECUTIVE SUMMARY

The 8th Edition of National Integrated Guidelines comprehensively address HIV, Hepatitis, and STIs through seventeen chapters. These guidelines provide detailed guidance on diagnosis, antiretroviral (ARV) drug use, and holistic care for people living with HIV (PLHIV). Spanning the entire continuum of care—from testing and prevention to treatment and support—they emphasize tailored interventions for specific groups. The chapters, outlined in the table below, collectively serve as a valuable resource for healthcare providers and professionals involved in HIV, Hepatitis, and STI care and treatment in Tanzania, ensuring high-quality services for PLHIV across all stages of their condition.

Chapter	Content
Chapter I: Overview of HIV and AIDS Service Provision in Tanzania	This chapter is about the overview of HIV and AIDS service provision in Tanzania, covering basic facts about HIV. It elaborates HIV transmission, pathophysiology, and the national epidemic situation.
Chapter II: HIV service delivery	Outlines how care and treatment should be provided through scheduled and management of treatment interruptions. Differentiated service delivery models for HIV prevention, testing and treatment.
Chapter III: Prevention and Testing of HIV	Details prevention strategies, including combination and biomedical prevention interventions. Covers HIV testing services, referral and linkage systems, and differentiated approaches to testing for diverse populations.
Chapter IV: Antiretroviral Therapy for Adolescents and Adults	Covers initiation of ART for adults and adolescents, first-line and second-line regimens, therapy adjustments, and monitoring PLHIV on ART. It also addresses clinical management for elderly individuals living with HIV.
Chapter V: Paediatric HIV and AIDS-Related Conditions	Focuses on diagnosing HIV in children, initiating ART, monitoring paediatric patients, and improving care models for children living with HIV.
Chapter VI: Prevention of Vertical Transmission of HIV, Syphilis, and Hepatitis B	Explains a comprehensive approach to preventing vertical transmission, including testing and counselling for pregnant and breastfeeding women, care during labor and delivery, postnatal care, and management of exposed infants.
Chapter VII: Adherence Preparation, Monitoring, Retention, and Support	Discusses factors influencing adherence, barriers, and counselling strategies to enhance adherence. It also introduces community-led adherence support programs.
Chapter VIII: Community Based HIV Services (CBHS)	Explores the roles, training, and contributions of community-based healthcare providers in delivering HIV-related services at the community level.

Chapter	Content
Chapter IX: TB/HIV Co-Infection	Focuses on collaborative approaches to managing TB/HIV co-infections, diagnostic tests for TB, preventive measures, infection control, and managing drug-resistant TB.
Chapter X: Management of Advanced HIV Disease and Opportunistic Infections	Covers screening, diagnosis, and treatment of common opportunistic infections in PLHIV with advanced disease.
Chapter XI: Hepatitis B and C Co-Infections	Details screening, clinical evaluation, treatment, and monitoring of Hepatitis B and C co-infections in PLHIV.
Chapter XII: Screening and Management of Sexually Transmitted and Reproductive Tract Infections Other Than HIV	Provides guidance on screening, diagnosis, and management of STIs and reproductive tract infections.
Chapter XIII: Integrated Management of Non-Communicable Diseases (NCDs) in HIV	Focuses on screening, diagnosis, and management of non-communicable diseases among PLHIV.
Chapter XIV: Mental Health Conditions in HIV and AIDS	Addresses mental health complications, including primary disorders and substance use disorders, associated with HIV.
Chapter XV: Pharmaceutical and Laboratory Services	Covers the rational use of HIV commodities, supply chain management, pharmacovigilance, and HIV-related laboratory services.
Chapter XVI: Nutrition in HIV and AIDS	Highlights the importance of nutrition for PLHIV, discussing dietary requirements, nutritional assessment, and the role of healthy eating in care and treatment.
Chapter XVII: Monitoring and Evaluation	Outlines the critical components of monitoring and evaluation for HIV/AIDS services, including data collection, recording, and reporting.

1 THE OVERVIEW OF HIV AND AIDS SERVICE PROVISION IN TANZANIA

1.1 Basic Facts about HIV

1.1.1 Aetiology of HIV and Transmission

In Tanzania, HIV infection is primarily caused by HIV-1, with predominant subtypes A, C, D, and their recombinants. The main route of transmission is unprotected sexual intercourse with an infected person. Other transmission routes include exposure to infected blood and blood products, as well as mother-to-child transmission during pregnancy, delivery, or through breastfeeding.

Although rare, HIV transmission can also occur through contact with infected body fluids such as cerebrospinal fluid, pleural fluid, and amniotic fluid. However, fluids such as saliva, sputum, sweat, tears, feces, nasal secretions, urine, and vomit carry a very low risk of transmission unless they are visibly contaminated with blood.

HIV can also be transmitted through the sharing of unsterilized sharp instruments, needles, and shaving tools. Importantly, there is no evidence of HIV transmission through insect bites, such as those from mosquitoes or bed bugs.

1.1.2 Pathophysiology of HIV Infection

HIV is an RNA retrovirus that infects and replicates within host cells through a multi-step process, as illustrated in Figure 1.1. The virus attaches (binding) to the CD4 receptor and co-receptors (CCR5 or CXCR4) on the surface of T lymphocytes and macrophages to enter host cells. The HIV envelope fuses with the CD4 cell membrane (fusion), releasing the HIV capsid into the host cell.

Once inside, the virus uses reverse transcriptase to convert its RNA into DNA (reverse transcription). This viral DNA is then integrated into the host cell's DNA (integration) with the assistance of the integrase enzyme. Following integration, the virus exploits the host CD4 cell's infrastructure to produce HIV RNA and proteins (replication). These newly formed HIV particles assemble and exit the host cell (budding), ultimately causing the infected CD4 cell to die. Infected CD4 T lymphocytes have a significantly shorter half-life (approximately two days), leading to their rapid depletion and decline in numbers.

For adults who are not on antiretroviral therapy (ART), a viral set point is reached earlier compared to children. In adults, the median time to develop AIDS is approximately 10 years. In contrast, children typically have higher viral set points and experience faster disease progression. In some cases, the time to AIDS in children can be as short as less than two years (Barbra et al., 2003).

Antiretroviral Therapy is directed at one or several of the following stages essential for viral replication: (1) attachment of the HIV virus to the host cell; (2) reverse transcription of viral RNA to DNA; (3) integration of the pro-viral DNA into the host cells' DNA; or (4) expression of the viral gene after it has been integrated into host cell DNA, including the transcription of more viral RNA and the translation of viral proteins (See Figure 1.1:1). It is recommended to combine antiretrovirals that target more than one part of the virus life cycle to minimize development of resistance. ARVs function by inhibiting HIV replication which results in reduction of viral load and a consequent increase in CD4+T lymphocytes recovery. In the absence of ART, disease progression goes through several stages as described by the WHO Clinical Staging Criteria in Annexes 18.1.

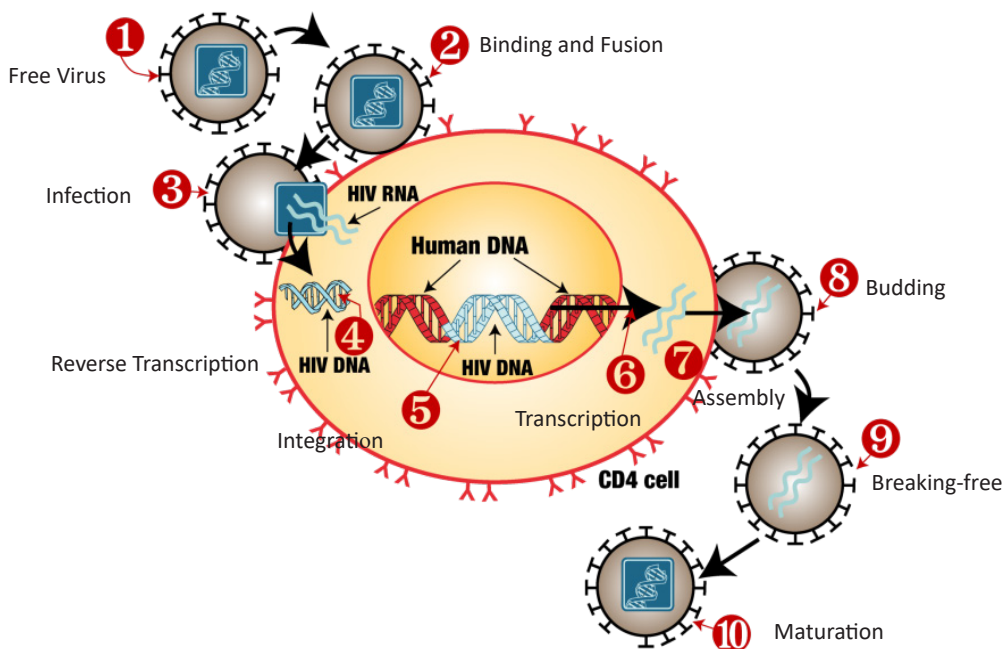


Figure 1.1:1 HIV attachment, entry, replication and release from host cell

1.2 HIV Epidemic in Tanzania

The summary of key epidemic indicators is depicted bellow

Table 1.2:1 National HIV Epidemic summary of key indicators

HIV Prevalence	
Adult General Population	4.4%
Adult Male (15 - 24 yrs)	1.0%
Adult Female (15 - 49 yrs)	3.8%
Number of People Living with HIV	
Total	1,548,000
Children (<15 yrs.)	73,610
New HIV Infections per year	
Total	60,000
Children (0-14 yrs)	4,621
AIDS-related Deaths	
Total	21,697
Children (<15 yrs.)	2,921
Mother to Child Transmission of HIV	
MTCT Rate at 6 wks	2.6%
Final MTCT Rate (Total)	6.5%

Table 1.2:2 National summary of HIV in Key and Vulnerable populations

KVP Group	Population Estimate	Estimated HIV prevalence
Female at high risk	155,450	26.0%
Men at high Risk	49,700	25.0%
People who inject drugs (PWID)	30,000	36.0%

Table 1.2:3 National Progress towards UNAIDS 95 - 95 - 95

	All Ages	0-14 yrs	10-19 yrs	Men (15+)	Women (15+)
1st 95	83%	68.2%	68%	79%	85%
2nd 95	98%	98.2%	99%	97%	98%
3rd 95	94%	90.3%	85%	93%	95%

2 HIV service delivery

2.1.1.1 Scheduled visits

a) The first six months of ART

Recipients of care (RoC) should attend the Care and Treatment Clinic/ PMTCT clinic monthly for the first six months for clinical, laboratory evaluation and drug refills.

b) After Six months on ART

Multi-month ARV refill appointments should be offered to RoC who have been on ART for more than six months and are established on ART. Pregnant women already established on ART and are on MMD, will continue with their MMD as standard of care (See Table 1.6).

2.1.1.2 Unscheduled visits

In addition to their regularly planned checkups, it is crucial for RoCs to show up at the clinic for a clinical assessment if they develop new signs and symptoms of opportunistic infections or adverse drug reactions. The proper management needed and re-categorization of clinical stability of the RoC will be determined after clinical evaluation.

2.1.1.3 Interruption in Treatment

Proactive follow-up by clinic team members, in collaboration with home-based care providers, is essential to track recipients of care (RoC) who miss their scheduled visits. It is important to implement measures to identify and address potential triggers for treatment interruptions during follow-up.

The use of tools such as reminder lists, promised-to-come diaries, appointment blocks, and strong ties between healthcare providers and RoC is encouraged, as these methods have been shown to improve adherence to treatment.

Appointment and tracking registers should be effectively utilized to identify and monitor RoC who have interrupted their treatment. Additionally, a well-functioning referral mechanism, supported by a comprehensive community service directory, should be established to ensure seamless linkages between clinics, other levels of healthcare delivery, and support services.

2.1.1.4 Procedures for RoC returning to care after Interruption in Treatment (IIT)

Interruption in Treatment (IIT) is defined as failure to attend to a Care and Treatment Clinic for ARV pick-up or clinical assessment, or a documented community visit with a community health worker, or peer from an ART refill group for more than 28 days since the last scheduled visit.

If <6 months of Interruption in Treatment

- Offer welcoming back messages
- Perform thorough clinical review and WHO staging.
- Re-initiate ARV regimen last used by the RoC

If ≥ 6 months of Interruption in Treatment.

- Offer welcoming back messages
- Perform thorough clinical review and WHO staging
- Conduct CD4 test
- If CD4 count <350 cells/mm³ initiate CPT
- Screen for TB and initiate TPT if eligible
- If the CD4 is < 200 cells/mm³ offer AHD package of care (refer to Chapter IX)
- Re-initiate ARV regimen last used by the RoC
- After six months of ART, HVL should be conducted together with recategorization of the RoC as either established or not established on ART.

2.1.2 Differentiated Service Delivery Models (DSDMs)

Differentiated HIV Care is a person-centered approach that simplifies and adapts HIV services across the care cascade to meet the needs of people living with and vulnerable to HIV while reducing unnecessary burdens on the health system. The approach aims to align with the preferences and expectations of diverse groups of People Living with HIV (PLHIV).

The principles of differentiated service delivery can be applied throughout the HIV continuum of care, including HIV prevention services, HIV Testing Services (HTS), linkage to care, ART initiation, follow-up, integration of HIV care, as well as the management of co-infections and co-morbidities.

Differentiated Service Delivery Models (DSDMs) are developed using a building blocks approach with four key components: (i) the types of services delivered; (ii) the location of service delivery; (iii) the provider of services; and (iv) the frequency of services as in Figure 2.1:1

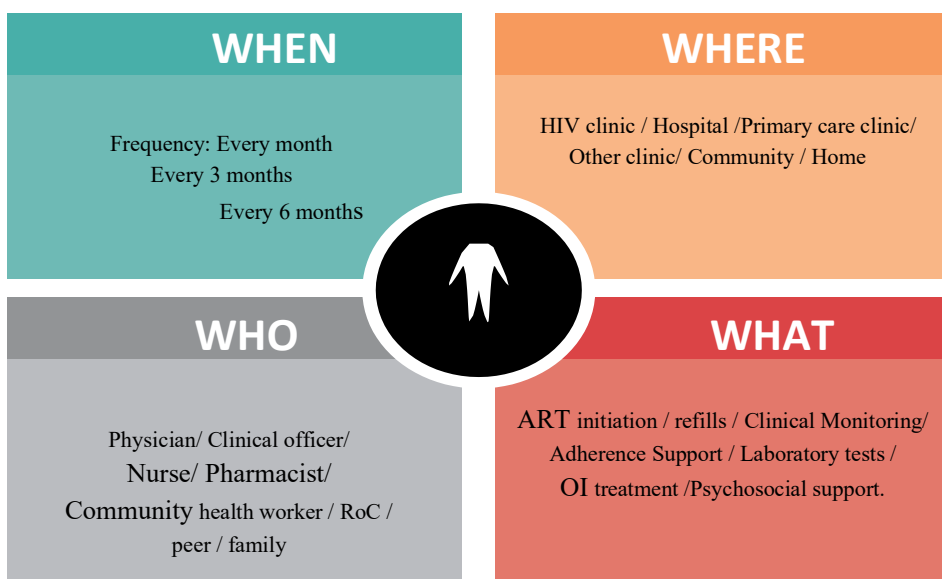


Figure 2.1:1 Building blocks of differentiated service delivery model

2.1.2.1 Differentiated HIV Prevention and Testing Services

The Table 2.2:1 below shows the when, where, who and what needs to be considered in differentiated HIV prevention, and testing services

Table 2.2:1 Differentiated HIV Testing Services

General Population	
When	- HTS should be available for 24 hours daily for facilities providing maternity and in-patient care. - For other facilities HTS should be available in all facilities beyond official working hours depending on local needs.
Where	- Targeted testing (PITC/CITC) should be offered in all entry points of the health facilities. These include general OPD, IPD CTC, TB, STI, RCH/PMTCT, and in specialized clinics. - Facility and community-based index client testing should be offered from all facilities. - Targeted testing should also be offered as community-based outreach testing from all facilities.
Who	- Trained Peers should mobilise communities to access HTS. - Trained and certified HCP to perform HTS. - Capacitated non healthcare worker's cadres (CHW, CDOs, SWOs, VHWS and CBHS providers) to conduct education and testing (including HIV self-testing) - Every facility to ensure that there is always a HCP on duty who has been trained to perform HTS.

What	• Index testing CITC • HIV self-testing PITC • Integrated approaches should be implemented in community testing strategies. This may include HIV testing, Viral hepatitis testing, TB and STI screening, blood pressure measurement, blood glucose checks, and nutrition assessments.
Special Considerations for Children and Adolescents	
When	• Extended hours, weekends and public holidays
Where	• Targeted outreach testing to schools, colleges, street children and orphanages should be included in the outreach planning. • PITC should be offered in all entry points of the health facilities including; RMNCH, VMMC clinics adolescent-friendly settings, malnutrition, and paediatric wards.
Who	• Trained peer adolescent should mobilise adolescents for testing. • All cadres of existing healthcare providers should be trained to prepare DBS samples for EID testing. • All facilities should ensure there is always a HCP on duty, certified to provide HTS and trained to prepare DBS samples for EID testing. Trained healthcare providers should perform HTS and EID DBS during mobile outreach activities.
What	• Index testing CITC • HIV self-testing • Optimized PITC • Social Network Services • Integrated approaches should be implemented in community testing strategies. • This may include HIV testing, TB and STI screening, blood pressure measurement, blood glucose checks, and nutrition assessments.
Linkage to Care	
All HIV-positive RoC identified at a facility should be guided (with their consent) to the CTC for enrolment into ART care. This should ideally be done by the HCP who has performed the test or a community health worker. Status Neutrality: All HIV-positive RoC identified should be linked, with their consent, with a community health worker, CBHS providers or other peer networks. A duplicate referral form should be completed for anyone testing positive in the community. A copy of the referral will remain with the community-based HIV services providers for follow up of effective referral. The community-based providers should encourage the client to attend the facility of their choice. All eligible HIV negative individuals should be linked to appropriate HIV preventive services according to preventive services guidelines. More details available at section 2.2.4. Any RoC tested positive and has not been linked to care after one month should be traced. Tracing should initially be by phone followed by a home visit.	

2.1.2.2 Differentiated service delivery for HIV Treatment

Differentiated service delivery refill models reduce the burden on the healthcare system by optimizing scarce resources, improving treatment adherence, and enhancing recipients of care (RoC) satisfaction and quality of life. These improvements lead to better health outcomes, reduced costs, and increased care continuity. Prioritizing the adoption of these models is essential to ensure that all RoCs receive optimal care.

Recipients of care should be assessed and categorized at the initiation of ART and six months thereafter, based on clinical characteristics, population group, and contextual factors, as outlined below.

i. Clinical characteristics of recipients of care:

At ART initiation, RoC should be assessed based on their clinical characteristics and categorised as either early presenters or late presenters (with or without advanced HIV disease) as Table 2.2:2.

Table 2.2:2 Recipients of care categories for use during ART initiation

Category	Characteristic
1. Early presenter – Requires additional adherence support to commit to lifelong ART	
	WHO stage 1 or 2 and/or CD4 $\geq$ 350 cells/mm ³
2. Late presenter	
a) Without Advanced HIV Disease	WHO stage 1 or 2 and/or CD4 $\geq$ 200 to <350 cells/mm ³
b) With Advanced HIV Disease	WHO stage 3 or 4 and/or CD4 < 200 cells/mm ³ All children below 5 years

WHO defines advanced HIV disease for adults and adolescents (and children five years and older) as having a CD4 cell count of less than 200 cells/mm³ or WHO clinical stage 3 or 4 disease. All children younger than five years living with HIV are considered to have advanced HIV disease. Children older than two years who have been receiving ART for more than one year and are clinically stable should not be considered to have advanced disease.

At six months of ART, RoC should be assessed based on viral load and clinical characteristics and categorised as established on ART (Clinically Stable), or not established on ART (Clinically Unstable).

Multi-month ARV prescription and dispensing should be offered to RoCs who are established on ART/stable (including pregnant and breastfeeding women, and those on TPT) (Table 2.2:3)

Table 2.2:3 Client's categories after six months on ART

Recipients of care category	Characteristics
Established on ART (Clinically Stable)	- Age $\geq$ 2 years. - No adverse drug reactions that require regular monitoring. - No current illnesses (OIs and uncontrolled co-morbidities) - Good understanding of lifelong adherence of 95% and have kept clinic visit appointments for the past six months. - On First, Second-line or Third line ART with viral load of less than 50 copies/ml.
Not established on ART (Clinically unstable)	- Age below 2 years - Presence of an active OI (including TB) or uncontrolled co-morbidity/ies in the past six months - Poor or suboptimal adherence to scheduled clinic visits in the past six months. - Recent detectable HVL above 50 copies/ml, - People Who Inject Drugs (PWID)

ii. **The sub-population:** ART delivery should also be differentiated based on the challenges of different sub-populations such as adults, children, adolescents, pregnant and breastfeeding women, men, key and vulnerable populations.

iii. **The context:** In order to maintain quality cART delivery, specific modifications are required when dealing with challenging settings such as conflict, urban/rural, high migration and low prevalence areas.

Table 2.2:4 Recommended refill models

	At the Facility	At Community
Group Models	- Facility Based HCP Managed Groups (Psychosocial support groups of RoC) - Teen/Adherence Clubs (Group of teens/youths formed to enhance adherence, facilitated by HCP)	Community ART Refill Groups (CARG) - Facility-based outreach to smaller health facilities - Facility-based groups ART refill at selected places such as WEO, VEO, churches - Group ART delivery by expert patients or CBHSPs to stable patients

	At the Facility	At Community
	Family Clubs/Family Centered Approach (Family of RoC receiving care together with synchronized appointments)	
Individual Models	Fast Track model Treatment supporter Refill Mult month refill model Expanded hours (evening, weekends and public holidays)	Community based Individual ART delivery through facility led Mobile Outreach services. Community ART Distribution Points (PODI) Include: Private pharmacies, MSD Community Pharmacy, Private hospitals, Home delivery.

2.1.3 Quality of Care and Treatment Services

Healthcare services at the CTC should be characterized by excellence, value, superior performance, and adherence to established standards, requirements, and specifications. The quality of care and treatment services is guided by global principles, including effectiveness, safety, people-centeredness, timeliness, equity, integration, and efficiency.

2.1.3.1 Quality Assurance

Quality Assurance (QA): Quality assurance involves planned, step-by-step activities to ensure that health services are delivered correctly, results are accurate, and errors are identified and corrected to prevent adverse outcomes. QA can be conducted either internally or externally.

Internal Quality Assurance: The facility administration and CTC in-charge are responsible for planning regular quality checks to ensure the delivery of high-quality care. This includes establishing and facilitating Work Improvement Teams (WITs).

External Quality Assurance: National, regional, and district QA teams should conduct supervision visits to CTCs to assess the quality of services. During these visits, areas needing improvement will be identified, and the supervising team will collaborate with CTC staff to develop an action plan. This plan must be documented for future reference and follow-up.

Quality Assurance for CTC Services: Quality assurance for CTC services is implemented by the WIT at the CTC. The health facility management is responsible for ensuring that the CTC remains functional at all times.

The WIT should assess the health facility's performance on all indicators monitored as per the M&E framework and guidelines. Performance of the assessed indicators should be documented using the Standard Evaluation System (SES) forms, and all criteria for data management, as outlined in the national guidelines for managing quality HIV and AIDS data, must be fulfilled.

2.1.3.2 Quality Improvement

Quality Improvement (QI) is a systematic process for assessing the performance of a health system and its services, identifying gaps and their root causes, and implementing measures to improve procedures to achieve the desired outcomes.

For HIV and AIDS QI, the PDSA cycle model (Plan, Do, Study, Act) has been adopted as the primary reference model. However, other QI models, such as the 5S approach (Sort, Set, Shine, Standardize, and Sustain) and the Improvement Collaborative approach, can also be used to enhance the quality of HIV and AIDS services.

At regional, district, and health facility levels, Quality Improvement Teams (QIT) and Work Improvement Teams (WIT) should be established to fulfill their roles and responsibilities as outlined in the QI guidelines. The National Quality Improvement Framework (NQIF) further describes the specific roles and responsibilities of QITs and WITs and should be referred to as a guiding document.

.

3 PREVENTION AND TESTING OF HIV INFECTION

3.1 Combination Prevention Intervention

The general community, including PLHIV, should be provided with combination prevention approaches that address factors contributing to HIV transmission. These approaches are categorized into three main areas: (i) Social and Behavioural Change, (ii) Biomedical, and (iii) Structural Prevention.

3.1.1 Social and Behaviour Change Communication (SBCC)

SBCC facilitates interactive engagement between practitioners and communities. Through the exchange of information, ideas, and emotions, SBCC promotes understanding and builds skills that empower communities to adopt positive behavioural changes. To effectively influence behaviour change, SBCC providers should combine interventions that focus on advocacy, communication, and social mobilization.

- i. **Advocacy:** Aims to influence leaders to create and enforce laws and policies that drive positive change.
- ii. **Communication:** Promotes behavioural change through various communication channels.
- iii. **Social Mobilization:** Encourages communities to actively support change while addressing barriers that contribute to the spread of diseases such as HIV and AIDS.

Social and environmental drivers of behaviour can either support health promotion or contribute to the spread of illnesses.

Essential Behavioural Interventions include; Peer education, Targeted IEC (Information, Education, and Communication) materials, Demand creation, Condom programming, Harm reduction and risk assessment

Table 3.1: Structural and Behavioural Drivers

Structural drivers	Behavioural drivers
• Migration and mobility • Disempowerment through poverty • Unemployment and economic inequality • Gender inequality • Social and cultural norms • An absence of services	• Stigma, discrimination, and lack of open communication around HIV and sex • Multiple concurrent sexual partnerships • Improper and inconsistent condom use • Intergenerational and transactional sex • Gender based violence • Alcohol and drug abuse

Source: Adapted from Situation and Response Analysis Report on SBCC for HIV and AIDS, TB and STIs in SADC Member States.

SBCC agents, including Healthcare providers, should be able to use SBCC as a tool to persuade and influence health behaviours in the continuum of care.

3.1.2 Integrate Positive Health, Dignity and Prevention (PHDP) into Care and Treatment of HIV RoC

“Positive Health, Dignity and Prevention” promotes shared responsibility for preventing HIV transmission, the meaningful involvement of people living with HIV (PLHIV) in responding to the epidemic, and building resilience to enable PLHIV to live healthy, fulfilling lives despite being HIV-positive. The PHDP package must include a synergistic combination of interventions integrated into routine care. The following prevention services will be available to PLHIV:

- Condom promotion and distribution
- Messaging and counselling support for social and behavioural change, including sexual risk reduction, treatment continuity, adherence to medication, and social and sexual partner HIV testing and counselling
- HIV testing and counselling
- ART as prevention
- Voluntary Medical Male Circumcision (VMMC)
- Screening and treatment of STIs and RTIs
- Prevention of Mother-to-Child Transmission (PMTCT)
- Safer pregnancy counselling and integration of family planning services
- Identification of social needs, referral, and linkage to community-based services
- Pre-Exposure Prophylaxis (PrEP)

Healthcare providers will use the appropriate CTC2 Code to guide regular discussions on these topics.

3.1.3 Condom Promotion and Distribution

Both male and female condoms are highly efficacious and cost-effective in preventing the sexual transmission of HIV and other STIs among both high-risk groups and the general population.

Condoms should be made acceptable, accessible, and readily available. Condom distribution should follow the principles outlined in the Tanzania Condom Distribution Guide. Healthcare providers and trained experts should deliver education and practical demonstrations on the consistent and correct use of condoms, ensuring options are available for both males and females.

To increase accessibility and utilization, mass media marketing and condom promotion should be utilized. Technologies such as **digital vending machines** have demonstrated effectiveness in reaching key and vulnerable populations and can serve as a model for differentiated HIV prevention strategies.

3.1.4 HIV Testing for Prevention

Healthcare providers should encourage individuals visiting service delivery points to test for HIV and know their status. Both provider-initiated and client-initiated testing approaches should be utilized as described in the **HIV Testing Guidelines**. **HIV self-testing** should also be promoted at all testing points.

PLHIV should be encouraged to bring their social and sexual partners for HIV testing and counselling. Individuals who test **positive** for HIV and start ART can maintain good health and, if they attain and sustain an undetectable viral load, will not transmit the virus. Individuals who test **negative** can make informed decisions regarding their health behaviours and adopt measures to avoid HIV risk. HIV-negative recipients of care (RoC) eligible for prevention services should be enrolled into programs such as PrEP, STI services, VMMC, and consistent condom use.

3.1.5 Screening for Sexually Transmitted and Reproductive Tract Infections (STIs/RTIs)

STIs/RTIs facilitate the sexual acquisition and transmission of HIV and have socio-economic impacts on families. Both the general community and PLHIV should undergo regular screening for STIs/RTIs, and those diagnosed must be promptly initiated on effective treatment. Syndromic management of STIs, based on defined symptoms and recognizable clinical signs, should be applied where appropriate (Refer to Chapter XI).

3.1.6 Family Planning and Safer Pregnancy Counselling Services

Healthcare providers should support PLHIV in planning pregnancies safely and preventing unwanted pregnancies while respecting their right

to express sexual desires. Providers should assess the fertility intentions of both male and female RoC.

HIV care facilities must promote **male partner involvement** to ensure shared decision-making regarding pregnancy and contraception. Comprehensive family planning services, including **dual protection** (condoms plus another method) and safer pregnancy counselling, should be integrated into CTCs to support informed reproductive decision-making for HIV-positive individuals and discordant couples.

3.2 Biomedical Prevention of HIV

HIV and other blood-borne pathogens (BBPs), such as Hepatitis B and Hepatitis C, can be transmitted in healthcare settings from patient to patient, patient to healthcare worker, or vice versa. To prevent accidental transmission of HIV and other BBPs, the following IPC measures must be implemented in healthcare facilities:

- Adherence to **standard precautions**, including proper hand hygiene
- Use of **Personal Protective Equipment (PPE)**, such as gloves
- Proper **healthcare waste management**
- **Decontamination of instruments**
- **Cleaning and sterilization** using High-Level Disinfectants (HLDs)
- Observing **safe work practices**

3.2.1 Post-Exposure Prophylaxis (PEP)

PEP must be provided **as soon as possible, but no later than 72 hours** after a high-risk event for HIV transmission. **High-risk events** include: exposure to blood or body fluids, Rape, unprotected sex, including condom failure during intercourse

Timely use of PEP within the recommended timeframe can significantly reduce the risk of HIV transmission. Effective post-exposure management includes the following steps:

- **Management of the exposure site**
- Exposure reporting
- Assessment of infection risk
- Prompt provision of a PEP starter pack (**preferably within 30 minutes**)
- Initiation of an appropriate PEP regimen and/or treatment, or linkage to care
- **Follow-up and counselling**

When an exposure occurs, the circumstances and the post-exposure management steps taken must be recorded in the exposed person's confidential form. This documentation ensures proper follow-up and continued care.

Recommended PEP Regimen:

For adult and adolescents: TDF 300mg +3TC 300mg +DTG 50mg once a day for four weeks and for children
AZT + 3TC + DTG (based on body weight) for four weeks.

Note:

- HIV antibody tests should be performed (preferably before and at least after) 4-6 weeks' post-exposure (i.e. at 6 & 12 weeks). Initiate ART if the test is positive.
- Monitor for drug toxicity at baseline and 2 weeks after starting PEP (full blood picture (FBP), renal function test (RFT-Serum creatinine and urinalysis) and hepatic function tests (LFT-ALT).)

3.2.2 Pre-Exposure Prophylaxis (PrEP)

PrEP for HIV prevention is the use of optimum dosage of antiretroviral (ARV) drugs by HIV-uninfected individuals to prevent HIV acquisition. PrEP minimizes but does not eliminate risk of HIV transmission. PrEP is recommended for people at substantial risk of acquiring HIV to lower their chances of infection. PrEP should be offered **voluntarily** as an additional prevention option to individuals at substantial risk, as part of combination prevention interventions. Other newer PrEP options, such as **Long-Acting PrEP** and **Event-Driven PrEP**, should be introduced through phased demonstration programs. These programs aim to assess the feasibility, acceptability, and effective implementation strategies in the country while building robust evidence of effectiveness.

i. Eligible Populations for PrEP

PrEP should be offered to individuals with substantial risk of acquiring HIV, including:

- Men and women at high risk
- People who inject drugs
- Vulnerable adolescent girls (15-19 years)
- Young women (20-24 years)
- High-risk pregnant and breastfeeding women
- HIV-negative partners of sero-discordant couples
- Mobile populations (e.g., long-distance truck drivers, fisherfolk and fishing communities, miners and mining communities, construction and plantation workers)

While PrEP is targeted toward these population groups, **not all individuals within these groups are automatically eligible**. Healthcare providers must educate and counsel potential PrEP users on the risks and benefits of PrEP. An **individualized risk-benefit assessment** should be conducted to determine eligibility. Therefore, PrEP should be offered only to those meeting the eligibility criteria as stipulated in Figure 3.2:1 and Table 3.2:1 Eligibility criteria for PrEP.

Providers are strongly encouraged to use **robust and consistent eligibility criteria screening** for individuals who both actively seek or passively receive PrEP services. Providers must ensure that all PrEP clients meet the established eligibility criteria.

The **Ministry of Health through NASHCOP** has adopted the 2022 WHO implementation guidance for simplified and differentiated service delivery of PrEP to align with Tanzania's HIV epidemiology, ensuring it is both suitable and practical for the local context. Figure 3.2:1 illustrates the screening process for identifying target populations of interest.

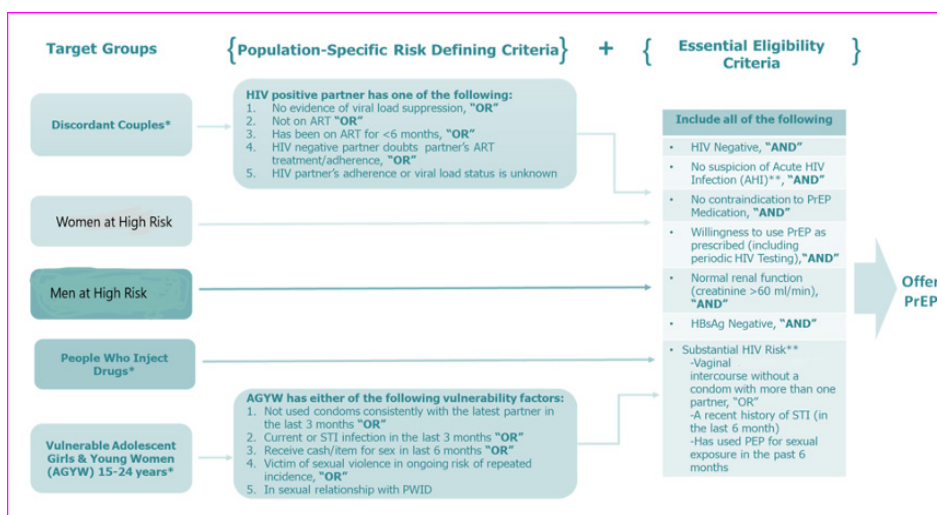


Figure 3.2:1 Population - specific and other key essential criteria for PrEP initiation

Note: Pregnant and breastfeeding women falling in the above groups fulfil the above criteria should be given PrEP accordingly

Note:

- Eligible pregnant and breastfeeding women (PBW) belonging to these groups can also be initiated on PrEP. Additionally, an individual who conceives while taking PrEP should continue with medications unless there is another reason to discontinue PrEP.

Note:

- Eligible pregnant and breastfeeding women (PBW) belonging to these groups can also be initiated on PrEP. Additionally, an individual who conceives while taking PrEP should continue with medications unless there is another reason to discontinue PrEP.
- Signs and symptoms of acute HIV infection (AHI) include:
 - » Symptoms: Anorexia, malaise, fatigue, myalgia, headache, sore throat, and sore glands
 - » Signs: Fever, rash, sweating, oral/oesophageal candidiasis, pharyngitis, Truncal rash, cervical adenopathy, viral meningitis, hepatosplenomegaly, and generalized lymphadenopathy.



In clients presenting with symptoms and signs suggestive of acute HIV infection, PrEP should be postponed until symptoms subside and a repeat rapid HIV test after 4 weeks remains negative

Ensuring widespread awareness of PrEP, particularly among eligible groups, is a critical component of a successful PrEP program.

Table 3.2:1 Eligibility criteria for PrEP

Eligible recipients of care for PrEP

- Aged 15 years and older.
- HIV sero negative
- At substantial risk* of HIV infection
- No suspicion of acute HIV infection
- Willingness and consent for PrEP use as prescribed

*** Substantial risk of HIV infection is defined as:**

- Engaging in sexual intercourse without a condom with multiple partners
- A history of a new sexually transmitted infection (STI)
- Use of post-exposure prophylaxis (PEP) for sexual exposure
- Having a known HIV-positive sexual partner who:
 - o Is not on antiretroviral therapy (ART).
 - o Has been on ART for less than six months, or
 - o Has not achieved viral suppression (less than 50 copies/mL)
- Refusing to disclose a risk category but still requesting PrEP

Recommended PrEP regimens in Tanzania

Oral PrEP	Emtricitabine (FTC) 200mg/Tenofovir Disoproxil Fumarate (TDF) 300mg (Truvada) PO Daily. Once daily for seven days to achieve protective concentrations and continue until 28 days after the last exposure
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Note:

If a client intending to stop PrEP, the last prescription of PrEP should cover for at least 28 days after the last possible exposure to HIV. After completing the final prescription, the client should return for an HIV test to confirm their status. Clients should also be referred to other relevant prevention services. Recipients of care (RoC) should continue using PrEP until there is no ongoing risk of HIV acquisition. (Refer to the PrEP Framework for more guidance).

ii. Contraindications for PrEP

Below are the main contraindications for PrEP. Individuals identified with any of these attributes should not be initiated on PrEP:

- Confirmed HIV infection (positive HIV test result)
- Signs or symptoms of acute HIV infection (probable recent exposure)
- Abnormal renal function (e.g., estimated creatinine clearance rate <60 ml/min)
- Allergy to any medicine in the PrEP regimen
- Contraindication to any medicine in the PrEP regimen (e.g., concurrent use of acyclovir or weight loss drugs)
- Kidney-related comorbidities (e.g., chronic heart disease or diabetes)

iii. PrEP Initiation and Follow-Up

During the initial and subsequent visits, a healthcare provider (community-level or facility-based) should conduct counselling sessions and provide detailed explanations on:

- What PrEP is
- The duration of taking PrEP
- How PrEP should be taken
- Importance to adherence to PrEP and its relation to reduction of HIV transmission
- Possible side effects including adverse events
- Conception while on PrEP
- Conditions for stopping PrEP



It should be emphasized that PrEP, when taken correctly and consistently, reduces the risk of HIV acquisition but does not protect against STIs or pregnancy.

Before initiating PrEP services, providers must conduct a **PrEP readiness assessment**. Providers should also counsel clients on risk reduction strategies and the use of **HIV combination prevention package**, including:

- HIV self-testing
- Condom use
- Addressing gender-based violence (GBV)
- STI prevention and treatment
- Alcohol and drug use reduction
- Cervical cancer screening
- Family planning methods

iv. HIV Seroconversion:

HIV seroconversion can occur after initiating PrEP due to discontinuation or poor adherence (2.2 per 100 person-years, 95% CI 1.2, 3.5⁴) If an HIV-positive result is confirmed, the individual should be initiated on ART or referred for ART initiation as soon as possible. Failure to detect acute HIV infection during the initial test is a common cause of pseudo-seroconversion after starting PrEP.

v. Basic Investigations Prior to PrEP Initiation

Before initiating PrEP, healthcare providers must perform a **behavioural risk assessment** along with a baseline clinical and laboratory evaluation. Table 3.2:2 outlines the mandatory and recommended laboratory tests.

Table 3.2:2 Recommended laboratory tests for PrEP

S/N	Test	Type of Sample	Frequency	Action to be taken
1	Conventional HIV rapid test* (refer to the algorithm in the national HTS guideline)	Whole Blood Serum Plasma (2-5 mls of blood)	• Initial visit • At 1-month visit • Every 3 months	To assess HIV status -If the client is HIV Positive, initiate ART -If the client is HIV Negative. Screen for PrEP.
2	Serum Creatinine	Serum (2-5 mls of blood)	• Three (3) months after PrEP initiation	If Creatinine Clearance** is less than 60ml/min,

S/N	Test	Type of Sample	Frequency	Action to be taken
			• After every 12 months for all clients who continue with PrEP. • As per clinician indication on follow up	DISCONTINUE PrEP; repeat test in 2 weeks, and if Creatinine Clearance is above 60 ml/min, RESTART PrEP <i>**Creatinine clearance testing is guided by age and assessment of Kidney related comorbidities</i>
3	Hepatitis B surface antigen* (using a rapid diagnostic test)	Serum (2-5 mls of blood)	At initial visit • As per clinician indication	If HBsAg is Negative: start PrEP and vaccinate, if available If HBsAg is Positive: start PrEP and refer to a doctor for liver function monitoring and management of Hepatitis B infection
4	Random Blood Glucose (RBG)	Whole blood	• Initial visit • As per clinician indication	Assess if the client is diabetic
5	Liver Function Tests (LFT) e.g., ALAT / ASAT	Serum (2-5 mls of blood)	• Initial visit • As per clinician indication	To monitor drugs toxicity
6	Haemoglobin Level	Whole blood (2-5 mls of blood)	• Initial visit • As per clinician indication	Assess Anaemia
7	STI screening and Testing	Serum (2-5 mls of blood) HVS	• At every visit	To diagnose and treat STI
8	Hepatitis C (using a rapid diagnostic test)	Serum (2-5 mls of blood)	• At initial visit • As per clinician indication	To assess HCV, particularly in PWID.

vi. **Guidance on Serum Creatinine Testing for PrEP Eligibility**

1. Routine Testing for Individuals Without Known Comorbidities

Clinician should determine the risk and benefit of initiating PrEP when renal function test is not available at PrEP initiation. Serum creatinine testing should be conducted within three months for all individuals eligible for pre-exposure prophylaxis (PrEP) who do not have known comorbidities. Importantly, the initiation or continuation of oral PrEP should not be delayed while awaiting serum creatinine test results.

2. Testing for Individuals with Comorbidities or Previous Abnormal Results

For individuals with comorbidities or those whose previous kidney function tests suggest a mild reduction in renal function (eGFR <90 mL/min per 1.73 m²), serum creatinine testing is recommended before starting PrEP. This ensures safe use by identifying potential risks related to impaired kidney function.

3. Contraindication to TDF-Based Oral PrEP

Impaired kidney function, characterized by an estimated glomerular filtration rate (eGFR) <60 mL/min per 1.73 m², is a contraindication for the use of tenofovir disoproxil fumarate (TDF)-based oral PrEP. Patients with such renal impairment should be evaluated for alternative PrEP options.

3.2.3 Voluntary Male Medical Circumcision (VMMC)

Medical male circumcision provides substantial protection against heterosexual HIV transmission from females to males, with studies demonstrating a risk reduction of approximately 60%. Healthcare providers should strongly recommend medical male circumcision to all uncircumcised males as part of comprehensive HIV prevention efforts to reduce the risk of transmission. It is essential to ensure the procedure is performed safely and with informed consent from the recipient of care (RoC).

Early Infant Male Circumcision (EIMC) is ideally performed during infancy, within 60 days of birth, as it is associated with lower rates of adverse events compared to circumcision performed later in life.

Table 3.2:3 Minimum package for VMMC and EIMC services

Minimum Package for VMMC Services	• Educate recipients of care on the link between VMMC and HIV prevention • Offer HIV testing and counselling and refer those who test positive to a care and treatment clinic • Screen for STIs and RTIs (and offer treatment, where indicated) • Counsel on risk reduction • Promote and distribute male and female condoms together with the promotion of their correct and consistent use • Provide surgical care that is safe and of high quality in settings that are adequately equipped and environmentally suitable for minor surgical procedures • Provide appropriate post-operative care and care of any associated adverse events
Minimum Package for EIMC Services	• Provide information to parents or guardians on advantages and risks of EIMC. • Offer HIV testing and counselling to parents or guardians to ensure identification of HIV- exposed infants. • Link HIV-positive parents to HIV care and treatment services and offer prevention services for those who are negative. • Counsel on the post-operative care of circumcised infants and identification of related complications, danger signs and where to go for follow-up care, if required. • Provide surgical care that is safe and of high quality in settings that are adequately equipped and environmentally suitable for minor surgical procedures. • Provide appropriate post-operative care and care of any associated adverse events. • Refer RoC to appropriate services such as immunization, well baby care, and HIV care and treatment for HIV-exposed infants and/or those infants found to be HIV-positive through Early Infant Diagnosis (EID).

3.2.4 Blood Safety

Unsafe blood transfusion can result in the transmission of HIV and other blood-borne infections. Poor donor recruitment and selection practices, along with the use of inadequately screened blood, pose significant risks to blood recipients and expose them to transfusion-transmissible infections.

The National Blood Transfusion Services (NBTS) plays a vital role in ensuring the availability of safe blood and blood products through a nationally coordinated blood transfusion system.

Healthcare workers must ensure that individuals in need of blood or blood products receive safe blood, thoroughly screened for all transfusion-transmissible pathogens in accordance with WHO guidelines at zonal blood centers. Access to safe blood transfusion is a critical component of delivering quality healthcare.

3.2.5 HIV Prevention Services for Key and Vulnerable Populations (KVP)

Key Populations (KPs) are defined as groups that, due to specific high-risk behaviors, are at an increased risk of HIV infection regardless of the type of epidemic or local context. These groups often face legal and social challenges related to their behaviors, which heighten their vulnerability to HIV and limit their access to healthcare services. Examples of key populations include: Men and women engaged in high-risk behaviors, People who inject drugs, and Individuals in prisons and other closed settings

Key populations play a significant role in the dynamics of HIV transmission and are essential partners in an effective response to the epidemic.

Vulnerable Populations (VPs) are groups of people who are particularly susceptible to HIV infection in specific situations or contexts. These groups include adolescents, orphans, street children, people with disabilities, and migrant or mobile workers. Vulnerable populations are not uniformly affected by HIV across all countries and epidemics. Examples of vulnerable populations include:

- **Adolescent Girls and Young Women (AGYW):** Adolescents aged 15–19 and young women aged 20–24
- **Mobile Populations:** Long-distance truck drivers, fisherfolk and fishing communities, miners and mining communities, and construction or plantation workers
- **People with Disabilities:** Individuals with any form of disability
- **Street Children and Displaced People:** Children living or working on the streets, as well as displaced populations

To effectively address the unique needs of key and vulnerable populations, healthcare providers must deliver inclusive and impartial services that extend beyond the healthcare setting. This requires identifying and addressing the specific needs of these groups in a non-judgmental and non-discriminatory manner, fostering trust and encouraging access to care.

The following list summarizes the key services to be offered to KVP:

- Promote and provide male and female condoms
- Provide VMMC service
- Provide HTS
- Provide ART to HIV infected individuals
- Provide pre-exposure prophylaxis (PrEP)
- Screen and manage STIs, RTIs and cervical cancer
- Counsel and offer Reproductive Health Services (RHS) inclusive of family planning services and dual protection as well as counselling and PMTCT
- Link to facility providing medication-assisted treatment (MAT) and other drug dependence treatments (i.e. harm reduction)
- Provide behavioural change and communication service
- Screen for Hepatitis B and C and provide vaccination for Hepatitis B as appropriate
- Screen for Tuberculosis and manage accordingly
- Screen for sexual violence and provide PEP along with other interventions for gender-based violence (GBV)
- Link with psychosocial support services through targeted campaigns in identified key and vulnerable population settings, use community based outreach, mobile phone technologies, social networking and develop key population friendly services at health facilities; this will facilitate dissemination of behavioural messages, promote follow-up and referral to services, improve adherence to treatment, and increase client participation in their own health care.
- Sensitize and educate health care providers, community health workers, peers, supportive staff and management on issues of specific key and vulnerable populations and on non-discriminatory practices and eliminating stigma, using pre-service and in-service training, job-aids, supportive supervision, and training follow up.
- Ensure confidentiality: Attention should be devoted to protecting privacy and confidentiality, e.g. closing the consultation room door or finding a private place to talk. RoC should be reassured of confidentiality.
- Proper linkage and referral mechanisms to community support programmes (e.g. psychosocial support, income generating group, spiritual support and legal support, etc.)

For further details on management of HIV for KVP, refer to the National Guidelines on Comprehensive HIV Interventions for Key and Vulnerable Populations, 2017 Edition.

3.3 HIV Testing Services

HIV testing services aim to:

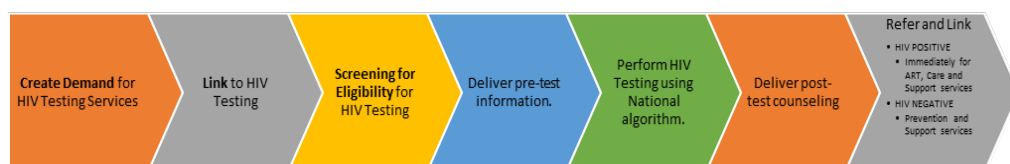
- **Identify individuals at substantial risk of HIV** and link them to prevention services to reduce HIV transmission, including voluntary medical male circumcision (VMMC), prevention of mother-to-child transmission (PMTCT), pre-exposure prophylaxis (PrEP), and post-exposure prophylaxis (PEP).
- **Detect individuals living with HIV** by providing quality testing services for individuals, couples, and families.
- **Facilitate effective linkage to appropriate care and services**, ensuring individuals and their families access HIV treatment, care, support, and prevention services tailored to their HIV status.
- **Support the scale-up of interventions** to reduce HIV-related morbidity and mortality through the implementation of antiretroviral therapy (ART).

HIV testing services, whether conventional or self-testing, must prioritize the needs and interests of the recipient of care (RoC) while adhering to national and international healthcare standards and human rights principles. Testing should be tailored to the unique risks and needs of the RoC.

To achieve these objectives, all HIV testing services must conform to the five foundational “C” principles:

- **Counseling** – Ensure pre- and post-test counseling is available.
- **Consent** – Obtain informed consent before testing.
- **Confidentiality** – Maintain strict confidentiality of the individual’s information.
- **Correct Results** – Provide accurate and reliable test results.
- **Connection (Linkage)** – Link individuals to appropriate care, treatment, and prevention services based on their needs.

Note: Provision of HTS in both settings should comply with Tanzania National Guidelines for HIV Testing Services, 2021.



3.3.1 HTS Package

The term “HIV Testing Services (HTS)” refers to the entire set of services that should be given in conjunction with HIV testing. When a RoC is diagnosed with HIV, they get the ability to make informed decisions regarding HIV prevention, treatment, care, and support services. Tanzania’s HTS package comprises demand generation, pre-test information, HIV testing, post-test counselling, and effective links to prevention, treatment, care, and support services for both HIV positive and HIV negative persons. (Table 3.3:1)

Table 3.3:1 HTS care cascade

Intervention	Description
Demand creation	HTS should be promoted through SBCC interventions in mass and social media, information, education, and communication materials (IEC material), signboards, combination, prevention, intervention, advocacy, sensitization and mobilization.
Link to HIV testing point	Peer educator should link clients to the nearby testing point or client to seek HIV testing service by self-referral.
Screening for eligibility for HIV testing	HTS provider should screen all clients using national HTS screening tool. Clients meeting the HTS eligibility criteria should be offered HTS in accordance with national testing algorithm. Screening should be conducted in a safe and private area while ensuring confidentiality of clients. Clients who do not meet the eligibility criteria but who insist to be tested for HIV should not be denied the service.
Pre-Test information/ counselling	HTS provider should conduct age-appropriate pre-test session either through one to one (individual) or group sessions aiming to obtain an informed consent or assent for testing. The sessions can also be conducted through a short, recorded pre-test video clip, or printed materials such as posters and brochures. HTS provider shall assess whether children under 18 years testing is in the best interest of a child and if it promotes the child’s physical and emotional welfare.
HIV Testing	Rapid HIV Testing should be done with privacy, maintaining confidentiality, adhering to the National HIV Testing Algorithm and good laboratory practices and quality assurance standards for quality results.
Post-test counselling	HTS provider should provide post-test counselling to clients regardless of the outcome of the test. This may be delivered to individuals, couples or families, depending on what they agreed to, during pre-testing counselling.

Note:

- *HTS providers should encourage couples to receive post-test counselling together to facilitate mutual disclosure.*
- *All HIV positive individuals should be retested for verification before ART initiation. The test should be conducted by a different provider using a different sample and the same testing algorithm. The individual is confirmed positive if the test results remain positive after verification. However, if the test results change, their sample should be referred for additional testing at a nearby higher-level testing site.*
- *Start treatment as soon as possible (preferably within 7 days after a positive result).*
- *Individuals who test negative for HIV should be counselled and linked to prevention services as described in the national HIV and testing guideline 2021.*

3.3.2 HIV Testing Services for Different Populations

- a) HIV-Exposed Infants (HEIs), Pregnant and Breastfeeding Women (PBFW), and Their Partners.** Testing is recommended for all infants and children under 18 months born to HIV-positive mothers, regardless of the mother's ART status. HIV testing for PBFW and their partners is detailed in Chapter V, which addresses the prevention of vertical transmission from mother to child.
- b) HIV Testing for Children Older Than 18 Months and Up to 15 Years.** HIV Testing Services (HTS) providers should identify children at risk of HIV infection using the HIV Risk Assessment Tool. Testing should be conducted using an HIV rapid test. Parental or guardian consent is required before testing children.
- c) HIV Testing for Key and Vulnerable Populations:** HIV testing and counseling should be provided to key and vulnerable populations either in health facilities or through community-based testing. For individuals who test HIV negative, re-testing is recommended after four weeks and subsequently every six months. Prisoners should be offered voluntary HIV testing services as part of general health services, whether in prison or other settings.
- d) HIV Testing for Men:** Men represent a critical population for HIV testing to increase awareness of their HIV status and reduce new infections in women. Differentiated HTS models, adapted for the Tanzanian context, should be implemented with fidelity to optimize service uptake among men. Strategies to identify men

include hotspot testing in locations such as fishing communities and among long-distance truck drivers, as well as initiatives like Mwanaume Jitambue (“Man, Know Yourself”).

- e) **HIV Testing for Sexual Partners and Biological Children of Index Recipients of Care:** Individuals who test HIV positive or are known positive should be encouraged to notify their sexual partners and biological children under 18 years for HIV testing. Those identified should be linked to prevention, care, and treatment services as appropriate. All PLHIV enrolled in HIV care should receive disclosure counseling and support for assisted disclosure of their HIV status.
- f) **HIV Testing for Social Network Partners:** Individuals who are HIV positive, known positive, or at high risk but test negative should be encouraged to bring their social network partners for HIV testing. These partners should then be linked to prevention, care, and treatment services as needed.
- g) **HIV Testing for Individuals Aged 50+ Years:** Limited consideration of HIV testing in older adults contributes to delayed initiation of ART among PLHIV aged 50 years and above. Healthcare providers should screen for HIV risk factors (e.g., multiple sexual partners, injectable drug use) and offer testing using recommended modalities in the HIV Testing Service Guideline 2021. HIV-positive older adults should be linked to care, while HIV-negative individuals at risk should be counseled and referred to HIV prevention services such as PrEP, condoms, PEP, and gender-based violence (GBV) support, as shown in Figure 3.3:1
- h) **Recency Testing:** Recency testing is a non-population-based test that differentiates between recent and long-standing HIV infections by detecting an immunologic marker of disease progression in newly identified PLHIV. Integrating routine recency testing into HTS can help identify areas of ongoing transmission, support national surveillance efforts, and guide prevention strategies. It is important to note that recency testing should not be used for clinical decision-making.

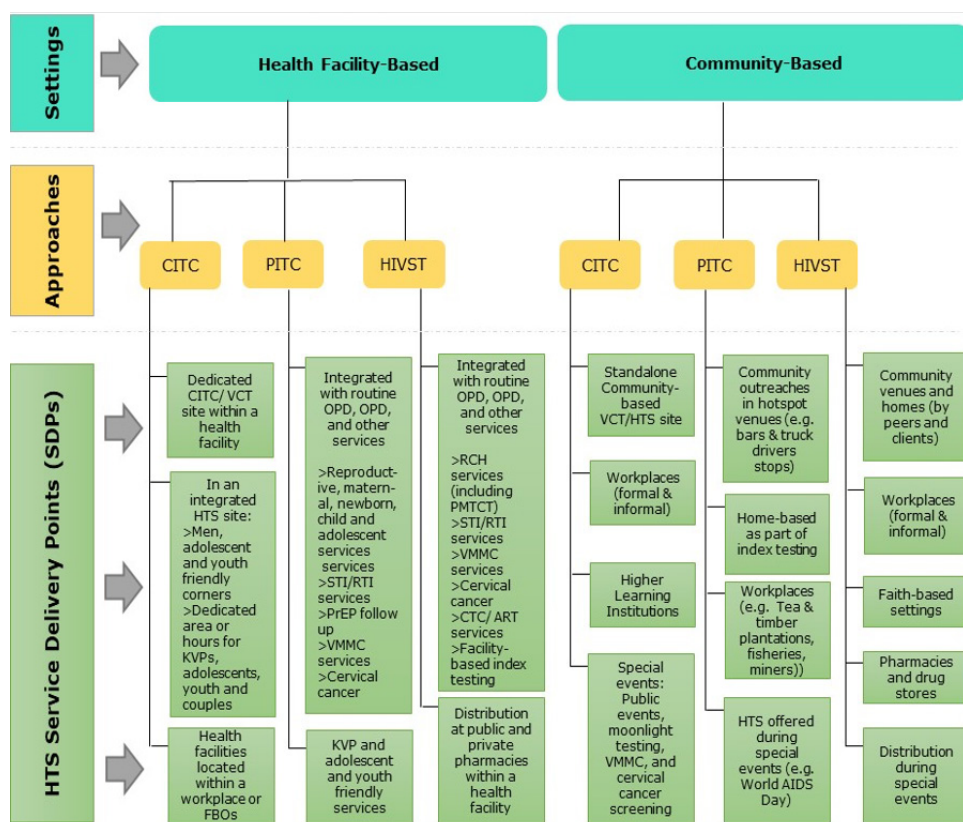


Figure 3.3:1 Summary of the HIV testing services

3.3.3 HIV Testing Algorithm

The declining HIV positivity rate in Tanzania necessitates a revised approach to HIV testing strategies. HIV-positive diagnoses depend on the number of individuals testing positive, and with a population prevalence now below 5%, achieving a 99% positive predictive value requires at least three consecutive reactive tests. To support this transition and enhance access and cost-effectiveness, the use of dual HIV/Syphilis rapid diagnostic tests (RDTs) as the initial test for pregnant women during antenatal care (ANC) is recommended. This approach aligns with efforts to streamline testing while addressing both HIV and syphilis in a single step. Given the low HIV prevalence, three reactive test results are required to confirm and report a positive HIV diagnosis, as illustrated in Figure 3.3:2 below.

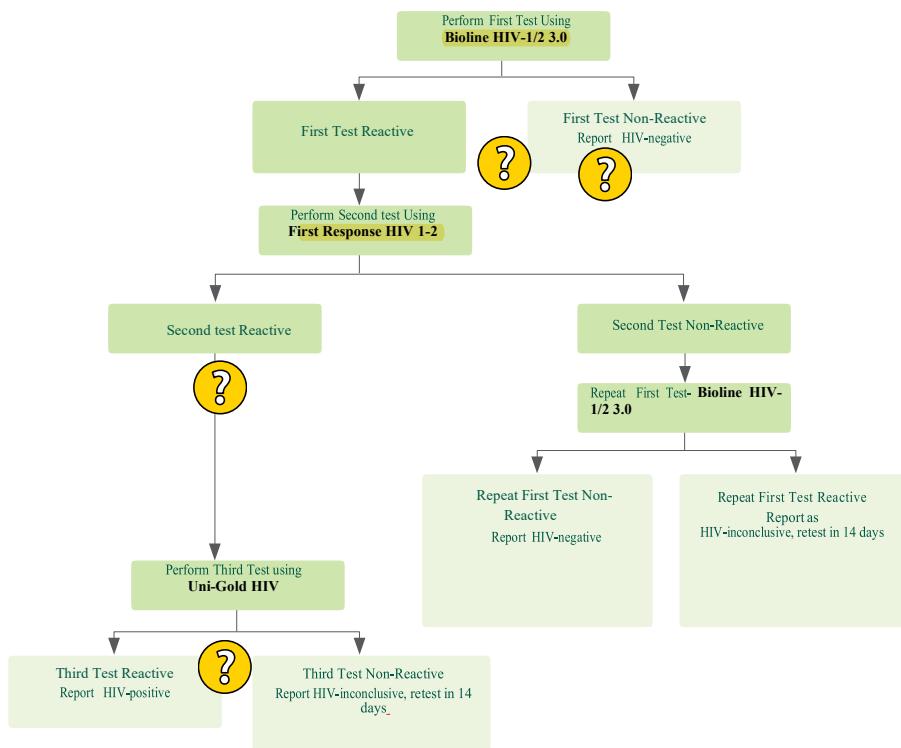


Figure 3.3:2 National HIV testing algorithm for general population

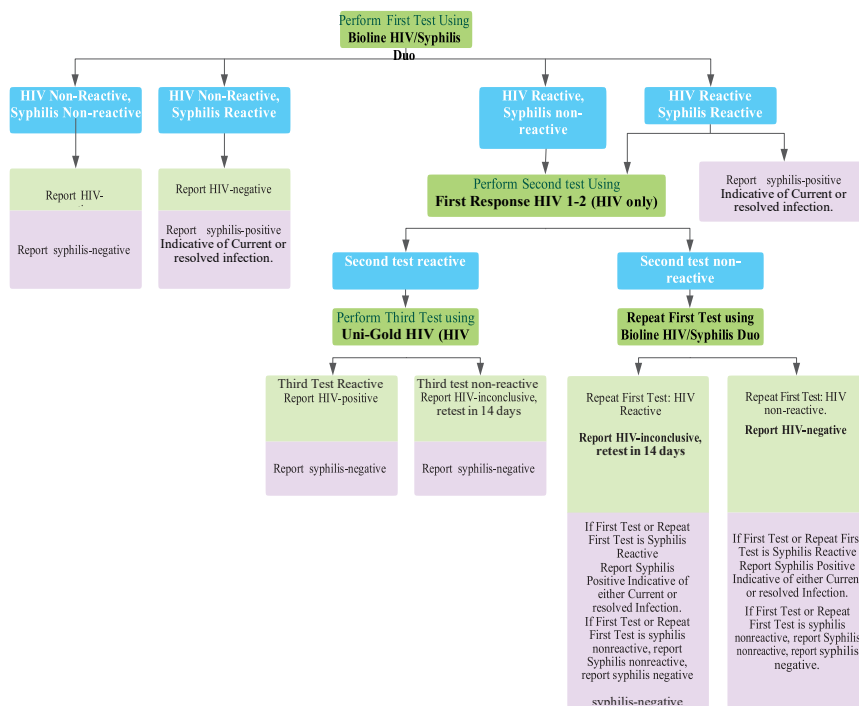


Figure 3.3:3 National HIV testing algorithm for ANC clients

3.4 Successful Referral and Linkage Services

Status Neutrality

In an HIV Testing Services (HTS) setting, individuals may test either positive or negative for HIV. Those who test positive are promptly linked to care and treatment services. Individuals who test negative are connected to appropriate prevention services following a comprehensive risk assessment.

The status-neutral approach emphasizes a whole-person approach to HIV prevention and care. This ensures the provision of high-quality services designed to engage and retain individuals, regardless of whether they require HIV treatment or prevention. This concept is illustrated in Figure 2.4 below.

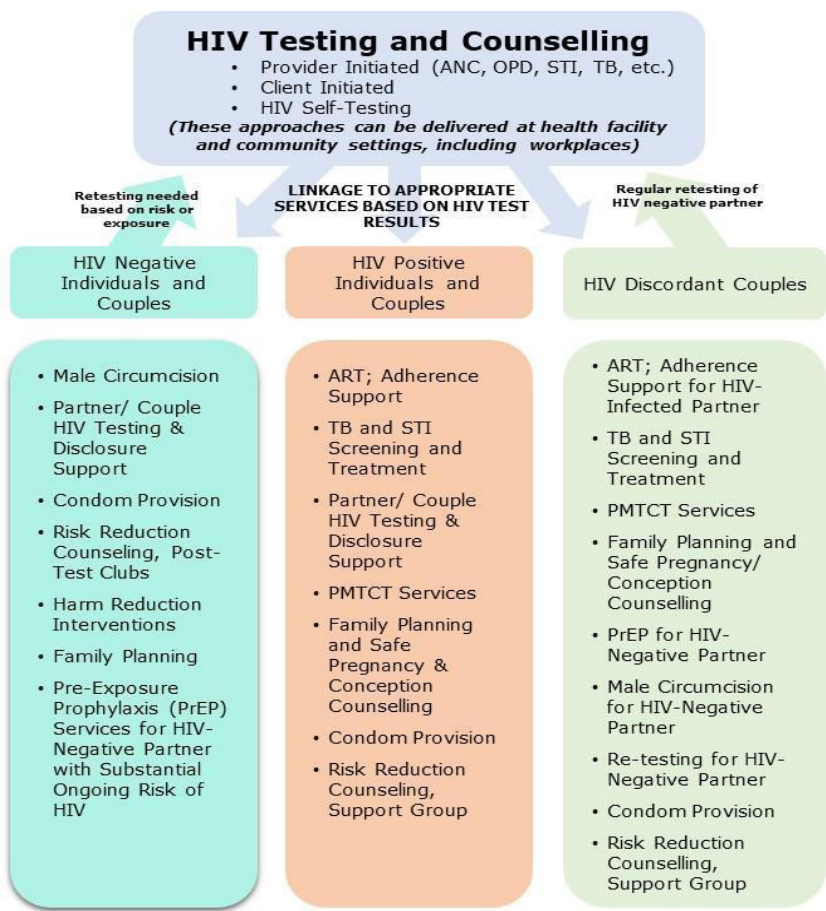


Figure 3.4:1 HIV Post-test linkage to prevention, care and treatment services

3.4.1 Strategies for Effective Referral and Linkage

- Refer all people who test positive at facility and in community settings to a health facility-based care and treatment services for confirmation of HIV test results and further assessment for care and treatment services.
- Sites offering facility-led community ART initiation for KVP with a 1-month starter pack, should link KVP to the nearest convenient ART site for continued care and follow up.
- The following strategies have demonstrated benefit in improving linkage to care following an HIV diagnosis:
 - Streamlined interventions to reduce time between diagnosis and engagement in care, including use of Linkage Case Management (LCM) and follow up for the first 6 months;
 - Follow up support to HIV positive RoC opted to care and start ART at other facility of their choice either via physical visit or virtually.
 - Support for HIV disclosure,
 - Training staff to provide multiple services,
 - Peer support and navigation approaches for linkage,
 - Using quality improvement approaches using data to improve linkage and RoC tracing.

3.4.2 Differentiated HTS services

Differentiated models of HTS delivery should be designed and implemented as a direct response to specific challenges or barriers identified for individuals seeking testing and/or healthcare workers as described in Chapter I.

4 ANTIRETROVIRAL THERAPY FOR ADOLESCENTS, ADULTS AND THE ELDERLY

4.1 Introduction

This chapter discusses the management of ART in adults and adolescents aged 15 years and above. All individuals confirmed to have HIV infection are eligible for ART, regardless of CD4 count, WHO clinical stage, or any other criteria, provided they are willing and ready to initiate ART and adhere to follow-up visits.

ART should be initiated within 7 days (same-day initiation), except in cases where there is an increased risk of immune reconstitution inflammatory syndrome (IRIS). Recipients of Care (RoC) with advanced HIV disease should be closely screened for opportunistic infections (OIs). ART initiation may need to be delayed (see Table 3.2) to prevent life-threatening IRIS.

RoC should be encouraged to identify and designate a treatment supporter to assist them during treatment and care. Healthcare providers should refer newly registered RoC, as needed, to additional programs such as PLHIV support groups, Medically Assisted Therapy (MAT), and adolescent clubs.

4.2 Goals of Antiretroviral Therapy

The goals of combination antiretroviral therapy include:

- Achieving an **undetectable viral load** (<50 copies/ml)
-  Restoring and/or preserving immunologic function by achieving CD4 recovery to normal thresholds (≥ 500 cells/mm³)
- **Reducing HIV-related morbidity and mortality**
- Improving **survival and quality of life**
- Preventing **onward new HIV infections** (Undetectable = Untransmittable)

4.3 Initiating ART for Adults and Adolescents Aged 15 Years and Above

ART should be initiated for all adults and adolescents living with HIV by qualified healthcare providers in both facility and community settings. This initiation should occur after conducting a comprehensive medical history and physical examination, irrespective of CD4 count and WHO clinical staging.

Key Steps for ART Initiation:

- RoC should be reviewed within 14 days of ART initiation.
- Baseline investigations should be performed to inform the choice of ART regimen and any necessary additional medications.

- A list of required baseline investigations (conducted on-site or via sample referral) is provided in Table 4.3:1 Baseline laboratory tests for adolescents and adults at ART initiation.

Table 4.3:1 Baseline laboratory tests for adolescents and adults at ART initiation

Laboratory test	Rationale
CD4 count	Categorization of RoC into Early and Late Presenters To determine eligibility for AHD package of care To determine severity of immune deficiency To determine Eligibility for CPT and CrAg screening
Serum creatinine	To determine kidney dysfunction by eGFR*
Liver enzyme markers (ALAT, ASAT)	To determine liver dysfunction
HBsAg	To determine co-infection with HBV
Haemoglobin (Hb)	To rule out anaemia
Random Blood Glucose	To rule out diabetes mellitus

$$*_e \text{ GFR (ml/min/1.73m}^2 = \frac{\text{Height (cm)} \times 40}{\text{Creatinine (umol/L)}}$$

Table 4.3:2 Conditions that require delay of ART initiation

	Opportunistic Infection	Immunological Status	Action
?	TB Meningitis	Any CD4 cell count	• Initiate ART within 4 weeks of starting TB treatment, when the client's symptoms are improving, and TB treatment is tolerated. • Corticosteroids should be given for 4 weeks as adjuvant treatment for TB meningitis
?	Pulmonary TB and Extra pulmonary TB other than TB Meningitis	Any CD4 cell count	• Initiate ART within 2 weeks of starting TB treatment, when the client's symptoms are improving, and TB treatment is tolerated
?	Cryptococcal Infection (CI)	CD4 <200 cells/mm ³	• Initiate ART 2 weeks after completion of induction phase of Pre-emptive Flucanazole treatment
?	Cryptococcal Meningitis (CM)	CD4 <200 cells/mm ³	• Initiate ART 5 weeks following initiation of treatment for Cryptococcal Meningitis (beginning of 4th week of consolidation phase)
?	Pneumocystis Jirovecii Pneumonia	Any CD4 count	• Do not delay ART initiation

Note:

- Delay of ART initiation prevents development of IRIS (a paradoxical inflammatory disorder characterised by worsening of pre-existing infections following the initiation of ART)
- Switching a failing client to a new regimen who has concurrent TB/Cryptococcal diagnosis requires delay of ART as per recommendations to prevent development of paradoxical worsening IRIS.
- IRIS definition: Refer section 4.7

To provide person centred care, RoC should be categorized to different groups based on clinical and laboratory findings as shown in Table 4.3:3.

Table 4.3:3 Recipients of care categories during ART initiation and management

Category	Management package
1. Early presenter	• Screen for TB • Provide TPT if no active TB
2. Late presenter	
a) Without Advanced HIV Disease	• Provide CPT • Screen for TB • Provide TPT if no active TB • Perform CD4 testing after 6 months
b) With Advanced HIV Disease	• Offer AHD package of care that includes: - o Provide CPT - o Screen for TB - o Provide TPT if no active TB - o Screen for cryptococcal infection/meningitis - o Perform CD4 testing after 6 months - o Screen for other severe infections (toxoplasmosis, CMV) if feasible - o Offer other components of AHD package as appropriate (refer to Chapter IX)

Note: TPT should be initiated two weeks after ART initiation

4.3.1 First Line ART

Dolutegravir in combination with NRTI backbone is recommended as first-line regimen for adults and adolescents 15 years and older living with HIV, regardless of previous ART exposure (PrEP, PEP, Neonatal HIV prophylaxis) as shown in Table 4.3:4:

Table 4.3:4 Recommended first line regimens for adult and adolescents

Recipients of care group	Default 1 st line Regimen	Alternative Regimen
Adults and adolescents (>15 years), Pregnant and Lactating mothers, HIV and HBV co-infection ≥30kg	TDF/3TC /DTG (TLD) 	ABC/3TC+ DTG *TDF/3TC/EFV (TLE400) Special situation **TDF/FTC + LPV/r 
Elderly (≥ 50 years), RoC with CKD, co-morbidities (DM, HTN) with increased risk of renal dysfunction & post-menopausal women	TAF /(3TC or FTC) + DTG 	ABC/3TC+ DTG
HIV and TB co-infections	TDF/3TC/DTG (Double dosage of DTG) 	TDF/3TC/EFV (TLE400)
		ABC/3TC+ DTG (Double dosage of DTG)
People who Inject Drugs (PWID)	TDF + 3TC +DTG	ABC/3TC+ DTG TAF /(3TC or FTC) + DTG TDF/FTC +ATV/r

Note:

- *TDF/3TC/EFV (TLE400) should only be used in cases of DTG intolerance because NNRTI based transmitted drug resistance is reported to be ≥10% in East Africa which is above the recommended threshold for using NNRTIs.
- **LPV/r should be used in the first line if DTG can't be used due to intolerance and when there is resistance to EFV
- TAF is contraindicated in TB/HIV co-infection as Rifampicin lowers TAF concentration below the therapeutic level.
- TAF use in pregnancy is currently not recommended due to limited data on its safety.
- For recipients of care with weight <30kg, avoid TDF and use ABC based regimen.
- Consider PI based ARV if unable to use DTG and EFV
- DTG does not interact with methadone; therefore, it is a suitable drug for regimens in PWID. Initiate treatment after the RoC has been stabilized, usually 2 to 3 months after starting MAT.
- Administer DTG 50mg twice a day for RoC on Rifampicin based regimen to reach DTG therapeutic levels.
- ARV drugs have the potential to decrease the bioavailability of hormonal contraceptives especially with oral contraceptives. Dual contraception with condoms and injectable contraceptives is therefore recommended.

4.3.2 Second-line antiretroviral therapy in adults and adolescents

A combination of a Protease Inhibitor (PI) and at least one new NRTI not previously used in first-line therapy are recommended for second-line treatment for adults and adolescents PLHIV aged 15 and above, as shown in Table 4.3:5. Lamivudine (3TC) or Emtricitabine (FTC) should be continued in subsequent regimens as it lowers viral fitness and increases susceptibility to AZT.

Table 4.3:5 Recommended second line regimens for adults and adolescents

Population	Preferred 2 nd line regimen	Alternative 2 nd line regimen
Adults, adolescents (>15 years) Pregnant women / breastfeeding mothers and PWID	AZT/3TC+ATV/r	AZT/3TC+LPV/r
HIV and TB co-infection	AZT/3TC+ LPV/r* *Double dose of LPV/r to 800/200 mg	

Note:

- Addition of DTG to failing PI based 2nd line regimen is not recommended because the client is considered to be on DTG monotherapy leading to development of DTG resistance, which compromises DTG efficacy and its future use
- Delayed diagnosis of treatment failure results in accumulation of resistance associated mutations (RAMs). Some of these RAMs compromise efficacy of drugs with similar resistance patterns for future use such as cross resistance between Abacavir (ABC) and TDF, and Atazanavir (ATV) and Darunavir (DRV). HCP should therefore exercise early detection and manage treatment failure.
- RoC who received a PI-based regimen as their first line of treatment can be switched to a DTG-based regimen as a potential option for their second line of treatment.

4.3.3 Third-line Antiretroviral Therapy

RoC failing second line may have substantial NRTI and NNRTI-associated resistance mutations (RAMs), limiting their usage in the third line regimen. Third-line regimens should include at least two, preferably three, effective drugs, constructed using second-generation formulations of preceding antiretrovirals with low or no cross-resistance. These second-generation drugs have a greater genetic barrier to resistance, and their efficacy is rarely jeopardized by RAMs associated with first-generation formulations.

Third line ARVs recommended regimens include:

- Integrase Strand Transfer Inhibitors (INSTIs): Dolutegravir (DTG) and Raltegravir (RAL)
- Second generation PI: Boosted Darunavir (DRV/r)
- Susceptible NRTI after Genotypic Antiretroviral Resistance Testing and
- Recycled Lamivudine or Emtricitabine regardless of presence and level of RAMs selected by these ARVs

Table 4.3:6 Recommended third line regimens for adults and adolescents.

Population	Preferred 3 rd line regimen	Alternative 3 rd line regimen
Adults, adolescents (>15 years), and Pregnant women / breastfeeding mothers and PWID	DRV/r + DTG + 3TC or FTC+1NRTIs (Based on genotypic antiretroviral resistance testing results)	DRV/r + DTG /3TC(Dovato) +1 NRTI (Based on genotypic resistance results) 2 NRTIs + DTG e.g.: TDF/3TC/DTG
HIV and TB co-infection	LPV/r + DTG +3TC or FTC +1NRTI (Based on genotypic antiretroviral resistance testing results) *Double dose of LPV/r to 800/200mg	LPV/r + RAL + 3TC or FTC +1NRTIs (Based on genotypic resistance results) *Double dose of LPV/r to 800/200mg 2NRTIs + DTG e.g.: TDF/3TC/DTG (Double dose of DTG)

Note:

- For TB and HIV co-infected RoC on LPV/r should be switched back to DRV/r after completion of TB treatment
- For second- and third-line regimens which are non TDF/TAF based, in case of new Hepatitis B co-infection TDF/TAF should be added to the new regimen as treatment of Hepatitis B. If single tab TDF or TAF is not available, a fixed dose combination of TDF/FTC or TAF/FTC can be used.

4.4 Changing Antiretroviral Therapy

There are multiple reasons that may prompt the need to change antiretroviral therapy. These can be grouped into two major categories:

- ARV specific adverse events (toxicity)
- Treatment failure

When changing treatment, the following should be observed:

If changing due to toxicity

- Change/substitute only the ARV suspected to be causing the toxicity.

If changing due to treatment failure

- It is never recommended to switch to monotherapy
- Change at least two ARVs, preferably change all three ARVs (DO NOT RETAIN THE BACKBONE)
- When making ARV selection, opt for drugs that haven't been previously used and don't have cross-resistance, overlapping toxicities, or drug-drug interactions
- Lamivudine has the advantage of reducing HIV viral fitness and increasing susceptibility to AZT and therefore should be retained when changing the failing regimen

4.4.1 Changing antiretroviral therapy due to treatment failure

Individuals receiving ART should be regularly monitored in order to detect and address treatment failure promptly, identify factors that may hinder adherence, and assess the client's eligibility for a regimen switch.

Treatment failure can manifest as virological, immunological, clinical, or a combination of these factors (Table 4.4:1). However, virological criteria should be used as the earliest marker for diagnosing treatment failure. Early detection of treatment failure through virological criteria prevents accumulation of RAMs, thus preserving future treatment options of ARVs in the same class. On the other hand, late diagnosis of treatment failure based on immunological (CD4) or clinical criteria can result in the accumulation of RAMs, which reduces future treatment options of ARVs in the same class.

Table 4.4:1 WHO definitions of treatment failure in chronological order of occurrence

Failure	Definition	Comments
Virological	Plasma viral load above 1000 copies/ml with a log drop <0.5 based on two consecutive viral load measurements after three months, with enhanced adherence support.	An individual must be taking ART for at least six months before it can be determined that a regimen has failed.
Immunological	CD4 count at 250 cells/mm ³ with stage 3 or 4 WHO clinical condition. OR CD4 cell count consistently below 100 cells/mm ³	Without concomitant or recent infection or steroid use to cause a transient decline in the CD4 cell count

Failure	Definition	Comments
Clinical	New or recurrent clinical manifestations/conditions indicating severe immunodeficiency (WHO clinical <u>stage 3 or 4 conditions</u>) after six months of effective treatment	The condition must be differentiated from IRIS

Note:

- **Virologic blip** is an isolated detectable transient rise in viral load after attaining virologic suppression, followed by a return to virologic suppression. It is not due to treatment failure and should not raise any concern.
- Treatment failure should be distinguished from IRIS which is a paradoxical reaction which occurs in RoC starting or restarting ART with advanced disease. In IRIS there is clinical deterioration while on ART but improving CD4 count.

4.4.2 Switching to Second-Line ART

Switching to second-line ART should be considered for all recipients of care (RoC) on first-line ART who meet the following criteria:

- **Unsuppressed viral load** ($\geq 1,000$ copies/ml) after three or more successful Enhanced Adherence Counseling (EAC) sessions, as per the established algorithm. Before switching, recipients must demonstrate good adherence ($>95\%$).
- **Routine Genotypic Antiretroviral Resistance Testing (GART)** is not required when switching from a first-line to a second-line antiretroviral regimen, as it does not significantly improve HIV viral re-suppression rates.

4.4.3 Switching to Third-Line ART Regimens

4.4.3.1 Pre-Switch Assessment for Second-Line Virological Failure

Before confirming virological failure of a second-line regimen, a Multidisciplinary Team (MDT) must convene to:

- Rule out other causes of high viremia, including suboptimal adherence.
- Plan for Enhanced Adherence Counseling (EAC) and provide necessary support to address adherence barriers.
 - o If adherence barriers are fully resolved, viral re-suppression may occur, avoiding an unnecessary switch to a third-line regimen.

4.4.3.2 Criteria for Switching to a Third-Line Regimen

When the decision to switch to a third-line regimen is made:

- i. **Genotypic Antiretroviral Resistance Testing (GART)** should be performed to:
 - o Rule out cross-resistance between first- and second-generation ARVs.
 - o Confirm drug resistance as the cause of treatment failure.
 - o Inform the selection of an effective third-line regimen, including the potential recycling of drugs from previous regimens.

A comprehensive evaluation must be conducted to determine the cause of treatment failure, including:

- Assessing and optimizing adherence.
- Ruling out drug interactions and adverse effects.

If treatment failure is confirmed, a new regimen should be initiated, consisting of at least two effective antiretroviral drugs.

ii. **The Central Third Line Committee (CTLC)** will review GART results and provide third-line regimen recommendations to the respective health facilities.

a) Requirements for Switching RoC to Third-Line ART

- RoC must be failing second-line ART.
- RoC must have completed EAC sessions and demonstrate good adherence.
- There must be documented virological failure (VL $\geq$ 1,000 copies/ml) post-EAC.

b) Process flow for Third line ART Initiation

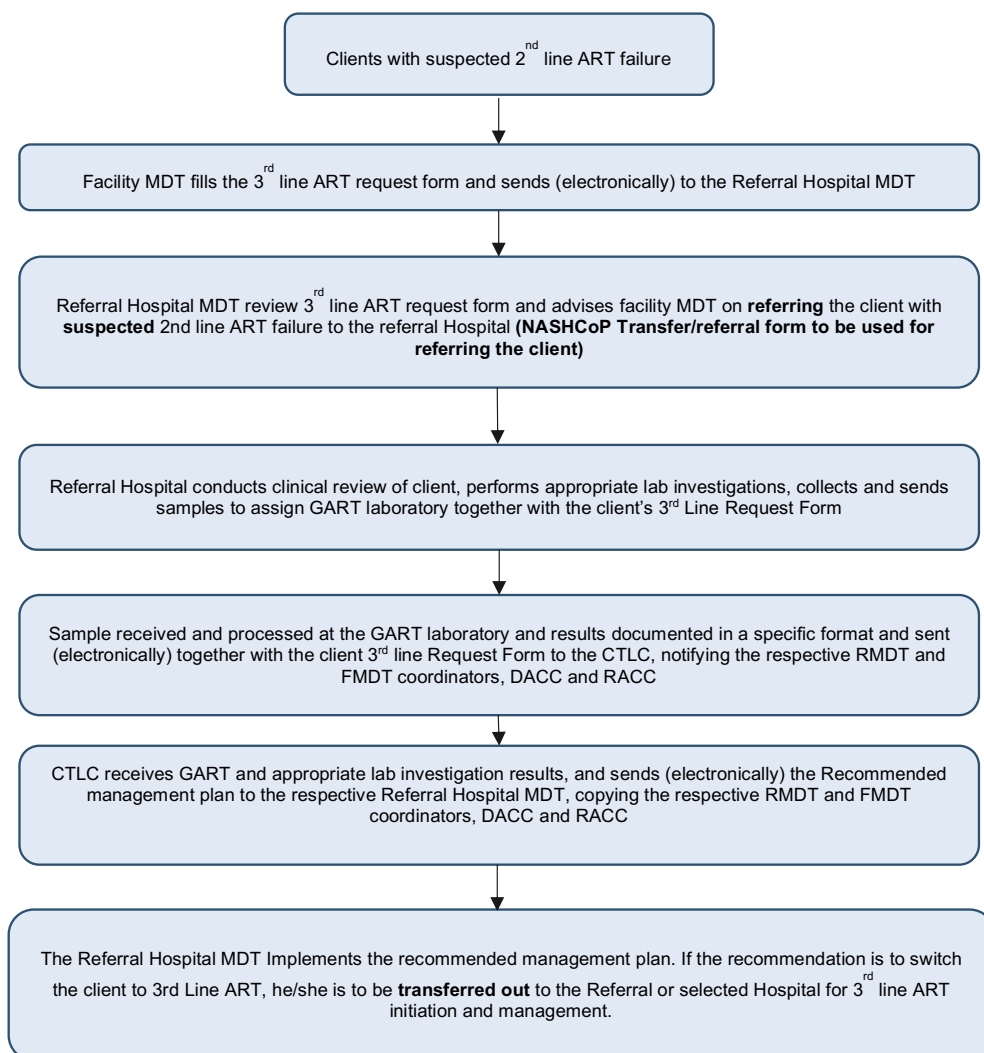


Figure 4.4:1 Process flow for third line ART initiation

NB: GART is recommended for RoCs on a second line regimen aged more than 2 years with Persistent Low-Level Viremia (> 500copies/ml) and good adherence.

4.5 Monitoring of Recipients of Care (RoC) on ART

4.5.1 Clinical Monitoring

All Recipients of Care (RoC) on ART should receive comprehensive clinical monitoring during each visit, which includes:

- A detailed history and physical examination, with all findings documented in their medical records.

- Thorough evaluation of any new or persistent opportunistic infections, as these may indicate treatment failure, as outlined in Table 4.4:1

Note: Successful treatment reduces the morbidity associated with opportunistic infections and enhances the quality of life for RoC.

4.5.2 Laboratory Monitoring

i. Viral Load (VL) Testing

Viral load testing is the gold standard for monitoring treatment failure as it provides an early and accurate indication of treatment outcomes. Its benefits include:

- Minimizing the accumulation of drug resistance mutations.
- Facilitating timely switching to appropriate antiretroviral (ARV) regimens, thereby improving clinical outcomes and preserving future ART options.

All RoC should have access to VL testing in accordance with the national HVL testing algorithm.

- Successful treatment is defined as achieving a VL of less than 50 copies/ml.

ii. CD4 Cell Count Monitoring

- Baseline Testing: CD4 cell count should be performed at baseline to assess for advanced HIV and monitor immunological response.
- Follow-Up Testing:
 - o For RoC with a baseline CD4 cell count of less than 350 cells/mm³, the test should be repeated after six months and periodically thereafter until the CD4+ T lymphocyte count exceeds 350 cells/mm³ and the RoC is stable on ART.
 - o CD4 testing may also be conducted at any time in cases of suspected Immune Reconstitution Inflammatory Syndrome (IRIS) or treatment failure.

iii. Other Laboratory Investigations

For additional laboratory monitoring requirements, refer to Table 4.5:1.

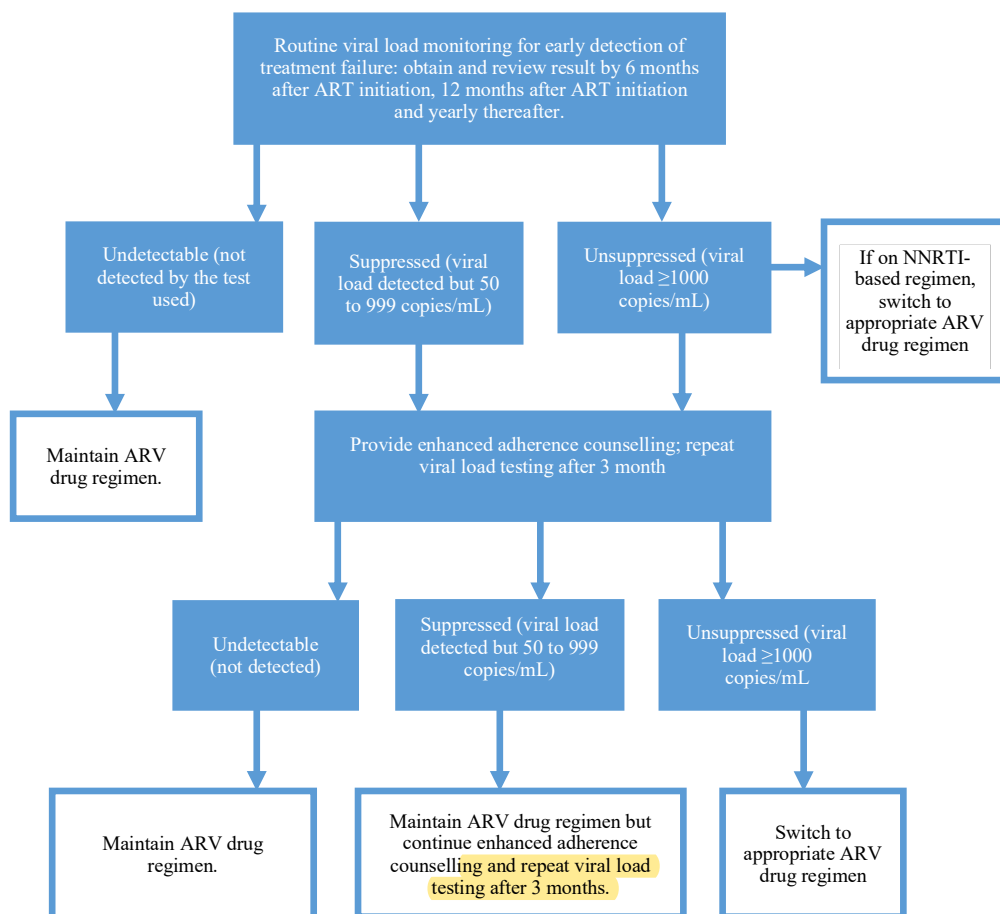


Figure 4.5:1 HIV viral load testing algorithm

Note:

- **Persistent Low-Level Viremia on Second-Line Regimen:**
 - a) For RoC on a second-line regimen for more than 2 years with **persistent low-level viremia** (>500 copies/ml) after two consecutive Enhanced Adherence Counseling (EAC) cycles and demonstrating good adherence, Genotypic **Antiretroviral Resistance Testing (GART)** is recommended.
 - b) The choice of the subsequent regimen will be informed by GART results.
 - c) If high viral load (HVL) is less than 500 copies/ml, RoC should continue with the same regimen and undergo HVL monitoring every 6 months.

- **Persistent Low-Level Viremia on First-Line Regimen:**
 - a) For RoC on a first-line regimen with **persistent low-level viremia** after two consecutive EAC cycles, **they should continue with the same regimen and undergo viral load monitoring every 12 months.**
 - b) **No regimen change should be made** until virological failure is confirmed.
- **Multi-Month Dispensing (MMD) Post-HVL Testing:**
 - a) RoCs on MMD should be dispensed **one month of ART** after HVL testing while awaiting results.
 - b) This allows for re-evaluation, proper categorization of care status, and prompt initiation of EAC if HVL is ≥ 50 copies/ml.
- **Same-Day Viral Load (VL) Testing:**
 - a) **Same-day VL testing using point-of-care testing is encouraged, where available, to expedite clinical decision-making for repeat viral load tests.**
 - b) If point-of-care testing is unavailable, priority should be given to viral load specimens and results across the laboratory referral process, including specimen collection, testing, and return of results.
- **PROBE Assessment:**
 - a) Refer to Chapter VI: “**Adherence Preparation, Monitoring, Retention, and Support**” for guidance on conducting the PROBE assessment.

Table 4.5:1 Summary of laboratory monitoring of adolescent and adults on first and second line ART regimens

Monitoring Tests	Frequency	Rationale
HVL (All Recipients of care)	• Refer to HVL algorithm	ART monitoring
CD4 (All Recipients of Care)	• At Baseline (All) • After every six months till CD4 count is above 350 cells/mm³ • Suspected treatment failure • Development of AHD	ART monitoring
FBP/Hb (All Recipients of Care)	• At Baseline,	Hb monitoring

Monitoring Tests	Frequency	Rationale
	If a RoC has Hb,8.5g/dl • At one month, thereafter • Every Six months	Avoid AZT if Hb<8.5g/dl
Serum Creatinine (for Recipients of Care on TDF or TAF)	• At Baseline, after every six months and whenever symptomatic	Screening for early renal toxicity
ALT (For recipient of cares on DTG)	• At Baseline, one month, after every six months and whenever symptomatic	Screening for early Liver toxicity
Cholesterol and RBG	• At 6 months then yearly	Identify/detect TAF, DTG or PI adverse effects
Bilirubin (for all Recipients of Care on ATV/r	• At 6 months • Whenever symptomatic	Indirect hyperbilirubinemia
Urinalysis (proteinuria, glycosuria)	• Every six months	Monitor Renal dysfunction

Note: -

- Frequent laboratory tests will be determined by clinical evaluation.
- The frequency of CD4 and viral load monitoring may be less than six months when IRIS or treatment failure is suspected.

iv. Discordant Viral Load and CD4 Cell Count Response to Antiretroviral Therapy (ART)

a. Concordant Response

A concordant response to ART occurs when the client achieves:

- Optimal HIV viral load suppression (<50 copies/ml).
- Robust CD4 cell count gain.

b. Discordant Response

A discordant response occurs when there is optimal HIV viral load suppression without a corresponding robust CD4 cell count gain. This condition is associated with increased morbidity and mortality.

c. Risk Factors for Immuno-Virological Discordance

The following factors are associated with an increased risk of a discordant response:

- Age >50 years at ART initiation.
- Comorbidities such as Hepatitis B, Cytomegalovirus (CMV), or Tuberculosis (TB).

- High viral load at ART initiation.
- Low baseline CD4 cell count.
- Late ART initiation (WHO clinical stage 3 or 4).
- Treatment-experienced RoC with drug-resistant HIV.

d. Management of Discordant Response

Healthcare providers (HCPs) should:

- Ensure the RoC is receiving appropriate prophylaxis for opportunistic infections (OIs), including Cotrimoxazole.
- Review the RoC's medication list for potential bone marrow-suppressing drugs. For example, consider switching from an AZT-containing regimen to one without AZT if necessary.
- Evaluate for clinical manifestations (e.g., systemic symptoms or pancytopenia) that might suggest a marrow infiltrative process.
- Continue ART, even if the RoC does not exhibit a significant CD4 cell count response.

4.6 ARV Refill Models for RoC

The table below outlines the recommended refill models for RoC on ART. Healthcare providers should guide RoC in selecting the model that best suits their care needs and lifestyle.

Table 4.6:1 Refill models for RoC who are established on ART (Stable)

	At facility	At Community
Individual models	• One monthly ART refill tied with clinical review, adherence counselling and lab tests according to algorithm. • Monthly ART refill • 3MMD ART refill fast-tracked. • 6MMD ART refill to be done at health facilities and it's tied with clinical reviews, adherence counselling and annual viral load test • Expanded hours (evening and weekend) • Synchronized VL sample collection with ARV refills and VLs collected. • Case management using an early warning dashboard (e.g. early or late missed appointments) • Welcome back to care after interruption in care	• Facility-based mobile outreaches to some individual RoC who have special needs such as those who are bedridden. • Community ART Distribution Points, Private pharmacies, MSD community pharmacies, Private hospitals, Vending machines

	At facility	At Community
Group models	• Family-centred models with alignment of children and caregiver appointments • Adolescent clinics • Operational triple zero clinics (Peer-based)	• Facility-based outreaches to smaller health facilities • Facility-based groups ART refill at selected places such as WEO, VEO, churches • Group ART delivery by expert patients or CBHSPs to stable patients

4.7 Immune Reconstitution Inflammatory Syndrome (IRIS)

IRIS is a phenomenon associated with the occurrence or worsening of opportunistic infections/malignancies which can occur early after initiation of ART or at later (several months to years) in the course of ART. IRIS occurs in 10 to 25% of people living with HIV starting ART.

RoC who are treatment naïve, have advanced HIV disease with CD4 cell count <50 cells/mm³, and/or have undiagnosed and untreated opportunistic conditions. Any OI, malignancy and autoimmune diseases may present as IRIS

IRIS has been reported to occur in as many as 30% of TB-HIV Co infected RoC. Although IRIS in PLHIV who have TB is self-limiting, severe cases may require the use of a brief course of corticosteroids to reduce inflammation for those CNS manifestation or those with severe respiratory symptoms.

Cryptococcal IRIS occurs among 10–50% of people with cryptococcal disease in 3 to 12 weeks (range of 1 to 10 months) of ART initiation and is associated with high mortality.

Initiation of ART can also unmask previously undiagnosed infections such as hepatitis B or C viral infections as it improves the inflammatory response while repairing the immune system.

Graves's disease can occur 8 to 33 months as an autoimmune IRIS after initiation of ART in certain HIV-1-infected individuals. It should be suspected in individuals who present with clinical features suggestive of hyperthyroidism or worsening of clinical features despite good virological and immunological response to ART.

Table 4.7:1 Clinical presentations of IRIS

Organism	Clinical features
Mycobacterium Tuberculosis	Pulmonary TB: Cough, shortness of breath, increased work of breathing. Extra pulmonary TB: Tender lymphadenopathy, cutaneous lesions. Systemic symptoms: Malaise, fever, chills, unintentional weight loss, night sweats.
Pneumocystis Jirovecii	Systemic symptoms: Fever, chills, malaise, unintentional weight loss. Pulmonary symptoms: Worsening shortness of breath, cough.
Cryptococcus Neoformans.	The central nervous system (CNS): Confusions, seizures, raised intracranial pressure. Pulmonary symptoms: Cough, shortness of breath.
Mycobacterium Avium Complex	Tender lymphadenopathy.
Cytomegalovirus (CMV)	Ocular symptoms related to immune reconstitution mediated uveitis such as blurred vision, decreased visual acuity, ocular pain. Extraocular symptoms: Shortness of breath due to pneumonia, diarrhoea due to CMV colitis.
Hepatitis B and C Viruses	Fever, chills, lack of appetite, unintentional weight loss, nausea, jaundice.
Kaposi Sarcoma	Worsening of skin and mucosal lesions with increased tenderness and swelling, peripheral edema.
Herpes Zoster	Cutaneous manifestations: Herpetic lesions in typical dermatomal distributions. Ocular lesions: Eye pain and red eye due to keratitis or iritis.
Graves' disease	Development or worsening of clinical features i.e. exophthalmos, excessive sweating, heat intolerance, irritability and weight loss

The criteria for making a diagnosis of IRIS are described in Table 4.7:2 below:

Table 4.7:2 IRIS diagnostic criteria

Diagnosis of IRIS would require both major (A plus B) criteria or Criterion A plus 2 minor criteria
Major criteria A. A typical presentation of “opportunistic infections or tumours” in RoC responding to ART includes: • Localized disease e.g. lymph nodes, liver, spleen • Exaggerated inflammatory reaction e.g. severe fever, with exclusion of other causes of painful lesions • Atypical inflammatory response in affected tissues e.g. granulomas, suppuration, necrosis, perivascular lymphocytic inflammatory cell infiltrate

• Progression of organ dysfunction or enlargement of pre-existing lesions after definite, clinical improvement with pathogen specific therapy prior to commencement of ART and exclusion of treatment toxicity and new diagnoses • Development or enlargement of cerebral space occupying lesions after treatment for cerebral Cryptococcus or toxoplasmosis • Progressive pneumonitis or the development of organizing pneumonia after treatment of pulmonary-TB or PJP • New onset or worsening of uveitis/ vitritis after resolution of CMV retinitis. • Fever and cytopenia after treatment for disseminated Mycobacterium avium complex (MAC) disease
• Enlargement of Kaposi's sarcoma lesions and subsequent resolution or partial regression without • Commencement of radiotherapy, systemic chemotherapy or intralesional therapy
B. Significant drop in HIV viral load count
Minor criteria • <u>Increased blood CD4+ cell count after initiation of ART</u> • <u>Increase in immune response specific to the relevant pathogen</u> e.g. delayed type hypersensitivity to mycobacterial antigens (PPD conversion) • Spontaneous resolution of disease without specific antimicrobial therapy or tumour chemotherapy with continuation of ART.

Table 4.7:3 Management of IRIS

Mild to moderate forms: • Reassure the patient • Do not stop ART • Provide specific treatment for the opportunistic infections/malignancies or other diseases
Severe life-threatening IRIS • Reassure the patient • Stop ART temporarily • Provide high doses of Prednisolone 1mg/kg for 4 weeks then taper down the dose. In case of severe cryptococcal meningitis IRIS, a short course of Steroid may be prescribed (Refer to cryptococcal meningitis management, in Chapter IX). • When using high-dose steroids, it is important to rule out Strongyloides stecalis infection to avoid disseminated strongyloidiasis. • Provide other appropriate supportive measures such as management of fever, oxygen therapy, i.e. fluids • Restart ART when the RoC is clinically stable

4.8 HIV and AIDS among old persons/Elderly

4.8.1 Epidemiology

Globally, the burden of HIV and AIDS among the elderly is increasing due to two main factors:

- i. Increased survival rates among individuals living with HIV, leading to an aging population (“aging with HIV”).
- ii. Lack of tailored prevention interventions for older adults, resulting in new infections in individuals over 50 years of age.

Many older individuals acquire HIV due to misconceptions that they are not at risk, and diagnoses in this population are often delayed or missed entirely. Men aged 50 and above frequently engage in sexual activity, including intergenerational sex, while postmenopausal women face an increased biological risk of acquiring HIV.

Approximately 21% of all people living with HIV (PLHIV) globally are aged 50 years or older.

4.8.2 Characteristics of HIV in the Elderly

Elderly PLHIV often present with rapid disease progression and delayed diagnoses. They also experience high rates of comorbidities, including:

i. Infectious Comorbidities:

- o Tuberculosis (TB), atypical pneumonias, *Pneumocystis jirovecii* pneumonia (PJP), Cryptococcal meningitis, and CNS toxoplasmosis.

ii. Non-Communicable Comorbidities:

- o AIDS-related malignancies (e.g., cervical cancer in women, prostate cancer in men), lymphomas (e.g., primary CNS lymphoma), hypertension, stroke, and diabetes mellitus.

iii. Neurological Conditions:

- o HIV-Associated Neurocognitive Disorders (HAND), depression, psychosis, and frailty.

Age-related declines in organ function further complicate management, affecting medication dosages, the risk of toxicity, and treatment responses.

4.8.3 Clinical Management Challenges in the Elderly

When managing elderly PLHIV, the following considerations are critical:

- i. **Early ART Initiation:** Initiate ART promptly, provided there are no opportunistic infections requiring deferral. Due to fast disease progression and late presentation, early ART initiation reduces mortality. However, elderly individuals are at higher risk for immune reconstitution inflammatory syndrome (IRIS), slow immunological response, and discordant ART response (optimal viral suppression with low CD4 cell count).
- ii. **Minimizing Toxicity:** ARV prescriptions should aim to minimize toxicity and adverse effects, accounting for renal and hepatic dysfunction, drug-drug interactions, and overlapping toxicities from polypharmacy.
- iii. **Screening for Non-Communicable Diseases (NCDs):** Regular screening for conditions such as hypertension, cardiovascular diseases, diabetes mellitus, and mental health issues is essential. Diagnosed comorbidities should be managed effectively to enhance quality of life.
- iv. **Laboratory Monitoring:** Laboratory work-up for elderly PLHIV should include standard PLHIV investigations along with screening and monitoring parameters for NCDs.
- v. **Adherence Challenges:** Polypharmacy and neuropsychiatric disorders increase the risk of poor adherence. To optimize adherence, it is essential to:
 - a) Screen for and treat mood disorders (e.g., anxiety and depression).
 - b) Provide adherence support through trusted family members or a buddy system.

4.8.4 Choice of ARVs for Elderly PLHIV

4.8.4.1 Preferred ARVs:

- a) **Tenofovir alafenamide (TAF)** is preferred due to the increased risk of renal dysfunction, osteoporosis, and pathological fractures in this population.
- b) **If TAF is unavailable, abacavir (ABC)** may be prescribed but with caution in individuals with coronary artery disease, where TAF is the preferred alternative.

- i. **Neuropsychiatric Considerations:** Efavirenz (EFV) should be avoided in individuals with neuropsychiatric conditions or a history of mental illness.
- ii. **Metabolic and Hepatic Conditions:**
 - a) For patients with metabolic syndrome or severe hepatic dysfunction, **dolutegravir (DTG)** should be used cautiously.
 - b) If needed, replace DTG with **EFV** in first-line therapy or **atazanavir/ritonavir (ATV/r)** in second-line therapy.

4.8.4.2 Co-Medications with ARVs

- i. **Diabetes Management:** For PLHIV with type 2 diabetes mellitus on metformin while using a DTG-based regimen, the dosage of metformin should not exceed 1g per day. Blood sugar levels should be monitored regularly, as these patients are at higher risk of hypoglycemia.
- ii. **Caution with DTG:** DTG should be used cautiously in individuals with diabetes. If necessary, replace DTG with **EFV** in first-line therapy or **ATV/r** in second-line therapy.

For more information on comorbidities and co-medications, refer to Chapter 12.

5 PAEDIATRIC HIV AND AIDS-RELATED CONDITIONS

5.1 Diagnosis of HIV Infection in Children below 18 Months

Children under 18 months of age born to HIV-infected mothers should undergo DNA PCR testing to confirm HIV infection. Maternal antibodies can persist in a child's blood for 9 to 18 months after birth, making it challenging to confirm HIV infection using rapid HIV antibody tests. However, children older than 9 months can be screened for HIV infection using antibody tests, with positive results confirmed by DNA PCR according to the established algorithm Figure 5.1:1.

Summary of Diagnosis of HIV infection in children <18 Months who are being breastfed by a mother known to be HIV Positive

- Do HIV DNA-PCR at birth to all exposed babies; if positive, take another sample immediately for DNA PCR confirmatory test and initiate the child on ART.
- Children with negative results for HIV should have DNA-PCR test at 6 weeks, 9 months of age. Another test should be performed 12 weeks after cessation of breastfeeding using antibody test. At 18 months of age, a final rapid antibody test should be done. Any results for a child below 18 months of age should be confirmed by DNA-PCR test and initiated on ART if found positive.

Note:

- *If the second (confirmatory) DNA PCR test is negative following a first positive DNA PCR result, a third DNA PCR test should be performed before considering the interruption of ART. (Refer to the PMTCT chapter on “Managing Discordant Test Results and Treatment Interruption in HIV-Exposed Children”).*
- *For a child who has never breastfed and will never breastfeed, a second negative DNA PCR test at 6 weeks of age excludes HIV infection.*
- *For a child who has completely stopped breastfeeding for more than 12 weeks before undergoing DNA PCR testing, a negative DNA PCR test excludes HIV infection.*
- *All positive antibody tests should be confirmed with a DNA PCR test. However, if a DNA PCR test is not readily available, and the child is symptomatic and classified as WHO Stage 3 or 4 (Table 4.2), a presumptive HIV diagnosis should be made based on a positive HIV antibody test, and ART should be initiated. The diagnosis must be confirmed when the DNA PCR test becomes available.*

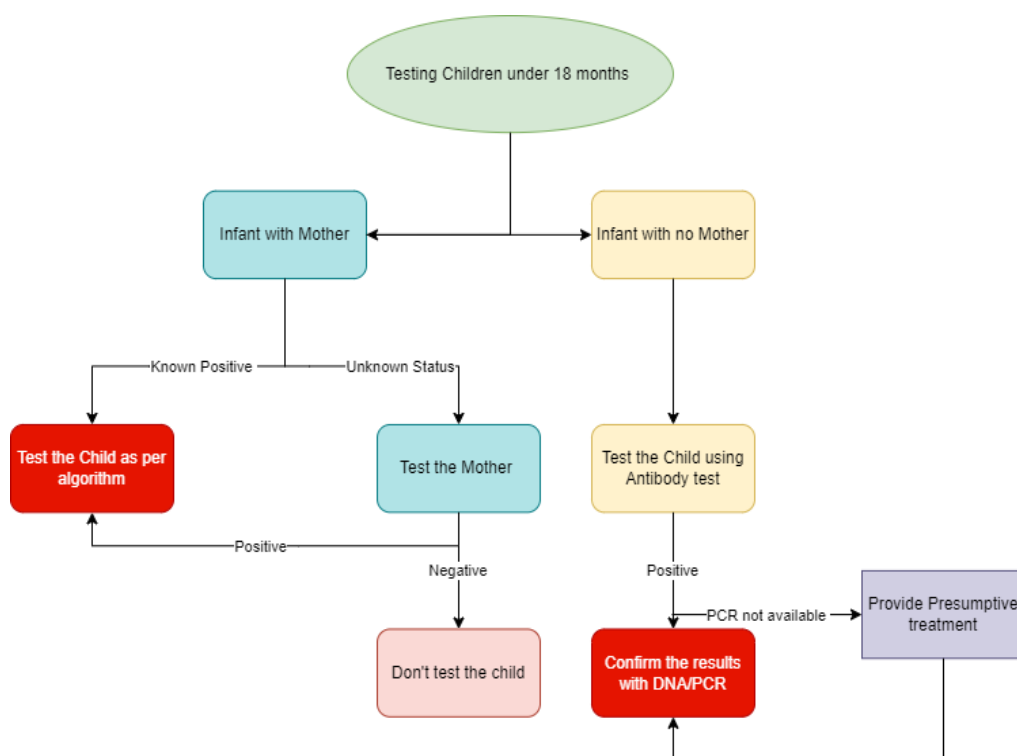


Figure 5.1:1 Flowchart for diagnosis of HIV in children <18 months

Note: *HIV Exposed Infants (HEI) should receive routine medical check-ups during their first year of life and thereafter, and healthcare providers should closely monitor their growth and developmental milestones. Poor growth and delayed development may indicate HIV infection*

5.1.1 Presumptive Diagnosis and Treatment

A presumptive diagnosis of HIV infection should be made if the child fulfils the criteria in Table 5.1:1.

Table 5.1:1 Criteria for presumptive diagnosis of severe HIV infection in infants and children <18 months

A presumptive diagnosis of severe HIV should be made if:		
1. A child has a positive rapid HIV antibody test result	AND	2 a. The child is symptomatic with two or more of the following: Oral thrush, Severe pneumonia, or Severe sepsis. OR 2 b. Any child who is fulfilling WHO stage 3 or 4 criteria

History of maternal high viral load and /or an HIV-related maternal death supports the presumptive diagnosis of HIV infection in a child with appositive antibody test. ART should be initiated immediately, and HIV confirmed as soon as possible.

5.1.2 Managing Discordant Results and Treatment Interruption in HIV-Exposed Children

A confirmatory DNA PCR test should be collected immediately after initiating ART following an initial positive DNA PCR result. If the second (confirmatory) DNA PCR is negative, a third DNA PCR should be performed before considering the interruption of ART.

When assessing patients for ART interruption after discordant test results (a positive first test, a negative second test, and a third test with a negative result), the following considerations should be made:

- The infant must have no clinical signs or symptoms suggestive of HIV infection.
- An active follow-up plan should be agreed upon with the family/ caregiver(s) and the healthcare provider to ensure that a potentially infected infant is retained in care and promptly re-initiated on treatment if virological rebound occurs.
- Contact information (e.g., phone number, address) for the family/ caregiver(s) must be collected and confirmed.
- Infants who develop signs or symptoms indicative of HIV infection should undergo immediate testing.

Note:

Virological rebound in HIV-infected infants who started treatment early is expected to occur within 8 months of interruption in more than 99% of cases.

5.1.3 Follow-Up of an Infant with ART Interruption

In settings where both early infant diagnosis (EID, qualitative) and viral load (VL, quantitative) tests are available, testing should be performed at 4 weeks, 4 months, and 8 months after treatment interruption.

Infants who test positive during follow-up should be promptly re-initiated on treatment according to current guidelines, and a confirmatory sample should be collected.

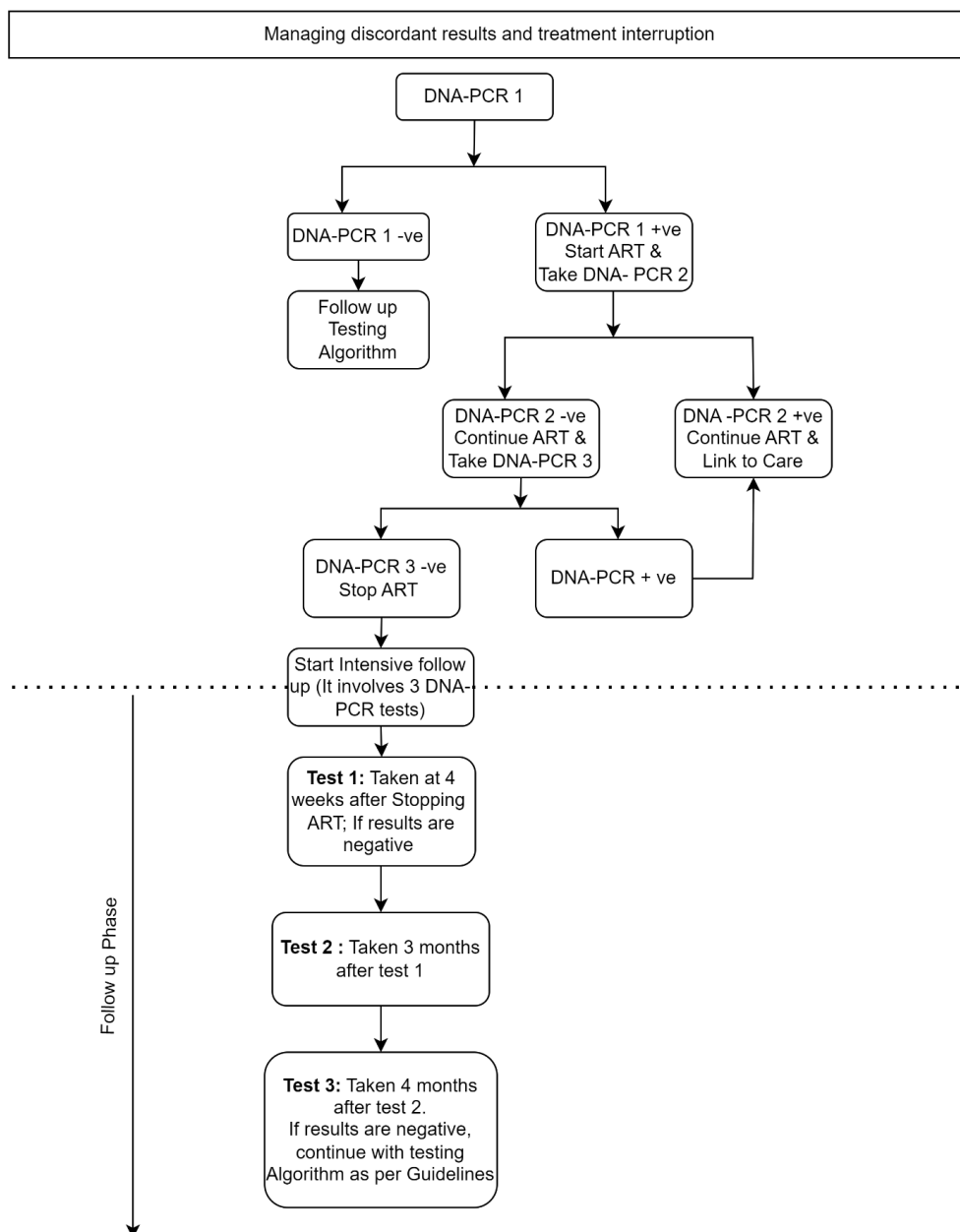


Figure 5.1:2 DNA-PCR test before and after ART interruption

Follow up on ART interruption should consider the continuous risk of transmission resulting from breastfeeding and, once the intensive follows up is completed (that is 8 months after treatment interruption), the national infant testing schedule for HIV-exposed infants should be applied to ensure an appropriate final diagnosis Figure 5.1:2.

If breastfeeding has stopped prior to the end of the intensive follow up, final HIV status can be defined with DNA-PCR performed at least 12 weeks post cessation of breastfeeding.

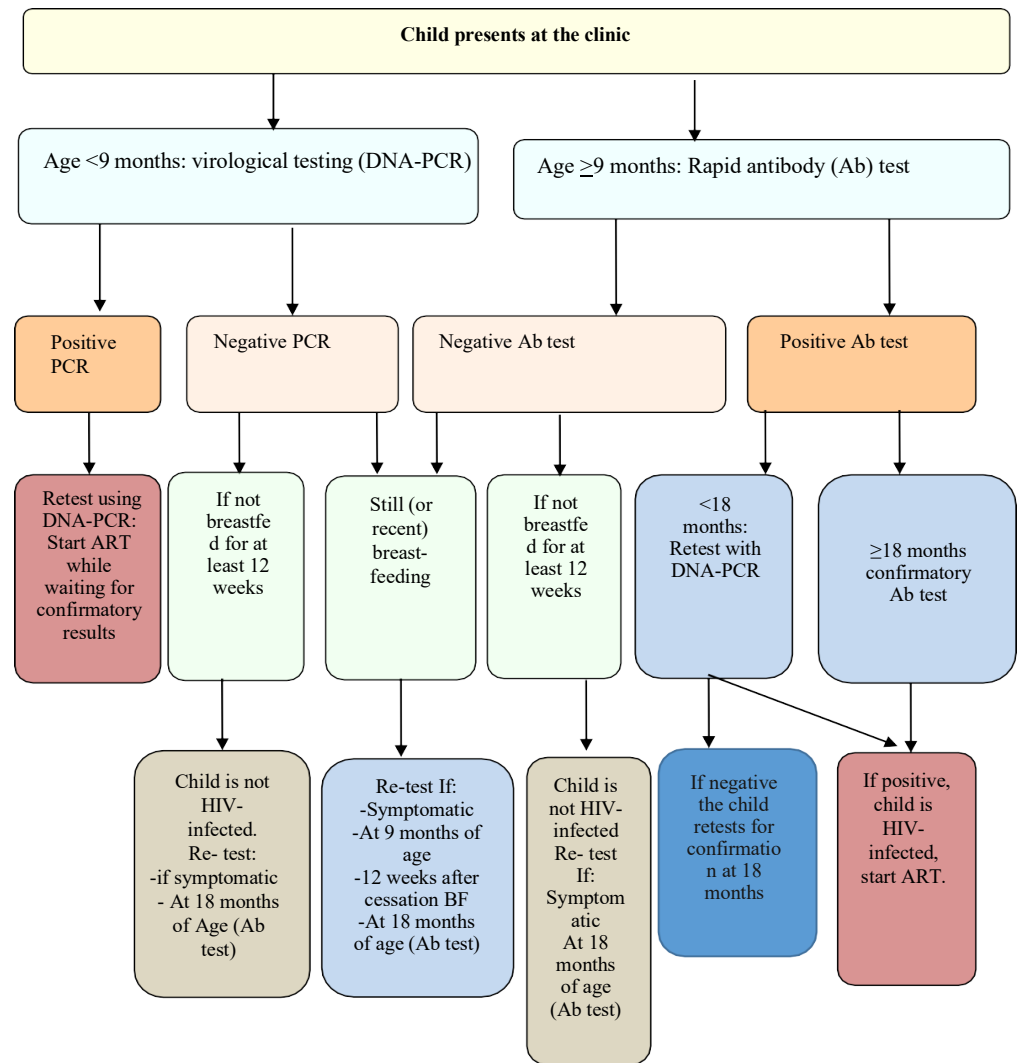


Figure 5.1:3 HIV Testing Algorithm for Infants and Young Children

Note: If the HIV DNA-PCR results are inconclusive, a repeat HIV DNA-PCR and viral load should be done the earliest possible.

5.2 Antiretroviral Therapy in Children Living with HIV

Goals of Antiretroviral Therapy in Children

The goals of ART are to:

- Achieve maximal and durable suppression of plasma HIV RNA (to <50 copies/mL).
- Restore and preserve immunologic function.
- Reduce HIV-associated morbidity and extend both the duration and quality of survival.
- Prevent HIV transmission.

5.3 Initiation of ART for Children Under the Age of 15 Years

All children with a confirmed or presumptive diagnosis of HIV infection are eligible for ART, regardless of age, CD4 count/percentage, clinical stage, co-infection status, or risk group. ART should be initiated as soon as possible, preferably within 7 days of confirmation of HIV status, provided the caregiver or parent is willing, supportive, and ready to adhere to follow-up care recommendations.

Note:

Severe acute malnutrition, opportunistic infections (e.g., tuberculosis, cryptococcal infection), and other acute medical conditions should be screened for and appropriately treated before initiating ART.

5.4 First-Line ARV Regimens for Children Under the Age of 15 Years

When determining the appropriate dosage for children under 15 years of age, both weight and age should be considered, with weight being the preferred criterion (refer to Table 5.4:1). For detailed dosing information on ARV regimens, refer to Annex 18.4: Paediatric Dosing Charts 1–3.

Table 5.4:1 Summary of first line ART regimens for children under the age of 15 years

Age	Weight	Preferred first line	Alternative
Neonates	<3kg	AZT/3TC + NVP syrup (Titrate according to weight)	
Infant ≥ 4 weeks	3-5.9Kg	ABC/3TC +DTG10mg	ABC/3TC+ LPV/r (Granules)
	6-9.9Kg	ABC/3TC +DTG10mg	ABC/3TC+ LPV/r (Granules)
	10 – 19.9Kg	ABC/3TC +DTG10 mg	ABC/3TC + LPV/r (Granules/Tabs) In Special Circumstances: AZT/ 3TC + LPV/r (Granules/Tabs)

Age	Weight	Preferred first line	Alternative
	20 – 29.9Kg	ABC/3TC+DTG50 mg	ABC/3TC + LPV/r (Granules/Tabs) In Special Circumstances: AZT/3TC +LPV/r (Granules/Tabs) AZT/3TC +DTG
	≥ 30Kg	TDF/3TC/DTG (TLD)	ABC/3TC+ DTG *TDF/3TC/EFV (TLE400) Special situation **TDF/FTC + LPV/r (Granules/Tabs)
HIV and TB co-infections		Double dosage of DTG and or LPV/r	

Note

- *Children weighing 6 to 24.9 kg currently on the pediatric regimen of pDTG + pABC/3TC should transition to the fixed-dose combination (FDC) dispersible tablet of pediatric ABC/3TC/DTG 60/30/5 mg (pALD) when available.*
- *TAF is contraindicated in TB/HIV co-infection because Rifampicin lowers TAF concentrations below therapeutic levels.*
- *Consider a PI-based ARV regimen if neither DTG nor EFV can be used.*
- *For children on a Rifampicin-based TB regimen, the DTG dose should be administered twice daily to achieve therapeutic levels.*
- *TDF/3TC/EFV (TLE400) should be used only in cases of DTG intolerance, as NNRTI-associated transmitted drug resistance exceeds the recommended threshold (≥10%) for not using NNRTIs.*
- *LPV/r should only be used as a first-line regimen if both DTG and EFV are not tolerated.*

Special Considerations for LPV/r Granules and Tablets

- LPV/r granules should be prescribed for infants with HIV infection who develop drug reactions to DTG.
- LPV/r tablets must be swallowed whole and should not be split or crushed, as this reduces their effectiveness.

5.5 Second-Line ARV Regimens in Children Under the Age of 15 Years

Children generally experience lower viral load suppression rates because they respond more slowly to ART. However, with optimized ART and good adherence, they can achieve and maintain effective viral suppression.

Healthcare providers must ensure that caregivers/parents and children are well-educated about ART frequency, dosage, and the benefits of good adherence to avoid treatment failure.

Treatment Failure in Children

- Virological treatment failure is confirmed when a child adherent to their current ART regimen for six months or more has two consecutive viral load measurements over 1,000 copies/mL taken three months apart.
- For children under 2 years, the viral load threshold is 0.7 log, and for children over 2 years, it is 0.5 log.
- Before switching to a new ART regimen, healthcare providers should assess adherence, as poor adherence is the most common cause of virological failure.

Note:

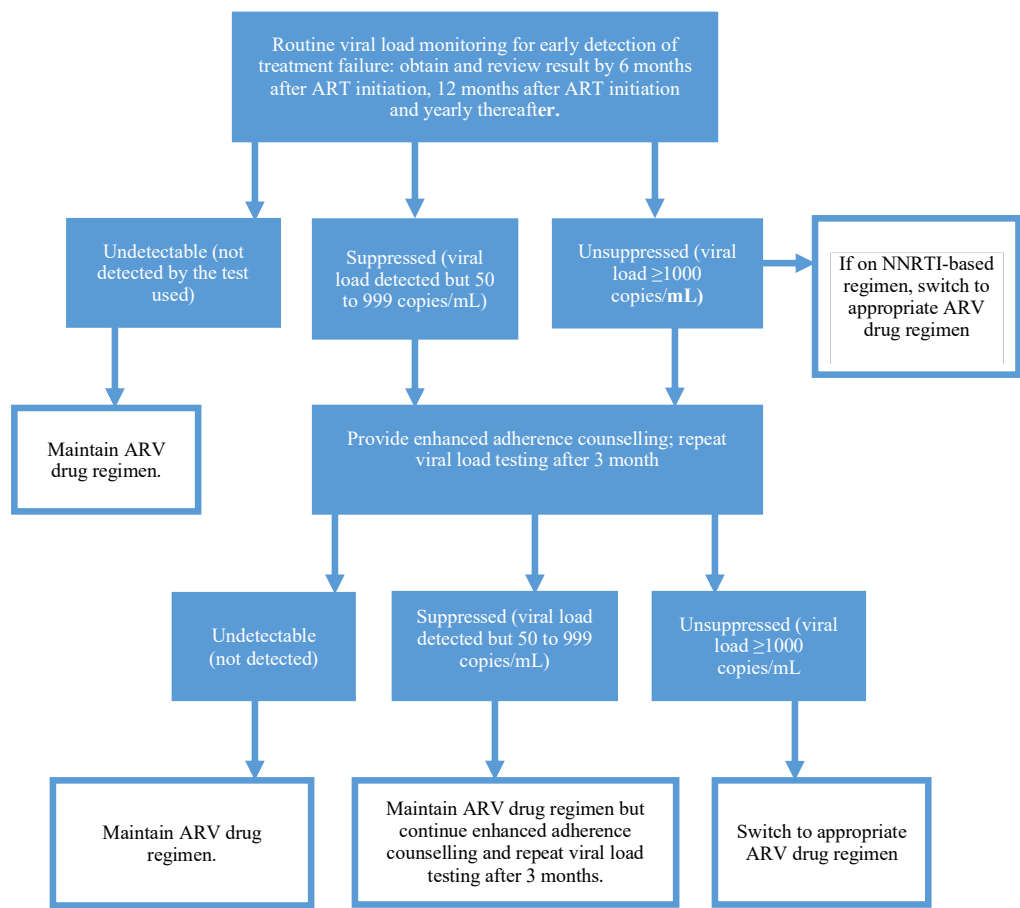


Figure 5.5:1 HIV viral load monitoring algorithm

5.5.1 EAC for Recipients of Care with Low-Level Viremia

The goal of Enhanced Adherence Counseling (EAC) for recipients of care with low-level viremia is to help them achieve an undetectable viral load of ≤50 copies/mL.

- The use of point-of-care viral load testing is encouraged, where available, to facilitate same-day delivery of results.
- If point-of-care testing is not available, repeat viral load specimens and results should be prioritized throughout the entire laboratory referral process, including specimen collection, testing, and result delivery.

5.5.2 Immunological Treatment Failure

In cases where adherence is good and a viral load (VL) test is not available, immunological criteria that indicate a need to switch to second-line therapy include the following:

- [Include criteria here, if provided in subsequent content or guidelines.]

Table 5.5:1 CD4 criteria for immunological failure

Immunological failure is recognized as developing or returning to the following age-related immunological thresholds after at least six months on ART, in a treatment-adherent child	
<5 years of age (Use Percentages)	CD4 count of <200 cells/mm ³ or CD4 <10%
≥5 years of age (Use absolute Counts)	CD4 count of <100 cells/mm ³
a Preferably, at least two CD4 measurements should be available Use percentage CD4 in children <5 years and absolute CD4 cell counts in those ≥5 years of age. If serial CD4 values are available, the rate of declines of CD4 cell count from the peak, CD4 cell count reached should be taken into consideration. CD4 cell percentage should not be measured during an inter-current infection but can be determined when the child has recovered. If there is a modest decline in CD4 cell count or percentage (<5%); and if there is no failure to thrive do not change medication, instead maintain close monitoring.	

5.5.3 Clinical treatment failure

Occurs when new or recurrent clinical event indicating advanced or severe immunodeficiency (WHO clinical stage 3 and 4 with the exception TB) after six months of effective treatment. Healthcare providers should consider switching to a second-line treatment after evaluating the viral load results and in the presence of good adherence to treatment.

Table 5.5:2 Recommended second-line ART regimens for children under the age of 15 years

Body weight	If is on the following regimen	Preferred 2 nd line regimen	Alternative 2 nd line regimen
3-5.9Kg	ABC+3TC +DTG-10mg	AZT+3TC +LPV/r (granules)	
6-9.9Kg	ABC+3TC +DTG-10mg	AZT+3TC +LPV/r (granules)	
10 – 19.9Kg	ABC+3TC +DTG-10mg	AZT+3TC +LPV/r (Granules/Tabs)	
20 – 29.9 Kg	ABC+3TC+DTG-50mg	AZT+3TC +LPV/r (Granules/Tabs)	AZT+3TC +ATV/r (if weight ≥25Kg)
≥ 30Kg	TDF+3TC+DTG-50mg (TLD)	AZT+3TC+ATV/r	AZT+3TC +LPV/r (Granules/Tabs)
			ABC+3TC +ATV/r
			ABC+3TC +LPV/r (Granules/Tabs)
For TB/HIV co-infection already on LPV/r	ABC + 3TC +LPV/r (Granules/Tabs)	If < 20Kg give AZ-T+3TC+ LPV/r (Granules/Tabs) (double dose LPV/r)	
		If ≥ 20Kg AZ-T+3TC+DTG (Double the dose of DTG)	
			If ≥ 30Kg TD-F+3TC+DTG (Double the dose of DTG)
For TB/HIV co-infected on TLD	TDF+3TC+DTG	AZT + 3TC + LPV/r (Granules/Tabs) (double LPV/r dose)	ABC+3TC+LPV/r (Granules/Tabs) (double LPV/r dose)

NB: Children on rifampicin based TPT medication should receive double dose of DTG or Lopinavir.

5.6 Third-Line ARV Regimens for Children Under the Age of 15 Years

Children on second-line ART should be closely monitored both clinically and through laboratory investigations to identify treatment failure. The criteria for diagnosing treatment failure in children on second-line therapy are similar to those for first-line failure.

Efforts must be made to assess and optimize adherence and to rule out any significant drug interactions. If these steps have been taken and there is still evidence of failure, children should be switched to a third-line regimen.

5.6.1 Initiation and Management of Third-Line Regimens

- Third-line therapy initiation and management will be conducted at referral hospitals for the first three months.
- Afterward, consultations and monthly refills can be provided at a nearby health facility, with the third-line regimen dispensed from the referral hospital to the client's facility.

5.6.2 Eligibility Criteria for Switching Children to a Third-Line Regimen

Children on LPV/r or DTG-based regimens are eligible for a third-line regimen if they meet all the following conditions:

- Confirmed HIV genotypic drug resistance through GART.
- Good ARV adherence, including consistent dosing, correct frequency, and proper administration.
- Unsuppressed viral load (>1,000 copies/mL) after completing three Enhanced Adherence Counseling (EAC) sessions.
- A log drop in viral load of <0.5 for children aged 2 years and older, or <0.7 for children under 2 years.

Table 5.6:1 Recommended 3rd line ART regimen in children

Client group	Third line options
Children < 20kg	DTG 10mg + DRV/r* + 3TC + other NRTI**
Children > 20kg and above	DTG 50mg + DRV/r* + 2NRTI**

*Darunavir/ritonavir (DRV/r) should be used for children ≥ 14 kg

**2NRTI will be guided by GART

Practice points

Confirmed HIV-positive infants born to mothers with confirmed PI or DTG resistance mutation should have GART done to exclude transmitted mutations to formulate an appropriate first-line ART regimen. While awaiting GART results, newly confirmed positive infants should be initiated on the default first-line regimen.

The genotypic resistance test should be used to determine the third-line regimen to be used.

5.6.3 Follow-Up of Children on Third-Line Regimens

Children on third-line regimens require extensive support due to the complexity of the regimens, increased pill burden, potential for drug intolerance and side effects, and the likelihood of treatment fatigue and emotional stress.

- HVL Testing: Conduct HVL testing at 6 and 12 months after initiating third-line therapy and then annually or as clinically indicated.

- **CD4 Monitoring:** If the CD4 count is ≤ 350 cells/mm³ at the start of third-line therapy, monitor the CD4 count every 6 months until it exceeds 350 cells/mm³. After that, follow the recommended schedule for CD4 and HVL monitoring.
- **HVL Algorithm:** If, after 6 months of third-line ART, the HVL remains $>1,000$ copies/mL, follow the HVL algorithm from the beginning.

5.7 Monitoring of Children Receiving ART

Children living with HIV (CLHIV) on ART should be assessed clinically and monitored through laboratory investigations to evaluate their response to therapy.

Drug Toxicity

When toxicity is linked to a specific drug in the regimen, the offending drug should be replaced with an alternative that does not have similar side effects.

Clinical Monitoring

- **Routine Assessments:** Children living with HIV require regular evaluations of nutrition, growth, and developmental milestones at every visit. These assessments include monitoring weight, height, mid-upper arm circumference, head circumference, feeding practices, nutritional status, neurologic symptoms, and developmental milestones.
- **Medication Plans:** Discuss medication regimens, including OI prophylaxis and ARV therapy, thoroughly with parents or guardians.
- **Toxicity and Adverse Events:** Monitor children closely for signs of drug toxicity and adverse events while providing age-appropriate counseling and psychosocial support, including disclosure sessions.

5.7.1 Laboratory Monitoring

- Conduct regular laboratory investigations as outlined in Table 5.7:1 to guide decision-making for the child's HIV care management.
- **Point-of-Care (POC) Testing:** Where possible, use POC platforms (e.g., GeneXpert) for viral load testing. POC testing is particularly recommended for monitoring treatment in specific populations, including children, pregnant women, and breastfeeding women suspected of treatment failure.

Table 5.7:1 Laboratory parameters for monitoring children under 15 years at baseline and during ART

Laboratory tests for diagnosis and monitoring	Baseline	At Switching to 2 nd and 3 rd line	Every 6 months	Symptom directed
HIV diagnostic testing	√ ^a			
Haemoglobin	√	√		√
WBC and differential count	√			√
%CD4+ or absolute CD4 cell count	√	√	√ ^b	√
Pregnancy testing in adolescent girls		√		√
Full chemistry (including, but not restricted to, liver enzymes, renal function, glucose, lipids, amylase, lipase and serum electrolytes)		√	√ ^c	√
HIV VL measurement		√	√ ^d	√
OI screening (where possible)	√	√	√	√

a. CD4 Monitoring for Children Under 5 Years

- Continue CD4 monitoring every six months for children under 5 years of age.
- A CD4 cell count should also be performed when there is an emergence of WHO Stage 3 or 4 disease.

e. Regular Monitoring of Chemistry for Infants and Children on ART

- Regular monitoring (every six months) of full chemistry, including lipid levels, liver enzymes, and renal function, should be considered for infants and children on ART.

f. Viral Load Monitoring

- Viral load monitoring should be done annually if the first two viral load results, taken 6 months apart, are <50 copies/mL.

5.8 Differentiated Service Delivery Models in Children

Children experience lower rates of HIV diagnosis, treatment uptake, and viral load suppression. To address these challenges, child-specific Differentiated Service Delivery Models (DSDM) should be used to meet the evolving needs of children according to their age.

The Child-Centered Family Care Approach (CCFCA) provides an opportunity to involve parents, caregivers, and other family members in the care of children living with HIV. This approach enhances access to

HIV services for families, improves uptake of care, facilitates disclosure, and reduces stigma and discrimination.

5.8.1 Differentiated HIV Testing and Linkage for Children

- Early identification of children exposed to or living with HIV is crucial to ensure they receive timely HIV testing and treatment services.
- Healthcare providers (HCPs) should emphasize **Early Infant Diagnosis (EID), index testing, and optimized Provider-Initiated Testing and Counseling (PITC)** for children.
- Parents and caregivers should be educated on the benefits of timely diagnosis.

5.8.2 HIV Testing Services (HTS):

- HTS should be available 24/7 in facilities with outpatient (OPD) and inpatient (IPD) departments.
- Community-based HIV testing services should adapt their timing based on the needs and context of the target population.

5.8.3 Pediatric Index and Family Testing:

- HIV testing services should be offered to all individuals under 19 years of age (biological or non-biological) living with an HIV-positive caregiver, parent, or caretaker.

5.8.4 Pediatric Enhanced PITC (Universal Testing for Children):

- All children seeking medical care at health facility entry points should be offered HIV testing, regardless of screening status.
- A **risk assessment tool** should be used for all children visiting OPD to guide testing decisions.

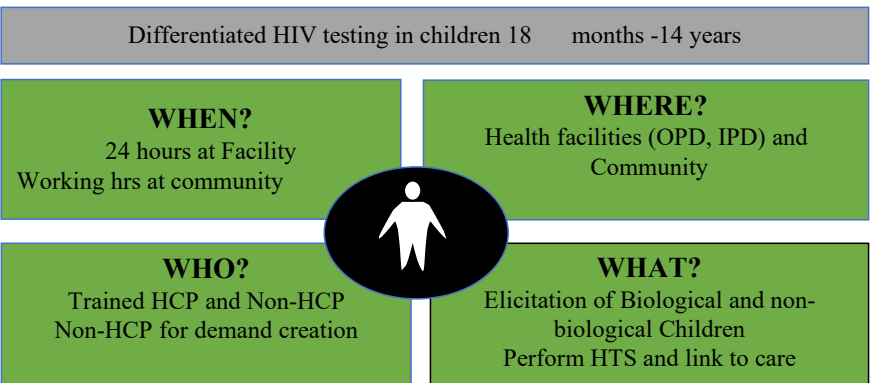


Figure 5.8:1 Differentiated testing for children <18 months of age

If possible, HIV-positive children should be escorted to the care and treatment centre (CTC) for immediate linkage, and those testing negative should be referred to relevant prevention services based on age. A referral form must be completed.

5.8.5 Linkage to OVC Services

All children infected with or affected by HIV should be linked to the **Orphans and Vulnerable Children (OVC) program** and other psychosocial services. These services provide essential community-based support, including psychosocial care and interventions targeting household economic strengthening. These efforts aim to enhance the socio-economic capacity of OVC families.

5.8.6 Differentiated ART Delivery for Children Not Established on ART

To promote a **family-centered approach to care**, children and their HIV-positive parents should have the same appointment dates for clinical review, regardless of the stability status of the parents. Scheduling families for the same appointment date fosters a supportive environment for both children and their caregivers, facilitating better adherence and comprehensive care.

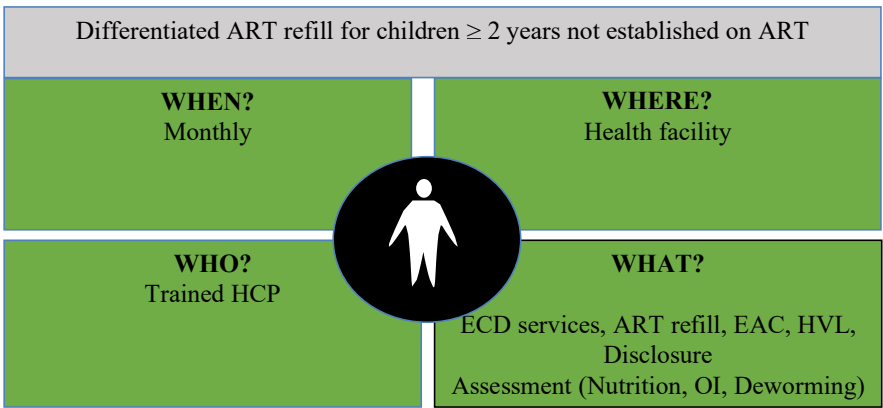


Figure 5.8:2 Differentiated ART for children

Note

- *All children below 2 years of age are considered not established on ART. Regardless of the caregiver’s ART status, appointments for both the parent/caregiver and the child should be scheduled monthly. Community models for ART delivery should be selected as outlined in Chapter III.*

5.9 Models to Support Improvement in the Quality of Care for Children

5.9.1 Clinical Audit, Morbidity, and Mortality Review

Healthcare providers should conduct regular morbidity and mortality audits and reviews to enhance the quality of care for children. This includes: Identifying cases, Reviewing clinical data, conducting interviews with caregivers and staff, identifying contributing factors and making recommendations for improvement. Children living with HIV (CLHIV) under 5 years should be prioritized due to their high risk of mortality. If resources allow, the deaths of all CLHIV should be reviewed.

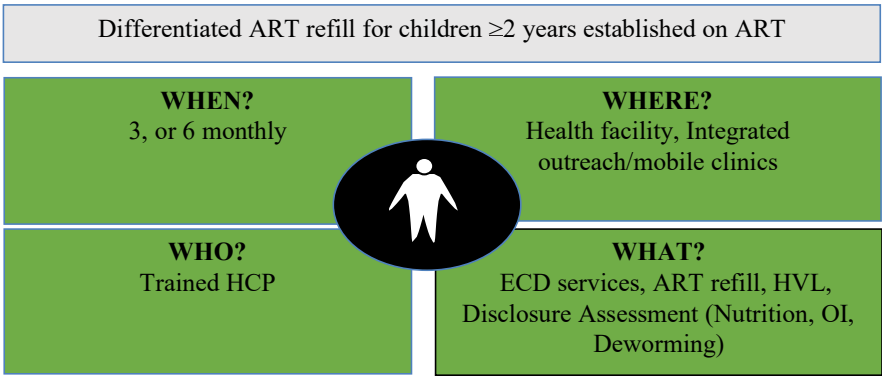


Figure 5.8:3 Differentiated ART refill for children >5 years established on ART

5.9.2 Transitioning of CLHIV in Care and Treatment Clinics

As children living with HIV grow into adulthood, transitioning to adult care settings becomes essential. This process requires flexibility, interaction, and prior planning by health teams, CLHIV, and their caregivers to ensure successful outcomes.

The transition occurs in two steps:

- **Step 1:** Transition from pediatric to adolescent HIV services (ages 8–10 years).
- **Step 2:** Transition from adolescent to adult HIV services (19 years old).

Note: *For further details, refer to the National Standard Operating Procedures on Transitioning Children Living with HIV in Care and Treatment Clinics.*

5.10 Prophylactic Treatment of Common Opportunistic Infections in Children

- Children under 5 years of age should receive Cotrimoxazole prophylaxis, regardless of symptoms or CD4 percentage, until they turn 5 years old. After age 5, prophylaxis should continue for symptomatic children or those with a CD4 count below 350 cells/mm³.
- Discontinuation of prophylaxis can be considered for children over 5 years with a CD4 count above 350 cells/mm³ and good adherence to ART.
- All children should be assessed for tuberculosis and offered Tuberculosis Preventive Therapy (TPT) as appropriate (refer to Chapter VIII for details).

5.11 Clinical Manifestations of Pediatric HIV Infection/AIDS-Related Conditions

5.12 Respiratory Conditions in Children with HIV Infection

Pneumonia and chronic lung diseases are common in immune-suppressed children, contributing to high morbidity and mortality.

5.12.1 Bacterial Pneumonia

Common Causes:

- *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Staphylococcus aureus*, and gram-negative bacteria such as *Klebsiella pneumoniae*.

Diagnosis:

- **History:** Fever, cough, and tachypnea (fast breathing).
- **Physical Examination:** Severe pneumonia signs (chest indrawing, cyanosis, lethargy), crepitations (crackles), decreased breath sounds, bronchial breathing, and hypoxia (<90% oxygen saturation at room air).
- **Laboratory Tests:** Blood count, blood culture, chest X-ray, sputum induction, or nasopharyngeal aspirate (to exclude tuberculosis or *Pneumocystis jirovecii* pneumonia).

Management of Pneumonia in OPD:

- Oral **Amoxicillin** 40 mg/kg/dose twice daily for 5 days.
- For atypical pneumonia in children >5 years: Azithromycin 10 mg/kg orally once daily for 5 days or **Erythromycin** 12.5 mg/kg orally four times daily for 5 days.
- Administer **paracetamol** for fever.

- High-dose Cotrimoxazole is recommended only for suspected *Pneumocystis jirovecii* pneumonia (PJP).
- Cough syrups are not indicated.

Management of Severe Pneumonia:

- **Supportive Care:**
 - o Monitor oxygen saturation with pulse oximetry and administer oxygen if levels are <90%.
 - o Ensure adequate hydration and nutrition.
 - o Provide paracetamol for fever and pain.
- **Specific Therapy:**
 - o **Ampicillin** 50 mg/kg IV/IM every 6 hours for 7 days and **Gentamicin** 7.5 mg/kg IV/IM once daily for 7 days.
 - o If no improvement within 48 hours, switch to Ceftriaxone 80 mg/kg IV/IM once daily.
 - o High-dose Cotrimoxazole is indicated for suspected PCP cases.
- For pneumonia with *Staphylococcus aureus* or complications like pneumatoceles, add **Clindamycin or Vancomycin**.

5.12.2 Lymphocytic Interstitial Pneumonitis (LIP)

Clinical Presentation:

- Chronic cough, difficulty breathing, cyanosis, digital clubbing, and chest X-ray may indicate diffuse bilateral reticulonodular infiltrate similar to miliary TB consolidation. Often mistaken for tuberculosis.

Management:

- Initiate or continue ART.
- Administer **Prednisone** 2 mg/kg/day for 2 weeks, then taper based on response. Monitor for TB reactivation.
- Provide oxygen therapy during hypoxia episodes.
- Use bronchodilators like salbutamol for wheezing.
- Treat concurrent infections with antibiotics.
- Offer supportive care, including nutritional support and correction of anemia.
- Refer to specialist care if the child shows poor response to treatment.

5.12.3 Pneumocystis Jirovecii Pneumonia (PJP)

PJP is a leading cause of severe pneumonia and mortality among HIV-infected infants, particularly in the first year of life. This can occur even in infants who appear healthy and show no symptoms of HIV infection.

Diagnosis

- Diagnosing PJP requires a high index of suspicion and relies primarily on clinical presentation. Early therapy initiation alongside bacterial pneumonia treatment is critical.
- Common symptoms include respiratory distress (chest indrawing, cyanosis, inability to drink), with or without fever. Infants may exhibit persistent cyanosis/hypoxia ($\text{SpO}_2 < 90\%$), hepatosplenomegaly, oral thrush, and lymphadenopathy.
- Chest X-ray findings may show hyperinflation, diffuse infiltrates, or appear normal. Diagnostic tests include sputum induction with nasopharyngeal aspirate stained with Giemsa or silver stains, or bronchoalveolar lavage if available.

Management of PJP

Specific Management:

- High-dose Cotrimoxazole: TMP 8 mg/kg and sulfamethoxazole 40 mg/kg every 8 hours for 21 days.
- Prednisone: 1–2 mg/kg/day for 7–14 days, tapered if given beyond 7 days.
- Secondary prophylaxis with Cotrimoxazole after acute PJP treatment.

Supportive Management:

- Oxygen therapy.
- Hydration monitoring and maintenance.
- Antipyretics for fever.
- Nutrition support.
- Continued therapy for bacterial pneumonia.

5.12.4 Tuberculosis in Children

HIV-infected children should be evaluated for tuberculosis (TB) at HIV diagnosis and whenever they present with symptoms suggestive of TB or recent exposure to an infected individual. The overlap of TB with other HIV-related lung diseases complicates diagnosis. For detailed guidelines, refer to Chapter VIII.

5.12.5 Diarrheal Disease

Diarrheal disease is a major cause of mortality in children under 5, particularly in HIV-infected children, where it tends to be more severe and prolonged (>14 days).

Types of Diarrhea

- **Acute watery diarrhea:** Non-bloody, lasting <14 days.
- **Dysentery:** Bloody diarrhea.
- **Persistent diarrhea:** Lasting >14 days.

Management

- Follow IMCI guidelines, prioritizing rehydration with ORS.
- Zinc supplementation: 10 mg/day for infants <6 months; 20 mg/day for older infants and children for 10–14 days.
- Continued feeding during and after the episode.
- Ciprofloxacin (15 mg/kg twice daily for 3 days) for bloody diarrhea.
- Micronutrient supplementation for 2 weeks in persistent diarrhea cases.

5.12.6 Oral Candidiasis

Persistent or recurrent oral candidiasis outside the neonatal period is a WHO Clinical Stage III condition.

Management:

- Miconazole oral gel 5 mL twice daily for 2 weeks.
- Nystatin suspension (Infants: 100,000 units every 6 hours; Children: 400,000–600,000 units every 6 hours) for 2 weeks.

5.12.7 Esophageal Candidiasis

Clinical Features:

- Associated with extensive oral thrush.
- **Infants:** Refusal to feed and crying during feeds.
- **Older children:** Painful swallowing.

Management:

- Fluconazole 3–6 mg/kg orally once daily for 2 weeks.
- IV Fluconazole for severe cases or those unable to tolerate oral medications.

5.12.8 Acute Otitis Media

- Symptoms: Ear pain, discharge, fever, inflamed tympanic membrane.
- Management:
 - Amoxicillin 40 mg/kg twice daily for 5 days or Azithromycin 10 mg/kg once daily for 3 days.
 - Paracetamol 15 mg/kg three times daily for 3 days.

5.12.9 Chronic Otitis Media

- Symptoms: Persistent ear discharge with tympanic membrane perforation.
- Management:
 - Ciprofloxacin ear drops 0.3% three times daily.
 - Cephalexin 12.5 mg/kg twice daily for 10 days.

5.12.10 Herpes Zoster

- Management:
 - Acyclovir 20 mg/kg orally or IV every 6 hours for 7 days.
 - Pain relief: Paracetamol or Gabapentin.

5.12.11 Kaposi Sarcoma

- Management includes ART, radiotherapy, or systemic chemotherapy based on disease severity. Refer to Chapter XII for details.

5.12.12 Identifying and Managing Malnutrition in Children

Clinical Presentation of Severe Malnutrition:

- Weight/height Z-score <-3.
- Severe visible wasting or bilateral pitting edema.
- MUAC thresholds: <11.5 cm (6–59 months), <13.5 cm (5–9 years), <16.0 cm (10–14 years).

Management:

- Use therapeutic milk (F75, F100) and ready-to-use therapeutic food (RUTF).
- Assess and counsel on nutrition status at every visit.
- Measure and document anthropometric data (weight, height, MUAC) regularly.

Refer to the Guidelines for Integrated Management of Severe Acute Malnutrition for more details.

6 PREVENTION OF VERTICAL TRANSMISSION OF HIV, SYPHILIS AND HEPATITIS B

6.1 Introduction

The **PMTCT program** aims to eliminate vertical transmission of **HIV, Syphilis, and Hepatitis B** through effective interventions such as antenatal screening, treatment, and vaccination. Without intervention, the risks of vertical transmission are significant, and multiple factors increase the likelihood of a mother transmitting these infections to her child. The program’s goals include: Achieving a **mother-to-child transmission (MTCT) rate** of less than 4% for HIV, reducing congenital syphilis to fewer than 50 cases per 100,000 live births and decreasing the prevalence of **HBsAg** among children to less than 0.1%.

To minimize the risk of vertical transmission of **HIV, Syphilis, and Hepatitis B**, it is essential to assess viral, maternal, and obstetric factors, as outlined in Table 6.1:1.

Table 6.1:1 Viral factors, maternal conditions, and obstetric interventions that may increase the risk of vertical transmission of HIV, Syphilis and Hepatitis B

HIV	Syphilis	Hepatitis B
• High maternal viral load and low CD4 cell count • Viral, bacterial, or parasitic placental infections (e.g. Malaria) • HIV serotype 1 • Chorioamnionitis from untreated STIs or other infections • Premature Rupture of membranes (more than 4 hours before delivery) • Prolonged labour • Oral disease in the infant (e.g. mouth sores) • Breast abscesses, nipple fissures, and mastitis • Duration of breast-feeding • Mixed feeding (i.e. breast-feeding combined with other foods or fluids) before 6 months of age	• Late maternal Syphilis diagnosis • High maternal syphilis titre • Maternal co-infection with HIV and other STIs	• Mothers with high HBV viral load • HBeAg positivity in the mother • Co-infection with HIV or Hepatitis C virus • Invasive procedures during delivery

1.1 Four elements of a comprehensive approach to Prevention of Vertical Transmission of HIV, Syphilis, and Hepatitis B Virus

Healthcare Providers should consider comprehensive approach to PMTCT consisting of four elements that guide interventions:

Four elements of a comprehensive approach

1. *Primary prevention of HIV, Syphilis and HBV infections among women of childbearing age and their partners*
2. *Prevention of unintended pregnancies amongst women living with HIV and HBV and linkage with other Sexual Reproductive Health services including STIs*
3. *Prevention of vertical transmission of HIV, Syphilis and HBV from infected women to their infants*
4. *Provision of treatment, care and support to women infected with HIV, Syphilis and HBV and their partners, infants, and families.*

6.1.1 Primary Prevention of HIV, Syphilis, and HBV Among Women and Their Partners

Primary prevention is the most effective way to control the spread of **HIV, Syphilis, and HBV**, minimizing their impact on individuals, families, and communities. Preventing these infections in women of childbearing age is the cornerstone of preventing vertical transmission.

Primary prevention begins with **HIV, Syphilis, and HBV testing**, achieved through various testing approaches across different settings. **Combination prevention**, which includes PrEP for pregnant and breastfeeding women (PBFW), safer sex practices, behavioral change, and timely diagnosis and treatment of STIs, is essential for achieving primary prevention.

6.1.2 Prevention of Unintended Pregnancies Among Women Infected with HIV and HBV

Family planning is an integral part of a public health strategy to prevent vertical transmission of HIV and HBV. Women living with these infections and their partners should receive family planning counseling and be empowered to access and use effective contraceptive methods to prevent unplanned pregnancies.

The choice of contraceptive methods should be based on the woman's or couple's health status and informed preferences.

6.1.3 Interventions to Prevent Vertical Transmission of HIV, Syphilis, and HBV

The PMTCT program offers a variety of services to reduce the risk of vertical transmission of HIV, Syphilis, and HBV, including:

- Education, counseling, and testing for PBFW and their partners.
- Antiretroviral treatment (ART).
- Maternal HBV mono-prophylaxis.

- Antibiotic treatment for syphilis.
- Prophylaxis for HIV-exposed infants.
- Universal Hepatitis B vaccination with a timely birth dose and/or **Hepatitis B immunoglobulin**.
- Safer delivery practices.
- Counseling on safer infant feeding practices.
- Postnatal care for exposed infants.

6.1.4 Treatment, Care, and Support for HIV, Syphilis, and HBV-Infected Women and Their Families

Providing treatment, care, and support is crucial to address the health needs of pregnant women infected with HIV, Syphilis, and HBV and to ensure the well-being of their children and families.

- **Integrated Services:** PMTCT programs should strive to provide integrated care within RCH clinics. When this is not possible, coordinated referral systems must ensure women, and their families access appropriate care.
- **Monitoring:** Women diagnosed with HIV and HBV should have their viral load, CD4 count (for HIV), and renal and liver function regularly monitored to evaluate disease progression. Achieving viral suppression before delivery and prompt syphilis treatment are critical to minimizing MTCT risk.
- **Follow-Up for Infants:** Exposed infants require close monitoring for growth, development, immunizations, prophylaxis against infections, and early diagnosis and treatment of congenital syphilis and HIV.

Essential Triple eMTCT Services:

- **Testing:** HIV, syphilis, and HBV testing for all pregnant women at RCH clinics, including maternal retesting as per guidelines.
- **Treatment:** Prompt and effective interventions for women who test positive to prevent transmission to their children.
- **Couple Counseling:** Reduces transmission risks and ensures proper treatment.
- **Safe Obstetric Practices:** Ensure pregnant women are attended to appropriately.
- **Infant Follow-Up:** Includes timely HBV vaccine birth dose, HBV immunoglobulin (when available), and ARV prophylaxis for HIV-exposed infants.
- **Infant Feeding Counseling:** Optimal feeding practices and routine immunizations.

- **Lifelong Care:** Lifelong treatment and care for eligible women living with HIV, HBV, or syphilis.
- **PrEP Services:** Offered to PBFW as per the national PrEP implementation framework.

6.1.5 Integrating PMTCT Into Routine Reproductive and Child Health Services

Integrating PMTCT into routine RCH services enables national healthcare programs to enhance pregnancy care and outcomes for all clients.

- Pregnant women should be tested for **HIV, Syphilis, and HBV** upon their first ANC visit.
- Expanded **obstetric and medical care** should address the specific needs of women living with these infections.
- Pregnant women living with HIV or infected with Syphilis and Hepatitis B should attend **monthly ANC clinic visits**, following national RMNCAH guidelines

Table 6.1:2 Essential package of Integrated ANC services for pregnant and breastfeeding women living with HIV and/HBV and / Syphilis infection

Components of essential packages	What needs to be done by health care providers and pregnant and breastfeeding women
Understanding Client's context	Assess need of psychosocial support, mental health support and tailored scheduling of visits
Disease Screening (History and Physical Examination)	Assess the current signs or symptoms of illnesses including HIV, HBV, TB, malaria, cervical cancer and STIs
Laboratory testing	Conduct routine tests and HIV/ HBV-specific laboratory tests (viral load, LFT and RFT and CD4 cell count for HIV) VDRL if indicated, Confirmatory HIV testing (if indicated), Urinalysis, Full Blood Picture (FBP) (or haemoglobin)
HIV staging	Conduct clinical and immunological staging according to the WHO clinical staging categories
Treatment readiness and support	Assess readiness to start ART and assist to resolve barriers of care including disclosure and plan for close follow up
Antiretroviral Treatment (ART)	Provide life-long ART to all HIV positive pregnant women and those eligible for HBV treatment or prophylaxis to prevent vertical transmission i.e Pregnant women testing positive for HBsAg with HBV DNA $\geq$ 200,000 IU/ml (or HBeAg positive) need to receive tenofovir prophylaxis from 28th week of pregnancy until at least birth

Components of essential packages	What needs to be done by health care providers and pregnant and breastfeeding women
Tuberculosis (TB)	Conduct TB screening, initiate TPT and treat TB accordingly
Opportunistic infection (OI) prophylaxis	Provide cotrimoxazole preventive therapy (CPT) to PBFW with CD4 cell count ≤ 350 cells/mm ³
Malaria	Intermittent Presumptive Treatment of malaria with Sulfadoxine Pyrimethamine should be provided. PBFW on Cotrimoxazole Preventive Therapy (CPT) do not need Sulfadoxine - pyrimethamine prophylaxis for malaria. Treat malaria accordingly
STI prevention and treatment	Assess risk, diagnose, and treat STIs according to the national guidelines. Counsel on preventing STIs. Recommend the use of condom throughout pregnancy and breastfeeding period
Adherence to ART, CPT and TPT	Provide counselling and education on healthy pregnancy, HIV and/or HBV care and treatment and PMTCT. Provide adherence sessions accordingly
Nutrition	Conduct nutritional and dietary assessment and provide counselling and supportive services. Give iron, folic acid and multivitamin supplements according to national guidelines.
Planning Delivery	Plan with the client on the mode and place of delivery. Explain the importance of delivering at health facility and birth preparedness.
Tetanus Toxoid	Administer immunization according to national guidelines.
Safe Motherhood	Encourage care seeking behaviour especially in case of symptoms of pregnancy complications such as bleeding, fever, signs and symptoms of pre-eclampsia, severe pallor, or abdominal pain. For pregnant women who present with miscarriage and abortion, comprehensive post-abortion care should be provided accordingly.
HIV/HBV-exposed Infant	Educate about the importance of timely HIV testing, infant ARV prophylaxis, immunoglobulin and vaccine birth dose for HBV
Infant feeding in the context of HIV	Encourage exclusive breastfeeding for first 6 months of life, followed by complementary feeding with continued breast-feeding until 12 months of age. Counsel for gradual cessation of breastfeeding at 12 months of age
Signs or symptoms related to HIV	Provide information and instructions on seeking healthcare for symptoms of HIV disease progression, such as frequent and recurrent illnesses, chronic persistent diarrhoea, oral and oesophageal candidiasis, fever, severe weight loss or signs of any opportunistic infection. Manage client accordingly.

Components of essential packages	What needs to be done by health care providers and pregnant and breastfeeding women
Mental health, Psychological and social support	Assess and address needs for mental health, psychological and social support and provide bidirectional referrals with existing community structures. Encourage joining community support groups
Effective family planning and safer sex	Counsel about consistent use of condoms during pregnancy and breast-feeding period to avoid acquisition of new HIV infection, re-infection and STIs Discuss long-term family planning options with male partner when possible.
Provision of PrEP	Provide PrEP as per PrEP implementation framework

6.2 HIV, Syphilis, and Hepatitis B Testing and Counseling for Pregnant and Breastfeeding Women

All pregnant women and their partners should be counseled and tested for HIV (unless already known to be HIV-positive), Syphilis, and Hepatitis B during their first antenatal care (ANC) visit. If a client is diagnosed with any of these infections, appropriate management should be initiated based on the specific disease.

6.2.1 Maternal Retesting for HIV and Syphilis-Negative Pregnant Women

For women who test negative for HIV and Syphilis, repeat testing should be conducted:

- During the third trimester, between 32 and 36 weeks of gestation.
- Follow-up HIV tests should be performed at 3 months, 6 months, and 18 months post-delivery.

Afterward, testing should follow the guidance provided for the general population.

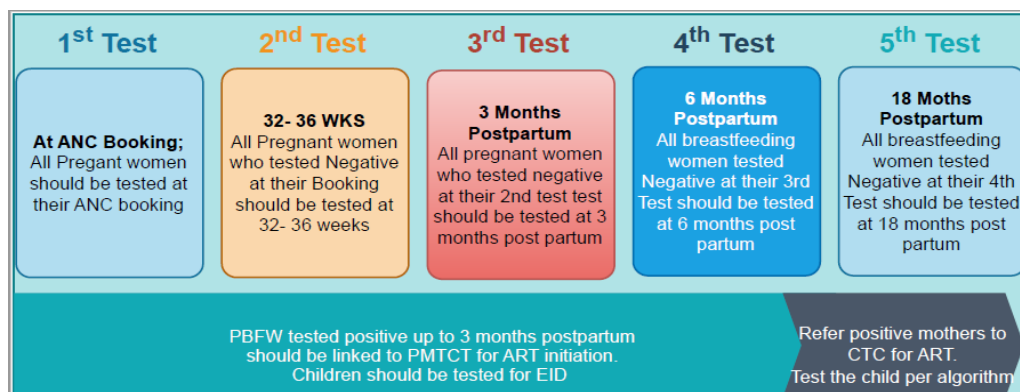


Figure 6.2:1 Maternal HIV testing schedule

Note:

All Pregnant women who did not receive their second test during the third trimester should be tested during labour and delivery. If testing at labour and delivery was not done, testing should be done during the immediate postpartum period (within 42 days). In case the first test was done at gestation age of 32 weeks or beyond, the second test will be conducted at labour and delivery.

6.2.2 Categories of Status of PMTCT Clients According to Risk of Vertical HIV Transmission

HIV-exposed infants are categorized into two groups based on the risk of mother-to-child transmission

6.2.3 High-Risk Exposed Infants

These infants have an increased risk of transmission and include those whose mothers meet any of the following criteria:

- Diagnosed with HIV during pregnancy or breastfeeding.
- Known to be living with HIV but not on ART.
- Started ART less than four weeks before delivery.
- On ART with a high viral load (≥ 50 copies/mL).

6.2.4 Low-Risk Exposed Infants

These are infants whose mothers:

- Are already on ART.
- Have achieved a viral load of < 50 copies/mL (obtained at least 4 weeks apart or within 4 weeks prior to delivery).

Note: *Low-risk status can change to high-risk and vice versa based on the availability of HVL test results. For ARV prophylaxis and the management of HIV-exposed infants, refer to Chapter IV on immediate post-delivery care.*

6.3 Care of Women and Newborns During Labour and Delivery to Prevent Vertical Transmission of HIV and HBV

All labour and delivery services should incorporate interventions to prevent MTCT, including:

- HIV testing for women with unknown status or those not retested during the third trimester (32–36 weeks of gestation).
- Hepatitis B testing for women without prior testing.
- Administration of ART to pregnant women living with HIV and immediate ARV prophylaxis to HIV-exposed infants.
- Universal DNA-PCR birth testing for all HIV-exposed infants.

- Universal Hepatitis B vaccine birth dose for all newborns within 24 hours.
- Hepatitis B Immunoglobulin (where available) for babies born to mothers with high HBV-DNA viral loads ($\geq 200,000$ IU/mL) or HBeAg positive, within 24 hours.
- Safe obstetric practices.
- Appropriate labour and delivery care.

6.3.1 Best Practices for Labour Management

Healthcare providers must follow obstetric best practices and use standard precautions during labour and delivery, as highlighted below:

1. **Use Standard Precautions** (good infection prevention practices) for all patient care.
2. Avoid unnecessary trauma during labour and delivery, such as episiotomies and instrumental vaginal deliveries (e.g., forceps, vacuum).
3. Prevent trauma in the immediate postpartum period by using separate scissors for cord separation and performing gentle suctioning of the newborn's airway.
4. Minimize vaginal examinations and perform them only when necessary, using sterile techniques.
5. Avoid prolonged labour and artificial rupture of membranes.
6. Minimize the risk of postpartum hemorrhage (PPH).
7. Ensure the use of safe, screened blood and blood product transfusion practices.
8. Avoid extensively delayed cord clamping.
9. Provide emotional support and reassurance to mothers.

Note: *All infants delivered at home should be brought to a health facility as soon as possible, preferably within 6 hours after delivery. HIV-exposed infants should receive ARV prophylaxis based on their risk category.*

6.3.2 Immediate Post-Delivery Care

Regardless of the mother's HIV, Syphilis, or Hepatitis B status, all infants should be:

- Kept warm after birth and carefully dried.
- Handled with gloved hands until maternal blood and secretions are washed off.

Healthcare providers must observe standard precautions when caring for newborns.

6.4 Care for HIV-Exposed Infants

6.4.1 Prophylaxis for Low-Risk HIV-Exposed Infants:

- Administer Nevirapine (NVP) syrup immediately after birth (preferably within 72 hours) and continue until six weeks of age.
- Nevirapine prophylaxis should not be initiated for infants presenting after four weeks (>4 weeks) of age.

Table 6.4:1 Infant prophylaxis dosing

Infant NVP dosing recommendations from Birth to 6-weeks	
Birth	NVP daily dosing
Birth weight <2000g	2mg/kg once daily as starting dose (0.2ml/kg)
Birth weight 2000–2499g	10mg once daily (1ml)
Birth weight ≥2500g	15mg once daily (1.5ml)

The recommended NVP dosing is based on the dosing required to sustain exposure in the infant of >100 ng/mL with the minimum dose changes.

6.4.2 Prophylaxis for High-Risk HIV Exposed Infants

- In case a high-risk HIV exposed infant is identified, give enhanced postnatal prophylaxis (ePNP) for a total of 12 weeks as described in the table Table 6.4:2 below

Table 6.4:2 Enhanced postnatal prophylaxis (ePNP) for high-risk HEI

Recommended Regimen	Infant Age	Dosage
AZT/3TC+NVP	Dose 0-6 weeks	AZT / 3TC (60/30mg); ¼ tab twice daily and NVP syrup Once daily (based on body weight).
	Dose 6-12 weeks NVP	NVP - once daily (based on body weight)

6.4.3 Care for Syphilis-exposed infants

All infants born to mothers with Syphilis infection should:

- Carefully be assessed for manifestations of congenital syphilis.
- Ensure proper treatment to infants diagnosed with congenital syphilis (Refer section 5.8.2.1 for management of Congenital Syphilis)

6.4.4 Care for Hepatitis B-exposed infants

All infants born to mothers with HBV infection should receive.

- A birth dose of HBV monovalent vaccine within 24 hours post delivery

- If available, a single dose of 100 IU HBIG intramuscularly should be given together with HBV vaccine birth dose. This should be given to all exposed infants born to mothers with high Hepatitis B DNA viral load (More than 200,000 IU/ML) preferably within 12 hours.
- Subsequent 3 doses of HBV vaccine during immunization schedule at 6, 10 and 14 weeks of age
- Verify if:
 - Mother with high HBV DNA viral load (>200,000IU/ml) is on Tenofovir prophylaxis.
 - Mother with HIV/HBV coinfection is on Tenofovir based regimen.

6.5 Management of HIV-infected women and their infants in the immediate postpartum period

6.5.1 Immediate post-delivery care:

Standard Precautions should be adhered by Healthcare Providers when assessing vaginal bleeding and should safely dispose the blood-stained linens and pads.

Postpartum care for women with unknown HIV status

Women whose status is unknown and those with negative results who were not retested previously should strongly be encouraged to test for HIV and be advised to breastfeed exclusively as per National infant and young children feeding recommendations. Partners and other siblings of HIV exposed infants should be encouraged to receive pre-test information, counselling and HIV testing.

6.5.2 Counselling for Safer Infant Feeding

- All women, regardless of their HIV status, should receive infant feeding counselling during pregnancy (ANC) and postpartum care, as outlined in the guidelines.
- All HIV-positive mothers should be encouraged to exclusively breastfeed for the first 6 months, followed by the introduction of complementary feeding while continuing breastfeeding up to 12 months of age.

Note:

Mothers who opt for **exclusive replacement feeding** must be assessed against the **AFASS criteria** (*Acceptability, Feasibility, Affordability, Sustainability, and Safeness*). They should be counselled that replacement feeding is a permanent option and cannot be alternated with breastfeeding.

6.5.3 HIV Testing and CPT Prophylaxis in HIV-Exposed Infants

Counselling and Testing:

HIV-positive women must be counselled on the importance of infant testing and scheduled for testing before discharge.

6.5.4 HIV Testing for HIV-Exposed Infants

- **Initial Test:** All HIV-exposed infants should have an initial DNA PCR test at birth (within 48 hours).
- **Follow-Up Tests:**
 - For **those who test** HIV-negative, a second DNA PCR test should be done at 6 weeks of age.
 - A **third DNA PCR test** should be performed at 9 months of age.
 - An **antibody test** should be conducted 3 months after breastfeeding cessation.
 - A **confirmation antibody test** should be done at 18 months of age.
- Infants under 18 months who test HIV-positive via DNA PCR should initiate ART immediately. A repeat DBS for DNA PCR should also be collected to confirm the diagnosis.
- For HIV exposed infants at 18 months of age and above, those who test HIV positive with antibody test initiate ART.

In settings where DNA-PCR is not available but antibody test is positive, and the infant has any of the following signs or symptoms.

- Oral thrush
- Severe pneumonia
- Severe sepsis
- Lymph nodes enlargement
- Stunted growth
- Any child fulfilling WHO stage 3 or 4 criteria,

Presumptive HIV diagnosis should be made and initiate ART to infant.

6.5.5 Cotrimoxazole Preventive therapy

All HIV-exposed infants should begin Cotrimoxazole Preventive Therapy (CPT) at the age of six weeks until the end of HIV exposure. Cotrimoxazole should continue up to seven days after the infants have been confirmed having HIV negative. Table 6.5:1 below provides recommended infant cotrimoxazole dosing:

Table 6.5:1 Infant cotrimoxazole preventive therapy dosing recommendations

Age	Co-trimoxazole syrup dose
6 weeks - 6months	2.5mls Once a day
6 months – end of exposure	5mls Once a day

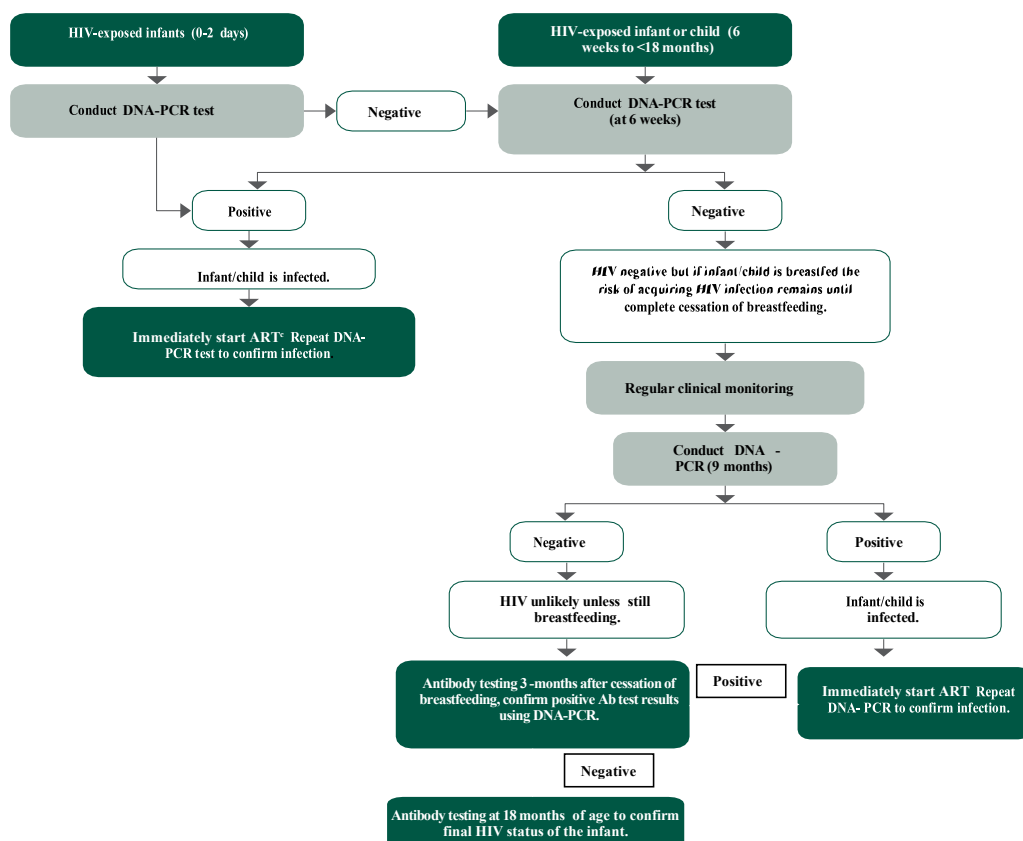


Figure 6.5:1 HIV Exposed infant testing algorithm

NB: *Point of Care testing for HEI using GeneXpert machine at birth using fresh whole blood is recommended whenever available to enable early detection for timely management. HIV exposed infants delivered at home or any other place outside the health facility should be tested within 48 hours.*

6.5.6 Comprehensive Care Schedule for the Mother and Infant

Mothers infected with HIV, HBV, and/or Syphilis, along with their families, require uninterrupted care, treatment, and support services. Healthcare workers should:

- Collaborate with the client to prepare a follow-up plan.
- Ensure the mother is informed of the time, location, contact person, and purpose of all follow-up appointments.
- Facilitate successful referrals and linkages to treatment, care, and support services if the required services are unavailable at the health facility.

6.6 Treatment of Mother to Prevent Vertical Transmission of HIV, Syphilis, and Hepatitis B

6.6.1 Maternal HIV Care and Treatment

Pregnant and breastfeeding women (PBFW) living with HIV should be started on lifelong ART immediately upon diagnosis. The recommended first-line regimen is:

- **Tenofovir disoproxil fumarate (TDF) + Lamivudine (3TC) + Dolutegravir (DTG)** as a once-daily fixed-dose combination.

In cases of **DTG side effects or adverse events**, the alternative regimen is:

- **Tenofovir disoproxil fumarate (TDF) + Lamivudine (3TC) + Efavirenz (EFV 400 mg).**

Before initiating ART, healthcare providers should:

- Explain the health benefits of lifelong ART, emphasizing its role in preventing mother-to-child transmission of HIV.
- Ensure the client fully understands the possible side effects of ARVs.

Clients already on second- or third-line ART regimens should continue their current treatment plans.

Table 6.6:1 : Maternal HIV Care and treatment

Pregnant and Lactating women	1 st line therapy	2 nd line therapy
Preferred Option	TDF + 3TC+ DTG	If TDF was used as first line, use ABC plus 3TC plus ATV/r or LPV/r
Alternative Options	ABC + 3TC + DTG TDF+3TC+EFV (400mg)	If TDF was used as first line, use ABC plus 3TC plus ATV/r or LPV/r If ABC was used as first line, use TDF plus 3TC plus ATV/r or LPV/r If TLE was used as first line, use ABC plus 3TC plus DTG

Note:

- *Tenofovir alafenamide (TAF) is not recommended in pregnancy due to unknown safety profile.*
- *It is recommended that all women on DTG should routinely receive high dose Folic acid (5mg) pre- conception and during pregnancy. Clients with DTG intolerance such as severe liver diseases should not be given DTG containing regimens.*

Whenever possible all women on ART should receive routine LFT and RFT as per guideline on monitoring of individuals on ART

6.6.2 Monitoring of Pregnant and Breastfeeding Women on ART

Successful ART leads to suppression of viral load, immune recovery, and an increase in CD4 cell count. This results in lower transmission risks, improved maternal survival, and enhanced HIV-free survival for the baby.

- CD4+ Lymphocyte Count:
 - o Conduct a baseline test to determine the immunological stage and establish the need for CPT.
 - o For clients with a CD4 count of <350 cells/mm³, repeat the test every 3 months.
 - o If CD4 count exceeds 350 cells/mm³, stop CD4 monitoring and continue with HIV viral load (HVL) monitoring.

HVL Testing Algorithm During Pregnancy and Breastfeeding

This algorithm aims to enable early identification and management of adherence challenges and treatment failure, thereby minimizing the risk of mother-to-child transmission (MTCT) due to high maternal HIV viral load.

a) For Pregnant and Breastfeeding Women Already on ART:

- Conduct viral load testing at the first ANC visit.
- Review results with the recipient of care (RoC) within 2 weeks of receiving them.

b) For Pregnant and Breastfeeding Women Not on ART or Newly Diagnosed:

- Conduct viral load testing 3 months after ART initiation.
- Review results with the RoC within 2 weeks of receiving them.

Figure 5.3 below illustrates the HVL testing algorithm for pregnant and breastfeeding women already on ART and those newly diagnosed

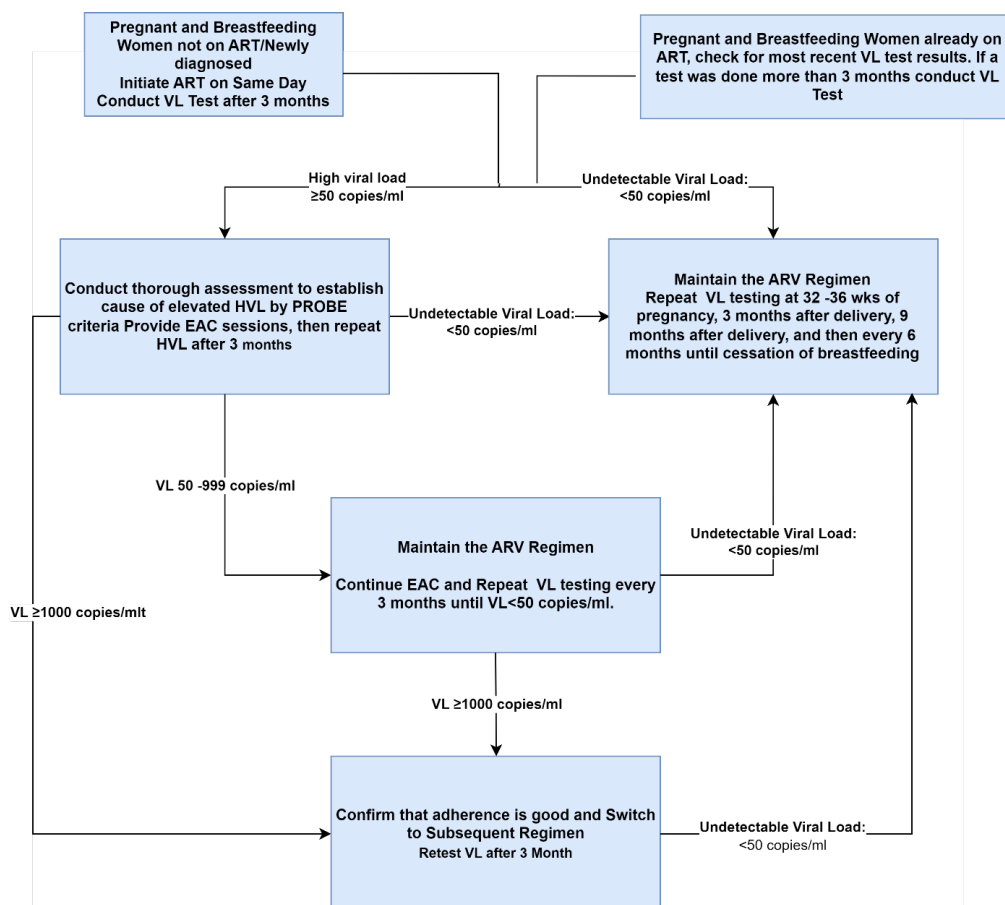


Figure 6.6:1 HVL testing for Pregnant and breastfeeding women already on ART and newly diagnosed.

*The goal of EAC in pregnant women is to achieve undetectable viral load (HVL <50copies/mL) to prevent mother to child transmission of HIV. EAC Sessions for Pregnant and Breastfeeding women should be conducted for three consecutive days followed by one session every two weeks for 3 months.

NB: *Point of Care HIV viral load testing is recommended to PBFW in addition to optimization of conventional platform.*

6.7 Maternal Care and treatment of Syphilis

Any pregnant woman identified positive for Syphilis should be treated for syphilis. The pregnant woman diagnosed with Syphilis should be managed as shown below:

Table 6.7:1 Maternal Care and treatment of Syphilis

Disease Stage	Preferred treatment	Alternative treatment
Early	Benzathine Penicillin 2.4MU I.M STAT OR Procaine Penicillin 1.2 MU OD 10/7	Tabs Erythromycin 500mg QID 14/7 OR IM Ceftriaxone 1G OD 14/7 OR Tabs Azithromycin 2G STAT
Late	Benzathine penicillin 3 doses of 2.4MU IM once weekly for three consecutive weeks	Erythromycin 500 mg QID 1/12 Doxycycline* 100 mg BD 28 days OR Tetracycline 500 mg QID 28 days
Neuro-Syphilis	Aqueous crystalline penicillin G 24 MU per day (administered as IV 4 MU four hourly or as continuous infusion for 14 days)	Procaine penicillin IM 2.4MU once daily PLUS Probenecid 500 mg orally 4 times a day 14/7

*Doxycycline is not recommended in Pregnancy.

6.7.1 Congenital Syphilis

All infants born to sero-positive mothers should be treated with a single intramuscular dose of benzathine penicillin, 50,000 IU/kg whether the mothers were treated during pregnancy (with or without penicillin).

Treatment regimens for early congenital syphilis:

Children under 2 years of age

- Aqueous benzyl penicillin 100,000–150,000 IU/kg/day administered as 50,000 IU/kg/dose IV 12 hourly, for the first 7 days and every 8 hourly thereafter for a total of 10 days.

Children 2 or more years of age

- Aqueous benzyl penicillin, 200,000–300, 000 IU/kg/day by intravenous or intramuscular injection, administered as 50,000 IU/kg every 4–6 hours for 10–14 days
- The alternative regimen for penicillin allergic patients, after the first month of life is Erythromycin, 7.5–12.5 mg/kg (PO) 4 times daily for 30 days.

6.7.2 Hepatitis B Virus Testing and management in pregnant women and exposed infants

All pregnant women should be tested for Hepatitis B and managed as in the flowchart below.

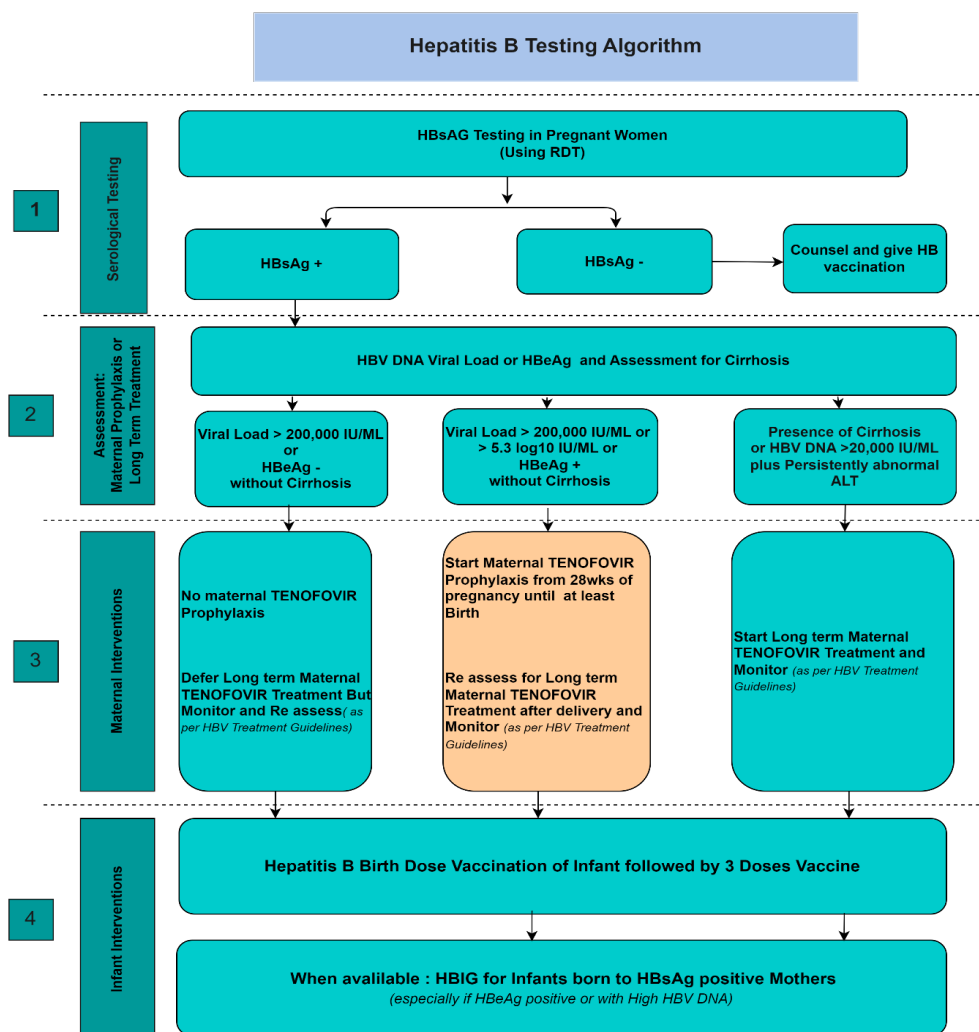


Figure 6.7:1 Testing and management of Hepatitis B in pregnant women and infants

Table 6.7:2 HBV disease progression tests

Tests	ALT, AST, FBP, Ultrasound and AFP	HBV DNA, HBeAg,	RFT and risk for renal diseases
Purpose and Frequency	Detection of HCC in patient with Liver cirrhosis or family history of cirrhosis	Adherence and staging of Liver disease	Toxicity Monitoring for RoC on treatment
Frequency	Every 6 months.	Every 12 Months	Annually

Note: For further details on HBV and HCV please refer chapter X.

7 ADHERENCE PREPARATION, MONITORING, RETENTION, DISCLOSURE AND SUPPORT

7.1 Introduction

Recipients of Care (RoC) are expected to adhere strictly to the prescribed medications, schedules, and recommendations provided and agreed upon with healthcare providers. Achieving a **95% adherence level** is essential for attaining the viral suppression required for people living with HIV (PLHIV) on ART.

Healthcare providers should employ and apply various strategies to support RoC in improving adherence. This chapter provides comprehensive guidance on adherence preparation, retention, monitoring, disclosure, and support across all age groups, including children, adolescents, youth, adults, and pregnant and breastfeeding women (PBFW).

7.1.1 Building blocks of adherence counselling

WHO will conduct counselling	WHAT are the counseling sessions	WHERE will counseling happen	HOW will it be conducted
Counseling shall be conducted by • Nurse Counselor • Trained Peer • Trained Mentor Mother • Trained CBHS provider • Any other trained provider	Two counseling Sessions • Adherence Counseling (in group or individual) • Enhanced Adherence Counseling (EAC)	Counseling sessions can be held • At the facility or Community • In person or Virtually- eg phone	• Enhanced Adherence Counseling should follow the EAC guide Annex 6.3.

7.1.2 Factors that influence Adherence

Healthcare providers should work with the recipients of care to facilitate adherence by emphasizing the following factors.

Table 7.1:1 Factors that enhance adherence.

Domain	Factors that influence adherence
Recipients of care related factors	Disclosure of HIV status and involvement of a treatment supporter/ family member, including couple counselling.
	Readiness to be on life-long medication and following instructions
	living environment
	Age (adults have better adherence)
	support provided, especially for children and adolescents
Health service providers related factors include:	A good treatment plan agreed upon by both the provider and RoC
	A trusting relationship with the Recipient of care
	Supportive, non-judgmental attitudes and behaviour
	Counselling knowledge and skills on adherence and possible side effects associated with ART

Domain	Factors that influence adherence
Regimen-Related Factors	Use of fixed drug combinations that have a low pill burden
	Minimal drug interactions and side effects (rational drug selection)
	Food requirements and timing of taking ART medications (e.g. with food, without food, etc.)
Family/caregiver related factors	Reliability, education, and socioeconomic status of the caregiver
	Family cultural beliefs and practices with regards to HIV
	HIV status of the partner, parents and/or caregivers
	Ability of parents/caregiver to disclose
	Relationship between the caregiver and the recipient of care
Health System related factors	Relationship between the caregiver and the facility staff
	Availability of medications at all refill times
	Community support services and continuum of care

7.1.3 Consequences of Poor Adherence

Poor adherence to ART can result in the following consequences:

- Treatment failure.
- HIV drug resistance.
- Increased morbidity and mortality.
- Growth and developmental faltering.

7.2 Adherence Preparation, Retention Monitoring, and Support

Adherence preparation should be tailored to the Recipient of Care (RoC) based on their age, gender, needs, and clinical status. Counselors should use the following guidance to prepare RoC for ARV initiation, retention, and follow-up.

Adherence Preparation and Support for Children, Adolescents, Adults, Pregnant and Breastfeeding Women

Refer to Table 7.2:1 for detailed guidance on adherence preparation and support for these groups.

Table 7.2:1 Adherence counseling and support guidance

Visit	Standard of Care
At enrolment into care	• Conduct a psychosocial assessment to identify potential psychological, emotional, and social factors influencing adherence and barriers to it. • Identify the primary caregiver or treatment supporter as soon as possible after the diagnosis of HIV. • If no caregiver is available, link the Recipient of Care (RoC) to a community health volunteer or treatment advocate while working on a more permanent solution.

Visit	Standard of Care
	• Discuss the benefits of disclosing the HIV status to the RoC and develop a disclosure plan for children aged 4 years and above. • Conduct a readiness assessment for initiating ART; ART should be started on the same day or within 7 days. • Review ART dosing and timing, including having the caregiver demonstrate how they measure and administer the ART. • Conclude the session by agreeing on a treatment and follow-up plan. • For RoCs starting ART at enrolment, schedule a return visit within 14 days of ART initiation. • Link the RoC to community-based health services (CBHS) and support groups for ongoing assistance. • Document the session outcomes in the Linkage Case Management (LCM) Register. <i>For children and adolescents</i> • Assess growth and developmental milestones to rule out growth retardation, developmental challenges such as autism, deafness and any other physical challenge. Any child with developmental challenges should be referred for appropriate care • The child/adolescent and their caregiver, if also infected, should be enrolled in the same clinic as that of the parent/caregiver • Provide HIV education and counselling to the caregiver (and child as appropriate for age) • Identify and establish appropriate adherence support interventions (Table 6.4), including linkage to paediatric and caregiver support groups and pediatric and adolescent clubs. • Adolescents older than 15 years and emancipated minors may not have or may not want the presence of a caregiver. In this case, the clinical team should explore alternative options to support the adolescents until they are ready to disclose to their caregivers/ guardians or identify someone to disclose to. The alternative options include linkage of the RoC to adolescent mentors, peer educators, social workers, nurses or community health workers, CBHS providers as may be appropriate. An adolescent can have a one on one counselling session with the counsellor or one that includes the caregiver when deemed appropriate • The healthcare provider should explore Sexual and Reproductive health (SRH) understanding, fears and needs of the adolescent and prioritize interventions as appropriate. For adults • Identify a treatment supporter (family member, friend, peer educator, community health worker) and involve them in HIV education and adherence counselling For pregnant and breastfeeding women • Discuss with the mother, how HIV transmission occurs from a mother living with HIV to her baby. Highlight that HIV transmission can occur during Pregnancy (Maternal), Labour and Delivery (L&D) and Breastfeeding (post-partum)

Visit	Standard of Care
	• Discuss with a mother on the importance of life long ART services • Address family planning especially Dual method (condom) • Provide assessment and nutritional support. • Discuss exclusive breast-feeding as an option to minimize HIV transmission
Two weeks after ART Initiation	• Review and reinforce the messages delivered at enrolment; • confirm the caregiver's understanding of key messages • Review the ART dosing, timing and reminders (including having the caregiver demonstrate how they measure and administer the medication) • Explore any barriers to adherence • Revisit benefits of disclosure and the individualized age-appropriate disclosure plan • Document session in the LCM Register
Four weeks after ART Initiation, and further follow- up visits	• Review and reinforce the messages delivered in previous sessions • Confirm the caregiver's understanding of key messages • Review ART dosing, timing and reminders (including having the caregiver demonstrate how they measure and administer the ART) • Explore any barriers to adherence • Revisit benefits of disclosure and the individualized age-appropriate disclosure plan • Document session in the Linkage Case Management (LCM) Register

7.2.1 Common adherence barriers

Table 7.2:2 Common Challenges and Strategies to Improve Adherence in Children

Challenge	Strategies
Child not taking medications	• Obtain a detailed history aimed at identification of the specific causes of his/her broad complaint. • Explore with the parent/caregiver on ways to convince the child to take the medications. • Teach the parent/caregiver on the importance of adherence for the child's survival. • Simplify adherence information to ensure the parent/ caregiver understands the treatment regimen
Medicines make a child sick e.g. nausea and vomiting	• Administer medications with food • Administer medications with liquid to help reduce gastric irritation • Reassure the caregiver/parent that most nausea and vomiting will resolve • If symptoms are severe, seek expert advice on regimen change and timing of medication.
Fear of ART harming the child e.g. the	• Ensure that the child is taking the correct dose • Examine the child for other opportunistic infections • Examine the child for side effects of the regimen

Challenge	Strategies
child is clinically deteriorating despite good adherence	• Encourage the caregiver/parent to continue the regimen unless the child has severe side effects, in which case seek an expert advice about changing the regimen • Utilize visiting nurses/CBHS providers to assist with adherence assessments and follow up home visits.
Regimen dosing confusing to a caregiver/ parent	• Provide the caregiver/parent with a written schedule/ illustration of medications • A written calendar could include symbols for the times of the day to aid with understanding, or utilize colour- coded labels to match with drug regimen colours- coded calendar • Where possible, elicit additional support from another family member or other community resource person.
Parent/ caregiver is ill or absent and other family members cannot give medications to the child	• Treat the ill parent/caregiver • Probe and promote disclosure. • Identify another treatment supporter. • Address stigma and discrimination; provide health education to dispel myths on HIV and AIDS. Refer to CBHS.
Frequently changing or multiple simultaneous caregivers	• Address the effect of frequent changing of caregivers. • Emphasize on the permanent caregiver/treatment supporter
Lack of affection and support of caregiver to child	• Assist the bond between caregiver and child through psychosocial and emotional support • Consider changing caregiver/treatment supporter
Complexity of measuring paediatric formulations e.g. 2nd line regimen (LPV/r syrup)	• Provide the parent/caregiver with a written schedule/illustration of medication • Demonstrate procedures; if possible seek additional support from other family members or other community resource persons.
Unprepared Transition process of children living with HIV to adult HIV care clinic	• Assist the transition of children and adolescents start at the right age to ensure that there's provision of <i>uninterrupted, coordinated, developmental, age-appropriate and comprehensive care before, during and after the transition</i> • Transfer the children aged 8-10 years to adolescents HIV clinic by 10 years and by 19 years transitioned to adult HIV clinic • Refer to 'National SOP on Transitioning Children Living with HIV in Care and treatment Clinics'

Challenge	Strategies
Lack of knowledge and skills on Children/adolescent HIV services to teachers	• Re-assure the linkage between facility and school • Increase awareness of HIV services to school teachers in order to facilitate adherence in school • Strengthen school health clubs

Table 7.2:3 Barriers to Adherence and How to alleviate for adults

Key Barrier to adherence	Suggestions to Alleviate
Social economic problems e.g. transportation, food insecurity	Refer /Link to support groups for assistance Refer to other organizations for economic support e.g. Income Generating Activities (IGA) • Identify nearby CTC sites • Involve other family members
No disclosure	Counsel on benefits of disclosure
Travels frequently	Plan the travels before Carry pills Collect pills in advance for longer period Walk with your CTC 1 card Visit any nearby health facility for required services Enrol in MMD and DSDMs
Behavioural barriers e.g. Drinking alcohol regularly, not planning refill for travels or not having a reminder	Counsel to stop or reduce alcohol intake Involve other family members for support Plan for drug refill Address the use of reminders
Emotional barriers e.g. Depression or Mental illness	Counsel for psychosocial support Refer to clinician for treatment Involve other family members for support
ARVs issues e.g. side effects, pill burden	• Reassure Recipients of care and address concerns • Discuss effectiveness and safety of drugs
Myth and misconception	• Discuss misconceptions • Discuss drugs side effects and resistance
Unexpected hospital admissions	• Carry pills to hospital • Inform healthcare staff that you are on ARV treatment
Stigma and Discrimination	• Link Recipients of care with support groups • Create awareness in the communities • Refer to CBHS • Provide health education on HIV and AIDS, Positive Health, Dignity and Prevention (PHDP)
Cultural Beliefs	• Provide Health Education on HIV and AIDS • Create awareness in the communities including religious leaders and traditional healers

Key Barrier to adherence	Suggestions to Alleviate
Gender-Based Violence	• Provide couple and family counselling, educate on importance of reporting early to the facility for PEP, Emergency FP and management of other disease in case of rape • Link RoC to clinical psychologist, Social welfare, Human Rights and Legal issues organizations for support
Communication Problems	• Use colours, symbols and pictures for elaboration. • Usage of simple language and sign language where needed • Use treatment supporter

7.2.2 Common Adherence barriers

Table 7.2:4 Common Challenges and Strategies to Improve Adherence in Children

Challenge	Strategies
Child not taking medications	• Obtain a detailed history aimed at identification of the specific causes of his/her broad complaint. • Explore with the parent/caregiver on ways to convince the child to take the medications. • Teach the parent/caregiver on the importance of adherence for the child's survival. • Simplify adherence information to ensure the parent/caregiver understands the treatment regimen
Medicines make a child sick e.g. nausea and vomiting	• Administer medications with food • Administer medications with liquid to help reduce gastric irritation • Reassure the caregiver/parent that most nausea and vomiting will resolve • If symptoms are severe, seek expert advice on regimen change and timing of medication.
Fear of ART harming the child e.g. the child is clinically deteriorating despite good adherence	• Ensure that the child is taking the correct dose • Examine the child for other opportunistic infections • Examine the child for side effects of the regimen • Encourage the caregiver/parent to continue the regimen unless the child has severe side effects, in which case seek an expert advice about changing the regimen • Utilize visiting nurses/CBHS providers to assist with adherence assessments and follow up home visits.
Regimen dosing confusing to a caregiver/ parent	• Provide the caregiver/parent with a written schedule/ illustration of medications • A written calendar could include symbols for the times of the day to aid with understanding, or utilize colour-coded labels to match with drug regimen colours-coded calendar • Where possible, elicit additional support from another family member or other community resource person.

Challenge	Strategies
Parent/caregiver is ill or absent and other family members cannot give medications to the child	• Treat the ill parent/caregiver • Probe and promote disclosure. • Identify another treatment supporter. • Address stigma and discrimination; provide health education to dispel myths on HIV and AIDS. Refer to CBHS.
Frequently changing or multiple simultaneous caregivers	• Address the effect of frequent changing of caregivers. • Emphasize on the permanent caregiver/treatment supporter
Lack of affection and support of caregiver to child	• Assist the bond between caregiver and child through psychosocial and emotional support • Consider changing caregiver/treatment supporter
Complexity of measuring paediatric formulations e.g. 2nd line regimen (LPV/r syrup)	• Provide the parent/caregiver with a written schedule/ illustration of medication • Demonstrate procedures; if possible seek additional support from other family members or other community resource persons.
Unprepared Transition process of children living with HIV to adult HIV care clinic	• Assist the transition of children and adolescents start at the right age to ensure that there's provision of uninterrupted, coordinated, developmental, age-appropriate and comprehensive care before, during and after the transition • Transfer the children aged 8-10 years to adolescents HIV clinic by 10 years and by 19 years transitioned to adult HIV clinic • Refer to 'National SOP on Transitioning Children Living with HIV in Care and treatment Clinics'
Lack of knowledge and skills on Children/ adolescent HIV services to teachers	• Re-assure the linkage between facility and school • Increase awareness of HIV services to school teachers in order to facilitate adherence in school • Strengthen school health clubs

Table 7.2:5 Barriers to Adherence and How to alleviate for adults

Challenge	Strategies
Social economic problems e.g. transportation, food insecurity	Refer /Link to support groups for assistance Refer to other organizations for economic support e.g. Income Generating Activities (IGA) • Identify nearby CTC sites • Involve other family members
No disclosure	Counsel on benefits of disclosure
Travels frequently	Plan the travels before Carry pills Collect pills in advance for longer period Walk with your CTC 1 card Visit any nearby health facility for required services Enrol in MMD and DSDMs

Challenge	Strategies
Behavioural barriers e.g. Drinking alcohol regularly, not planning refill for travels or not having a reminder	Counsel to stop or reduce alcohol intake Involve other family members for support Plan for drug refill Address the use of reminders
Emotional barriers e.g. Depression or Mental illness	Counsel for psychosocial support Refer to clinician for treatment Involve other family members for support
ARVs issues e.g. side effects, pill burden	• Reassure Recipients of care and address concerns • Discuss effectiveness and safety of drugs
Myth and misconception	• Discuss misconceptions • Discuss drugs side effects and resistance
Unexpected hospital admissions	• Carry pills to hospital • Inform healthcare staff that you are on ARV treatment
Stigma and Discrimination	• Link Recipients of care with support groups • Create awareness in the communities • Refer to CBHS • Provide health education on HIV and AIDS, Positive Health, Dignity and Prevention (PHDP)
Cultural Beliefs	• Provide Health Education on HIV and AIDS • Create awareness in the communities including religious leaders and traditional healers
Gender-Based Violence	• Provide couple and family counselling, educate on importance of reporting early to the facility for PEP, Emergency FP and management of other disease in case of rape • Link RoC to clinical psychologist, Social welfare, Human Rights and Legal issues organizations for support
Communication Problems	• Use colours, symbols and pictures for elaboration. • Usage of simple language and sign language where needed • Use treatment supporter

7.2.3 Adherence, monitoring, counselling and support among Key and Vulnerable Populations

In many settings, key vulnerable populations face multiple challenges related to stigma and discrimination that can affect access to health services, all of which may impact negatively on adherence and clinical outcomes.

a) Key factors to consider when providing services to key and vulnerable populations

Assuring access: Create demand for HIV testing and counselling and prevention services through targeted campaigns in identified key and vulnerable population settings, use community-based outreach, online services, mobile phone technologies, social networking and develop friendly key population services at health facilities; this will facilitate dissemination of behavioural messages, promote follow-up and referral to services, improve adherence to treatment, and increase Recipients of care participation in their own healthcare.

Ensure confidentiality: Attention should be devoted to protecting privacy and confidentiality, e.g. closing the consultation room door or finding a private place to talk. Recipients of care should be reassured of confidentiality.

b) Adherence Monitoring, Counselling and Support for Recipients of care with Undetectable Viral Load

Once a Recipient of care has confirmed viral suppression (with VL below 50 copies/ml) it is a confirmation of adequate adherence to ART

c) Counselling and support to achieve undetectable viral load

- Elicit any concerns the Recipient of care/caregiver has about ART, other medications, visit schedule, or health
- Address any concerns or engage another care team member who can address them
- Explore any major recent or expected changes in the Recipient of care's/caregiver's life or daily routine that could disrupt adherence
- Update Recipient of care locator and contact information
- Encourage the Recipient of care/caregiver to continue with the supportive system

d) Adherence and support to ROCs with Interruption in Treatment (IIT) and Stopped ART

RoC with interruption in treatment is one of the following:

- Who have not received ARVs within four weeks (i.e. 28 days) of their last missed ART pick-up.
- Those who have not received ARVs within 28 days from the last missed documented community visit with a community health worker, HCPs or peer from a scheduled ART refill group.

- Those who previously opted out/stopped ART by any named reason(s). With exceptional of those who reported to be on ART and had ART refill from other facility or were using spouse/partner's drugs should be excluded.

Table 7.2:6 Adherence monitoring and support approach to RoC who IIT and stopped ART

Adherence	Monitoring and support
Step 1: Provide the welcome Back Package	• Warmly welcoming the RoC, including congratulating them for choosing to come back to care (Avoid scolding or being judgmental)
	• Pledging support: 'I am here to support you through your ART journey'
	• Identify the reason for Interruption In Treatment
	• Ascertain which drugs the RoC was taking, and for how long, the reasons for stopping treatment, check if they had any side-effects
	• Assess presence of chronic and/or Non-Communicable diseases, OIs and link to appropriate services (e.g. cervical cancer screening, AHD etc.)
	• Request for a CD4 if RoC has Interrupted Treatment for >6 months. If CD4 is less than 200, conduct CrAg testing, start CTX and offer other AHD services as needed
	• Review RoC's CTC1 and CTC2 Card/files particularly but not limited to CD4 Cell count, HIV Viral Load test and results, TPT, and update the new contact information if any
Step 2: Adherence and monitoring sessions	• Verbal consent for linkage case management Services
	• Enrol the RoC to Enhanced Adherence counselling (EAC) monitoring and support for the period of 3 months and test for HIV Viral load (Depending on the HVL Results follow the National HVL Monitoring algorithm)
	• Discharge the RoC from EAC through community based HIV and AIDS services Program
	• Whenever necessary, encourage the RoC to enrol into PLHIV Support Groups

e) **Monitoring Adherence**

Effective communication between facility (CTC) and community structures is important to facilitate RoC tracking and follow up to mitigate any reasons for missed visits and Interruption in Treatment.

The following are the strategies for adherence monitoring.

Table 7.2:7 Adherence Monitoring Strategies

Adherence Monitoring Strategy	Technique	Frequency
Morisky Medication Adherence Scale-4 (Annexes 18.8)	Assess adherence using a standardized questionnaire, and act as required	Every recipient of care, every visit
Morisky Medication Adherence Scale-8	Assess adherence using a standardized questionnaire, and act as required	Any time a healthcare worker suspects adherence problems (e.g. RoC with suspected or confirmed treatment failure, or who misses an appointment)
Pill counts	Ask the recipient of care to bring all their pills with them to follow-up visits. Calculate how many pills should be remaining based on the previous prescription date and amount prescribed and compare to how many pills are actually remaining. Excess pills are assumed to be missed doses	At every visit until confirmed viral suppression Any time a healthcare worker suspects adherence problems
Pharmacy refill records	Compare drug pick-up date with expected date of pick-up (based on number of pills dispensed at last visit). If drug pick-up date is later than expected, it is assumed the recipient of care is missing doses equivalent to the number of days late	At every drug pick-up Any time a healthcare worker suspects adherence problems
Viral load	Follow the viral load monitoring algorithm. Undetectable VL is the the best affirmation of optimal adherence	Age 0-24 years: every 6 months Age ≥ 25 years: at month 6 after ART initiation and month 12 then annually For pregnant and breastfeeding women: at first ANC visit if already on ART, or 3 months after ART initiation if starting ART during pregnancy, and then every 6 months (refer to Chapter IV; Refer to HVL Algorithm)

Adherence Monitoring Strategy	Technique	Frequency
Urine Tenofovir Rapid Assay (UTRA) Has shown good results and high acceptability among healthcare providers and PLHIV.	UTRA assesses tenofovir presence in urine at point-of-care and can be used to identify PLHIV with short-term poor adherence to preexposure prophylaxis (PrEP) or TDF containing treatment regimens. Distinguishes high or low adherence patterns within the last 48 hours, as well as non-adherence extending over one week prior to measurement	The frequency of urine assay for tenofovir is determined based on individual client needs and at provider discretion
Home visit	Observe where and how a recipient of care stores their medications and assess if they have extra medications signifying missed doses. Home visits may also provide a better understanding of a RoC living situation and specific barriers to adherence. Unscheduled home visits may be more revealing, however, should only be conducted if the RoC consented to home visits previously (preferably at the time of enrolment or initiation).	For RoC with suspected or confirmed treatment failure, RoC who interrupt treatment, or any time the MDT feels a home visit will contribute to RoC management

Formula for Calculating % Adherence:



$$\% \text{ of pills missed} = \frac{\text{Number of Pills Remaining}}{\text{Total number of pills prescribed}} \times 100$$

$$\text{Percentage adherence} = 100 - \% \text{ Pills Missed}$$

Table 7.2:8 Adherence Support and Retention Interventions

Standard Adherence	Support Interventions
Structural interventions	• Conduct a baseline psychosocial assessment to explore the various aspects of the RoC life that may influence their adherence to treatment and prevention, and their general well-being and identify issues that need to be explored in detail during the counselling session e.g. disclosure, living circumstances • Use a multidisciplinary team approach to develop and implement treatment plans for each RoC • Engage peer educators to lead HIV education and support services

Standard Adherence	Support Interventions
	• Adequately prepare and assess the RoC's readiness to initiate and continue with ART • Implement a system for identifying and taking action when RoC misses an appointment • Formalize a system for providing health talks and treatment literacy classes for RoC. • Formalize a system for linking RoC to community-based resources, including: community support groups, religious groups, CBOs, groups supporting income-generating activities, organizations providing food support, child welfare societies, community health volunteers/units, schools, children's homes
HIV education and counselling	• Remind the RoC about HIV disease, how ART works, the importance of high-level adherence and the consequences of non-adherence • Risk disease progression, poor quality of life and death • Role of ART in restoring and maintaining good health • Link between adherence and viral load, CD4 and health • Side effects of medications and how to minimize, recognize and manage them. • Address misconceptions and beliefs about HIV and ART • Discuss and agree on a treatment plan with the RoC. Gain commitment from the recipient of care to follow through • Discuss use of alcohol and drugs and how to prevent these from affecting the treatment plan • It is important to maintain a non-judgmental attitude, establish trust with parents/caregivers, and involve the child as they mature
Disclosure and stigma	• Respect RoC privacy and confidentiality • Discuss with the RoC the role of disclosure to close family members/trusted friend in promoting adherence • Offer to facilitate disclosure • For children/adolescents, discuss age-appropriate disclosure with the caregiver and offer to support the process • Conduct stigma assessment and support appropriately
Treatment supporter	• Encourage the RoC to identify a treatment supporter/ buddy who will provide RoC encouragement and social support and even remind the RoC to take medication • Invite the treatment supporter to at least one of the adherence counselling sessions • Obtain consent from the RoC to contact the treatment supporter if needed
Support groups	• Develop population-specific support groups when possible (youth groups with peer educators for adolescent's children clubs, caregiver support groups)

Standard Adherence	Support Interventions
	• Link the RoC to psychosocial support groups and other community-based support mechanisms (preferably through direct introduction) • MDT members should be patrons to the support groups to guide activities in line with intended objectives. More references can be found in National Guidelines for the Formation and Management of Support Groups, 2013
SMS reminder system	• Enrol RoC into an automated SMS reminder system with their consent. • Review the type of messages the RoC may receive, the frequency of messages, and any actions the RoC should take when receiving the message. • Ensure the system and messages maintain RoC privacy and confidentiality
Other reminder strategies	• Encourage RoC/caregiver to set a specific time of day to take ART, and to associate the ART time with a specific event/s in their daily schedule. • Encourage RoC/caregiver to set an alarm on their phone.

7.2.4 Readiness to Start Treatment

Before initiating medications, healthcare providers (HCPs) should conduct a readiness assessment (refer to Annex 18.5).

7.3 Enhanced Adherence Counselling (EAC)

Enhanced Adherence Counselling (EAC) is an intensive counselling session aimed at addressing adherence-related issues, poor treatment response, or treatment failure. It is conducted for Recipients of Care (RoC) who have been on ART for at least six months and have an HIV viral load (HVL) of ≥ 50 copies/mL. A minimum of three sessions over a 12-week period is recommended. For continuity and effective follow-up, it is preferable for the same counsellor to conduct all EAC sessions for a Recipient of Care. All identified issues should be documented for timely follow-up.

7.3.1 The Goals of EAC

- Assess and understand barriers affecting adherence.
- Develop strategies to overcome these barriers.
- Provide education on clinical assessment outcomes and HVL results.
- Understand the RoC's knowledge and perception regarding treatment and the importance of viral load suppression.
- Correct misconceptions and provide flexibility to address common adherence barriers, such as alcohol or drug use or rigid schedules causing missed doses.

7.3.2 Conducting EAC for Different Population Groups

a) EAC for RoC with Low Viremia (≥ 50 copies/mL to < 1000 copies/mL)

HIV transmission and treatment failure can occur at low levels of viremia. Therefore, EAC is essential for this group to help achieve an undetectable viral load (HVL < 50 copies/mL) and untransmittable status.

- Upon receiving HVL results, all RoC with HVL 50–999 copies/mL should begin EAC sessions as per the HVL algorithm.
- EAC should be conducted using PROBE (refer to Annex 18.6).

b) EAC for RoC with High Viremia (≥ 1000 copies/mL)

- Initiate at least three EAC sessions over three months.
- Retest HVL one month after the third session.
- If HVL remains ≥ 1000 copies/mL, continue monthly EAC and retest after three months.
- If HVL is persistently ≥ 1000 copies/mL, refer the RoC to the facility Multi-Disciplinary Team (MDT) for discussion and consider switching to subsequent ART lines.

c) EAC for Pregnant and Breastfeeding Women (PBFW)

- For PBFW with HVL 50–999 copies/mL or ≥ 1000 copies/mL, initiate the first EAC session over three consecutive days starting from the day of results.
- Follow up with one session every two weeks for three months.
- Repeat HVL testing on the day of the last session.
 - o If HVL is 50–999 copies/mL, continue the EAC cycle and retest HVL after three months.
 - o If HVL remains ≥ 1000 copies/mL despite good adherence, assess the log drop, switch to a second-line regimen, and retest HVL three months after switching.
 - o For RoC with HVL persistently above 500 copies/mL for 12 months, conduct GART and switch to an appropriate regimen.

Special Considerations During EAC Sessions

For Pregnant Women:

- Continue EAC until HVL becomes undetectable.
- Discuss the modes of HIV transmission, emphasizing that transmission can occur during pregnancy, labour, delivery, and breastfeeding.
- Provide counselling on proper infant feeding practices.

For Children and Adolescents:

- Involve the child in the EAC process; for younger children, involve parents or caregivers in planning and implementation.
- Implement a **Child Care Family-Centered Approach**.
- Ask the caregiver about the child's medication tolerance (e.g., does the child refuse to swallow, spit out, or vomit medication?).
- Review disclosure criteria and encourage the parent or caregiver to disclose the child's HIV status:
 - o Partial disclosure at ages 5–9 years.
 - o Full disclosure by age 12 years.
- Link the child to **age-appropriate peer groups** and teen clubs.
- Provide **adolescent-friendly** services tailored to their unique needs.

Table 7.3:1 Components of Sessions for Enhance Adherence

Component	Enhance Adherence Counselling
Session 1	Explain the purpose of your session, define terms and • Review understanding of viral load (VL) and explain the benefits of VL suppression. • Conduct PROBE assessment. • Assess any barrier result into an increase in viral load • Address any reported barriers to adherence and discuss effective strategies to overcome. • Update or develop an adherence plan with the RoC • Discuss risk reduction (e.g. for substance abuse) • Mental health screening (screen for depression using PHQ-9, GAD-7, dementia scale etc.) • Discuss Recipient of care's support systems, referrals and linkage to different support groups • Assist Recipient of care to develop adherence plan to address the identified issues
Session 2	• Education on the assessment result • Review adherence plan from the first session and discuss any challenges • Identify other possible gaps and issues emerging. • Referrals and networking • Assist Recipient of care to modify the adherence plan to address the identified issues
Session 3	• Review adherence plan from the first and second session and discuss any challenges. • Identify other possible gaps and issues emerging • Assist the Recipient of care to modify the adherence plan to address the identified issues • Decision on repeat VL based on current adherence

Component	Enhance Adherence Counselling
	• If the adherence is good: plan repeat VL testing after three months of good adherence and explain possible ways forward, emphasizing role of the Recipient of care and the health facility • If adherence challenges persist: plan further Enhanced Adherence Counselling sessions before repeating the VL
Assessment session	• Assess the level of adherence • Restart EAC to RoCs with existing barrier repeat HVL test for those with the eligibility criteria's
Session to Discuss Repeat Viral Load Results	• Discuss result of the second VL test • Plan the way forward: If VL now undetectable: continue current regimen with enhanced adherence, repeat VL after 6 months. - If VL $\geq$ 1,000: prepare Recipient of care for change of regimen. - If VL is detectable but $<$ 1,000: may continue to monitor or may prepare for change of regimen, pending MDT discussion and consultation with Regional or National HIV Clinical TWG
Case management	• Assign a CBHS provider to all children and adolescents (those not achieving optimum treatment outcomes); pregnant women, orphans, Recipients of care with alcohol and substance abuse, Recipients of care with mental illness, Recipients of care with suspected or confirmed treatment failure, and any Recipients of care who the healthcare team feels has poor adherence or is at high risk of defaulting from care • The case manager is the link between the Recipient of care and the MDT • Roles of the case managers include: - Coordinating multidisciplinary management for Recipients of care under case management - Following up on appointment-keeping for their Recipients of care - Organizing Recipient of care reminders (SMS, calling the day before) and other support systems - Ensuring appropriate defaulter tracing - Coordinating home visits to their Recipients of care.

7.4 Community-Led Adherence Support

Recipients of care (RoC) who are stable on ART and established in their treatment do not require intensive care models. For community-led adherence support to be effective, it should be monitored as follows:

- **Attachment to CBHS Providers:**
Each RoC should be linked to Community-Based Health Services (CBHS) providers, Community Health Workers (CHWs), and peer educators.
- **Roles of Treatment Advocates:**
 - o Provide adherence support and monitor treatment outcomes.
 - o Record progress and share it with healthcare providers.
- **Roles of Health Facilities:**
 - o Monitor the progress of PLHIV using records shared by the community-based team.
- **Roles of Family/Family Members:**
 - o Encourage and remind RoC to take medication regularly.

Note: *RoC whose stability status changes should be referred back to the health facility for appropriate care.*

7.5 Disclosure

Disclosure refers to sharing one's HIV status with someone else to reduce the stress of living with HIV. It enhances adherence to ART, and healthcare workers (HCWs) and lay workers should be equipped with adequate skills, knowledge, and tools to support the disclosure process and provide post-disclosure support.

For PLHIV with adherence issues, their disclosure status should be assessed based on their age category and understanding. Several factors influence the decision to disclose an HIV diagnosis to a child, adolescent, or spouse. Similar to other chronic illnesses, **age and developmental maturity** are critical in determining the appropriate timing and method for disclosure.

Disclosure is generally categorized as follows:

1. **Partial Disclosure:**
 - o Begins at ages **4–6 years**.
 - o At this stage, the child or adolescent should understand why they attend the clinic and take medications.
 - o HCWs and caregivers can explain that ARVs help “put germs to sleep” to keep the child healthy and strong.

2. Full Disclosure:

- o Begins at ages **8–10 years**.
- o At this stage, the child should receive full information and knowledge about HIV.

3. Accidental Disclosure:

- o Strongly discouraged.
- o It is essential to avoid telling someone their HIV status without preparing them or disclosing by accident.

Reference: For more detailed guidance, refer to the **Job Aids for Comprehensive Differentiated Delivery of HIV and AIDS Services (2019)**

8 COMMUNITY BASED HIV AND AIDS SERVICES

8.1 Introduction

Due to advancements in the management of HIV and AIDS and the evolving needs of Recipients of Care (RoC), the scope of **Community-Based HIV and AIDS Services (CBHS)** has shifted significantly. It has transitioned from primarily caring for bedridden RoC to providing ambulatory, differentiated services tailored to individual needs.

This chapter outlines the scope of **Community-Based HIV and AIDS Services** as an integral part of the comprehensive continuum of HIV care. It identifies and addresses the diverse needs of people living with HIV (PLHIV) and their family members, offering recommendations to support them in coping with lifelong ART.

8.1.1 Overall goal

To harness community resources to support Recipients of Care (RoC) throughout the HIV care and treatment cascade, as well as to promote health and prevent non-communicable diseases (NCDs).

7.1.1 Objectives

- To promote continuum of care through enhanced ART adherence and retention of PLHIV and prevention of NCDs
- To strengthen PLHIV structures at community level for enhanced case identification, bi-directional referrals, and linkage mechanisms.
- To increase demand and utilization of quality HIV services and prevention of NCDs
- Facilitate effective implementation of ART refilling option for PLHIV and prevent NCDs

8.1.2 Scope of CBHS

CBHS ensures continuity of care provided to PLHIV. Continuum of care means a set of comprehensive treatment, care, and psycho-social support services provided from facility to community level. CBH services are provided by the government, NGOs, Community-Based Organizations (CBOs), Faith-Based Organizations (FBOs), and community members. A continuum of care is achieved through a combination of well-defined strategies and intervention packages to improve health and social well-being through home- and facility-based services throughout the lifecycle of a health or social welfare issue. In provision of CBH services it is important to recognize that PLHIV can play an important role in delivering

continuum of care services through sharing lived experiences and strong desire to ensure their peers are linked to treatment and care services.

PLHIV and their affected families and households have a variety of other needs beyond mere clinical needs. Such needs include access to post-GBV services, psycho-social, nutritional, educational, economic, and legal services. It is the CBHS approach that caters for these needs and creates mechanisms for commonly addressing them.

8.1.3 Beneficiaries of CBH Services

Community-Based HIV Services (CBHS) target various populations, including:

- People living with HIV (PLHIV), including older people, children, pregnant women, breastfeeding mothers, Adolescent Boys and Young Men (ABYM), Adolescent Girls and Young Women (AGYW), People Who Inject Drugs (PWIDs), People Who Use Drugs (PUDs), high-risk women, fisherfolk, miners, long-distance truck drivers, prisoners, and chronically ill Recipients of Care (RoC).
- **Others includes**, Caregivers of PLHIV to enhance the continuum of care, treatment, and disclosure to PLHIV.

8.2 Community-Based HIV Services Providers (CBHSP)

The Ministry of Health (MoH) is committed to formalizing the **Community Health Worker (CHW) cadre**, though various CHW cadres currently focus on specific areas and are often donor-funded. These cadres include Home-Based Care (HBC) Providers, Treatment Advocates (TAs), Peer Educators, Liaison Persons, Client Trackers, Volunteers, and Community-Based Distributors (CBDs).

The lack of coordination in recruiting, placement, payment, or a uniform stipend has caused confusion and duplication across programs. This guideline establishes standards to harmonize CBHS provision at facilities and within communities.

The Ministry formally designates CHWs as a **formal paid cadre**, trained using the approved national training package and certified to support community activities. CHWs are categorized into two groups:

- 1. Licensed CHWs (Community Health Professionals)**
- 2. Non-Licensed CHWs**

CHWs will generally work on a voluntary basis but are eligible for financial and non-financial incentives. Licensed CHWs will collaborate with non-licensed CHWs to achieve the goals set by the National Health Sector HIV and AIDS Strategic Plan.

8.2.1 Selection Criteria for CBHS Providers (Non-Licensed CHWs)

To ensure the selection of competent individuals capable of working under challenging field conditions and maintaining confidentiality, non-licensed CHWs (CBHS) will be selected based on the following criteria:

- Willingness to do volunteer work.
- Residency in the community for at least three years.
- Completion of at least primary school education with the ability to read and write.
- Familiarity with local language, culture, and social norms.
- No record of child abuse, mistreatment, or other unlawful conduct.
- Commitment, hard work, and passion for serving the community.
- Trustworthiness and ability to maintain client confidentiality.
- Priority for those experienced in community-based health and social welfare.
- Ability to build strong interpersonal relationships.
- Reliability and flexibility to engage with various community groups, particularly youth and children.

8.2.2 CBHS Providers Training

Only CHWs trained and certified using the approved national training package can serve as CHWs. Training will be conducted by national trainers as outlined in the National Operational Guideline for Community-Based Health Program 2021.

a) Roles and Responsibilities of CBHS Providers

CBHS providers must adopt new roles to align with advancements in HIV clinical management, ensuring treatment continuity and quality ART services:

- Provide education on condom use, disclosure, gender-based violence (GBV), TB, and STIs.
- Distribute condoms at the community level.
- Promote stigma and discrimination reduction through peer support, escorted referrals, treatment literacy sessions, and disclosure support.
- Encourage male involvement in HIV response through targeted approaches.

- Identify pregnant and breastfeeding women in the community and refer them to health facilities for HIV testing and post-test services.
- Track mothers enrolled in PMTCT who have delivered but have not returned for newborn DBS results.
- Refer key populations, such as PWIDs, PUDs, AGYW, ABYM, and others, for HIV services, including testing, prevention (e.g., PrEP and PEP), ART, and sexual reproductive health services.
- Facilitate index contact elicitation and link contacts to HIV testing.
- Conduct health education, treatment literacy, and adherence counselling for PBFW living with HIV.
- Establish and manage Post-Test Clubs/Support Groups at the community level.
- Promote youth-friendly services to increase treatment uptake among adolescents and youth.
- Provide nutritional support, including education, assessments, and counselling.
- Deliver treatment literacy messages to caregivers, parents, school administrators, and religious leaders.
- Refer PLHIV with pain or distressing symptoms to palliative care services.
- Establish and manage community-based PLHIV support groups for enhanced treatment literacy and peer support.
- Conduct community visits for adherence counselling and health education to ensure treatment retention.
- Track treatment interrupters via CTC Desk and return them to ART through line-listing, follow-up visits, or phone calls.
- Advocate for CBHS in community meetings.
- Facilitate disclosure support for PLHIV to choose a primary caregiver for treatment assistance.
- Link newly diagnosed PLHIV to peer empowerment groups offering psychosocial support and other services.

b) Bi-Directional Referral

Bi-directional referral involves the exchange of information between health facilities and the community to strengthen the continuum of care.

Key steps include:

- Filling and issuing referral forms to RoC.
- Ensuring feedback forms are completed and returned to the initial provider.
- Referring RoC based on their specific needs, such as spiritual, legal, economic, and nutritional support.

- Developing and updating the local referral services directory.
- Facilitating referrals for HIV-positive individuals identified through self-testing kits.

8.2.3 Contribution of CBHS Programs in the HIV Response

CBHS programs serve as a bridge between primary health facilities and communities, contributing significantly to the HIV response by:

- Providing access to services for hard-to-reach populations.
- Reducing inequalities in HIV service delivery, increasing service utilization.
- Decongesting health facilities by shifting some services to the community.

a) Role of Educational Institutions in the HIV Response

- Ensure students receive HIV prevention and treatment services at schools and higher learning institutions.
- Create an enabling environment for treatment adherence among students and pupils living with HIV.

b) Role of Religious Institutions in the HIV Response

- Promote and address HIV treatment adherence.
- Engage in reducing stigma and discrimination against PLHIV.
- Align with national HIV management guidelines and prevention policies.
- Integrate HIV response activities into religious structures and practices to support identification, treatment adherence, and viral suppression.

8.2.4 Community-Led Monitoring (CLM)

Community-Led Monitoring (CLM) is a PLHIV-centered approach designed to enhance the availability, accessibility, acceptability, affordability, appropriateness, and quality of HIV services. It empowers PLHIV to take leadership roles in collecting feedback on enablers and barriers to service delivery from their peers and health facilities. This feedback is used to generate recommendations and plans for tracking HIV service provision at the facility level.

The purpose of CLM is to encourage health workers and stakeholders at all levels of the HIV response to develop innovative approaches for quality improvement, implement them effectively, and institutionalize high-quality healthcare at various levels based on recommendations from service users, national interests, and the vision for HIV service delivery.

CLM is guided by:

- The **United Nation-Meaningful Involvement of PLHIV (MIPA) Strategy**.
- The **Tanzania Quality Improvement Framework in Healthcare (TQIF)**, released in October 2011.
- The **Community Quality Improvement Framework (CQIF)**, First Edition, 2018.

Community-Led Monitoring helps identify issues of concern for PLHIV and propose measures based on their experiences with ART service delivery. By raising these concerns through their structures, PLHIV can amplify their voices, improving the quality of HIV services and mobilizing others for **HTS, ART**, prevention, and **HVL monitoring**.

Engagement of PLHIV in CLM

PLHIV are engaged to:

- Conduct meetings with peers to gather feedback on the availability, accessibility, acceptability, affordability, and appropriateness of services.
- Meet with facility staff and health management teams at council and regional levels to clarify and collect quantitative data related to peer feedback.
- Engage policy and decision-makers at the ministerial level to track policy responses to long-term HIV challenges highlighted in CLM reports.
- Monitor changes agreed upon during feedback sessions with health facilities, districts, regions, and policymakers.
- Disseminate progress reports to key stakeholders on improvements in HIV service delivery.

8.2.5 Community ART Refill

The Community ART Refill Group (CARG) model is a differentiated service delivery approach for stable Recipients of Care (RoC) on ART. In this model, group members voluntarily form a group, and one member collects ART for all members.

Community ART refills improve retention in ART services, leading to effective HIV management. This approach addresses the **Human Resource for Health (HRH)** gap, allowing health staff to serve a reasonable number of RoC more efficiently.

Implementation of the CARG Model

- Stable RoC from the same geographic area form a group and take turns visiting the health facility to collect ART for the group.
- The designated group member distributes the ART to other group members.
- Collection and distribution of ART are done rotationally, with group members taking turns during each collection period.
- All PLHIV in the group must attend the facility for scheduled clinical consultations and HVL testing.
- Facility staff, in collaboration with Konga structures (RoC organizations at district and ward levels), coordinate the model.

8.2.6 Retention and Tracking of RoC

Retention refers to the continuous use of services, adherence to clinic schedules, and compliance with healthcare provider (HCP) instructions. It is essential for achieving care and treatment goals. **Tracking** is a critical role of Community-Based Health Services (CBHS) providers in ensuring retention.

Steps for Retention and Tracking

- Attend **CTC sessions** and work at the **CBHS desk**.
- Obtain a list of RoC for tracking.
- Identify the locations of individual RoC for follow-up.
- Conduct follow-ups and provide counselling on the importance of attending clinic appointments.
- Document outcomes using designated tracking tools

9 TB AND HIV CO-INFECTION

9.1 Introduction

TB is the **most common opportunistic infection** and a leading cause of death among people living with HIV (PLHIV). HIV significantly increases the risk of acquiring TB, reactivating latent TB, and progressing from TB infection to active disease. Individuals with HIV have a fourfold increased risk of developing active TB.

The Ministry of Health (MoH) updated the **TB and HIV Collaborative Policy Guidelines** in 2022, emphasizing priority interventions to address TB/HIV co-infection effectively.

9.2 Collaborative TB and HIV Activities

Addressing TB and HIV co-infection requires a collaborative approach involving healthcare providers, community organizations, and government agencies. This collaboration should aim to prevent, identify, diagnose, and treat TB and HIV in individuals and communities. The goal is to achieve joint management and deliver integrated TB and HIV services through coordinated planning and resource mobilization.

At the **Care and Treatment Clinic (CTC)** level, healthcare providers should focus on reducing the burden of TB among PLHIV by ensuring regular screening and timely initiation of ART. Providers should be trained in the case management of both TB and HIV and offer comprehensive care to co-infected Recipients of Care (RoC). Emphasis should also be placed on TB infection control measures in HIV care settings and congregate environments. These measures include:

- Ensuring proper ventilation.
- Encouraging the use of masks.
- Regularly screening healthcare workers to prevent TB transmission within healthcare facilities.

Community organizations play a key role by providing education, reducing stigma, and supporting individuals living with TB and HIV. They should also encourage treatment adherence and, where necessary, refer individuals for diagnosis, treatment, and other support services.

For more detailed guidance, refer to the **TB/HIV Collaborative Policy Guidelines, Third Edition, 2022**.

9.2.1 Intensified TB Case Finding Among PLHIV

- **Routine Screening:** All PLHIV should be screened for TB at every clinic visit to ensure early identification and treatment of TB. PLHIV without symptoms or signs of TB should be provided with **TB Preventive Therapy (TPT)** to reduce the risk of developing active TB.
- **Atypical Presentation:** PLHIV, particularly those with advanced HIV disease, may present with **atypical clinical symptoms**, complicating the diagnosis of pulmonary and extrapulmonary TB.
- **Diagnostic Approaches:** Both clinical (signs and symptoms) and laboratory methods (e.g., microscopy, culture, or WHO-recommended rapid diagnostic tests) should be used to assess TB infection.

9.2.2 Clinical evaluation

Table 9.2:1 Clinical diagnosis involves history taking and physical examination.

Component	Enhance Adherence Counselling
History Taking	• Ask the RoC about any existing symptoms suggestive of TB disease among adults (Such as cough of any duration (sputum colour and quantity), night sweats, fever or weight loss) • Ask about history of TB disease and the outcome of treatment. • Ask about history of <u>TB among contact(s) of TB patients.</u> • Ask about history of other risk factors such as <u>tobacco smoking, drugs and alcohol use or occupational history (miners)</u>
Physical examination	Assess the RoC for • Fever • Anaemia • Lymphadenopathy, • Presence or absence of chest and neurological abnormalities • <u>Hepato-splenomegaly in order to screen for co-morbidities and rule out EPTB in all RoC.</u>

Table 9.2:2 Laboratory diagnosis

Molecular tests	Sputum smear microscopy	Culture and sensitivity	Additional supportive diagnostic tests
WHO recommended molecular diagnostics tests (mWRD) • Xpert MTB RIF/ULTRA • Truenat Note: collect spot sputum sample from All presumptive TB case for molecular testing. Line Probe Assay (MTBDRplus (1st line DST) and MTBDRsl (2nd line DST)	• Spot-Morning Light microscopy (ZN staining) • Spot-Morning LED Fluorescent microscopy Note: • Diagnose people with active TB in the absence of molecular tests. • Monitor the progress of treatment. • Confirm whether cure has been achieved	Solid culture (LJ medium) or liquid culture	• Chest X ray • Hematology tests (ESR) • Chemistries

**Refer to the National Manual for Management of TB and Leprosy, 2020 for more on TB diagnosis*

Diagnosis of Tuberculosis in Children

The diagnosis of TB in children can be challenging. Sputum can hardly be obtained from children and is often negative even on culture; and signs and symptoms are atypical to TB. The diagnosis is therefore based on the use of TB score chart. Older children who can cough up sputum should go through the same assessment as adults using Gene-Expert as the “gold standard” test.

9.2.3 New recommended Tests for TB screening and diagnosis

Tests	Recommended for	Description
Urine for TB Lateral Flow Lipoarabinomannan (LAM-Test)	TB screening	Urine lateral flow (LF-LAM) may be used to assist in the diagnosis of active TB in adult patients with advanced HIV disease, with or without signs and symptoms of TB (pulmonary and/or extra-pulmonary).
Computer – aid- ed detection (CAD4TB) of Chest X ray.	TB Screening	This use artificial intelligence technology (CAD4TB) to read chest X ray images (This should not be used for initiation of TB treatment on its own).

Tests	Recommended for	Description
TrueNat TB test	TB diagnosis	This is a rapid molecular test for detecting TB and resistance to rifampicin as PoC at lower levels.
Use of stool sample in TB diagnosis among children	TB diagnosis	This is a non-invasive detection of Mycobacterium tuberculosis complex (MTBC) in stool specimens recommended alongside sputum (expectorated or induced), nasopharyngeal aspirate or gastric aspirate (GA) for both Xpert MTB/RIF and Xpert Ultra as the initial diagnostic test for TB and rifampicin resistance in children under 10 years with signs and symptoms of pulmonary TB.

9.2.4 Standard TB Treatment Regimens

New or Previously Treated Recipients of Care:

- All new or previously treated Recipients of Care (RoC) should receive a six-month regimen containing rifampicin: **2RHZE/4RH**. Treatment should be administered as **daily observed treatment (DOT)** by a healthcare provider or another designated individual, such as a family member or friend, throughout the six-month period, as outlined in Table 9.2:3.
- **Previously Treated TB Recipients of Care:**
All previously treated TB RoC should provide a specimen for rapid molecular testing (GeneXpert MTB/RIF or TrueNat MTB/RIF), where available, as well as culture and drug susceptibility testing (DST).
 - o **Rifampicin-resistant cases:** RoC with rifampicin resistance should be referred to a designated health facility for **multidrug-resistant TB (MDR TB)** treatment.
 - o **Rifampicin-susceptible cases:** RoC who are rifampicin susceptible should be treated with a **first-line treatment regimen** containing all four drugs (2RHZE/4RH) while waiting for DST results.
- **Treatment Failure:**
RoC who have experienced treatment failure should have a **sputum sample** collected for culture and DST. If a molecular test is not immediately available, they should be started on a **first-line TB treatment regimen** while awaiting culture and DST results.

Note:

Samples for molecular testing can be referred to the nearest facility with molecular testing capacity.

Table 9.2:3 Summary of first-line TB treatment regimens for adults

Duration of treatment	Drugs	ADULT Pre-treatment weight			
		21 – 30kg	31 - 50kg	51 – 74 kg	≥75kg
2 months intensive phase, daily observed	{RHZE} 150/75/400/275	2	3	4	5
4 months continuation phase, daily observed	{RH} 150/75	2	3	4	5

Table 9.2:4 Summary of first line TB treatment regimens for Paediatric.

Weight (Kg)	Intensive Phase (2 months)		Continuation Phase (4 months)
	RHZ (paediatric) 75/30/150	Ethambutol 100 mg	RH (paediatric) 75/50
2 - 2.9 kg	½ tablet	½ tablet	½ tablet
3 - 3.9 kg	1 tablet	½ tablet	1 tablet
4 – 7.9 kg	1 tablet	1 tablet	1 tablet
8 – 11.9 kg	2 tablets	2 tablets	2 tablets
12 – 15 .9kg	3 tablets	3 tablets	3 tablets
16 – 24.9 kg	4 tablets	4 tablets	4 tablets
25 + kg	Use Adult regimen		

*H – Isoniazid; R – Rifampicin; Z – Pyrazinamide; E - Ethambutol

Note:

30 percent of children with a miliary picture on chest radiography have central nervous system involvement and should be treated with a 12-month regimen (See Manual for the Management of TB and Leprosy, 2020).

9.2.5 Tuberculosis-associated Immune Reconstitution Syndrome

- In RoCs experiencing signs of Immune Reconstitution Inflammatory Syndrome (IRIS), (occurrence of features of active TB or a temporary exacerbation TB signs and symptoms, with or without an aggravated radiographic manifestation after the initiation of ART), TB treatment failure should be excluded before IRIS is confirmed.

- If IRIS is diagnosed, continue with both ART and anti-TB treatment. In severe cases, prednisone 1-2 mg/kg for 1-2 weeks can be given (thereafter gradually decreasing dosage).

9.2.6 Monitoring response to TB treatment

A monthly TB smear microscopy should be used to monitor response to treatment of pulmonary TB. TB culture is also recommended in monitoring TB treatment in line with TB smear microscopy. Details on how and when to collect specimens, transportation, and results feedback are prescribed in NTLP National Manual for Management of TB and Leprosy, 2020.

9.3 TB Preventive Measures

9.3.1 BCG vaccination to Children

As recommended by WHO all neonates should be given BCG vaccine immediately after birth, regardless of HIV status. BCG vaccine rarely causes disseminated infection of *M. bovis* in HIV neonates and if it occurs it should be treated with 2RHZE/4RH. The benefits of BCG (including prevention of severe form of TB), outweigh the possible disadvantages. However, BCG vaccine should not be given to children who are immunosuppressed (with clear signs and symptoms of HIV disease or AIDS).

9.3.2 TB Preventive Treatment (TPT)

- To prevent the progression of TB infection to active TB disease, all PLHIV should receive TB Preventive Treatment (TPT) once in a lifetime after the exclusion of active TB.
- The recommended TPT regimens are three months of Isoniazid plus rifapentine (3HP) weekly; three months of Isoniazid plus rifampicin (3HR) daily and six months of Isoniazid monotherapy (6H) daily to eligible RoC as shown in Table 9.3:1 below:

Table 9.3:1 Recommended TPT regimen

Age	Preferred Regimen and Duration	Alternative
Children <15 years, pregnant and breastfeeding women, and adult who cannot tolerate 3HP	Daily 3HR (Isoniazid plus rifampicin for 3 months)	Daily 6 months Isoniazid Monotherapy
PLHIV ≥ 15 years (excluding PBFW)	Weekly 3HP (Isoniazid plus rifapentine for 3 Months)	Daily 6 months Isoniazid Monotherapy

Note:

For children <15 years and PBFW who are on 3HR, the dose of Antiretroviral Therapy (ART) should be double because Rifampicin has potential drug–drug interactions with protease inhibitors-based regimens (lopinavir – ritonavir), nevirapine or integrase inhibitors (dolutegravir).

6H is the preferred regimen among HIV infected children on protease inhibitor-based regimen (lopinavir–ritonavir), nevirapine, or integrase strand transfer inhibitors (dolutegravir) due to potential drug–drug interactions with other regimens.

Post – treatment TPT among PLHIV

RoC who have successfully finished their TB treatment should promptly begin a course of TPT right after completing their TB treatment as they are at higher risk of recurrence (5 to 7 times) with over 90% of these recurrences associated with reinfection.

9.3.3 Eligibility for TPT among children, adolescents and adults

Age	Preferred Regimen and Duration
For people with no history of TB treatment	• All HIV positive individuals who screen negative for active TB. • <5 years household contacts of bacteriologically confirmed TB. • ≥5years house contacts of bacteriologically confirmed TB clients, prisoners, PWUDs and miners.
For people with history of TB treatment	• People living with HIV who successfully completed their TB treatment should immediately receive TPT. It has been shown that PLHIV who receive TPT immediately after completion of TB treatment have less risk of TB recurrence, relapse and re-infection and mortality.
Pregnant and breastfeeding	• TB preventive treatment is not contraindicated in pregnancy and it can be given during any trimester.

Note:

- *TPT should not be offered to PLHIV who abuse alcohol, are non-adherent to long term treatment, have current/past history of hepatitis and/or have medical contra-indication to INH.*
- *Offer a recommended dose for TPT as described in the National Tuberculosis Infection Manual.*
- *Follow up to detect and treat side effects as early as possible. Refer Figure 9.3:1 below for more clarity.*
- *TPT should be provided where there is quality supportive counselling, capacity for follow-up and monitoring of RoC to encourage adherence and manage side effects.*

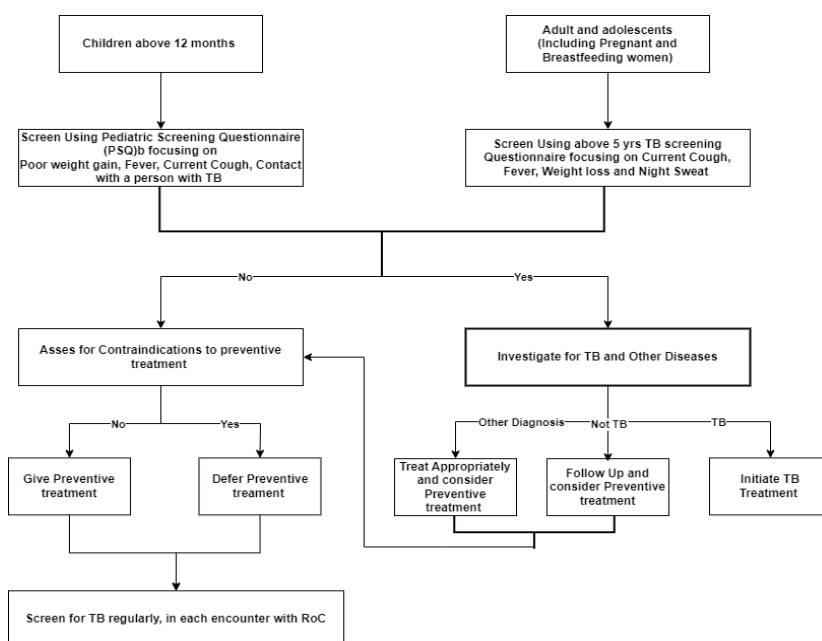


Figure 9.3:1 Algorithm for ruling out TB; for children above 12 months, Adult and adolescent including pregnant women and breastfeeding mothers living with HIV.

Table 9.3:2 Recommended dosage for the treatment of tuberculosis infection

Drug regimen	Population	Dose per kg body weight per day	Maximum daily dose
Daily isoniazid Plus rifampicin for three months(3HR)	Children < 10 years	Rifampicin: 15 mg (range 10-20 mg) Isoniazid: 10 mg (range 7-15 mg)	Rifampicin: 600mg Isoniazid: 300 mg
	Children ≥ 10 years	Rifampicin: 10 mg Isoniazid: 5 mg	
Three months of rifapentine plus high dose isoniazid (3HP) weekly (12 doses)	≥ 15 years	Medicine, formulation	Number of tablets for body weight more than 30 kg
		Isoniazid 300 mg	3
		Rifapentine 150 mg	6
		Isoniazid + rifapentine fixed dose combination (300 mg/300 mg)	3
Isoniazid alone, daily for 6 months (6H)	All population groups.	Children 5mg Adult 10 mg (range, 7–15 mg)	300 mg

Note: For Weight and Age Based Dosing Charts: Refer Annexes 18.9.

Table 9.3:3 Management of potential adverse reactions following treatment with TPT

Adverse event	Stop and consider reintroduction with caution	Stop and do not reintroduce
Flu-like syndrome with fever, chills and malaise, sometimes with headache, dizziness or bone pain	If mild and not increasing continue treatment and observe closely	If moderate to severe symptoms, consider alternative TPT options without a rifamycin (6H)
Drug-associated fever only	Consider re-introduction if fever settles below 39°C, but stop permanently if fever recurs	If fever more than 39°C after the previous episode of drug – associated fever
Persistent nausea, frequent vomiting and/ or persistent episodes of unformed watery stools	Administer an antiemetic or anti-diarrheal medication and consider reintroducing TPT with caution once the symptoms have resolved	If there is nausea, vomiting or diarrhoea which requires aggressive rehydration
Cutaneous reactions	Diffuse rash (no vesicles) Diffuse rash with limited vesicles	If there are extensive bullous lesions/ulceration of mucous membranes/ Stevens Johnson or toxic epidermal necrolysis, contact a specialist and use steroids
Other hypersensitivity reactions (hypotension, acute bronchospasm, conjunctivitis, thrombocytopenia)	Assess the clinical severity of the symptoms and if severe consider alternative TPT options without a rifamycin (6H)	
Hepatitis (early symptoms of weakness, fatigue, loss of appetite, persistent nausea)	ALT/AST < 5 times upper limits of normal and absence of symptoms	ALT/AST ≥5 times the upper limit of normal in the absence of symptoms ALT/AST ≥3 times the upper limit of normal in the presence of symptoms.
Psychosis	Psychiatric evaluation, antipsychotic therapy, pyridoxine.	Attributable to Isoniazid
Seizures	Withhold isoniazid pending resolution of seizures, evaluate possible causes of seizures.	Attributable to Isoniazid

9.3.4 Provision of TPT in DSDM

All PLHIV on ART, and receiving TPT are eligible for MMD provided they are established on ART/Stable.

Table 9.3:4 MMD for TPT

RoC Category	Regimen	MMD Plan for TPT		
		Month 0	Month 1	Month 2
Not Established on ART/ Unstable	3HR	1 Months	2 months	None
	3HP	1 Month	2 Months	None
	6H	1 month	2 months	3 months
Established on ART/Stable (Missed on opportunity)	3HR	3 months	None	None
	3HP	3 months	None	None
	6H	6 Months		

- Healthcare providers should monitor RoC enrolled on TPT at health facilities or community settings working with CHWs/CBHS.
- PLHIV using TPT and ART should be educated to return to health facilities symptoms of TB or development ADRs.
- Monthly TB screening will be conducted by community HIV based service providers.

NOTE:

- *For TBHIV co-infected individuals who are not on ART at TB diagnosis, TB treatment should be started first, followed by ART as soon as possible, within the first 2 weeks after starting TB treatment.*
- *For PLHIV who are already on ART at TB diagnosis, TB treatment should be started immediately.*
- *In individuals who need TB treatment and who require an ART regimen containing a boosted protease inhibitor (PI), LPV/r double dose (i.e., LPV/r 800mg/200mg twice a day) should be used or with an adjusted, super-boosted dose of RTV (i.e., LPV/r 400mg/400mg twice a day). However use of this regimen is frequently associated with toxicity and RoC on this regimen require regular clinical and laboratory monitoring.*
- *When TB is diagnosed in PLHIV who are already on ART and Medical Assisted Therapy using Methadone, Rifampicin decreases Methadone level by 33% to 68% and hence an increase in the methadone dose increase may be required.*

9.3.5 TPT Treatment completion

A number of endpoints are proposed that could be used to trigger a review of persons on TPT and in some instances, changes to treatment. A RoC on TPT is considered to have completed treatment if she/he takes the minimum required number of doses in the specified time.

Key Note:

Treatment completion. A person initiated on TPT who completed at least:

- **80% of recommended dose (68/84) consumed within 4 months of planned TPT duration for 3HR.**
- **90% of recommended dose (11/12) consumed within 4 months of planned TPT duration for 3HP.**
- **80% of recommended dose (144/180) consumed within 9 months of planned TPT duration for 6H.**

9.3.6 TPT Treatment Interruption; 3HP (Weekly Schedule)

Missed One Dose:

- **Remembered Within 2 Days:**
If the missed dose is remembered within 2 days, the person should take the dose immediately and continue the schedule as originally planned (i.e., take the remaining doses on the same weekly schedule).
- **Remembered After 2 Days:**
If the missed dose is remembered more than 2 days later, the person should take the missed dose immediately and adjust the weekly schedule to the day the missed dose was taken. This ensures that two weekly doses are not taken less than 4 days apart.

Missed More Than One Weekly Dose:

- **1–3 Weekly Doses Missed:**
Continue treatment until all 12 doses are taken, extending the treatment duration to a maximum of 16 weeks.
- **4 or More Weekly Doses Missed:**
Offer counselling and restart the full TPT course under Directly Observed Therapy (DOT). If adherence to a weekly schedule is not feasible, discontinue 3HP and offer a daily regimen instead.

9.3.7 3HR and 6H (Daily Schedule)

Missed Less Than 2 Weeks:

- Resume preventive treatment immediately upon return and add the number of missed days to the total treatment duration.
- Do not change the scheduled date of the next follow-up visit; however, the final follow-up visit will be postponed by the number of extra days to compensate for the missed doses.

Missed More Than 2 Weeks:

- If More Than 80% of Doses Taken:
 - o For 6H: 144 doses.
 - o For 3HR: 72 doses.No action is required. Continue and complete the remaining treatment as per the original plan.
- **If Less Than 80% of Doses Taken and Course Can Be Completed Within Additional Time:**
 - o Add up to 33% extra time for treatment completion:
 - For 6H: Total duration = 239 days.
 - For 3HR: Total duration = 120 days.No action is required. Continue and complete the remaining treatment as per the original plan.
- **If Less Than 80% of Doses Taken and Course Cannot Be Completed Within Additional Time:**

Restart the full TPT course.



9.3.8 TPT Drug Interaction with ARV drugs

Name of ARV	Type of interaction	Recommended action
TDF	No Interaction	No dose adjustment
3TC/FTC	No interaction	No dose adjustment
EFV	Reduced concentration	No dose adjustment
NVP	Reduced concentration	Don't use
DTG or pDTG	Reduced concentration	Double the dose (50mg) and administer in 2 doses 12 hourly apart when using 3HR as TPT No dose adjustment required when using 3HP
AZT	No interactions	No dose adjustment
ATV/r	Reduced concentration	Don't use
LPV/r	Reduced concentration	Double the dose to 800mg/200mg

Name of ARV	Type of interaction	Recommended action
RTV	Reduced concentration	Don't use
ABC	No interaction	No dose adjustment
RAL	Reduced concentration	Double the dose to 800mg
DRV/r	No studies	Don't use

9.4 TB Infection control in healthcare facilities and congregate settings

TB infection control should be implemented in healthcare facilities and congregate settings where people with TB and HIV are frequently confined. Measures to reduce TB transmission include administrative, environmental, and personal protection measures, which are generally aimed at reducing exposure to *M. tuberculosis* among healthcare workers, prison staff, police officers and their clients and other persons in the congregate settings. Measures are explained in the Table 9.4:1 below.

Table 9.4:1 TB Infection control in healthcare facilities and congregate settings

Control Measure	Evaluation
Administrative measures	• Early identification of TB clients and reduction of TB transmission. • All RoC should be screened for TB as soon as they arrive at the facility to identify those with at least one TB symptom. • In outpatient departments, coughing patients should wait in well-ventilated areas. • TB suspects need to be examined in a well-ventilated room. • Have patients turn their heads and cover their mouths when they cough. • Avoid contact between TB and HIV positive patients by separating them. • If TB clinic is providing ART, channel PTB and HIV co-infected patients to the TB clinic where they should receive TB and HIV care i.e. anti TB treatment/CPT, ART, adherence counselling; • Refer them to CTC at the end of the TB treatment to ensure continuum of care (general HIV care, CPT, ARV provision, HBC, etc.) • If the TB clinic is not providing ART, evaluate PTB and HIV co-infected patients at the CTC on separate days to avoid sharing the same waiting area with PLHIV. • If volunteers living with HIV (e.g. peer educators) are working at the HF level (e.g. CTC), they should be informed about the risk of developing TB and they should avoid accompanying TB suspects/RoC.

Control Measure	Evaluation
Environmental control measures	• Open doors and windows to allow cross air ventilation. • Waiting places and examination rooms designed in a manner that they have maximum natural ventilation. Fans may also assist in the process of air circulation. • Collection of sputum for TB should be done in an open environment and away from other people, not in small rooms or other enclosed places.
Personal protective measures	• Respiratory protective equipment is an additional measure to protect HCPs from inhaling infectious droplet nuclei expelled out into the air by a patient with infectious TB disease. • Personal protective measures should be used by healthcare providers to protect from risk of TB transmission. • Respirators are among the equipment and interventions used to protect personnel who must work in environments with contaminated air. In Tanzania they are recommended to be used when providing care to infectious DR-TB patients or people suspected of having infectious smear positive DR-TB. • The primary way to prevent transmission of TB to health workers and others at the health facility is for TB patients to take their medicines regularly. By doing so, they will become non-infectious in a week or two. Proper ventilation of the place where treatment is provided is also very important.

TB infection control plan:

Every health facility needs to have TB infection plan which is to be reviewed at least once every year. TB infection control plan should contain information regarding TB control in the respective health facility. Moreover, every health facility needs to have TB focal person to oversee implementation of TB infection control measures.

Note:

- ***Increased Risk for Healthcare Workers:***
All healthcare workers who are HIV-positive are at an increased risk of developing TB. Those working in departments where TB patients are admitted should be advised to test for HIV. If found HIV-positive, they should avoid contact with presumptive and confirmed TB cases.
- ***Limitations of Normal Masks:***
Standard masks do not protect medical staff against TB infection.

9.5 Drug-Resistant TB

Drug-Resistant TB (DR-TB) is a form of TB in which first-line anti-TB drugs have limited or no efficacy. Diagnosis is confirmed through **drug susceptibility testing (DST)** using molecular and/or phenotypic methods on *M. tuberculosis* strains.

All cases of MTB detected as **rifampicin-resistant (RR-TB)** should have specimens sent to a designated zonal TB laboratory for first- and second-line line probe assay (LPA), culture, and DST.

9.5.1 Identification of Presumptive Drug-Resistant TB Patients

The following patient groups are at high risk for DR-TB and should have their sputum specimens sent immediately for molecular testing, culture, and DST:

- Patients with treatment failure after using first-line anti-TB medicines.
- Close contacts of known DR-TB cases.
- Patients who remain sputum smear-positive after completing the intensive phase of the first-line anti-TB drug regimen.
- Patients experiencing relapse or returning after loss to follow-up, without recent treatment failure.
- Healthcare providers (HCPs) presenting with TB symptoms.
- Vulnerable groups in congregate settings (e.g., prisoners, urban poor, miners, people who use drugs).
- Patients who had contact with someone who died while on DOT for TB.
- Residents or migrants from high MDR-TB burden settings presenting with TB symptoms.

Management of RR-TB or MDR-TB Cases:

- Patients diagnosed with RR-TB or MDR-TB should be started on an appropriate second-line drug regimen.
- Treatment involves standardized and individualized approaches, consisting of intensive and continuation phases.

For further details, refer to Chapter IX in the TB and Leprosy Manual, 7th Edition, 2020.

9.5.2 TB/HIV and COVID-19

There is no evidence suggesting that TB/HIV co-infected patients who are clinically and immunologically stable, and on both TB and ART medications, have a higher risk of COVID-19 infection or complications compared to the general population. However, co-infected patients who contract COVID-19 are likely to experience more severe disease with poor outcomes.

Risk Factors:

- PLHIV with unsuppressed viral loads or low CD4 levels.
- People with active TB.

The **COVID-19 pandemic** caused significant disruption to essential health services, including TB/HIV care. Therefore, an integrated approach is essential to ensure continued access to TB/HIV and COVID-19 services.

Policy Statements

1. Adhere to COVID-19 protocols in care and treatment facilities.
2. Ensure the use of PPE in care and treatment facilities.
3. Properly manage COVID-19 in HIV and TB/HIV co-infected RoC.
4. Reduce the risk of COVID-19 infection among HIV and TB/HIV co-infected patients.
5. Support the COVID-19 response by leveraging existing resources.

Priorities in TB, HIV, and COVID-19 Response

Key strategies to ensure the continuation of essential services for HIV, AIDS, and TB/HIV co-infected patients include:

a) Rapidly Scale Up Differentiated Service Delivery Models (DSDM):

- Focus on eligible recipients of care using a family-centered approach.

b) Strengthen Facility Infection Control Readiness:

- Implement infection control practices, including bi-directional screening and separating those with COVID-19 symptoms.
- Fast-track vulnerable patients in facilities.
- Ensure chronic care service delivery points are separate from COVID-19 screening, testing, and care areas, ideally in open-air environments.

c) Expand Community-Based Services:

- Scale up community TB and HIV screening, DOT, and ART provision.

d) Strengthen Support for Adherence:

- Enhance in-facility and community processes to support TB and HIV clients in maintaining treatment adherence.

e) Emphasize Vaccination:

- Promote the importance of COVID-19 vaccination in TB and HIV/AIDS messaging.

f) Continue Treatment:

- Ensure continued treatment for HIV, TB, and other comorbidities.

10 MANAGEMENT OF ADVANCED HIV DISEASE AND COMMON OPPORTUNISTIC INFECTIONS

10.1 Introduction

All PLHIV aged five years and above with a baseline CD4 cell count <200 cells/mm³ or WHO HIV clinical stage 3 or 4 are considered to have Advanced HIV Disease (AHD). All children younger than five years who are infected with HIV are also considered to have AHD due to their increased risk of disease progression and mortality. Individuals who are ART-naïve, have documented ART failure, or are returning to care after six months or more of ART interruption should be evaluated for AHD. CD4 testing is essential for identifying Recipients of Care (RoC) with AHD among PLHIV aged five years and above.

Adults, adolescents, and children aged five years and older identified to have AHD based on CD4 count or WHO clinical staging should promptly receive the AHD package of care (Table 9.1). This package includes:

- Screening and diagnosis of opportunistic infections,
- Prophylaxis and pre-emptive therapy,
- Rapid initiation of ART, and
- Adherence support.

Additionally, the STOP (Screen, Treat, Optimize, and Prevent) AIDS approach has been shown to reduce morbidity and mortality among children with AHD.

Note:

Routine cryptococcal antigen screening and pre-emptive therapy are not recommended for children younger than 10 years due to the low prevalence of cryptococcal meningitis in this age group. However, if a child younger than 10 years presents with signs and symptoms of meningitis, cryptococcal meningitis should still be considered, and appropriate investigations and treatment must be conducted.

Definition of AHD

- Advanced HIV disease is defined as CD4 cell count < 200 cells/mm³ or WHO HIV clinical stage 3 or 4 in adults, adolescents and children aged five years and above.
- All HIV infected children younger than five years old are considered as having AHD.

Priority Groups for AHD Screening

- Newly diagnosed individuals
- Individuals returning after interruption of treatment for more than 6 months.
- Those failing treatment (VL > 1000 copies after EAC)
- RoC newly categorized as WHO stage 3 or 4

10.1.1 AHD Screening

CD4 cell count progressively decreases as HIV advances and the immune system deteriorates. It serves as an important immunological marker for identifying Recipients of Care (RoC) who are at an increased risk of developing opportunistic infections and death. **CD4 cell count should be used as the primary method to identify RoC with Advanced HIV Disease (AHD).** If access to CD4 testing is limited or unavailable, **WHO clinical staging** should be utilized. The steps for AHD screening are detailed in **Figure 9.1** below.

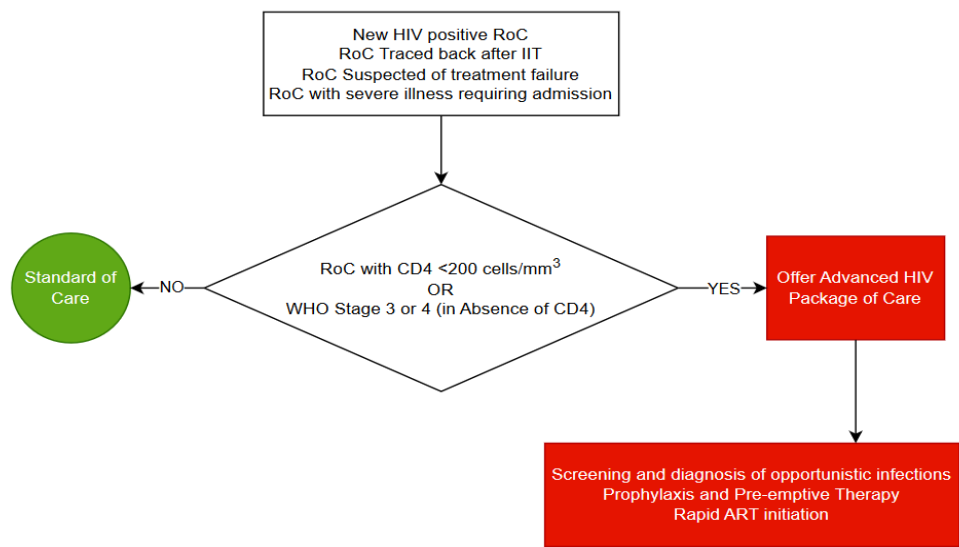


Figure 10.1:1 AHD screening algorithm

To optimize and expand access to AHD services, the **Hub and Spoke model** will be employed for both screening for AHD and delivering the AHD package of care.

Table 10.1:1 Advanced HIV Disease package in different age groups

AHD services	Interventions	CD4 Cell count/ WHO HIV clinical staging.	Adults and Adolescents (10 years and above)	Children (0-9 years)
Screening and Diagnosis	• TB screening questionnaires (TSQs) • Chest X-ray where available. • Sputum for microscopy	All PLHIV	YES	YES

AHD services	Interventions	CD4 Cell count/ WHO HIV clinical staging.	Adults and Adolescents (10 years and above)	Children (0-9 years)
	and molecular test (Xpert/ TruNat) for TB diagnosis among symptomatic PLHIV.			
	• Urine Lateral Flow-LAM for TB	CD4<200 cells/mm3 or WHO clinical stage 3 or 4	YES	YES
	Cryptococcal antigen Screening (CrAg)	<200 cells/mm3 or WHO HIV clinical stage 3 or 4	YES	NO (unless presents with signs and symptoms of meningitis)
Prophylaxis	Co-trimoxazole prophylaxis	≤350 cells/mm3 or WHO clinical stage 3 and 4.	YES	YES
	TB preventive treatment	All PLHIV	YES	YES
	Fluconazole pre-emptive therapy for serum CrAg – positive PLHIV without evidence of meningitis. N.B: <i>When Lumbar puncture services are available, Pre-emptive therapy among serum CrAg positive should be initiated upon confirmed CSF CrAg negative but should not delay initiation of Pre-emptive therapy.</i>	<200 cells/mm3	YES	NA

AHD services	Interventions	CD4 Cell count/ WHO HIV clinical staging.	Adults and Adolescents (10 years and above)	Children (0-9 years)
ART Initiation	Rapid ART initiation	All newly diagnosed HIV positive RoC.	YES	YES
	Delay ART initiation if clinical symptoms suggest TB or cryptococcal meningitis	• Start ART within two weeks after starting anti TB treatment/ Fluconazole Pre-emptive therapy. • <u>At least 5 weeks after starting treatment for cryptococcal meningitis</u>	YES	YES
Adherence support	Tailored counselling to ensure optimal adherence to the AHD package, including home visits if feasible	<200 cells/mm ³ or WHO Clinical stage 3 or 4	YES	YES

Note:

Lumber puncture should be done in all PLHIV with a positive +ve CRAG test in serum/plasma/blood with signs and symptoms of meningitis. Additionally, lumber puncture is recommended for all PLHIV with a CD4 count of <100cells/mm³ regardless of signs and symptoms.

Table 10.1:2 Management of children with AHD: Screen, Treat, Optimize and Prevent AIDS (STOP) among Children (0-9 years)

Interventions	Diseases	Description
Screen	Tuberculosis	• Screen for TB using available screening tools as indicated. Stages for those who screen positive use the following diagnostic tests to confirm TB as applicable. • Rapid molecular diagnostic on (induced) sputum, stool, gastric aspirate, or nasopharyngeal aspirate or other extra pulmonary samples if relevant - LF LAM assay

Interventions	Diseases	Description
	Cryptococcal infection among adolescents	• Serum or plasma or blood CrAg screening • Lumbar puncture if serum CrAg positive and symptomatic
	Malnutrition	• Weight for height • Height for age • Mid upper arm circumference among children 2-5 years old
Treat	TB, severe pneumonia, severe bacterial infections, cryptococcal meningitis and severe acute malnutrition	
	Optimize	In accordance with existing national guideline
Optimize	Rapid ART initiation	
	ART counselling	Preferably same day after the diagnosis
	Bacterial infections and P.Jirovecii Pneumonia	In accordance with National guidelines
	TB	TB preventive treatment
	Cryptococcal meningitis among adolescents	Fluconazole pre-emptive therapy if cryptococcal antigen positive or cryptococcal antigen unavailable
	Vaccinations	
Prevent	Bacterial infections and P.jirovecii pneumonia	Co-trimoxazole prophylaxis
	TB	TB preventive treatment
	Cryptococcal meningitis among adolescents	Fluconazole pre-emptive therapy if cryptococcal antigen positive or cryptococcal antigen unavailable
	Vaccinations	• Pneumococcal vaccine • Human papillomavirus • Measles • BCG

10.2 Screening, Diagnosis, and Management of Common Opportunistic Infections in RoC with AHD

Opportunistic Infections (OIs) occur more frequently and with greater severity among PLHIV due to their weakened immune systems, making OIs the leading cause of mortality in this population. The risk of OIs increases significantly with lower CD4 counts, particularly among individuals with CD4 <200 cells/mm³.

10.2.1 Screening and Diagnosis of Cryptococcal Infection and Disease

Cryptococcal infection (CI) can manifest as **pneumonitis or skin lesions**, but Cryptococcal meningitis (CM) is the most common and severe form of cryptococcosis. CM is a **leading cause of mortality among hospitalized adults living with HIV**, though it is less common in children under 10 years of age. The risk of CM increases as CD4 counts fall below 200 cells/mm³.

Note:

*Not all RoC with cryptococcal infection progress to disease (e.g., meningitis, pneumonitis, or skin lesions). Asymptomatic cases can be managed with **CrAg screening** and prompt initiation of pre-emptive therapy to prevent severe forms of cryptococcosis.*

a) Cryptococcal Screening

- **Serum CrAg Testing:**

Serum CrAg testing is recommended for all adults and adolescents living with HIV **with CD4 counts <200 cells/mm³** at any point during their care.

- **Sample or Client Referral:**

- o If CD4 testing is unavailable on-site, refer the sample or client to a nearby hub or facility for testing.
- o In seriously ill RoC **in WHO stage 3 or 4**, serum CrAg testing should be performed immediately, and management should proceed based on clinical presentation while arranging virtual consultation with a hub facility.

- **Management Based on CrAg Results:**

- o **Asymptomatic RoC** who are serum CrAg-positive should be managed as cryptococcal infection (CI).
- o **RoC with symptoms of CM** who are serum CrAg-positive should be evaluated further for CM.

Common Signs and Symptoms of Cryptococcal Meningitis:

- Fever
- Severe headache
- Neck stiffness
- Confusion (altered mental status)
- Cranial nerve palsy
- Focal neurological deficits, such as:

- o Blurred vision
- o Photophobia
- o Decreased hearing
- o Difficulty in walking

b) Diagnosis

The recommended diagnostic approach for **adults, adolescents, and children living with HIV** suspected of having CM includes:

- **Lumbar puncture (LP):**
 - o Measure cerebrospinal fluid (CSF) opening pressure.
 - o Conduct a CSF CrAg test.
- **Alternative Tests:**
 - o India ink test or CSF culture for *Cryptococcus neoformans*.
- **Referral for Further Investigation:**
 - o If a lumbar puncture cannot be performed or is contraindicated, promptly refer the RoC to a nearby health facility for further investigation and management following a positive serum CrAg test.

Common Contraindications for Lumbar Puncture:

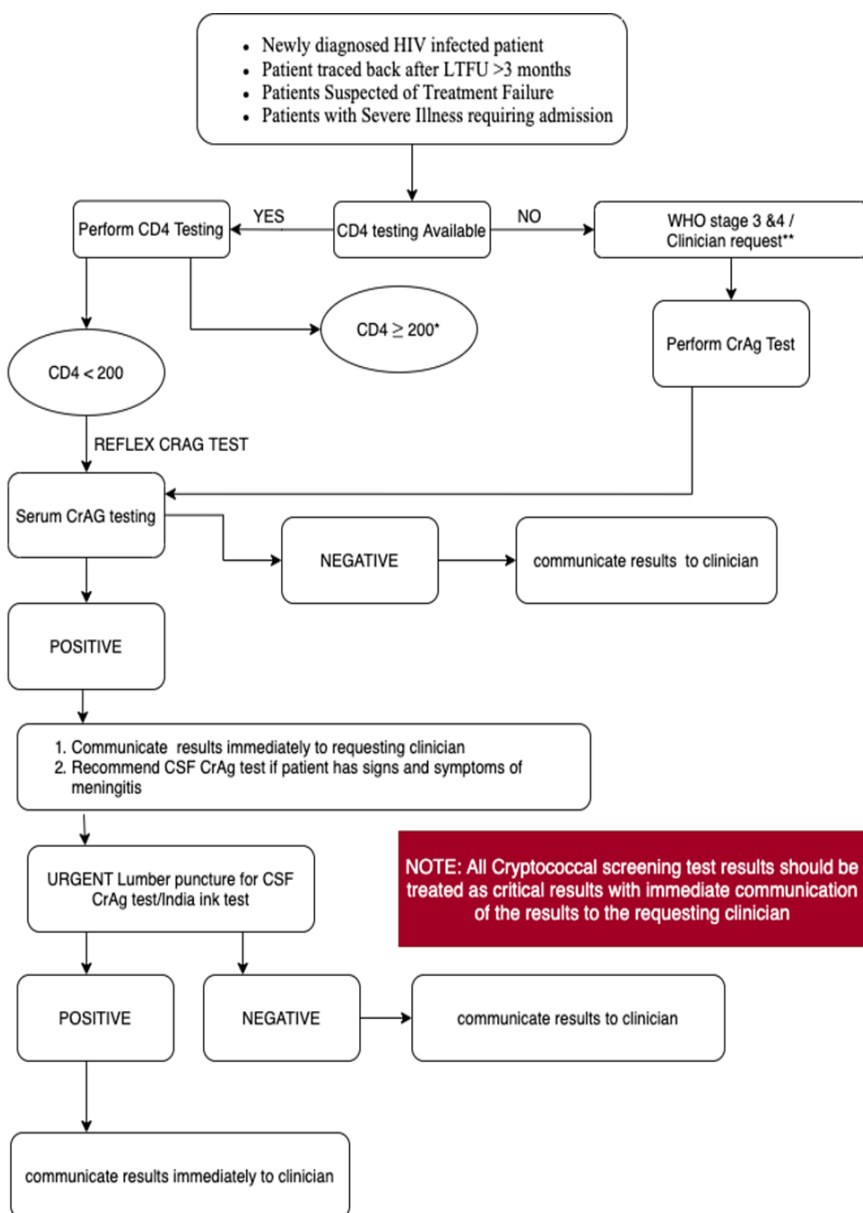
- Significant coagulopathy.
- Suspected CNS space-occupying lesion indicated by:
 - o Focal nervous system signs (excluding cranial nerve VI palsy).
 - o Recurrent seizures.
 - o Confirmation via computed tomography, where possible.

Note:

Signs of raised intracranial pressure do not contraindicate lumbar puncture in suspected cryptococcal meningitis once a space-occupying lesion is ruled out. Lumbar puncture can help relieve intracranial pressure and alleviate severe headaches in CM.

Differential Diagnosis:

Other conditions that present with similar signs and symptoms (e.g., viral, bacterial, or tuberculosis meningitis) should also be investigated to rule out alternative causes.



**Clinician may request for a CrAg test if the patient has signs and symptoms of meningitis regardless of WHO stage and/or CD4 count

Figure 10.2:1 Algorithm for reflex screening and diagnosis of cryptococcal infection and meningitis

a) Pre-emptive therapy for Cryptococcal Infection

Pre-emptive therapy is recommended for asymptomatic CrAg-positive RoC to prevent CM. Initiate fluconazole pre-emptive therapy as illustrated in Table 10.2:1, if the RoC has no clinical signs and symptoms of CM or has CrAg negative CSF.

Table 10.2:1 Fluconazole pre-emptive therapy

Infection	Prophylaxis
Cryptococcal infection (antigenemia) (pre-emptive therapy)	Pre-emptive therapy should be done in 3 phases: Phase 1: Induction phase Fluconazole 800 mg/day for adults for 2 weeks. Phase 2: Consolidation phase Fluconazole 400mg/ day for 8 weeks. Phase 3: Maintenance phase Fluconazole 200mg/day for at least 1 year. Discontinue if client is adherent to ART and fluconazole and $CD4 \geq 100$ and viral load is undetectable (<50 copies/mL) OR $CD4 \geq 200$ cells/mm³ (if viral load monitoring is not available) after being on treatment for at least one year.

b) Treatment of cryptococcal meningitis

Early treatment of CM is a key intervention to reduce mortality from cryptococcal meningitis. This manual recommends the use of triple therapy (amphotericin B, flucytosine, and fluconazole), antifungal medications shown to improve survival in RoC with cryptococcal meningitis.

Table 10.2:2 Treatment of cryptococcal meningitis

Disease	Treatment
Cryptococcal meningitis	The treatment should be done in 3 phases: Phase 1: Induction phase A single high dose (10 mg/kg) of liposomal amphotericin B with 14 days of flucytosine (100 mg/kg per day divided into four doses per day) and fluconazole (1200 mg/daily for adults; 12 mg/kg per day for children and adolescents up to a maximum of 800 mg daily) should be used as the preferred induction regimen for treating people with cryptococcal meningitis. Alternative induction regimens If liposomal amphotericin B is not available A seven-day course of amphotericin B deoxycholate (1 mg/kg per day) and flucytosine (100 mg/kg per day, divided into four doses per day) followed by <u>seven days of fluconazole (1200 mg daily for adults and 12 mg/kg per day for children and adolescents up to a maximum of 800 mg daily).</u>
	If amphotericin B is not available 14 days of Fluconazole (1200 mg daily, 12 mg/kg per day for children and adolescents) and flucytosine (100 mg/kg per day, divided into four doses per day). Note: Fluconazole and flucytosine is the only recommended oral combination regimen and has been associated with lower mortality compared with amphotericin B deoxycholate and Fluconazole.
	If flucytosine is not available 14 days of liposomal amphotericin B (3–4 mg/kg per day) and Fluconazole (1200 mg daily, 12 mg/kg per day for children and adolescents up to a maximum of 800 mg daily).

Disease	Treatment
	If liposomal amphotericin B and flucytosine are not available 14 days of amphotericin B deoxycholate (1 mg/kg per day) and Fluconazole (1200 mg daily, 12 mg/kg per day for children and adolescents up to a maximum of 800 mg daily). Note: flucytosine-containing regimens are superior, and steps should be taken to ensure access to this drug. Phase 2: Consolidation phase Fluconazole (800 mg daily for adults or 6–12 mg/kg per day for children and adolescents up to a maximum of 800 mg daily) is recommended for the consolidation phase (for eight weeks following the induction phase). Phase 3: Maintenance phase Fluconazole 200mg/day for at least 1 year. Discontinue if client is adherent to ART and Fluconazole and CD4 ≥ 100 and viral load is undetectable (<50 copies/mL) OR CD4 ≥ 200 if viral load monitoring is not available) after being on treatment for at least one year.

NOTE: Routine use of adjunctive corticosteroid therapy during the induction phase is not recommended in treating adults, adolescents and children who have HIV-associated cryptococcal meningitis.

Therapeutic lumbar puncture: Draining 20-30ml of cerebrospinal fluid is recommended to sufficiently reduce the raised intracranial pressure in setting where lumbar puncture is possible.

For further information on preparation, administration and management of drug toxicity of amphotericin B therapy, refer to Section 3.7 of the “Advanced HIV Disease Management, A Manual for Healthcare providers”.

c) Timing of ART

For new Recipients of Care (RoC) with Cryptococcal Meningitis, **ART initiation** should be delayed for at least **5 weeks** after starting treatment for Cryptococcal Meningitis and **2 weeks** for Cryptococcal Infection. Immediate ART initiation is not recommended due to the increased risk of mortality.

10.2.2 Screening and Diagnosis of Pneumocystis Pneumonia

Pneumocystis jirovecii Pneumonia (PJP) is caused by the fungus *Pneumocystis jirovecii*. Approximately 30%–40% of adult PLHIV are infected with PJP, and 13% of them require hospitalization.

Clinical Presentation of PJP:

- No or low-grade fever.
- Marked respiratory distress.

- Normal breath sounds or fine diffuse crepitations.
- Severe, persistent cyanosis/hypoxia (**SpO₂ <90%**).
- Poor response to standard antibiotic treatment.

a) Diagnosis

- Diagnosis is mainly through clinical examination.
- Chest X-ray may show hyperinflation, diffuse infiltrates, or normal findings.
- Diagnostic samples (e.g., nasopharyngeal aspirates or bronchoalveolar lavage) can be stained using Giemsa, Silver, or immunofluorescent stains to detect the presence of Pneumocystis jirovecii.

b) Treatment

Management of PJP includes both specific and supportive treatment. Refer to Table 9.5 for details.

c) Prophylaxis Therapy

Cotrimoxazole is used as prophylaxis against PJP among PLHIV.

10.2.3 Screening and Diagnosis of Toxoplasmosis

Toxoplasmosis is a parasitic infection caused by *Toxoplasma gondii* and is a leading cause of central nervous system lesions among adults living with HIV. The prevalence of toxoplasmosis co-infection is high in sub-Saharan Africa, and PLHIV with CD4 counts below 200 cells/mm³ are at increased risk of developing cerebral toxoplasmosis.

a) Toxoplasmosis Screening

- Serologic tests (IgG and IgM) are used for screening.
- Detection of IgG indicates a past infection, while IgM suggests an active infection, especially important for pregnant women.

b) Diagnosis

- Primarily based on clinical findings after ruling out other neurological causes.
- Diagnostic tools (if available):
 - o Serum IgG/IgM.
 - o CT or MRI with contrast.
 - o PCR or biopsy.
- The severity of the infection guides the choice of diagnostic methods.

c) Treatment

Empirical treatment reduces mortality among hospitalized PLHIV by 15%. Refer to Table 9.5 for detailed management guidelines.

d) Prophylaxis Therapy

Cotrimoxazole is used as prophylaxis against toxoplasmosis in PLHIV.

10.2.4 Screening and Diagnosis of Severe Bacterial Infections

PLHIV with **Advanced HIV Disease (AHD)** frequently present with severe **bacterial infections** affecting the bloodstream, respiratory, genitourinary, central nervous, and gastrointestinal systems.

Common Bacteria Isolated:

- **Salmonella, Shigella, and Campylobacter.**
- **Clostridium difficile-associated infection (CDI)** is also common in this population.

a) Common Signs and Symptoms

- Symptoms vary depending on the affected area.
- Fever is commonly present in severe bacterial infections.
- Patients with bacteraemia often appear critically ill.
- Screening for OIs should be conducted at every clinical visit to prevent complications and improve outcomes.

b) Diagnosis

- Use **full blood count** to support diagnosis.
- **Culture and sensitivity testing** are critical to isolate organisms and guide appropriate treatment.

c) Treatment

- Drug susceptibility patterns are essential for optimal antibiotic selection.
- In the absence of culture and sensitivity testing, empirical treatment targeting suspected organisms should be initiated.

d) Prophylaxis Therapy

Cotrimoxazole prophylaxis (CTX) is the recommended first-line regimen for primary prophylaxis against bacterial infections.

Table 10.2:3 Treatment of Choice for Common Opportunistic Infections among Adults and Adolescents

Interventions	Diseases	Description
Pneumocystis jirovecii Pneumonia (PJP)	This condition is common in Tanzania especially among HIV infected children. RoC usually present with non-productive cough, fever, chest tightness and shortness of breath that has evolved over 2-4 weeks. A Chest x-ray may show increased diffuse and symmetrical interstitial markings or diffuse alveolar pattern with infiltrations characterized by asymmetry, nodularity, or cavitations. Normally there is a “bat’s wing’s appearance”. Chest radiograph may appear normal in 10-30% of RoC. Usually in clinical circumstances diagnosis is based on clinical presentation and exclusion of other common causes of severe dyspnoea.	Cotrimoxazole 1920 mg 8 hourly for 21 days and in severe cases give IV cotrimoxazole 15–20mgTMP/75-100mgSMX/kg/day, administered 6-8 hourly, may switch to oral after clinical improvement. For those allergic to sulphur, and if available, give Trimethoprim 12- 15mg/kg/day + Dapsone 100mg/day for 21 days as well as Clindamycin 450mg 4 times/day or 600 mg three times daily + Primaquine 30 mg once daily for 21 days. Adjuvant therapy with steroids may also be beneficial in severe cases. Give Prednisolone 40 mg twice daily for days 1 to 5, then 40 mg once daily for days 6 to 10, and then 20 mg once daily for days 11 to 21 Supportive: • Oxygen therapy • Maintain and monitor hydration • Antipyretic if there is fever • Continue therapy for bacterial pneumonia • Nutrition support
Toxoplasmosis	Clinical features include: Seizures, new onset of focal neurologic findings like hemiparesis and signs of cerebral edema such as confusion, altered mental status and lethargy which can progress to coma. Diagnosis predominantly based on clinical findings after exclusion of other common causes of neurological deficit. If available, a CT scan or MRI with contrast is very useful for confirmation.	Treatment of Active infection Sulfadiazine tabs 1 gm 6 hourly + Pyrimethamine tabs 100mg loading dose, then 50mg /day + Folinic acid tabs 10mg /day for 6 weeks. OR Clindamycin capsules 450mg -600mg 6 hourly + Pyrimethamine tabs 100mg loading dose, then 50mg /day for 6 weeks. NOTE: It is a must to start altogether the three drugs on the same day to prevent bone marrow toxicity. Alternative therapy Trimethoprim– Sulphamethoxazole 5 mg TMP/kg bid iv/po, 25 mg SMX/kg bid iv/po for six weeks (consider IV route if oral route not possible)

Interventions	Diseases	Description
		Maintenance Therapy After six weeks of treatment (Secondary prophylaxis, Maintenance Therapy) Give prophylaxis therapy with Sulfadiazine tabs 500mg 6 hourly + Pyrimethamine tabs 25-50 mg /day + Folinic acid tabs 10mg /day. For those allergic to sulphur replace Sulfadiazine tabs with Clindamycin capsules 450 mg 6 hourly. Alternative therapy Trimethoprim– Sulphamethoxazole (TMP-SMX) tabs 160/800 mg orally twice a day Discontinue maintenance therapy when CD4 count is >200 cells/ml, initial therapy is completed, and the RoC is asymptomatic. Primary prophylaxis therapy for toxoplasmosis can be accomplished with Trimethoprim– Sulphamethoxazole (TMP-SMX) tabs 160/800 mg administered orally/day. For those allergic to sulphur, give Dapsone tabs 50mg/day + Pyrimethamine tabs 50mg per week + Folinic Acid tabs 10mg 3 times a week.
Severe bacterial infections	No specific signs and symptoms of severe bacterial infections since it depends on the area affected	The type of infection, the severity of the illness and the drug susceptibility patterns will guide the choice of antibiotics. In addition to antibiotics, supportive care may be necessary to manage symptoms and prevent complications

10.2.5 Primary Prophylactic Treatment of Common Opportunistic Infections in HIV and AIDS

Cotrimoxazole has a mortality benefit and must be provided in all RoC with AHD. For adults; one double strength tablet (160/800mg) or two single strength tablets once a day and continue until discontinuation criteria are met. For those whose weight is <60kg, refer to the dosing chart (Annex 18.9). Many opportunistic infections can be prevented by using cotrimoxazole prophylaxis, particularly in the case of: Pneumocystis Pneumonia (PCP), Toxoplasmosis, severe bacterial infections, skin infections.

Table 10.2:4 Indication for Cotrimoxazole Preventive Therapy Treatment

Population	Criteria for initiating cotrimoxazole prophylaxis	Criteria for discontinuing cotrimoxazole prophylaxis
Adults, adolescents and children > 5 years of age (including pregnant women) living with HIV	Initiate for everyone with advanced HIV disease (WHO clinical stage 3 or 4) or CD4 cell count ≤ 350 cells/mm ³	Stop when CD4 cell is above 350 cells/mm ³ and HIV Viral load is suppressed (< 50 copies/ml)
Children < 5 years of age living with HIV	Initiate for everyone regardless of WHO clinical stage or CD4 cell count	Continue until above 5yrs of age, then follow criteria for discontinuation for above > 5yrs of age
HIV-exposed infants	Initiate for everyone starting at 4–6 weeks after birth	Until the risk of HIV transmission ends, and HIV infection is excluded with age-appropriate test
People living with HIV and TB	Initiate for everyone with active TB regardless of CD4 cell count	Until the criteria for discontinuation for adults or children are met

Caution in Initiating Cotrimoxazole Preventive Treatment (CPT)

Caution should be exercised when initiating **Cotrimoxazole Preventive Treatment (CPT)** during the first trimester of pregnancy, as cotrimoxazole can cause folic acid deficiency. Pregnant women receiving CPT do not require **sulfadoxine-pyrimethamine (SP)** as an additional medication for malaria prevention.

a) Dosage:

- **Adults:** One double-strength tablet (160/800mg) or two single-strength tablets once daily.
- Use specific Co-trimoxazole formulations and dosages for infants and children living with HIV or exposed to HIV weighing less than 60kg (Refer to Annex 18.9).

b) Follow-Up and Monitoring:

- **Frequency:**
 - o Initially, follow up monthly for the first three months.
 - o If the medication is well tolerated, switch to every three months.
- **Mandatory Monitoring:**
 - o Assess for side effects and adherence.
 - o Monitor for skin reactions, hemoglobin levels, and white blood counts every six months or when clinically indicated.

- o Severe side effects, such as severe cutaneous reactions or fixed drug reactions, are criteria for stopping cotrimoxazole prophylaxis.

10.3 Management of Other Common Opportunistic Infections

Efforts must focus on identifying and managing treatable opportunistic infections in people with HIV and AIDS. These conditions should be addressed at various levels of the healthcare system, emphasizing:

- **Early detection.**
- **Prompt treatment.**
- **Proper referral when necessary.**

Refer to Table 10.3:1 for detailed recommendations on identifying and managing common opportunistic infections in HIV-infected individuals

Table 10.3:1 Management of Common Opportunistic Infections among Adults and Adolescents

Clinical Condition	Clinical Features and Investigations	Treatment
Skin conditions		
Scabies	Diagnosis of many skin conditions is usually made on clinical findings. Other diagnostic investigations include: Potassium hydroxide preparation microscopy, Skin scrapings microscopy	Benzyl benzoate Emulsion 25% (twice a day application after bath for 2-3 consecutive days) Crusted scabies use Ivermectin 20mcg/kg once then repeated in two weeks If secondarily infected Ampiclox 500mg TID for 5-7 days or Erythromycin 500mg TID for 5-7 days
Dermatomycoses	Pus swab for culture & sensitivity Skin biopsy	Whitefield's ointment twice a day for 2-4 weeks Griseofulvin tablets 15-25mg/kg once daily for 6 weeks for Tinea Clotrimazole or Miconazole cream for Candidiasis twice a day 2-4 weeks Terbinafine 250mg od for at least 2 weeks Fluconazole 150mg or 200mg od for at least 2 weeks
Impetigo		Localized –use topical mupirocin ointment 2% BD for 5 days Extensive – Cloxacillin 250mg TID for 5-7 days Erythromycin 500mg TID for 5-7 days

Clinical Condition	Clinical Features and Investigations	Treatment
Pruritic Papular Eruption (PPE)		Antihistamine, e.g., Cetirizine 10mg once daily for 3 days or Loratidine 10mg Topical steroids, e.g. hydrocortisone, Mometasone cream apply twice a day Antibiotics if there is a secondary bacterial infection, e.g. Cloxacillin or erythromycin
seborrheic Dermatitis		Antifungal Ketoconazole 2% lotion 2-3 times/ week for 4 weeks' Systemic antifungal if severe Topical Steroids (careful if concomitant TB is suspected) 3% salicylic acid ointment
Molluscum Contagiosum		ART should be considered as the primary treatment of extensive and/ or unusually distributed molluscum contagiosum in HIV-infected RoC. Individual lesion may be treated by: Curettage, Cryotherapy, Electro cauterization
Kaposi's sarcoma (KS)	Cutaneous biopsies using punch biopsy Diagnosis based on clinical criteria and chest X-ray, abdominal USS in cases of systemic KS	This depends on the extent and severity and the options include: Anti-retroviral therapy (preferably PI-based, especially when extensive) Referral for chemotherapy and radiotherapy.
Viral infections		
Herpes simplex	Diagnosis is usually based on clinical history and physical findings. The classical presentation of primary HSV infection includes:	Acyclovir 400mg orally 8 hourly for 7 days for mild and moderate cases of HSV (e.g. cold sores)
	Fever Lymph node enlargement Small painful vesicles	Acyclovir 800mg orally, five hourly for 5 days for severe and recurrent HSV (e.g. genital infection, gingivitis, pharyngeal tonsillitis)
	Painful ulcers on the mucosa and skin Pain along gluteal and upper thigh muscles (Sacral radiculomyelitis) may occur with genital/ rectal HSV	Antibiotics such as Erythromycin should be used when there is secondary bacterial infection. Analgesics when pain is severe.

Clinical Condition	Clinical Features and Investigations	Treatment
	Lesions that usually resolve within 10-21 days after primary infection. The HSV then becomes latent in trigeminal and sacral nuclei and may reactivate.	
Herpes Zoster or Shingles	Early symptoms include pain (often severe and radicular) and fever followed by vesicular rash over involved dermatome(s) 2-4 days later. Primary varicella-zoster virus (VZV) infection usually results in chicken pox. Herpes zoster in HIV infected individuals may be more severe, with more recurrences and may involve more dermatomes. The diagnosis of herpes zoster is usually based on findings of characteristic of painful skin lesions at different stages of evolution (e.g. erythema, papule, vesicles, and crusts) in a dermatomal distribution.	Acyclovir 800mg 5 hourly for 7-10 days for mild and moderate cases IIV/oral Acyclovir 10 mg/kg/day 8 hourly for 7 days for disseminated VZV or ophthalmic nerve involvement (If diagnosed within 72 Hrs) Erythromycin or Cloxacillin 500mg 8 hourly for 7 days for bacterial super-infection Amitriptylin 25-50mg nocte for post-herpetic pain (neuralgia) or Carbamazepine start 100mg od Analgesics, e.g., Paracetamol, Aspirin, or Diclofenac to relieve pain Note: Use of steroids (prednisolone) in herpes zoster is not recommended.
Human Papilloma Virus Infection (HPV)	HPV is a family of viruses that cause genital warts in men and women. HPV is also known to cause cellular changes that can lead to cancer of the cervix in women and anal cancers especially in gay men. The association between HIV and invasive cervical cancer is complex, due to a more rapid progression of cancer amongst HIV-infected women.	Primary prevention of cervical cancer involves prevention of infection with HPV, therefore it can be achieved through behavioural change approaches and the use of biological mechanisms, including HPV vaccination and consistent condom use can reduce the risk of HPV transmission. Annual cervical cancer screening is recommended to all sexually active women under the age of 50 years done at Care and treatment centres (CTC) using VIA (visual inspection of the cervix with acetic acid).

Clinical Condition	Clinical Features and Investigations	Treatment
		Treatment There is no cure for the virus (HPV) itself. There are treatments for the health problems that HPV can cause, such as genital warts, cervical changes, and cervical cancer. For more details for the treatment of genital warts, cervical changes and cervical cancer, refer to treatment guidelines.
Fungal Infections		
Oral, Oropharyngeal, Oesophageal, Trachea-Bronchial and Pulmonary Candidiasis	RoC with oropharyngeal and oesophageal candidiasis may complain of pain and/or difficulty in swallowing, which may be due to infection of the oesophagus with Candida. On examination white painless plaque ("curd like") on buccal or pharyngeal mucosa or tongue surface that can easily be scrapped off will be seen. Where available, barium swallow X- ray/ oesophago-gastro duodenoscopy (OGD) can be performed.	For treatment, any of the following may be used: Fluconazole oral/IV 150mg/day or 200mg/day for 2-3 weeks (for oropharyngeal candidiasis and others) Miconazole oral gel 3-4 times/day after meals for 7 days Nystatin oral suspension 4-6mls 3- 4 times/day continue for at least 2days after oral lesions have disappeared
	Symptoms for trachea-bronchial and pulmonary candidiasis may include fever, non-productive cough, dyspnoea, and tachypnea. Investigations include bronchio-alveolar lavage (BAL) for microscopy and biopsy using bronchoscopy.	Gentian violet solution Note: Treatment should be continued until symptoms resolve.
Vaginal Candidiasis	This is one of the common illnesses presenting with itching and curd-like genital discharge.	Clotrimazole pessaries 100mg Nocte 5 days Miconazole pessaries 400mg nocte 5 days Fluconazole taken orally (in case of pessaries failure)

Clinical Condition	Clinical Features and Investigations	Treatment
Cryptosporidiosis	Cryptosporidiosis (cryptosporidium parvum, cryptosporidium meleagridis and cryptosporidium hominis) is the common cause of chronic diarrhoea.	Nitazoxanide is the recommended treatment 500mg-1000mg twice daily for 14 days + rehydration, electrolytes replacement and optimized ART. Alternatively Paromomycin 500mg PO QID for 14-21 days can be used.
	Diagnosis can be made by microscopic examination of the oocytes in stool or tissue with acid-fast staining or direct immunofluorescence for better sensitivity.	

11 VIRAL HEPATITIS AND CO-INFECTIONS

11.1 Viral Hepatitis Overview

Viral hepatitis is an inflammation of the liver caused by five distinct hepatitis viruses: A, B, C, D, and E. Hepatitis A, C, D, and E have an RNA genome, while Hepatitis B (HBV) has a DNA genome. Viral hepatitis is emerging as a significant global public health challenge. It shares similar risk factors, transmission routes, and populations at risk with HIV and STIs, infecting approximately 5 million people annually worldwide.

In Tanzania, about 4% of adults aged 15–49 are estimated to live with chronic hepatitis B infection. This burden is notably higher among specific populations:

- 7% in young women,
- 8% in pregnant women,
- 5–16% in key and vulnerable populations.

Despite these prevalence rates—comparable to HIV in terms of affected populations and risk groups—viral hepatitis has yet to receive sufficient attention and resources for an adequate public health response.

Hepatitis B Virus (HBV) Transmission

HBV can be transmitted:

- **Vertically (perinatal):** From mother to child at birth.
- **Horizontally:** Through percutaneous or mucosal exposure to infectious blood or body fluids, sexual exposure, or contact with open cuts and sores—particularly among children in highly endemic areas.

Transmission Patterns:

- Approximately 90% of HBV infections occur through perinatal transmission.
- 25–50% of cases occur in children under age 5.
- Less than 5% of infections occur in adulthood.

Clinical Presentation

Most infected individuals are asymptomatic, with the diagnosis often made during routine investigations. Symptomatic Recipients of Care (RoC) may present with:

- Signs of stigmata of chronic liver disease.
- Signs of liver decompensation or liver failure (refer to Table 11.1:1 for details).

Table 11.1:1 Signs of stigmata of chronic liver disease, Liver Decompensation and Liver Failure

Chronic stigmata of Liver Disease	• Ascites • Testicular atrophy • Splenomegaly • Asterixis • Finger clubbing • Fetor hepaticus • Leukonychia • Caput medusa • Anemia/Pallor • Dupuytre contracture* • Axillary hair loss • Xanthelasma • Spider naevi/angiomas • Jaundice • Palmar erythema • Parotid enlargement* • Muscle Wasting • Scratch pruritic marks • Petechial hemorrhage • Body swelling • Gynecomastia in males or Breast atrophy in females
Signs of Liver decompensation	• Ascites • Bleeding tendency • Hepatic encephalopathy • Spontaneous bacterial peritonitis • Hepatorenal syndrome
Signs of liver failure	• Fatal Hepaticus • Asterixis • Bleeding diathesis • Mental confusion • Testicular atrophy

* Signifies presence of alcoholic liver disease

Clinical evaluation of RoC is based on the following approach as per Figure 11.1:1 below

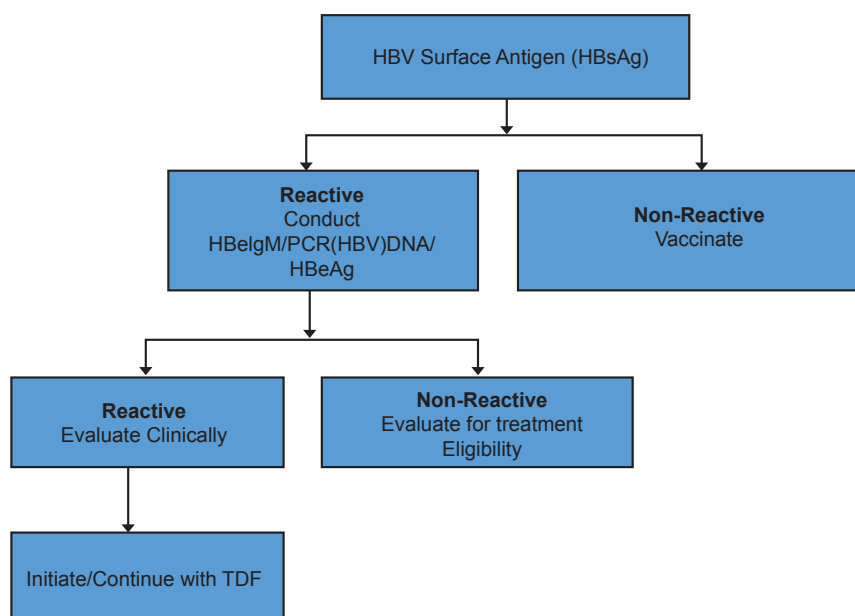


Figure 11.1:1 HBV Diagnostic Testing and Linkage algorithm

Laboratory diagnosis and interpretation of the viral markers are described in the Table 11.1:2

Table 11.1:2 Screening of HBV and its interpretation

Marker	Characteristics
Hepatitis B surface Antigen (HBsAg)/ antibody (HBsAb):	• HbsAg Positive (HbsAg +ve) indicates presence of acute or chronic infection
	• HbsAg Negative (HbsAg -ve) indicates no past infection or resolved infection
	• HBsAb Positive (HBsAb +ve) indicates clearance of the virus associated with HBsAg loss providing lifelong immunity
Hepatitis B core antibody (Anti-HBc IgM/ IgG)	• Anti-HBcIgM indicates acute infection that resolves within six months since the onset of acquiring the infection
	• Anti-HBc IgG Indicates past or chronic infection
Hepatitis B envelope antigen HBeAg+ve / Antibody	• HBeAg Positive (HBeAg+ve) indicates high viral replication with high viral loads
	• Its disappearance (HBeAg -ve) indicates immune control against the virus associated with low viral loads
	• HBe Antibody Positive (HBeAb+ve) indicates establishment of intermediate immune marker after HBe antigen loss
HBV DNA	• Is a confirmatory marker for disease presence, viral replication and disease progression • Is used for therapeutic monitoring and detection of resistant variants

HBV-Infected Recipients of Care (RoC)

HBV-infected RoC should receive a lifelong Tenofovir-based regimen.

- **HBV Monoinfected RoC:** Tenofovir monotherapy (300 mg daily) should be initiated based on eligibility criteria outlined in the National Hepatitis B Treatment Guidelines.
- **HBV/HIV Co-infected RoC:** A Tenofovir-based ART regimen should be initiated regardless of the eligibility criteria for HBV monoinfected individuals.
- **Alternative Regimen:** Entecavir (0.5 mg–1 mg daily) is recommended for RoC who cannot tolerate Tenofovir, especially those with kidney-associated diseases and children below 12 years.

Note:

- RoC with chronic liver disease, cirrhosis, or associated conditions such as diabetes, tuberculosis, and chronic kidney disease should be referred to higher levels of care.
- Treatment eligibility for monoinfected hepatitis B includes chronic liver disease with cirrhosis, an APRI score above 0.5, persistently elevated ALT levels (two values above the upper limit of normal within a 6–12 month period), and HBV DNA above 2000 IU/ml regardless of APRI score.

11.2 HBV Co-Infection

Recent local surveys indicate HBV co-infection among HIV-infected adults ranges from 6.2% to 7.3%. Younger males with low CD4 counts and advanced immune suppression are at higher risk. HBV/HIV co-infected RoC experience:

- Rapid clinical deterioration and poor outcomes compared to mono-infected counterparts.
- Increased rates of HBeAg seropositivity, HBV-DNA levels, liver fibrosis, and mortality.
- Higher risk of chronic hepatitis, reduced chances of spontaneous clearance, HBV replication or reactivation, and complications such as hepatotoxicity and immune reconstitution syndrome.

Recommendations:

- All HIV-infected individuals, including newly diagnosed patients, should be screened for HBsAg and vaccinated if not co-infected.
- Pregnant and breastfeeding women, people who inject drugs (PWID), and other key populations should be prioritized for screening, testing, and treatment due to higher co-infection rates.

11.2.1 HBV Treatment Among PLHIV

- **HBV/HIV Co-infected RoC:** Should receive a Tenofovir-based ART regimen.
- **Alternative:** Entecavir for those unable to tolerate Tenofovir.

11.2.2 HBV Prevention

- HIV-infected individuals without evidence of HBV infection (HBsAg-negative) should be vaccinated following the standard regimen.
- Neonates and Infants: Born to HBV co-infected mothers should receive a birth dose of HBV vaccine plus HBIG (when available) and follow the EPI schedule (6, 10, 14 weeks).
- Pregnant Mothers: HIV-positive mothers should be tested for HBsAg and follow the recommended immunization protocol for themselves and their infants.

For mono-infected pregnant mothers:

- With viral load $\geq 200,000$ IU/ml or HBeAg positive without cirrhosis, initiate Tenofovir or Entecavir at 28 weeks of gestation and continue until six months postpartum when the child is fully immunized.
- Postpartum mothers should be reassessed for treatment eligibility or follow-up protocols as per the National Viral Hepatitis Guidelines.

11.2.3 HBV Monitoring

- **Biannual Monitoring:** ALT, AST, creatinine, FBP, AFP, and abdominal ultrasound for hepatorenal toxicity and hepatocellular carcinoma surveillance.
- **Annual Monitoring:** HBV DNA viral load (PCR) and HBeAg loss to evaluate disease progression.

11.3 Hepatitis C Co-Infection

- **Global Prevalence:** HCV affects 2–3% globally, with higher rates (30–70%) among key populations like PWID.
- **HCV Transmission:** Primarily through percutaneous injections, unscreened blood, and less commonly via sexual, perinatal, or utero routes.

11.3.1 HCV Co-Infection Clinical Evaluation

- Diagnostic evaluation should include detailed history-taking, baseline laboratory tests (ALT, AST, creatinine, AFP), abdominal ultrasound, and confirmation through HCV testing algorithms.

Recommendations:

- **HCV Screening:** Rapid serological antibody testing (anti-HCV) or core antigen testing should be prioritized for high-risk groups (e.g., PWID) and those with clinical suspicion of chronic viral hepatitis.
- **Prevention:** As there is no HCV vaccine, focus on primary prevention through health education, blood and injection safety, and harm reduction.

HCV and Co-Infection Outcomes

- HIV/HCV co-infected patients show more rapid clinical deterioration compared to mono-infected individuals, with higher risks of liver fibrosis, decompensation, hepatocellular carcinoma, and mortality.

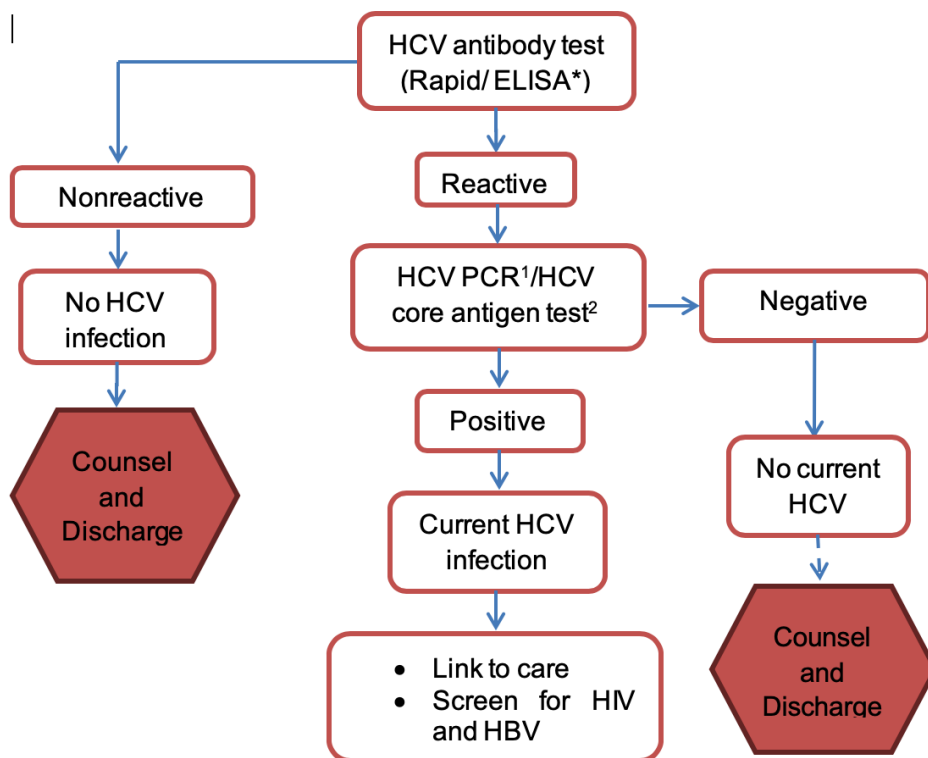


Figure 11.3:1 Diagnosis and management of Hepatitis C Virus.

- Linkage to care involves performing baseline assessment, staging of disease severity and planning for specific treatment.

11.3.2 HCV Treatment

- Treatment of HCV is curative irrespective of HIV status
- Use of fixed combination of anti-HCV based therapy is recommend based on age, weight (Table 10.2.1) and history of past treatment exposure.
- **Sofosbuvir 400mg+Velpatasivir 100mg** daily for 12 weeks up to 24 weeks is recommended regimen of choice for treatment naïve and treatment experienced RoC's respectively
- Depending on availability other fixed combination regimen (**Sofosbuvir 400mg+ Ledpasivir 90mg OR Sofosbuvir 400mg + Daclatasvir 60mg**) daily for 12 weeks up to 24 weeks may be used (Table 10.2.1)
- All RoC completing anti HCV dose duration should have their viral load (HCV RNA PCR) done at three (3) months after treatment completion to establish sustained viral remission (SVR).

Table 11.3:1 Anti-HCV based therapy

Recommended pan genotypic Direct acting antiviral agents (DAA) regimens		Non-pan genotypic DAA Regimen
SOF/VEL	SOF/DCV2	SOF/LED
>30 kg 400/100 mg od (FDC tablet)	>26 kg 400/60 mg od (film-coated tablets)	≥35 kg 90/400 mg od (FDC tablet)
17–29 kg 200/50 mg od (FDC tablet or granules)	14–25 kg 200 mg/30 mg2 (as single tablets, SOF preferred as smaller, 100 mg tablet)	17– 35 kg 45/200 mg (tablet)
<17 kg 150/37.5 mg od (coated granules)		<17 kg 33.75/150 mg (FDC granules packets)

Note

- HCV and HIV Co-infected patients should receive concurrent treatment regardless of CD4 count and fibrosis stage.
- Due to potential drug-drug interactions and presence of hepatorenal toxicity, selection of antiviral regimen should be considered, and duration of treatment extended from 24 up to 48 weeks (Table 10.2.3)
- For individuals with CD4 count less than 200cells/ml, it is recommended to defer HCV therapy until the individual stabilizes (CD4 count greater than 200cells/ml) on ART with HIV RNA suppressed levels of less than 1000 copies/ml.

Table 10.1: Drug–drug interactions between antiretroviral and direct-acting antivirals

DAA	ABC	ATZ/r	DTG	EFV	DRV/r	LPV/r	NVP	TDF	RAL	TAF	3TC	FTC
Sofosbuvir/ ledipasvir												
Sofosbuvir/ Velpatasvir												

Adopted and Modified from WHO guidelines for the care and treatment of person with chronic Hepatitis C virus infection Pg 34

Key:

	Do not co-administer
	Possible toxicity/interaction/dose adjustment, as specified
	No interaction; can be co-administered

ABC: abacavir; ATZ/r: atazanavir/ritonavir; DRV/r: darunavir/ritonavir; DTG: dolutegravir; EFV: efavirenz; LPV/r: lopinavir/r; NVP: nevirapine; RAL: raltegravir; ZDV: zidovudine; TDF: tenofovir disoproxil fumarate; XTC: emtricitabine/lamivudine; TAF: tenofovir alafenamide

11.3.3 HCV Treatment Monitoring

- Baseline and three-month post treatment HCV RNA PCR testing should be performed
- Evaluation of ALT of AST should be done at baseline and repeated every 4 weeks until the end of treatment
- If serum ALT and or AST increases up to 5X upper limit of normal (ULN) but with normal Bilirubin levels and without concomitant symptoms for overt liver disease (jaundice, weakness, nausea and vomiting) then ART can be continued with close monitoring of liver indices until clinical and biochemical remission is resumed.
- ART can be deferred where individual has ALT and or AST elevation is greater than 5XULN, with an increase in Total Bilirubin and or concomitant overt liver disease symptoms and referral for further expert consultation should be sought.

12 SCREENING AND MANAGEMENT OF SEXUALLY TRANSMITTED AND REPRODUCTIVE TRACT INFECTIONS OTHER THAN HIV

12.1 Introduction

Sexually Transmitted and Reproductive Tract Infections (STIs and RTIs) are major global public health concerns, closely linked to HIV transmission. Implementing comprehensive STI and RTI control and prevention programs is critical for reducing the risk of acquiring and transmitting HIV.

12.2 Screening for STIs and RTIs

All Recipients of Care (RoC) should be offered STI and RTI screening services using the screening tool provided (see Flow Charts Annex 18.12).

12.2.1 Approaches for the Management of STIs and RTIs

STIs and RTIs can be managed through the following approaches:

12.2.2 Aetiological Laboratory Approach

This approach identifies causative agents through laboratory tests, followed by disease-specific treatment. It is limited to a few health facilities with well-equipped functional laboratories and is typically applied in STI referral centers.

12.2.3 Aetiological Clinical Approach

This approach targets treatment based on suspected causative agents diagnosed clinically. However, it is not recommended due to its reliance on the clinical expertise of the service provider and the potential for incorrect diagnoses, leading to insufficient treatment.

12.2.4 Syndromic Approach

The syndromic approach identifies clinical syndromes through symptoms and signs, followed by targeted treatment for causative agents. This method, recommended by the Ministry of Health in Tanzania, has proven effective in various healthcare settings.

Common signs and symptoms of STIs and RTIs include:

- Painful urination
- Abnormal discharge
- Genital ulcerations
- Itching
- Swollen lymph nodes

- Scrotal swelling
- Lower abdominal pain
- Pain during sexual activity

Each syndrome may be caused by various agents. Healthcare workers must distinguish between RTIs and STIs, ensuring prompt and accurate diagnosis and treatment. This includes treating sexual partners and providing periodic screening and presumptive treatment to sex workers, as outlined in national guidelines.

12.2 Indications and Opportunities for Screening

Screening opportunities for STIs and RTIs include:

- Family planning services
- Voluntary medical male circumcision (VMMC)
- Speculum examinations
- Testing all male partners of females with STIs and RTIs, and vice versa

Pregnancy-Related Screening:

- Conduct STI and RTI screening early in pregnancy and, if possible, repeat during the third trimester.
- Perform screening at delivery for women who did not attend antenatal clinics.
- Screening for syphilis in pregnant women allows for early detection and treatment, reducing but not entirely preventing congenital syphilis.

Additional Screening:

- Women with a history of spontaneous abortion or stillbirth should be screened for syphilis to address significant causes of adverse pregnancy outcomes.
- Men and women presenting with STI symptoms other than genital ulcers should also be screened for syphilis.

12.3 Overview of STI Syndromes

Although many organisms cause STIs, they result in a limited number of syndromes. Table 11.1 below provides an overview of nine common STI syndromes and their etiologic agents.

13 INTEGRATED MANAGEMENT OF NON-COMMUNICABLE DISEASES (NCDs) IN HIV SERVICES

13.1 Introduction

With the advent of long-term antiretroviral therapy, Recipients of Care (RoC) are living longer, making HIV a chronic condition. However, this longevity increases the risk of developing non-communicable diseases (NCDs) due to the inflammatory effects of the HIV virus and the long-term impact of antiretroviral medications. Integrating NCD screening and treatment into HIV care can help reduce hospital visits, lower costs, and improve service efficiency. This integration will focus on screening, prevention, and control of common NCDs among people living with HIV (PLHIV). This chapter outlines the screening, prevention, and management of common NCDs such as hypertension, diabetes, cervical cancer, and Kaposi’s sarcoma among PLHIV.

13.2 Screening, Diagnosis, and Management of Hypertension

HIV infection and some antiretroviral medications increase the risk of hypertension and cardiovascular diseases in PLHIV. Regular screening for hypertension and its risk factors such as smoking, obesity, physical inactivity, and unhealthy diets—is critical. Blood pressure should be measured and recorded during clinic visits.

A diagnosis of hypertension requires an average of three elevated readings taken on separate occasions when the RoC is calm and comfortable (e.g., on days 1, 3, and 5). Guidance on diagnosing and managing hypertension among PLHIV is provided in the tables and algorithm below.

Table 13.2:1 Screening and diagnosis of Hypertension

Sn	Area of Intervention	Rationale for Intervention
1	Why screen for Hypertension	Uncontrolled hypertension is a preventable risk factor for stroke, heart attack, kidney disease and heart failure
2	Who to screen for Hypertension	All Persons living with HIV aged ≥ 18 years
3	When to Screen for hypertension	At baseline and on every clinic visit
4	What are requirements for measuring hypertension	• Digital Blood Pressure machine • A firm seat with back support • A quiet room or area • Recipient of care treatment card (CTC1) • Recipient of care Monitoring tools

Sn	Area of Intervention	Rationale for Intervention
5	When to diagnose hypertension	- RoC aged ≥ 18 years with: - An average systolic BP ≥ 140 and or diastolic BP ≥ 90mmHg on three separate readings on the same day. - A person on hypertension medicines even when the blood pressure is less than 140/90mmHg

Table 13.2:2 Classification of Hypertension

Class of HTN	SBP mmHg	DBP mmHg	Confirm diagnosis
Grade 1	140-159	90-99	After 3 Month of lifestyle modification
Grade 2	160-179	100-109	Same visit
Grade 3	≥ 180	≥ 110	Same visit

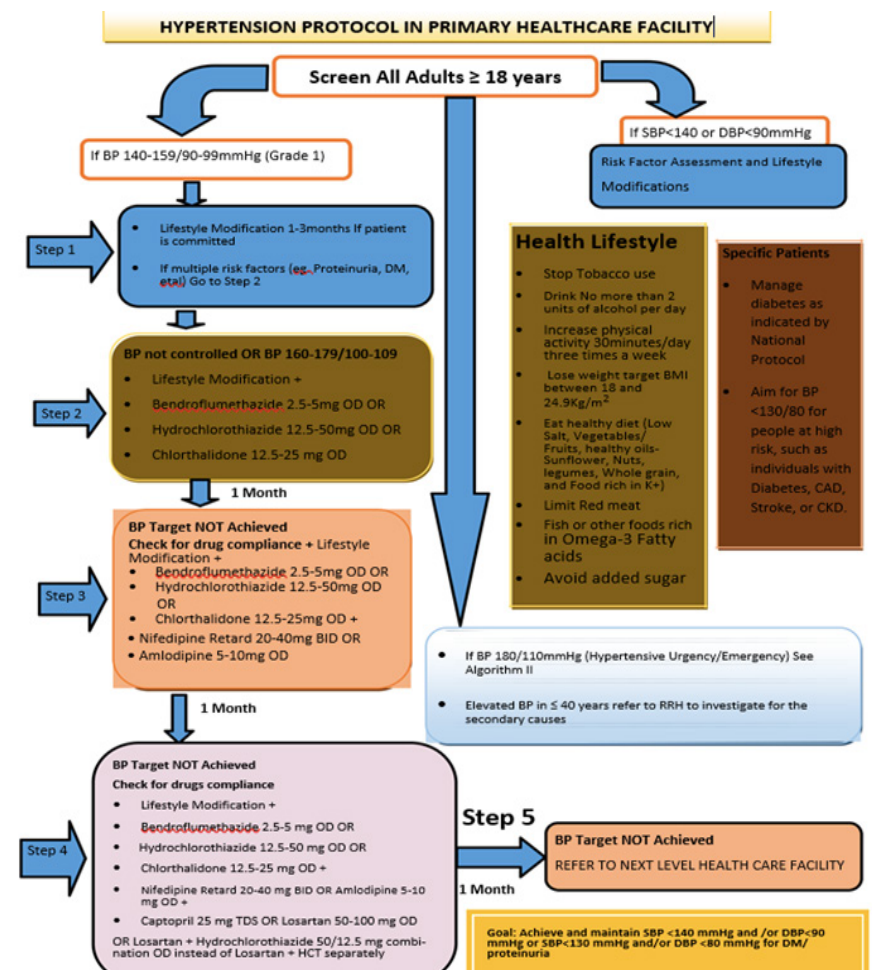


Figure 13.2:1 : Integrated Management of Non-Communicable Diseases (NCDs) IN HIV

13.3 Drug Interaction between Antihypertensive and ARVs

Thiazide diuretics and ACE inhibitors do not interact with ARVs. Calcium channel blockers and beta-blockers may have their drug levels affected by PIs and NNRTIs, requiring potential dose adjustments. However, calcium channel blockers and beta-blockers do not affect ARV drug levels, eliminating the need for dose adjustments.

Table 13.3:1 Drug interactions between antihypertensive and ARVs

Hypertension Drug	Interaction with ART	Action Required
Dihydropyridines (amlodipine and Nifedipine)	Efavirenz and Nevirapine could potentially decrease drug levels	No dose adjustments are required
	LPV/r, ATV/r and DRV/r increase drug levels	Use Nifedipine with caution
Losartan	Efavirenz may reduce formation of active form of losartan	No dose adjustments are required
	Protease inhibitors reduce elimination of losartan	Use with caution, in recipients of care with hepatic impairment
Valsartan (and other ARBs except losartan and lbesartan)	No clinically significant drug-drug interactions	None
All ACE inhibitors (including captopril)	No clinically significant drug-drug interactions	None
All diuretics (including Hydrochlorothiazide and Bendroflumethiazide)	No clinically significant drug-drug interactions	None

Note;

The goal is to achieve and maintain SBP <140mmHg and/or DBP <90mmHg

13.4 Screening, Diagnosis, and Management of Diabetes Mellitus Among PLHIV

Traditional risk factors such as age, Body Mass Index (BMI), and genetic predisposition contribute to the development of diabetes in people living with HIV (PLHIV). Some antiretroviral drugs (ARVs), such as lopinavir (LPV), have been shown to be diabetogenic, with stronger associations to diabetes compared to others like tenofovir (TDF), emtricitabine (FTC), lamivudine (3TC), darunavir (DRV), and atazanavir (ATZ).

Integrase strand transfer inhibitors (INSTIs), such as dolutegravir, have been associated with metabolic syndrome in some patients, characterized by central obesity, hypertension (HTN), and insulin resistance. Additionally, HIV infection itself, along with co-infections like hepatitis C virus (HCV), can further increase the risk of developing diabetes.

PLHIV should undergo regular blood glucose monitoring, starting with a baseline evaluation and continuing with assessments every six months. Recipients of Care (RoC) who experience symptoms such as polyuria (frequent urination), polydipsia (excessive thirst), polyphagia (increased hunger), or have a family history of diabetes or a BMI greater than 30 kg/m², should be prioritized for screening and management.

Table 13.4:1 Clinical Interpretation of Blood Sugar results

Test Category	Diagnosis		
	Normal	Prediabetic	Diabetic
Random Blood Sugar (RBG)	< 11mmol/L	7.8 – 11mmol/L	>11 mmol/L
Fasting Blood Sugar	< 6 mmol/L	6.1- 6.9mmol/L	>7 mmol/L

For further management of diabetes, including management of complications a RoC should be referred to high level facilities.

Table 13.4:2 Screening, diagnosis and management of Diabetes Mellitus among PLHIV

S/N	Area of Intervention	Rationale for Intervention
1	Why screen for Diabetes	• Undiagnosed and/or Uncontrolled Diabetes leads to Stroke, kidney disease, Peripheral Neuropathy, Blindness, Diabetic foot and amputation
2	Who to screen for Diabetes	• All RoC aged ≥15 years • All RoC on DTG and/or LPV based regimen. • All RoC before initiating ART
3	When to Screen for Diabetes	• At baseline • Present with signs and symptoms (Polyphagia, polydipsia, polyuria, polyneuropathy) • Every 6 months
4	How to diagnose Diabetes?	• Two glucose tests, one week apart, are advised for individuals with elevated glucose levels but no hyperglycaemia symptoms. For RoCs experiencing symptoms of high glucose, a single glucose test is recommended.
5	What are requirements for measuring blood glucose	• Screening tool for Diabetes • Glucometer (RBG Machine) with strips • SOP for measuring blood glucose

S/N	Area of Intervention	Rationale for Intervention
6	When to diagnose Diabetes	- Fasting blood glucose (8 hours without food or drink except water): Glucose reading $>7.0\text{mmol/l}$ (126mg/dl). - Random Blood Glucose (RBS) with classical symptoms of hyperglycaemia: Glucose reading $>11.1\text{mmol/l}$ (200mg/dl). - HBA1c: Glucose reading $>6.5\%$.

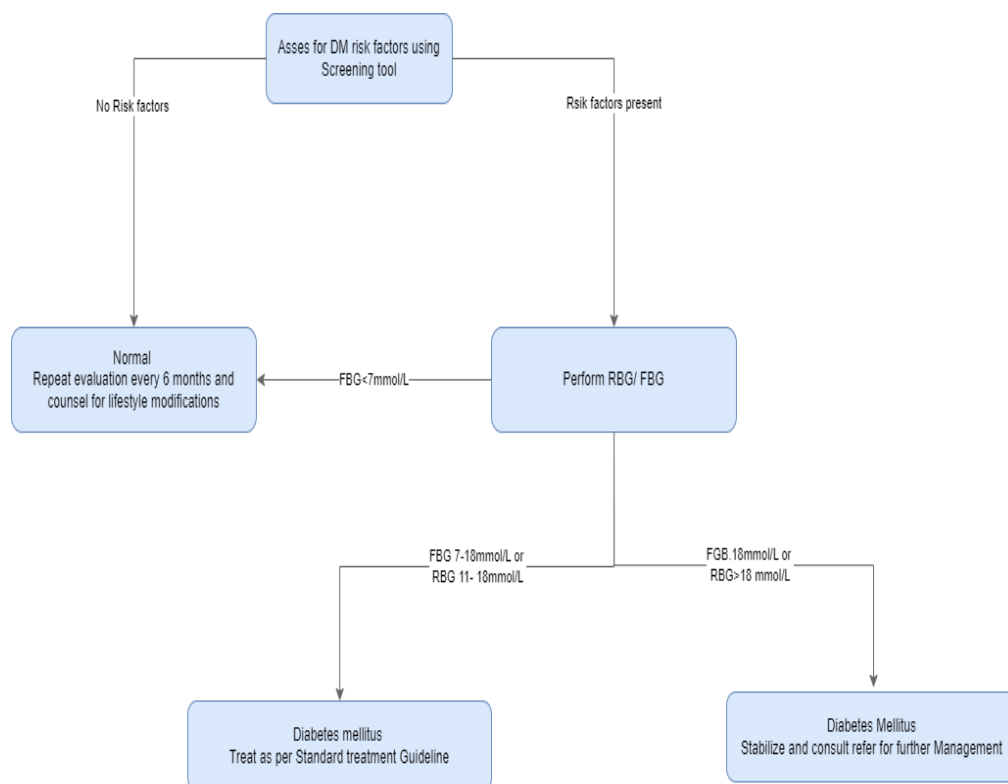


Figure 13.4:1 Algorithm for Screening, Diagnosis & Management Diabetes Mellitus among PLHIV

- FBG – Fasting Blood Glucose (Sugar)
 - RBG – Random Blood Glucose (Sugar)
- * Fasting: no food and only water for 8–14 hours before the test

13.4.1 ART among PLHIV with Diabetes Mellitus

If a RoC develops diabetes while taking potentially diabetogenic ARVs, particularly thymidine based NRTIs (zidovudine), DTG and first-generation PIs (LPV, IDV), it is advisable to switch them to an alternative ART regimen if available. ARVs with a safer metabolic profile, such as TDF, ATZ, EFV and DRV, may be considered part of the new regimen.

13.4.2 Drug Interactions of Diabetes Medicines with ARVs

Metformin and DTG**: a lower dose of metformin may be needed with closer monitoring of blood glucose.

** Do not initiate or Transition RoC to DTG if known or confirmed to have diabetes

13.4.3 Lifestyle Modification for prevention and control of complications of common NCDs

Conduct psychosocial education and counselling on the lifestyle modification for prevention and successful control of HTN based on the following guide.

Table 13.4:3 Lifestyle modification for prevention and control of complication of common NCDs

Risk Factor	Behaviour and Lifestyle Modification
Tobacco Smoking	Ceasing to smoke reduces the risk of uncontrolled hypertension, diabetes, heart disease, stroke, and chronic lung diseases
Physical Inactivity	Persons should be advised to have physical exercises for at least 30 minutes a day, 5 days a week. Healthcare providers should help RoC find activities that they enjoy because this increases adherence
Unhealthy Diet	Eat a diet high in fruits and vegetables and low in fat
	Limit processed and fast foods e.g canned foods, Fried fast foods
	Reduce refined sugar intake e.g Soda, Processed juices
	Reduce Salt intake to <1.5 g/day (less than one teaspoon)
Overweight/ Obesity	Maintain a normal body weight of a body mass index of 18 – 25kg/m ² and or a waist circumference of < 82cm for females and < 102cm for males
Alcohol and Drug use	Promote responsible alcohol drinking
	Avoid use of psychoactive drugs

13.5 Screening and Management of Dyslipidaemias Among PLHIV

Baseline screening of fasting lipid profiles, including total cholesterol, LDL, and triglycerides, should be conducted for all PLHIV. Dyslipidaemia is diagnosed when fasting total cholesterol exceeds 5.2 mmol/L, LDL is greater than 3.4 mmol/L, or triglycerides are above 2.2 mmol/L.

Lipid profile monitoring should occur six months after ART initiation and annually for those with normal results. PLHIV should receive counselling on lifestyle modifications as both a preventive and treatment measure for dyslipidaemia. Recipients of Care (RoC) on ARVs known to exacerbate

dyslipidaemia, such as LPV/r, should be switched to more lipid-friendly drugs like ATV/r before considering lipid-lowering medications. Persistent dyslipidaemia despite these interventions requires referral for further management.

13.6 Chronic Kidney Disease Among PLHIV

Chronic Kidney Disease (CKD) can develop as a complication of HIV, often linked to certain ART therapies. Comorbid conditions such as diabetes mellitus, hypertension, hepatitis B/C co-infection, and nephrotoxic drugs contribute to increased proteinuria, severely reduced eGFR, and a heightened risk of CKD progression. However, early initiation of ART has been shown to improve kidney function in cases of HIV-related CKD.

Routine screening for kidney disease among PLHIV is essential for early diagnosis and management. Baseline serum creatinine and urinalysis for protein, with regular follow-up, are recommended. In cases of abnormal results, dose adjustments or the avoidance of nephrotoxic ARVs like TDF should be considered. For RoC on a TDF-based regimen, switching to a non-TDF-based regimen, such as TAF or ABC, is advised once CKD is diagnosed.

13.7 Cervical Cancer

Tanzania faces a high burden of cervical cancer, with approximately 10,241 new cases annually and a mortality rate of 42.7 deaths per 100,000 women. It is the leading cause of morbidity and mortality among women in the country.

HIV-positive women are at an increased risk and experience an earlier onset of cervical cancer (by about 10 years) compared to HIV-negative women, primarily due to higher rates of persistent HPV co-infection. To address this, cervical cancer screening must be integrated into routine care for HIV-positive women.

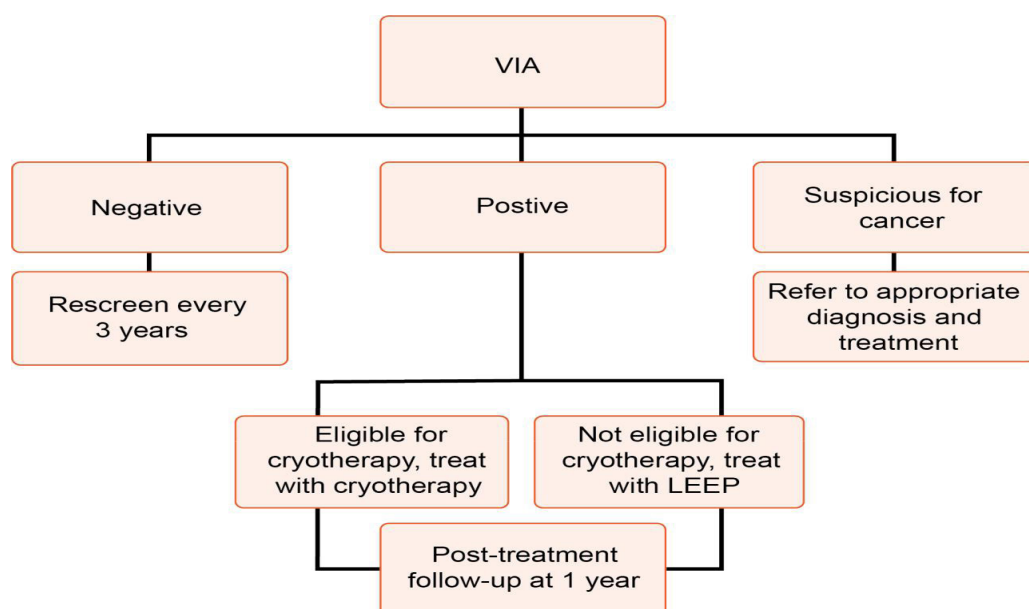


Figure 13.7:1 Flow diagram for cervical cancer screening with VIA

- **Screen-and-Treat Approach (SVA-Single Visit Approach):** Treatment is provided based on a positive primary screening test alone, without triage (i.e., no second screening test and no histopathological diagnosis) and treatment is offered on the same day.
- When the patient is eligible for treatment of small and moderate size pre cancer lesions, can use either Cryotherapy or Thermocoagulation (Ablative Treatment).
- Women who are not eligible for ablation can use Loop Electrosurgical Excision Procedure (LEEP). If LEEP is not available on-site, women should be referred.
- **Screen, Triage and Treat Approach:** The triage test is done if the primary screening test is positive, and the decision to treat is made when both primary

Table 13.7:2 Programmatic intervention to prevent and/or Control cervical cancer

Primary prevention	• Focuses on Girls 9- 14 years** • HPV vaccination for Girls RoC should also be offered, according to IVD guidelines. • Health information and education. • Sex education tailored to age and culture. • Condom promotion and provision for those engaged in sexual activity • Male circumcision
Secondary Prevention	• Focuses on women living with HIV ≥ 25 years of age (Screen every 3 years) • HPV DNA screening is recommended for women aged 25-60 years. • Screening with VIA for positive HPV DNA test, followed by immediate treatment as quickly as possible after identification of precancerous lesions. • Perform Pap smear test for women who are 60 years and above.

Tertiary Prevention	• All WLHIV as needed. • Treatment of invasive cancer at any age (Surgery, Radiotherapy, Chemotherapy, Palliative care)
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Girls aged 9–14 years living with HIV are given three doses of the HPV vaccine at months zero, two, and six.
 (Refer to the “Tanzania Service Delivery Guidelines for Cervical Cancer Prevention and Control (2023)” for detailed information and guidance.)

13.8 Kaposi’s Sarcoma (KS)

Kaposi’s sarcoma is a malignant tumor of angio-proliferative cells, typically starting in the skin but occasionally affecting other organs. It is caused by the Kaposi’s Sarcoma Herpes Virus (KSHV). Kaposi’s sarcoma is classified into four primary types:

1. Epidemic (AIDS-related) Kaposi’s sarcoma
2. Classic (sporadic) Kaposi’s sarcoma
3. Endemic (African) Kaposi’s sarcoma
4. Iatrogenic (immunosuppression-related) Kaposi’s sarcoma

In Tanzania, AIDS-related (epidemic) KS is the most common form. While ART has significantly reduced the prevalence of KS among PLHIV, the emergence of drug resistance poses a potential risk for an increase in AIDS-related KS.

Table 13.8:1 Clinical presentation, diagnosis, treatment and follow up

Clinical presentation	Lesions may involve skin, oral mucosa, lymph nodes, visceral organs, and mucous membrane (palate, gingiva, conjunctiva) May present as macular, papular, nodular, or plaque like appearances. Presence of B symptoms (fever, sweating and weight loss) is commonly associated with epidemic type.
Diagnosis	*Biopsy and Histology of KS lesions to confirm diagnosis
Treatment	Antiretroviral therapy together with Systemic chemotherapy* and/or Radiation Therapy
Follow up	Follow up Physical examination, CXR, Abdominal Pelvic USS every 3 months for 2 years, every 4 months in third year, every 6 months in fourth and fifth years, then annually.

**A referral to higher level facility should be issued if services are not available. Refer to the Tanzania Service Delivery Guidelines for Cervical Cancer Prevention and Control (2023) for detailed information and guidance.*

14 INTEGRATION OF MENTAL HEALTH SERVICES IN HIV AND AIDS CARE

14.1 Introduction

Mental health conditions are prevalent among individuals living with HIV. These conditions may arise due to pre-existing mental health issues, psychological consequences of chronic HIV infection, or the direct effects of HIV on the brain. HIV-infected individuals are at an increased risk of developing mental health conditions, including cognitive, mood, and anxiety disorders, as a result of social challenges, stigma, and discrimination. Healthcare workers (HCWs) should routinely assess and manage mental health conditions as part of the standard care for PLHIV during every visit.

Common mental health conditions among PLHIV include the following:

- **Mood Disorders** (e.g., depression and mania)
- **Anxiety Disorders** (e.g., adjustment disorders, panic disorders, generalized anxiety disorders, post-traumatic stress disorders, HIV- and AIDS-related phobias)
- **Psychotic Disorders** (e.g., schizophrenia, schizoaffective disorders)
- **Organic Disorders** (e.g., delirium and dementia)
- **Alcohol and Substance Use Disorders** (e.g., cannabis, heroin, and cocaine)

14.2 Mood Disorders Among PLHIV

Mood disorders, particularly depression and manic disorders, are common among PLHIV.

14.2.1 Depression

PLHIV face a high risk of developing depression. In Africa, it is estimated that 36% of PLHIV experience depressive conditions, which can negatively impact ART adherence and lead to poor treatment outcomes. Depression screening is particularly important in the following situations:

- When newly diagnosed with HIV or disclosing HIV status to family and friends
- Experiencing physical illness, recognizing new symptoms, progression of the disease, or hospitalization due to AIDS
- Initiating medication
- Persistent high viral load
- Returning to care after being lost to follow-up
- Post-gender-based violence (GBV)

- Death of significant others
- Major life changes (e.g., childbirth, pregnancy, job loss, or the end of a relationship)

Screening and Management

- **Screening:** HCWs should use the Patient Health Questionnaire No. 2 (PHQ-2, Annexes 18.14) as an initial tool for detecting symptoms of depression.
 - o For patients with mild symptoms (PHQ-2 score of 3–4), provide counseling using Mental Health and Psychosocial Support (MHPSS).
 - o For patients with a PHQ-2 score above 4, refer them for a psychiatric evaluation.
- **Management:**
 - o Psychosocial Support: Use the MHPSS counseling tool for depression.

Note

- For PLHIV receiving antidepressant medications, HCWs should monitor for ART/antidepressant interactions as highlighted in Table 14.2:1.

Table 14.2:1 Antidepressant dosage and possible ART interaction

Drug groups of antidepressants	Specific drugs registered in Tanzania	Dose range (mg)	Interactions with ARVs
1. Tricyclic Antidepressant	Amitriptyline Imipramine	25mg to 75mg per day	Lopinavir/r & ritonavir increase antidepressant levels in serum
2. Selective Serotonin (re-uptake inhibitors) Recommended in RoC on ART	Fluoxetine Citalopram	10mg to 20mg per day 10mg to 40mg per day	Nevirapine decreases level; AD increases levels of Amprenavir, Delavirdine, Efavirenz, Indinavir, LPV/r, Ritonavir, Saquinavir

14.2.2 HIV-Related Mania

Definition and Characteristic Features:

AIDS-related mania is a condition secondary to HIV Central Nervous System (CNS) involvement. It is characterized by the inability to control mood, presenting with elated or irritable moods, increased activity and energy regardless of physical status, decreased need for sleep, and an exaggerated sense of self-importance. This condition often occurs in individuals with advanced immunosuppression.

Management:

- Continue ART treatment as it alleviates symptoms of AIDS-related mania and refer the RoC for further psychiatric evaluation.

Possible Drug Interactions:

- Carbamazepine induces liver enzymes and increases the metabolism of ART drugs. If possible, avoid using Carbamazepine in RoC on ART.

14.2.3 Anxiety Disorders

Recipients of Care (RoC) with HIV infection may develop various anxiety disorders, with generalized anxiety, post-traumatic stress, and obsessive-compulsive disorders being particularly common.

Symptoms:

- **Physical:** Shortness of breath, chest pain, increased heart rate, dizziness, and gastrointestinal disturbances. These symptoms may overlap with other common medical conditions.
- **Psychological:** Fear, worry, insomnia, impaired concentration and memory, diminished appetite, compulsive rituals, and avoidance of anxiety-inducing situations.

Screening and Management:

- **Screening:** HCWs should use the Generalized Anxiety Disorder 2 (GAD-2) tool (Annex 18.14) as the initial approach. For patients with mild anxiety symptoms (GAD-2 score of 3–4), provide counseling using Mental Health and Psychosocial Support (MHPSS). For patients with a GAD-2 score above 4, refer them for further psychiatric evaluation.

Management:

- **Psychosocial Support:** Use the MHPSS counseling tool for anxiety.

14.2.4 Psychotic Disorders

RoC with psychotic conditions may present with schizophrenia or schizoaffective disorders.

Common Symptoms:

- Hallucinations
- Delusions
- Thought disorders

Screening and Management:

- **Screening:** HCWs should use the modified Psychosis Screening Questionnaire (PSQ, Annex 18.14) to identify psychotic conditions. For patients with a positive response to any symptoms in the PSQ, refer them for further psychiatric evaluation.

14.3 Organic Disorders

RoC with organic disorders may present with delirium or HIV-associated dementia.

14.3.1 Delirium

- **Definition:** Acute onset of impaired consciousness and memory loss.
- **Management:**
 - Ensure airway, breathing, and circulation support.
 - Refer immediately to the emergency department for further assessment and treatment.

14.3.2 HIV-Associated Dementia (HAD)

- **Definition:** A condition characterized by memory loss and psychomotor retardation.
- **Screening and Management:**
 - Use the International HIV Dementia Scale (Annex 18.14) for screening.

14.4 Alcohol and Drug Use Disorders

Alcohol and drug use disorders occur when an individual uses substances, leading to health and psychosocial impairments. RoC with substance use disorders are at risk of poor adherence to HIV care and treatment, drug-drug interactions, substance-related toxicities, and risk-taking behaviors like promiscuity.

Commonly Used Substances:

- Alcohol
- Benzodiazepines
- Cannabis
- Opioids
- Cocaine

Screening and Management:

- **Screening:** HCWs should use the modified AUDIT-3 or ASSIST tools (Annex 18.14) to identify alcohol and substance use disorders.

- **Management:**
 - o Provide appropriate interventions to manage substance use and support adherence to HIV care.

Management of Alcohol and other substance abuse

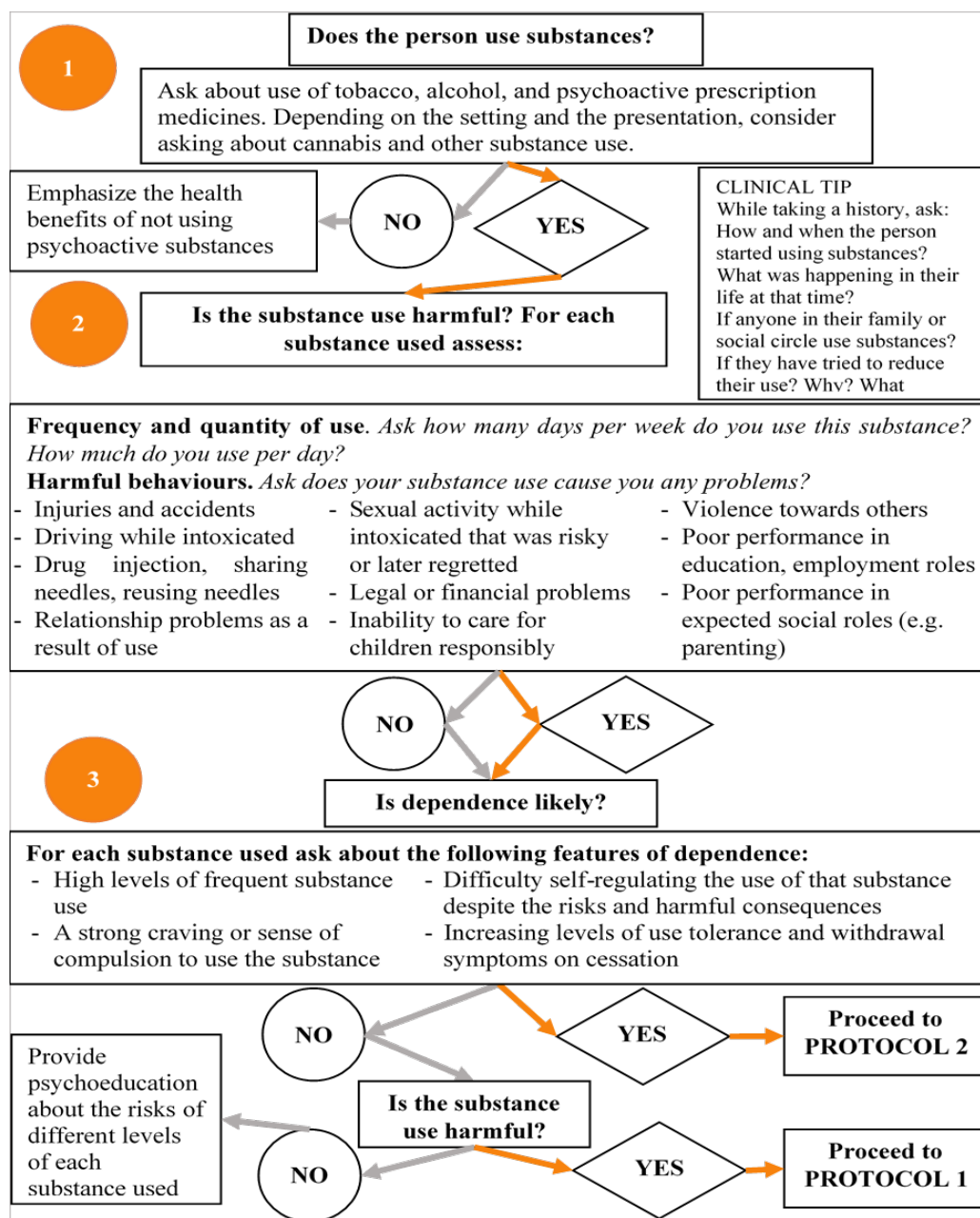


Figure 14.4:1 Algorithm for Assessment of Substance Use

Table 14.4:1 Management of PLHIV with Substance Use Disorder, Protocol 1 and 2.

Protocol 1: Harmful Use	
• Provide psychoeducation and emphasize that the level of substance use is causing harm to health • Explore the person's motivations for substance use. Conduct motivational interviewing. • Advise stopping the substance completely or consuming it at a non-harmful level, if one exists. Verbalize your intention to support the person to do this. Ask them if they are ready to make this change. • Explore strategies for reducing or stopping use and strategies for reducing harm. • Address food, housing, and employment needs • If the person is an adolescent or a woman of child-bearing age, pregnant, or breastfeeding, see special population.	
Protocol 2: Dependence	
If The Person Is Dependent on Opioids: • Assess severity of dependence and, if appropriate, REFER the person for opioid agonist maintenance treatment, also known as opioid substitution therapy (OST) such as Medical Assisted Treatment (MAT) If The Person Is Dependent On Benzodiazepines: • Sudden cessation can lead to seizures and delirium. Consider gradually reducing the dose of benzodiazepine with supervised dispensing or a more rapid reduction in an inpatient setting, refer RoC or consult mental health specialist addiction professional. If The Person Is Dependent On Alcohol: • Sudden alcohol cessation can lead to seizures and delirium; however, if the person is willing to stop using alcohol, facilitate this. Determine the appropriate setting to cease alcohol use, and arrange inpatient detoxification, REFER RoC, or consult mental health specialist. • Provide thiamine at a dose of 100 mg/day p.o. • Consider pharmacologic intervention to prevent relapse in alcohol dependence; medications include acamprosate, naltrexone and disulfiram.	For All Substances: • Advise stopping the substance completely and verbalise potential health effects on ART adherence. • Explore strategies for reducing or stopping use and strategies for reducing harm. • Consider referral to peer help groups or rehabilitation/residential therapeutic communities. • Address food, housing, and employment needs • Assess and treat any physical or mental health co-morbidity, ideally after 2-3 weeks of abstinence. In all cases: • Provide psychoeducation. • Arrange for detoxification services. • Treat withdrawal symptoms as needed. • Provide a brief intervention using motivational interviewing to encourage the person to engage in treatment of their substance dependence. • Provide psychosocial treatment for persons with ongoing problems related to their substance use, if they do not respond to the initial brief interventions

15 PHARMACEUTICAL AND LABORATORY SERVICES - PART 1: SUPPLY CHAIN MANAGEMENT AND RATIONAL USE OF HIV AND AIDS COMMODITIES

15.1 Introduction

A comprehensive HIV and AIDS program requires a diverse range of commodities to support interventions in prevention, care, and treatment. Effective supply chain management of HIV and AIDS commodities is critical to ensuring the implementation of national policies and the continuous availability of quality, affordable essential medicines, diagnostics, and other consumables. These must be delivered to service delivery sites in the correct quantities, at the lowest possible cost, and in a timely manner. Given the high cost of these commodities, proper handling is essential to ensure their effective use.

As all people living with HIV are initiated on ART, robust resources and a strong Procurement and Supply Management (PSM) system must be in place at all levels of the health system. Procurement teams and ART program managers must collaborate to ensure the national supply system functions efficiently, including forecasting, procuring, and distributing the required quantities of HIV and AIDS commodities to meet national demand and achieve the 95–95–95 targets.

Key components of procurement and supply management include:

- 1. Product selection**
- 2. Forecasting and supply planning**
- 3. Procurement**
- 4. Storage and distribution**
- 5. Logistics Management Information System (LMIS)**
- 6. Use or serving customers**
- 7. Quality monitoring**
- 8. Policy**

Management support is integral to each component. Activities span all levels of the healthcare delivery system, from the national level to service points where medicines are dispensed and diagnostics are used. Key activities include managing the Logistics Management Information System (LMIS), ensuring the timely flow of information between stakeholders, and securing financial and other resources for the procurement, storage, and distribution of essential medicines and diagnostics needed for the program.

15.2 Rational Use of Medicines (RUM)

Rational use of medicines ensures that healthcare providers prescribe and dispense appropriate medications tailored to the clinical needs of recipients of care (RoC). Doses should meet individual requirements, medications should be provided for an adequate period, and costs should remain as low as possible for both the RoC and the community.

ART involves a wide variety of drugs, making it a complex and lifelong treatment that evolves over time. The rational use of medicines is particularly important in the context of ART to prevent unwanted consequences at both individual and population levels. These include:

- **Treatment failure**
- **Rapid development of drug resistance**
- **Increased risk of toxicity**
- **Higher treatment costs due to the need for expensive medication after first-line regimen failure**
- **Spread of new HIV and Hepatitis infections**

Proper adherence to RUM principles ensures better treatment outcomes, minimizes risks, and optimizes resource use in HIV and AIDS programs.

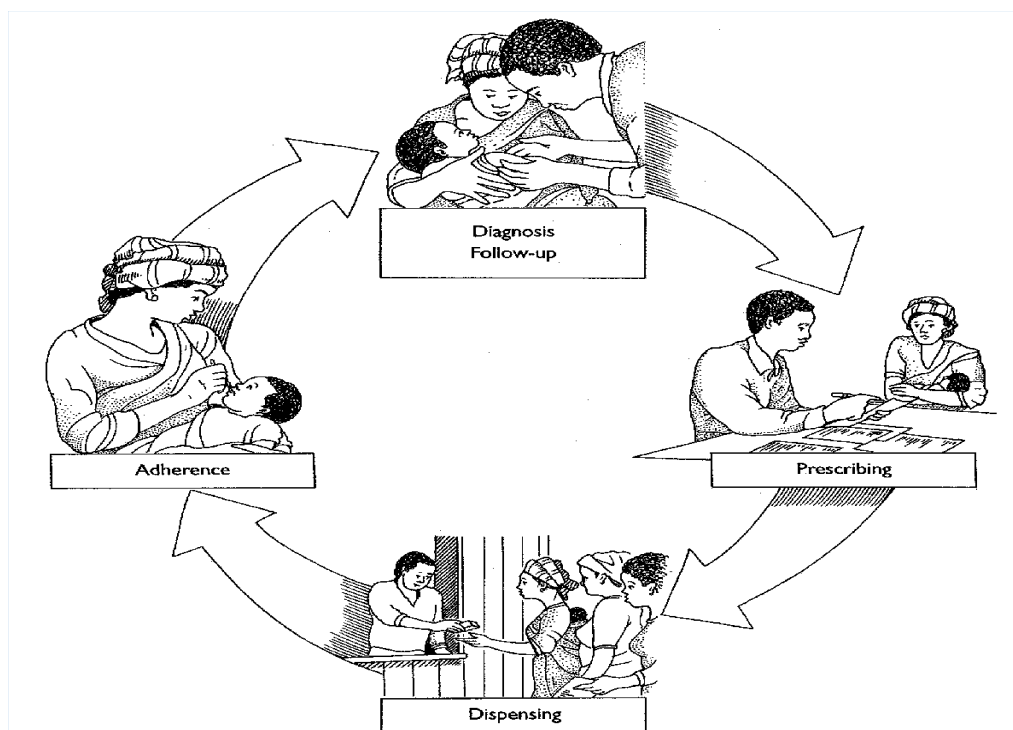


Figure 15.2:1 Medicine Use Process

Table 15.2:1 Aspects HCPs should observe to Avoid Irrational use of Medicine.

Diagnosis	• Inadequate examination of a RoC • Incomplete communication between a RoC and the doctor • Lack of documented medical history • Inadequate laboratory Resources.
Prescribing: Irrational prescribing is observed when there is:	• Incorrect prescribing • Diagnosis is inadequate Inappropriate medicines are prescribed • Under prescribing • Needed medications are not prescribed • Dosage is inadequate • Inadequate duration of treatment • Overprescribing • Prescribing inappropriate length of course • Prescribing very high dose • Extravagant prescribing • Prescribing a more expensive branded medicines when there is a less expensive generic medicine • Treating symptoms instead of treating the disease • Multiple prescribing • Two or more medications are prescribed when fewer would achieve the same effect.
Dispensing: Incorrect interpretation of the prescription	• The dispenser does not pick up errors or the dispenser sees the error but does nothing about it • Incorrect calculation of dosage • Retrieval of wrong medicines • Inaccurate counting • Inadequate labelling • Unsanitary procedures • Inability to effectively communicate with RoC on how to use the prescribed medicines and adherence to dose schedules.
RoC aspects of irrational Use of Medicines. This occurs when	• The RoC demands a prescription of more medicines than required • Not following the given instructions • Sharing medicines with others • Medicine misinformation • Lack of RoC readiness • There is stigma • The conflict between cultural values and therapy • Misleading beliefs about HIV and AIDS • RoCs' misunderstandings about the medicines and their uses • RoC concerns about side effects and ADRs

15.2.1 Prescriptions

Only trained and authorized prescribers in certified healthcare facilities are permitted to prescribe ARVs. ARV prescriptions must clearly indicate the name/RoC ID number, age, sex of the RoC, body weight, medications, dosage, and include the prescriber's name, signature, and prescriber code.

15.2.2 Dispensing

ARVs are prescription-only medicines and should be dispensed exclusively to treatment-ready RoCs with clear instructions and advice. The dispenser must verify that the prescription is appropriately written and signed by an authorized prescriber before dispensing. ARVs should only be provided to the named RoC or their appointed adherence assistant. Adequate time should be allocated for antiretroviral dispensing and counselling.

The pharmacist/dispenser must ensure the RoC understands the dosage, drug intake schedule, storage instructions, and food requirements. Additionally, they should caution the RoC about possible side effects and drug-drug interactions, addressing any specific questions or issues related to ARV treatment. The dispenser must also advise the RoC on how to manage side effects, including the importance of returning to the clinic if unwanted effects occur.

15.2.3 Recipient of Care Identification Cards

Recipients of Care (or their appointed adherence assistants) must present their identification cards to the dispenser each time they collect medications. All medications dispensed must be recorded on the identification card.

Note: ARVs dispensed at the community level must be prescribed only by authorized certified prescribers and dispensed by certified experts.

15.3 Supply Chain Management

15.3.1 Selection of Pharmaceuticals and Diagnostics

The selection of antiretroviral (ARV) drugs and regimens follows recommendations from the World Health Organization (WHO) and considers the specific context of each country's ART scale-up efforts. Laboratory diagnostic platforms and technologies will align with the National Diagnostic Selection Criteria, which may be periodically reviewed. The approved list of diagnostic equipment will guide the selection of technologies based on the country's testing landscape, capabilities, and environments.

Public health treatment guidelines provide comprehensive guidance to ensure universal access, standardization, and simplification of ARV regimens and laboratory diagnostics while minimizing the risk of drug resistance. Key considerations for the selection of ARV drugs, regimens, formulations, and laboratory commodities include procurement, forecasting, and distribution issues.

The guidelines also provide criteria for:

- Different treatment lines
- Monitoring and management of treatment failure or toxicity
- Specific subgroups, such as RoC with tuberculosis, pregnant women, children, and healthcare workers in need of pre-exposure and post-exposure prophylaxis.

15.3.2 Forecasting and Supply Planning (Quantification)

Program coordinators play a critical role in preparing medium-term forecasts to ensure the uninterrupted supply of HIV and AIDS commodities in Tanzania. These forecasts, typically covering a one-year period, are based on morbidity data, national strategies, and stakeholder input. Quantification is informed by data collected at service delivery points using a bottom-up approach. Health facilities rely on consumption data from the electronic Logistics Management Information System (e-LMIS) to determine their requirements for HIV and AIDS commodities.

Forecasts and procurement plans undergo regular revisions every six months to align with evolving pipeline needs and changing demand and supply chain data. This process allows for effective coordination of funding, procurement, and supply planning, ensuring continuous access to ARVs and laboratory services for Recipients of Care (RoC).

15.3.3 Procurement

A uniform and harmonized procurement system is essential for acquiring quality-assured and affordable health commodities, including ARV drugs and laboratory supplies. The procurement process should be guided by:

- The selection of appropriate products
- Forecasted needs based on consumption patterns
- Service expansion
- Formulation and technological changes
- WHO recommendations

The Medical Stores Department (MSD) is responsible for the procurement, storage, and distribution of these commodities. Transparent procedures should be implemented to ensure cost-effective procurement, and a quality assurance system should be in place.

Procurement should prioritize commodities meeting minimum quality standards as stipulated by in-country regulations. Collaboration with partners supporting the national HIV program and pooling demands under

a common tender system can enhance efficiency. Public access to price information and adherence to the United Nations interagency guidelines for donated drugs are key principles to follow in the procurement process.

15.3.4 Inventory Management

a) Ordering and Receiving HIV and AIDS Commodities

HIV commodities should be reported and ordered through the MSD on a bi-monthly basis (R&R forms) and reported monthly via the electronic Logistics Management Information System (eLMIS) under the redesigned Integrated Logistics System (ILS) and Laboratory Supplies Logistics System.

Upon receiving the commodities at facilities, custodians should record them in the store ledger and issue them to the dispensing area as per logistics Standard Operating Procedures (SOPs).

Steps for Report and Request Form Preparation:

1. Obtain stock-on-hand data by conducting a physical inventory at the end of the reporting month.
2. Collect consumption and usage data from dispensing registers, the CTC2 database (pharmacy module), the laboratory information system (LIS), and the national rapid test register.
3. Prepare orders by filling out bi-monthly R&R forms for ARVs, laboratory commodities, and related supplies.
 - o Submit forms to the Council by the 5th day of the ordering month for approval and ensure they reach the MSD by the 10th day for primary health facilities, dispensaries, and health centers.
 - o Hospitals should follow the same timeline for submission and approval.
4. Submit electronic reports and orders via eLMIS, ensuring they are endorsed and approved by healthcare facility authorities.
5. The MSD reviews orders, processes them, and delivers medicines and related supplies directly to health facilities by the 30th of the same month.

For Health Facilities with CTC and PMTCT Services:

- PMTCT should report and order ARVs and RTKs from CTC and labs, respectively.
- Stand-alone PMTCT sites should follow the ILS to order PMTCT commodities along with essential medicines from the MSD.

Receiving HIV and AIDS Commodities

Upon receiving commodities, the receiving officer must verify that the details on the delivery note and invoice match the delivered items in the following areas:

- **Strength and dosage form** (for ARVs)
- **Concentration** (for liquid-based lab reagents)
- **Pack sizes**
- **Batch numbers**
- **Expiry dates** (at least 3 months of remaining shelf life)
- **Specifications**
- **Quantities delivered**
- **Condition of the commodities** (e.g., undamaged items)

If all areas are satisfactory, the receiving officer should sign, stamp, and date the invoice and delivery note.

In Case of Discrepancies:

- Do not accept items with discrepancies.
- Sign against disputed items on the delivery note and note “item not accepted.”
- Record all discrepancies on the Verification and Claims Form (Form 7) and submit it to the supplier, copying it for facility records.

a) Storage and Distribution

Facilities must have adequate storage space with proper conditions, trained personnel, and appropriate logistics tools (e.g., store ledgers and bin cards for usable commodities, and unserviceable ledgers for expired items) to manage supplies effectively. Stock should be secured in a high-security storage area accessible only to a single pharmaceutical or laboratory personnel at a time, alongside other authorized individuals responsible for receipts and issues.

Stock Management Procedures:

- Arrange commodities to follow the First-to-Expire-First-Out (FEFO) and First-In-First-Out (FIFO) principles.
- Maintain accurate inventory records, creating a tracking system to monitor products entering and leaving the supply system, with a running balance and ledgers for each item.

Monthly Inventory Check:

- Conduct a physical inventory at the end of each month, comparing available stock against stock records.
- Enter inventory data into store ledgers/bin cards.

Handling Short Shelf-Life Stock:

- Redistribute stocks with short shelf life that cannot be used before their expiry dates to facilities in need using a redistribution form and issue voucher.

Damaged and Expired Commodities:

- Segregate damaged and expired commodities immediately from usable stock.
- Record them in a suspensory ledger.
- Follow government procedures for the disposal of expired and damaged products.

c) Assessing Stock Status

Adequate stock levels of Max-Min of 4/2 months of stock for each required item must be maintained at all times. If the stock level for a particular item falls below the emergency order point (1 month of stock), an emergency order should be placed to restore the stock to its maximum level.

For Laboratory Commodities:

- Different Max-Min levels should be observed based on the nature and shelf life of the commodities.
- Refer to the table below for specific stock level guidelines for laboratory commodities.

Table 15.3:1 Min- Max parameters according to shelf life of Laboratory Commodities.

		MSD	
S/N	Shelf Life	Min	Max
1.	≥ 24 Months	5	9
2.	≥ 6 months <24 Months	4	6
3.	>6 Months	0	0

Stock Monitoring and Months of Stock (MoS) Calculation

Facilities must determine the number of months (Months of Stock - MoS) their HIV commodities will last based on the current consumption/usage rate. This calculation ensures proper inventory management and helps prevent stockouts or expiries.

Months of Stocks (MoS) = Stock on Hand (SoH) ÷ Average Monthly Consumption (AMC)

The **MoS result** will guide providers to make informed decisions about:

1. Placing **emergency orders** if stock levels are critically low.
2. Conducting **redistribution** of commodities to other facilities with higher needs to avoid wastage or expiry.

Standardized national stock levels should be referred to when interpreting MoS calculations to ensure alignment with national guidelines and optimal inventory management.

c) Record Keeping

Efficient administration and management of HIV commodities requires thorough record-keeping. All information regarding ARVs and opportunistic infection (OI) medicines dispensed should be recorded in a dedicated dispensing register or the **CTC2 database** (pharmacy module) and **ART RoC card (CTC1)**. Information about the usage of RTKs and other laboratory diagnostics should be documented in the rapid test register and laboratory register, respectively. At the store level, all HIV commodity transactions must be logged in the store ledger or the CTC2 database (pharmacy module).

Close monitoring of consumption data and stock levels is crucial for ensuring the correct quantity and quality of supplies, responding to changes in demand, managing increased commodity volumes, and minimizing pilferage and misuse.

15.4 Logistics Management Information System (LMIS)

The LMIS collects, processes, and reports supply chain information, enabling accurate, timely, and appropriate logistics data for decision-making across the supply chain. LMIS can be either manual (paper-based) or electronic (pharmacy database).

Three essential LMIS data types must be documented and reported:

- Stock on Hand
- Losses and Adjustments
- Consumption Data

15.4.1 Logistics Management Tools for HIV and AIDS Commodities

Health facilities should use logistics tools for recording, reporting, and requesting (R&R) supplies, as well as for issuing and receiving commodities. These tools include:

- **Store's Ledger** (and suspensory ledger for expired commodities)
- **Form A1: ARV Daily Dispensing Register/Pharmacy Module**
- **Rapid Test Registers**
- **Laboratory Register**

- **Requisition and Issue Voucher**
- **Form 4:** MSD Sales Invoice
- **Form 6:** Goods Received Note
- **Form 7:** Claim and Verification for Redistribution Form

Facilities must ensure the availability and proper completion of these tools in a timely manner.

15.4.2 Supply Chain Monitoring

Monitoring and evaluation are essential for ensuring commodity security. National programmes must measure progress and outcomes to meet targets and address challenges. Monitoring will involve the use of early warning indicators, such as:

- **Stock availability**
- **Stock status** (e.g., percentage of overstock, understock, adequately stocked, or stock-out)
- **Expiry rate**
- **Reporting rate and timeliness**

These indicators are part of key performance metrics to prevent stock-outs and overstocking that could lead to expiries.

15.4.3 Supply Chain Data Quality Monitoring

Accurate and timely data are critical for effective supply chain operations. High-quality data enable accurate forecasts and ensure constant commodity availability at service delivery points (SDPs).

To improve data quality:

- Conduct **Data Quality Assessments (DQA)** and implement data improvement strategies.
- Utilize the **Information Mobilized for Performance Analysis and Continuous Transformation (IMPACT)** approach, scaling it up to all primary health facilities.

IMPACT Team Responsibilities:

1. Regular meetings to review performance against targets.
2. Identifying and prioritizing challenges.
3. Conducting root cause analyses.
4. Developing and implementing performance and action plans to address identified issues.

15.5 Pharmacovigilance

The monitoring and reporting of adverse drug events should follow the guidelines set by the **Tanzania Medicines and Medical Devices Authority (TMDA)**.

Adverse drug reactions (ADR) reporting guidelines:

- Use yellow forms or the electronic reporting system (via smartphones or ADR Report Tool).
- Reporting options include:
 - **USSD:** 15200#
 - **Toll-free number:** 0800110084
 - **Email:** info@tmda.go.tz

Health facilities should utilize reported ADR data to monitor RoC and make regimen adjustments when necessary. For example:

- **Drug-induced nephrotoxicity:** Switch to ABC-based regimens.
- **Severe hepatotoxicity induced by DTG:** Switch to PIs.

Healthcare providers must document such changes during follow-ups using TMDA's yellow form and adhere to the surveillance protocol.

Table 15.5:1 Types of toxicities associated with First, Second and Third-line ARV drugs.

ARV	Major types of toxicity	Risk factors	Suggested management
TDF	Tubular renal dysfunction, Fanconi syndrome	• Underlying renal disease • Older age • BMI <18.5 (or body weight <50kg) • Untreated diabetes mellitus • Untreated hypertension • Concomitant use of nephrotoxic drugs or a boosted PI	• If TDF is being used in first-line ART, substitute it with ABC or TAF in special circumstances (see the section on using TAF in first line ART) • Do not initiate TDF at an estimated glomerular filtration rate of <50mL/min, uncontrolled hypertension, untreated diabetes or kidney failure • If TDF is being used in second-line ART, substitute it with AZT or ABC.
	Decreases in bone mineral density	• History of osteomalacia and pathological fracture • Risk factors for osteoporosis or bone loss	
	Lactic acidosis or severe hepatomegaly with steatosis	• Prolonged exposure to nucleoside analogues • Obesity • Liver disease	

ARV	Major types of toxicity	Risk factors	Suggested management
	Exacerbation of hepatitis B (hepatic flares)	Discontinuation of TDF due to toxicity	No available alternative drug in the country for treatment of hepatitis B e.g. Entecavir
ABC	Hypersensitivity reaction	Genetic predisposition (HLA-B 5701 gene)	If ABC is being used in first or second-line ART, substitute with TDF or AZT
TAF	Dyslipidaemia Body weight gain	Female sex Concomitant use of DTG	Monitor body weight and promote anti-obesity measures (such as diet, physical exercise). If significant increase despite measures, consider substituting EFV or boosted PI
AZT	Anaemia, neutropaenia,	- Baseline anaemia or Neutropaenia - CD4 cell count $\leq$ 200 cells/mm³	If AZT is being substitute with TDF or ABC consider consider using low dose AZT
	Lactic acidosis or severe hepatomegaly with steatosis Lipoatrophy, lipodystrophy Myopathy	- BMI >25 (or bodyweight >75 kg) - Prolonged exposure to nucleoside analogues	Substitute TDF or ABC
EFV	Persistent central nervous system toxicity (such as dizziness, insomnia and abnormal dreams) or mental symptoms (anxiety, depression and mental confusion)	- Depression or other mental disorder (previous or at baseline) - Daytime dosing	For central nervous system symptoms, dosing at bedtime. Consider using EFV at a lower dose (400 mg/day or an integrase inhibitor (DTG) if EFV 400 mg is not effective in reducing symptoms.
	Convulsions	History of seizure	
	Hepatotoxicity	- Underlying epatic disease - Coinfection with hepatitis B or C - Concomitant use of hepatotoxic drugs	For severe hepatotoxicity on hypersensitivity reactions, substitute with another therapeutic class (integrase inhibitors or boosted PIs).
	Severe skin and hypersensitivity reactions	Risk factors unknown	

ARV	Major types of toxicity	Risk factors	Suggested management
	Gynaecomastia	Risk factors unknown	Substitute with another therapeutic class (integrase inhibitors or boosted PIs).
NVP	• Hepatotoxicity • Severe skin rash and Hypersensitivity reaction, including Stevens- Johnson syndrome	• Underlying hepatic disease • Coinfection with hepatitis B or C • Concomitant use of hepatotoxic drugs • High baseline CD4 cell count (CD4 count >250 cells/mm³ for women or >400 cells/mm³ for men)	• If hepatotoxicity is mild, consider substituting with EFV, including for children three years and older. • For severe hepatotoxicity and hypersensitivity, and for children younger than three years, substitute with another therapeutic class (integrase inhibitors or boosted PIs).
LPV/r	Hepatotoxicity	• Underlying hepatic disease • HBV and HCV co- infection • Concomitant use of hepatotoxic drugs	Replace it with ATV/r
	Pancreatitis	Advanced HIV disease	
	Electrocardiographic abnormalities (PR and QRS interval prolongation, torsades de pointes	People with pre-existing conduction system disease Concomitant use of other drugs that may prolong the PR or QRS intervals Congenital long QT syndrome Hypokalaemia	Use with caution for people with pre-existing conduction disease or taking concomitant drugs that may prolong the PR or QRS intervals
	Lipoatrophy or metabolic syndrome dyslipidaemia	Cardiovascular risk factors such as obesity and diabetes	Substitute another therapeutic class (INSTIs)
	severe diarrhoea and risk of prematurity	Risk factor is unknown	Substitute atazanavir/r, darunavir/r or INSTIs
ATV/r	Indirect hyperbilirubinaemia (clinical jaundice)	• Underlying hepatic disease • HBV and HCV co- infection	Indirect hyperbilirubinemia is usually transient and ATV/r can be

ARV	Major types of toxicity	Risk factors	Suggested management
		• Concomitant use of hepatotoxic drugs • presence of • UDPglucuronogly syltransferase 1-1 (UGT1A1) enzyme (UGT1A1*28 gene)	continued, however, if severe jaundice develops and is associated with significantly raised transaminases, then ATV/r should be replaced with LPV/r This phenomenon is clinically benign but potentially stigmatizing. Substitute only if adherence is compromised.
	Nephrolithiasis and risk of prematurity	History of Nephrolithiasis	Replace it with LPV/r or DR/r. If boosted PIs are contraindicated and NNRTIs have failed in firstline ART, consider substituting INSTIs
	Electrocardiographic abnormalities (PR and QRS interval prolongation)	People with pre-existing conduction system disease Concomitant use of other drugs that may prolong the PR or QRS intervals Congenital long QT syndrome	Use with caution for people with pre-existing conduction disease or who are taking concomitant drugs that may prolong the PR or QRS intervals
DTG	Increase in cholesterol levels; mild elevated liver enzymes; significant rises in creatinine levels; and headache may also be experienced	History of dyslipidaemia, diabetes, hypertension	• Monitor cholesterol levels; monitor Liver function especially in HBV and HCV. • Provide symptomatic treatment
	Insomnia Body weight gain or obesity	Older than 60 years Low CD4 or high viral load Female African ethnicity Concomitant use of TAF	• Consider morning dose or substitute EFV, boosted PI or RAL • Monitor body weight and promote anti-obesity measures (such as diet and physical exercise). • If significant increase despite measures, consider substituting EFV or boosted PI

ARV	Major types of toxicity	Risk factors	Suggested management
	Hepatotoxicity Hypersensitivity reaction	Coinfection with hepatitis B or C Liver disease	• Substitute another therapeutic class: EFV or boosted PIs
ETR	Common: Skin rash, allergic reactions, Nausea, increased low density Lipids, Gastrointestinal disorders and Fatigue Rare: Severe skin rash, Peripheral neuropathy and renal failure	No known risk factors	• Monitor severity and occurrence of fever and other symptoms. • Provide symptomatic treatment
RAL	Increased Cholesterol levels, Glucose, Aspartate Amino Transferase (AST), Bilirubin, Rash, Cough, Fatigue, dizziness and insomnia	History of dyslipidemia, diabetes, hypertension	In case of severe adverse effects, switch to DTG if RoC is >12 years old
	Rhabdomyolysis, myopathy and myalgia	Concomitant use of other drugs that increase the risk of myopathy and rhabdomyolysis, including statin	Stop ART. When symptoms are resolved, substitute another therapeutic class (NNRTIs, boosted PIs)
	Hepatitis and hepatic failure Severe skin rash and hypersensitivity reaction	Risk factor(s) unknown	
DRV/r	Increased Cholesterol levels, triglycerides; Diarrhoea, Headache, Rash, Abdominal pain and Nausea	History of dyslipidaemia	• Monitor severity and occurrence of fever and other symptoms. • Provide symptomatic treatment
	Hepatotoxicity	Underlying hepatic disease Coinfection with hepatitis B or C Concomitant use of hepatotoxic drug	• Substitute with ATV/r or LPV/r. When it is used in third-line ART, limited options are available For hypersensitivity reactions, substitute another therapeutic clas
	Severe skin and hypersensitivity reactions	Sulfonamide allergy	

15.6 Multidisciplinary team

Close collaboration between pharmaceutical, laboratory personnel, and clinical staff is critical to ensuring accurate diagnosis, appropriate prescription, and optimal ARV regimen selection. Pharmaceutical and laboratory personnel, along with other healthcare workers involved in supply chain management, must keep clinical staff informed about ARV and laboratory commodity stock levels, especially for items at risk of stock-outs or expiration.

In cases of nationwide supply shortages, effective communication between supply chain personnel and clinical staff is essential to determine the best course of action. Additionally, pharmacists should stay updated on new developments and changes in ARV regimens. They serve as a valuable resource for clinicians, providing guidance on drug-related side effects, formulation changes, available options, and drug combinations.

15.7 Ensuring Program Flexibility and Avoiding Commodity Expiry

Minimizing the risk of stock-outs and commodity expiries requires proactive measures, including:

- Programme managers and clinicians must align on treatment initiation rates, outcome measurements, screening of new-borns, and assessment of individuals with advanced HIV disease to ensure the availability of required commodities.
- Supply chain implications of changes in service delivery models, such as multi-month dispensing or community ARV distribution, should be well understood.
- ARV and diagnostic supply chain distributions should be allocated by facility, reflecting the geographical distribution of the epidemic.
- New recipients should initiate treatment with the preferred first-line regimen unless clinically contraindicated.
- Quantification and ordering should include a rotating safety buffer to account for forecasting errors and potential delivery delays.
- Orders must be placed at least 12 months before the required delivery date to allow adequate production time and reduce shipping costs by enabling sea freight where feasible.
- Deliveries should be staged rather than arriving as a large shipment to cover six months or more of stock.

15.8 Laboratory Services

Medical laboratories play a vital role in clinical decision-making by providing critical information to prevent, diagnose, treat, and manage diseases.

The Ministry of Health (MoH) has implemented Laboratory Management Systems (LMS) to strengthen Laboratory Information Systems (LIS), Planned Preventive Maintenance (PPM), Quality Management Systems (QMS), and sample transportation systems.

Aligned with the Diagnostic Network Optimization (DNO) concept, a network analytics approach will continue to improve and implement a recipient-centered, cost-efficient diagnostic system that ensures equitable diagnostic services for all.

Integrated HIV, STI, and Hepatitis programs rely on laboratory services for diagnostic and monitoring tests for recipients of care (RoC), following established algorithms for HTS, CD4, and HVL testing. Smooth transitions in testing methodologies and technologies should consider:

- Facility training needs.
- Existing national stock levels.
- Incoming shipment schedules.

These steps help eliminate supply chain interruptions. Laboratory commodity supply chain management follows the same principles and procedures as other HIV commodities described above.

15.8.1 Sample Collection and Management

Standard Operating Procedures (SOPs) for sample collection, handling, and storage should be in place for all sample types, including Whole Blood, Plasma, DBS, and Buffy Coat. Storage requirements for different sample types are outlined in Table 14.3 below.

Table 15.8:1 Storage requirements for whole Blood, Plasma, DBS and Buffy Coat

Storage time	Sample type	Storage temperatures			
		15-30oC (Ambient temperature)	2 to 8oC	-20oC	-70 or -80oC
	Whole blood	Max. 6 hours	24 hours	NA	NA
	DBS	Max. 2 weeks	NA	NA	Longer storage
	Plasma	NA	Max. 5 days	60 days	Longer storage
	Buffy coat	NA	Max. 5 days	60 days	Longer storage

15.8.2 Sample referral System

Existing levels of Laboratory services such as HEID, HVL, CD4 and HIV-DR facilities are linked together by hub and spoke model of Comprehensive sample referral system. Referral starts from the primary sample collection at facility, which acts as a spoke and then gathered at Hub level temporarily before transported to the testing laboratory. With near POC stationed at the Hubs/Spoke.

For sample referral, the minimum recommendations should be adhered as stipulated in Integrated Sample referral Manual including: timely transportation of the samples and return of results, use of manifest forms, use of triple package system that fully ensure Sample integrity and protect the courier against occupational exposure or accidental leaks, Samples should not exceed the maximum stay across all levels and installation of the interoperable electronic sample referral system (eSRS).

15.8.3 Health laboratory information management system

Health laboratory information management system (LIMS) in the country includes routine data collection at testing laboratories on sample details being processed. This include all information available and generated on the sample management process such as rejections, backlogs, turnaround time, sample results and error messages generated from testing machines. The country has multiple and different information management systems instituted within testing laboratories. Efforts, should be made to ensure integration and interoperability of these systems.

15.8.4 Near Point-of-Care testing

Testing near Point-of-Care (POC) technologies, have been introduced to allow a wide range of trained testers and placement to more HFs with no conventional platforms hence increasing access to diagnostics services. There should be a strategy to use both near POC testing and conventional platform in order to optimise service delivery. The selection criteria of technologies should not only rely solely on the technical specifications and characteristics of platforms but also sample type specificities and operational considerations, such as efficiency of sample transport system, expected testing volumes, human resources and technical support.

Note:

Other detail information is well elaborated in:

- *Redesign Logistics system manual 2019*
- *HVL and HEID testing guidelines.*
- *Sample referral system guidelines*
- *HTS guidelines*

16 NUTRITION IN HIV AND AIDS

16.1 Introduction

Malnutrition is prevalent among people living with HIV, adversely affecting their health and leading to varied responses to treatment. Malnutrition can result in treatment failure and increased susceptibility to infections. Children are the most impacted group. Addressing malnutrition is essential, particularly in the context of universal HIV testing and treatment.

16.2 Relationship Between Nutrition and HIV/AIDS

Malnutrition and HIV are interconnected in a vicious cycle. HIV can lead to malnutrition, while malnutrition exacerbates the effects of HIV. HIV weakens the immune system, making individuals more prone to infections that deplete vital nutrients, while malnutrition further impairs immunity. Malnourished individuals with HIV progress more rapidly to AIDS compared to their well-nourished counterparts.

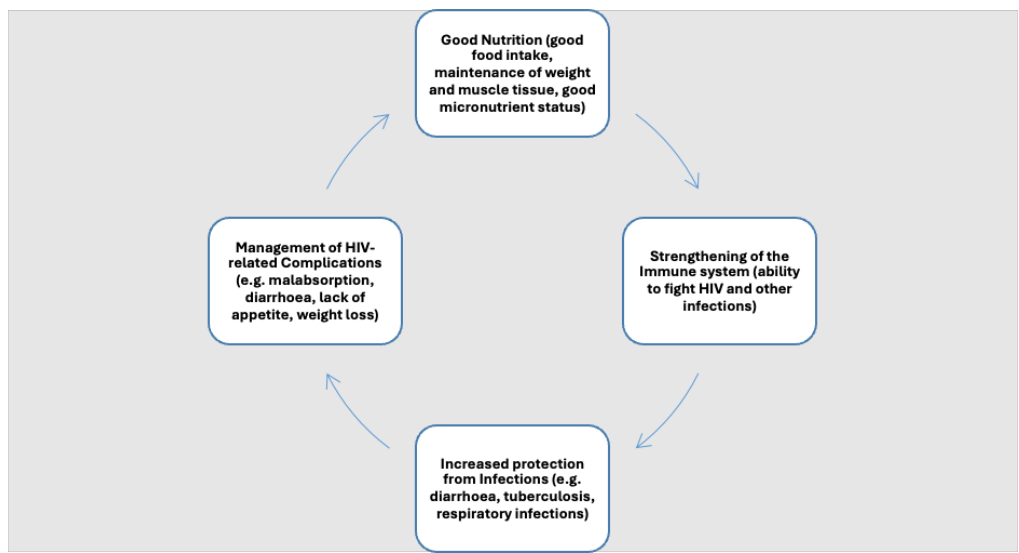


Figure 16.2:1 The Cycle of Nutrition and protection from infections in Context of HIV

HIV and AIDS reduce food intake, nutrient absorption, and food security. This cycle contributes to illness and death, with nutrient deficiencies impairing immune function and accelerating HIV progression. Conversely, good nutrition can delay the progression from HIV to AIDS, enhance the body's ability to combat opportunistic infections, and improve the effectiveness of antiretroviral therapy (ART), ultimately enhancing the quality of life (see Figure 16.2:1)

16.3 Healthy Eating for People Living with HIV

Healthcare provider should encourage PLHIV to include different food groups at each meal as directed in the National Food Based Dietary Guidelines of 2022 (FBDGs)

- Variety-Recommend choosing different types of food within each food group whenever possible.
- Balance- Recommend choosing foods from all food groups according to the recommended amounts.
- Moderation– Recommend controlling portion size so that balance and variety are possible. This is essential to avoid over-nutrition or under-nutrition.
- Although water is not part of the food groups it is recommended that a person should drink at least eight glasses (1.5 litres) of water a day.

Healthcare Providers should use the following tips during nutrition counselling, education sessions and monitoring as in Table 16.3:1

Table 16.3:1 Diet, lifestyle and monitoring

Diet and Lifestyle Tips	Monitoring Nutrition Status	Key Actions
• Eat variety of foods together to aid absorption of nutrients. • Use high nutrient-dense foods such as grounded nuts, milk, fats and oils, honey, coconut milk and eggs to enrich the meals especially starchy foods • Eat snacks of fruits, cooked or roasted groundnuts or porridge at least twice a day to increase energy and nutrient intake. • Be physically active to stimulate the appetite, reduce nausea, improves functioning of the digestive system • Avoid alcohol, which interferes with food intake, digestion and absorption and decreases the effectiveness of ARVs.	• Clinical assessment - History taking (intake, weight changes and growth, GI symptoms, and functional capacity) • Physical examination • Anthropometric measurements for growth monitoring in children - Height, Weight, BMI for age and Sex MUAC • Bodyweight changes in adults • (BMI in kg/m²) - underweight ≤ 18.5 - normal = 18.5 – 24.9 - overweight = 25 – 29.9 - obese ≥ 29.9 • Waist -Hip Ratio • Biochemical measurements of metabolic parameters, serum proteins, and micronutrients	• Monitor nutritional status. • Counsel to consume healthy diet • Promote physical activity to stay healthy and prevent excessive weight gain that might lead to overweight and obesity. • Promote good hygiene and sanitation, water and food safety to prevent food and water borne-diseases. • Provide micronutrient supplementations when necessary. • Discourage use of alcohol, tobacco, traditional herbs and others (charcoal, soil). • Provide therapeutic and supplementary foods to malnourished clients according to the national protocol. (Food by prescription)

Diet and Lifestyle Tips	Monitoring Nutrition Status	Key Actions
• Avoid consuming “empty calories” such as sodas and other artificial drinks. • Use spices for appetite and absorption: ginger, garlic, cardamom, lemon • Eat germinated and fermented foods such as togwa or yoghurt to improve taste and prevent the growth of diarrhoea-causing germs and enhance digestion and absorption in larger quantities. • Mash or grind food, facilitate swallowing, digestion and absorption of nutrients. • Eat fortified foods such as fortified wheat and maize, edible Oil with vitamin A and iodized salt to improve micronutrient intake. • Eat biofortified foods such as orange fleshed sweet potatoes (OFSP), iron and zinc rich beans (Jesca), Mung beans etc to increase micronutrients intake • Drink plenty of fluids, including boiled or treated water, natural fruit juice etc. to avoid dehydration and clear toxins from the body. • Observe food safety, improve cooking methods and hygiene principles. • Limit/ reduce sugar and salt intake • Eat adequate amount of fruits and vegetables daily	Dietary assessment • Use 24-hour dietary recall and food frequency questionnaire. • Probe information about the types, quality and amount of meals taken, appetite, food habits and eating behaviours. • Identify food availability, traditional food taboos and economic factors that affect food intake. • Ask about medications and alternative therapies that the RoC is taking. • Review food preparation methods to determine the amount of salt, sugar and oil/fat which when taken in excess is harmful to health as may increase risk of Non-Communicable Diseases	• Encourage use of ARVs and other drugs prescribed by medical personnel as per regimen. • Encourage use of iodized salt throughout pregnancy and lactating period to prevent iodine deficiency. • Encourage pregnant and lactating women to meet their additional energy requirements by eating at other home-prepared nutritious foods • Promote exclusive breastfeeding for the first 6 months of life; complementary foods for children aged 6 - 24 months with continued breastfeeding to prevent malnutrition. • Educate on food security and linkage to HIV sensitive social protection such as household food support, home-based care, agricultural extension services, and economic strengthening and livelihood support

Note: There is no single food that contains all the nutrients that the body needs, except breast milk for infants up to six months of age. For a balanced meal use at least one type of food from each food group.

Table 16.3:2 Daily Protein and Energy requirements for PLHIV

Energy and protein requirement	Energy (calories per day)	Protein (grams per day)	Micronutrients
2–3 years	1,000 - 1,400	25	Same to general population
4–8 years	1,400 - 1,600	35	
9–13 years	1,600 - 2,000 (girls)	62 (girls)	
	1,800 - 2,200 (boys)	65 (boys)	
14–18 years	2,000 (girls)	64 (girls)	
	2,400 (boys)	84 (boys)	
Adults	2,000–2,580	50–80	
Pregnant/ lactating women	2,460–2,570	55–68	
HIV+ adults (symptomatic)	10% extra (asymptomatic) 20%–30% extra (symptomatic)	50–80	
HIV+ children	10% extra (asymptomatic) 20%–30% extra (symptomatic) 50%–100% extra (losing weight)	Same as for uninfected children	

16.3.1 Nutritional consideration at different stages of HIV infection

Different HIV infection stages presents with different conditions such as mouth sores, sore throats, and diarrhoea. Infections increase energy requirements, potentially causing nutrient deficiencies and further stressing the already weakened immune system. Table 16.3:3 below describes Nutrition, Care, and Support Priorities categorized by stages.

Table 16.3:3 Nutritional, care and support priorities by WHO HIV stages

HIV stage	Features	Nutritional Advice
Early Stage (stage 1 & 2 of WHO clinical staging)	Asymptomatic or mild symptoms weight loss under 10% of presumed or measurable body weight	• Assess nutritional status regularly including anthropometric, biochemical, clinical and dietary history • Counsel on healthy diet and healthy lifestyle • Consider critical nutrition actions during counselling
Middle Stage (stage 3 of WHO clinical staging)	• Weight loss over 10% of presumed or measurable body weight	• Assess nutritional status regularly including anthropometric, biochemical, clinical and dietary history

HIV stage	Features	Nutritional Advice
	• Opportunistic infections	• Consider critical nutrition actions during counselling • Provide therapeutic food when severely malnourished
Late stage (stage 4 of WHO clinical staging)	Weight loss Symptomatic	• Assess nutritional status regularly including anthropometric, biochemical, clinical and dietary history • Advise on treating opportunistic infections. • Counsel to modify diet according to symptoms. • Consider critical nutrition actions during counselling. • Provide therapeutic food when severely malnourished

16.3.2 Key actions to ensure good lifestyle behaviour to PLHIV

Note: HIV infected individuals are encouraged to include a variety of foods in the diet to prevent deficiency. There is evidence that some micronutrient supplements such as vitamin A, zinc and iron at higher doses may produce adverse outcomes in HIV infected persons.

Do not give high dose of Vitamin A if the Recipient of Care (RoC) with SAM are already receiving F75, F100 or RUTF which already have sufficient Vitamin A. (only give to those who are not provided F75, F100 or RUTF according to the protocol).

16.4 Nutrition Care and Support for PLHIV

People living with HIV (PLHIV) require adequate care and support to improve their nutritional status at all stages of HIV infection. Nutrition care and support for PLHIV should aim to:

- Ensure adequate nutrient intake by promoting improved eating habits to build essential nutrient stores needed for effective immune system functioning.
- Prevent nutritional deficiencies and excessive weight gain.
- Prevent weight and muscle mass loss.
- Enhance uptake and adherence to antiretroviral therapy (ART).
- Manage HIV-related symptoms and medication side effects that impact food intake.
- Promote well-being, self-esteem, and a positive attitude, thereby improving the overall quality of life.

To achieve these goals, health service providers should integrate Nutrition Assessment, Counselling, and Support (NACS) services as part of the clinical HIV care and treatment package. These services should establish strong links with community-based programs. All PLHIV should receive nutritional assessment, counselling, and support tailored to their individual needs, including:

1. Nutrition Assessment, Classification, and Care Plans
2. Nutrition Education and Counselling
3. Nutrition Support, including therapeutic and/or supplementary feeding for malnourished HIV-infected individuals.
4. Referral to Follow-up Care and additional services such as food security, livelihood programs, and social safety net initiatives (e.g., food support and cash transfers).

16.4.1 Interpretation of Z-Scores, BMI, and MUAC

Healthcare providers should regularly measure and interpret the nutritional status of PLHIV using **Z-scores**, **Body Mass Index (BMI)**, and **Mid-Upper Arm Circumference (MUAC)**. Specific actions should be taken for each age group based on the results, as outlined in Table 16.4:1

Table 16.4:1 Interpretation of Z-scores in Children

Ratio	Indicator	Z-score	Severity
Weight/Age	Underweight	< - 3	Severe
Height/Age	Stunting	-3 to -2	Moderate
Weight/Height	Wasting*	> -2 to -1	Mild
		> -1	Normal
*Children with weight/height z-score of -2 or less should be supported with therapeutic/supplementary foods			

Table 16.4:2 Interpretation of BMI Results for Adults

BMI Level	Classification	Action to be taken
< 16	Severe malnutrition	Recommend for health facility based therapeutic intervention; rehabilitation with therapeutic foods; counselling on intake issues and possible metabolic issues. Screen for TB
16.0–18.4	Mild/moderate malnutrition	Conduct Nutritional counselling and supplementary feeding. Screen for TB
18.5–25.0	Normal/ recommended	Recommend nutritional counselling, consistent exercise to build muscles

BMI Level	Classification	Action to be taken
25.1–30	Overweight	Recommend nutritional counselling to reduce energy intake; aerobic physical activity to reduce weight
>30	Obesity	Recommend counselling to change lifestyle and reduce energy intake; aerobic physical activity to reduce weight. Recommend screening for NCDs for early management and treatment Counselling based on healthy eating tips including limiting sugar, salt and oil intake

Table 16.4:3 Interpretation of MUAC results for children, adolescents and PBFW

Interpretation of MUAC results for children, adolescents				
6-9 months	5-9 years	10-18 years	Classification	Action to take
< 11.5	< 13.5	< 14.5 cm	Severe acute malnutrition	Irrespective of clinical signs, admission (referral) for stabilization/therapeutic rehabilitation
11.5–12.5	13.5-14.5	14.5-18.5	Moderate acute malnutrition	Supplementary feeding is recommended
12.6–13.5			Mild acute malnutrition	Nutritional education and counselling recommended
> 13.5			Normal	Education and counselling of caregivers recommended
Interpretation of MUAC results for Pregnant and Breastfeeding Women				
≤ 23	Malnourished	Provide nutritional support		
> 23	Normal	Provide nutrition education and counseling		

16.5 Nutrition Education and Counselling

16.5.1 Nutrition Education

Nutrition education helps improve eating behaviors and nutritional status by providing accurate and relevant information to enable PLHIV to make informed nutritional choices and adopt healthier lifestyle behaviors. It is typically delivered to groups of people on topics of common interest, tailored to the audience's needs and characteristics (e.g., location, age, culture, socioeconomic status).

Possible topics for nutrition education for PLHIV include:

- The relationship between nutrition and HIV
- Nutritional requirements for PLHIV

- Promotion of healthy lifestyle behaviors for PLHIV
- Food and water safety and hygiene
- Dietary management of HIV-related symptoms and medication side effects

16.5.2 Nutrition Counselling

Nutrition counselling for PLHIV should be personalized based on the individual's nutritional status, stage of HIV infection, ARV use, conditions or complications, and socio-economic situation. Counselling can be provided at health facilities or during home visits and is instrumental in managing food-medication interactions, medication side effects, and improving medication efficacy, tolerance, and adherence.

16.5.3 Critical Nutrition Actions for PLHIV

Below are essential counselling messages for PLHIV during nutrition counselling:

- Get weighed regularly and have your weight recorded.
- Eat a variety of foods and increase your intake of nutritious foods.
- Drink plenty of boiled or treated water.
- Maintain food hygiene and water safety.
- Avoid habits that may lead to poor nutrition and health.
- Engage in regular exercise such as walking or light housework to build muscle, improve appetite, manage stress, and enhance overall health and alertness.
- Prevent infections and seek early treatment while managing symptoms through diet.
- Manage medication and food interactions and side effects through diet.
- Take all medications as prescribed by your healthcare provider.
- Avoid stress and seek psychosocial support, as stress, depression, and stigma can negatively affect appetite.

16.6 Nutrition Support for PLHIV

Providing nutritional support to PLHIV or individuals with TB can help reduce the risk of drug non-adherence, particularly among those who are food-insecure or acutely malnourished. Nutritional support may include therapeutic food products, micronutrient supplementation, and enteral or parenteral feeding.

16.6.1 Therapeutic Food Products

Therapeutic food products are specially designed to treat acute malnutrition. These products are prescribed to severely malnourished individuals to stabilize their health, improve nutritional status, and enhance treatment outcomes.

Recommended Therapeutic Food Products:

1. Therapeutic Milks

- o F-100 and F-75: Used for inpatient treatment of Severe Acute Malnutrition (SAM).

2. Ready-to-Use Therapeutic Food (RUTF)

- o RUTF is an energy-dense, mineral- and vitamin-enriched food requiring no preparation, designed for treating SAM in both inpatient and outpatient settings.

16.6.2 Therapeutic Foods for Managing Acute Malnutrition

Following nutritional status assessments:

- Children under five years of age who are severely malnourished and without medical complications should receive nutrition education and RUTF, such as Plumpy’Nut.
- Children with medical complications should not be given RUTF but referred for inpatient care and treatment.

Management of Acute Malnutrition should adhere to existing guidelines, as detailed in the conceptual framework and Annexes 18.15.

Note:

- *BMI is not used to assess nutritional status of pregnant women and women within the period of six months after delivery.*
- *Visual assessment is not recommended as the primary method for screening or nutritional assessment.*
- *MUAC is recommended as the primary method for screening or nutritional assessment for pregnant women.*

16.7 Medication-Food Interactions

ARVs can negatively change the way the body consumes fat, protein and energy. Some ARVs affect nutrient availability, absorption and utilisation. Consequently, some foods might reduce the effectiveness of certain ARVs and other medications by interfering with their absorption, metabolism, distribution and excretion. In addition, ARV side effects can reduce food intake, absorption of nutrients and adherence to the medications. People infected with HIV may take several medications, including antibiotics, ARVs, anti-malarial, anti-helminths, anti-fungal, etc. Foods and medications can interact in 4 major ways as shown in Table 16.7:1 below:

Table 16.7:1 Relationship between foods and medications

1. FOOD	(Affects)	MEDICATION ABSORPTION, METABOLISM, DISTRIBUTION, EXCRETION
2. MEDICATION	(Affects)	NUTRIENT ABSORPTION, METABOLISM, DISTRIBUTION, EXCRETION
3. MEDICATION SIDE EFFECTS	(Affects)	FOOD CONSUMPTION, NUTRIENT ABSORPTION
4. MEDICATION + CERTAIN FOODS	(Creates)	SIDE EFFECTS

Further, Health service provider should guide on medication-food interactions and approaches to be taken to mitigate such interactions and side effects. Annex 18.16 describes the common ARVs and other medications used to manage opportunistic infections in Tanzania along with guidance on how to take them and possible side effects:

17 MONITORING AND EVALUATION

17.1 Introduction

A comprehensive **Monitoring and Evaluation (M&E)** system is essential for assessing the response to HIV, Viral Hepatitis, and STIs interventions among recipients of care (RoC). This system must be practical, uncomplicated, and designed to collect current, useful, and actionable information while adhering to human rights principles.

The M&E system involves tracking RoC at each stage of service (monitoring), evaluating whether the program meets its planned targets (performance evaluation), and determining whether observed changes can be attributed to the program (impact evaluation).

- **Monitoring** refers to the continuous process of systematically collecting and analyzing information about HIV, Viral Hepatitis, and STIs programs. It compares results against set targets to assess how well interventions are being implemented. Monitoring uses program-generated data, such as individual participant characteristics, enrolment, attendance, outcomes, and program costs. It enables comparison across individuals, interventions, and locations. A reliable monitoring system is crucial for compiling and analyzing RoC data over time and across service points, using either paper-based or electronic records. Its primary purpose is to help clinical staff record and use RoC data to guide clinical management and ensure continuity of care. Monthly aggregated data are reported through DHIS2 to provide a snapshot of progress on selected performance indicators.
- **Evaluation** systematically and objectively assesses all program elements (e.g., design, implementation, and outcomes) to determine overall worth or significance. It provides credible information for decision-makers to identify ways to achieve better results. Evaluation can be categorized as:
 - o **Process/Performance Evaluation:** Focuses on service delivery quality and outcomes (e.g., TPT achievement levels by age, sex, location, or RoC on multi-month dispensing).
 - o **Impact Evaluation:** Examines changes directly attributable to the program, estimating what would have occurred had beneficiaries not participated.

Countries, including Tanzania, have adopted UNAIDS' 95-95-95 targets to end HIV epidemics by 2030, alongside WHO global health sector strategies for HIV, viral hepatitis, and STIs (2022–2030). Innovative, integrated interventions are needed to achieve these targets.

The shift from paper-based to electronic-based systems (e.g., electronic medical records, eMR) is essential for improving data quality, access, and usage for program improvement. M&E systems are evolving to focus on **longitudinal individual monitoring** and **Case-Based Surveillance (CBS)**, tracking HIV diagnosis and subsequent events across various health services. Unique identifiers ensure accurate tracking across service points. An **interoperability platform** is necessary for integrating electronic data from diverse sources while maintaining consistency in variable formats.

Digitalizing health information provides readily available data to inform programming for HIV, Viral Hepatitis, and STIs. However, appropriate governance and security measures are crucial to safeguard client privacy and ensure data security.

17.2 Key Components of HIV and AIDS, Viral Hepatitis, and STI Monitoring and Evaluation

17.2.1 Data Recording

Data collection for HIV, Viral Hepatitis, and STI interventions is carried out by healthcare and community health workers using standardized tools at health facilities (HFs) and community levels. Reporting is done monthly, with data aggregated at council levels and entered into the DHIS2 system.

At the national level, **NASHCoP** compiles data from HFs to generate national reports, informing programming for various stakeholders.

17.2.2 Tools for Recording:

- **RoC Identification Card (CTC1):** A card with a unique ID issued during the client's first visit to the care and treatment clinic (CTC).
- **RoC Encounters' Record Form (CTC2):** Used for recording clinical outcomes and management during care and treatment visits.
- **Registers:**
 - **Appointment Register:** Tracks clinic attendance for all RoC.
 - **Tracking Register:** Tracks RoC who missed appointments or were lost to follow-up.
 - **Pre-ART Register:** Records RoC not yet initiated on ART.
 - **ART Register:** Records RoC on ART.
 - **Cohort Analysis Register:** Compiles cohort reports at specific intervals.

In high-volume facilities, electronic systems (e.g., CTC2 database) generate registers automatically.

- **RoC Referral Form:** Transfers client information between HFs.

17.2.3 Data Storage:

Data are stored electronically (e.g., CTC2, pharmacy modules) or on hard copies, with appropriate measures to ensure confidentiality.

17.2.4 Data Analysis:

Analysis is conducted from HF to the national level using indicator-based and cascade analyses.

17.2.5 Data Reporting:

Facilities with electronic systems generate automated reports sent to DHIS2. Paper-based facilities aggregate data manually and submit reports to the DMO's office.

17.2.6 Data Presentation and Dissemination:

Data are presented as notes, tables, graphs, maps, or charts and disseminated through public noticeboards, conferences, and reports to inform planning and program improvements.

17.2.7 Data Quality:

Data must be complete, accurate, consistent, relevant, and timely. Routine Data Quality Assessments (DQAs) ensure high-quality data.

17.3 Roles and Responsibilities in M&E

17.3.1 National Level (NASHCoP):

- Develops M&E frameworks, guidelines, and tools.
- Coordinates data management, capacity building, and dissemination activities.
- Provides feedback and guidance to sub-national levels.

17.3.2 Regional and District Levels:

- Builds capacity for data management.
- Coordinates data reporting, analysis, and dissemination.
- Provides feedback to HFs.

17.3.3 Health Facility Level:

- Ensures availability and use of recording/reporting tools.
- Submits reports to council offices and conducts internal data quality assessments.

17.3.4 Implementing Partners:

- Comply with national M&E systems.
- Support data analysis, dissemination, and review processes.

17.4 Supportive Supervision and Mentoring

Supportive supervision involves helping healthcare providers improve their performance by strengthening communication, solving problems, and providing leadership. Mentorship fosters ongoing professional development, ensuring sustainable, high-quality outcomes.

All levels of health service delivery should adhere to guidelines in the Manual of Comprehensive Supportive Supervision and Mentoring of HIV and AIDS Services (2017 Edition).

18 ANNEXES

18.1 WHO Clinical Staging

Adult and Adolescents	Children
Clinical Stage 1	
Asymptomatic	Asymptomatic
Persistent generalized lymphadenopathy	Persistent generalized lymphadenopathy
Clinical stage 2	
Moderate unexplained weight loss (<10% of presumed or measured body weight) tonsillitis, otitis media, pharyngitis)	Unexplained persistent hepatosplenomegaly
Herpes zoster	Recurrent or chronic upper respiratory tract infections (Otitis media, otorrhea, sinusitis, tonsillitis)
Angular cheilitis	Herpes zoster
Recurrent oral ulceration	Lineal gingival erythema
Papular pruritic eruption	Recurrent oral ulceration
Fungal nail infections	Papular pruritic eruption
Seborrheic dermatitis	Fungal nail infections
	Extensive wart virus infection
	Unexplained persistent parotid enlargement
Clinical Stage 3	
Unexplained severe weight loss (>10% of presumed or measured body weight)	Unexplained moderate malnutrition not adequately
Unexplained chronic diarrhoea for longer than 1 month	responding to standard therapy
Unexplained persistent fever (intermittent or constant for longer than 1 month)	Unexplained persistent diarrhea (14 days or more)
Persistent oral candidiasis	Unexplained persistent fever (above 37.5°C,
Oral hairy leukoplakia	intermittent or constant, for longer than one 1 month)
Pulmonary tuberculosis	Persistent oral candidiasis (after first six weeks of life)
Severe bacterial infections (such as pneumonia, empyema, pyomyositis, bone or joint infection, meningitis, and bacteraemia)	Oral hairy leukoplakia

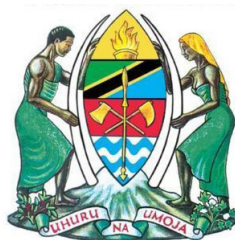
Acute necrotizing ulcerative stomatitis, gingivitis or periodontitis	Lymph node tuberculosis; pulmonary tuberculosis
Unexplained anaemia (<8 g/dl), neutropenia (<0.5 × 10 ⁹ /L) and/or chronic thrombocytopenia (<50 × 10 ⁹ /L)	Severe recurrent bacterial pneumonia
	Acute necrotizing ulcerative gingivitis or periodontitis
	Unexplained anemia (<8 g/dL), neutropenia (<0.5 × 10 ⁹ /L) or chronic thrombocytopenia (<50 × 10 ⁹ /L)
	Symptomatic lymphoid interstitial pneumonitis
Chronic HIV-associated lung disease, including	
bronchiectasis	

Adult and Adolescents	Children
Clinical Stage 4	
HIV wasting syndrome	Unexplained severe wasting, stunting or severe malnutrition not responding to standard therapy
Pneumocystis (jirovecii) pneumonia	Pneumocystis (jirovecii) pneumonia
Recurrent severe bacterial pneumonia	Recurrent severe bacterial infections (such as empyema, pyomyositis, bone or joint infection, meningitis, but excluding pneumonia)
Chronic herpes simplex infection (orolabial, genital or anorectal of more than one month in duration or visceral at any site)	Chronic herpes simplex infection (orolabial or cutaneous of more than 1 month's duration or visceral at any site)
Esophageal candidiasis (or candidiasis of trachea, bronchi or lungs)	Oesophageal candidiasis (or candidiasis of trachea, bronchi or lungs)
Extrapulmonary tuberculosis	Extrapulmonary tuberculosis
Kaposi sarcoma	Kaposi sarcoma
Cytomegalovirus infection (retinitis or infection of other organs)	Cytomegalovirus infection (retinitis or infection of other organs with onset at age older than one month)
Central nervous system toxoplasmosis HIV encephalopathy	Central nervous system toxoplasmosis (after the neonatal period)
Extrapulmonary cryptococcosis, including meningitis	HIV encephalopathy
Disseminated nontuberculous mycobacterial infection	Extrapulmonary cryptococcosis, including meningitis

Adult and Adolescents	Children
Progressive multifocal leukoencephalopathy	Disseminated nontuberculous mycobacterial infection
Chronic cryptosporidiosis	Progressive multifocal leukoencephalopathy
Chronic isosporiasis	Chronic cryptosporidiosis (with diarrhoea)
Disseminated mycosis (extrapulmonary histoplasmosis, coccidioidomycosis)	Chronic isosporiasis
Lymphoma (cerebral or B-cell non-Hodgkin)	Disseminated endemic mycosis (extrapulmonary)
Symptomatic HIV-associated nephropathy or cardiomyopathy	
Recurrent septicaemia (including nontyphoidal Salmonella)	
Invasive cervical carcinoma Atypical disseminated leishmaniasis	

18.2 Health Facility Assessment Tool

THE UNITED REPUBLIC OF TANZANIA



MINISTRY OF HEALTH

ASSESSMENT TOOL FOR HEALTH FACILITY TO PROVIDE HIV CARE AND TREATMENT SERVICES

Minimum Criteria for a Health Centre and Dispensary

Name of health facility:	
Type of Facility:	
Village/Street:	
Ward:	
District:	
Region:	
Date of Assessment:	

Baseline visit

re-assessment visit

Instruction

1. The answers to the minimum criteria should be derived from the health centre and Dispensary assessment tool (version 4, October 2021).
2. The numbers in brackets refer to the question numbers in the Health centre and Dispensary Assessment tool.
3. Circle the appropriate response (Yes/No) as recorded in the assessment tools
4. If the response is Yes, circle the appropriate score
5. Sum all circled Yes score to make a total of minimum score
6. Cut of point
 - a. If the facility scored 60% and above with a consultation room, at least three staff (Clinician, nurse and other health workers) the facility qualifies for initiation.
 - b. If the facility scored below 60% action plan for improvement should be in place before 2nd assessment.

1 Organization of HIV and AIDS care services within facility			
1.a One or more confidential consultation rooms			Score
One clinical consultation rooms available (4.2.1)	Yes	No	
Consultation rooms with visual privacy (4.2.4)	Yes	No	
Consultation rooms with Auditory privacy (4.2.5)	Yes	No	
1.b Space/room and register for registration of HIV and AIDS patients (1.8.1 and 1.8.2 = Yes, seen)	Yes	No	
2 Human resource capacity and training			
2.a At least one clinician (ACO, CO, AMO, MO) (2.3.1)	Yes	No	
2.b At least one nurse (NO,ANO,Nurse) (2.3.3)	Yes	No	
2.c At least one other health worker (2.3.4, 2.3.5, 2.3.6 or 2.3.7)	Yes	No	
2.d At least two staff (Clinicians and Nurses) trained on National Curriculum on ART (CTC/PMTCT) (2.2)	Yes	No	
3 HIV Testing Services			
3.a One confidential room for testing and counseling 3.2.1 and 3.2.2 + 3.2.3 = Yes, observed			Score
One confidential room for testing and Counseling (3.2.1)	Yes	No	
Testing & Counselling rooms with visual privacy (3.2.2)	Yes	No	
Testing & Counselling rooms with Auditory privacy (3.2.3)	Yes	No	
3.b At least one HTS provider (CITC/PITC) (3.3.3 & 3.4.4)	Yes	No	
4 Care and Treatment Services			
4.a Availability of PMTCT services (4.8.1)	Yes	No	

5 Patient records and Administration			
5.a An established and working medical record system (5.1.1-5.1.5,=Yes)	Yes	No	
5.b Locked area for medical records with limited access (5.2.1=Yes)	Yes	No	
6 Continuum of Care			
6.a Availability of CBHS (at least one = 6.2.1 & 6.2.2)	Yes	No	
6.b Functional system for patients tracking (5.5.2)	Yes	No	
7 Laboratory services			
7.a Adequate laboratory space (7.2.1= yes, 7.2.3 = Yes), at least one room	Yes	No	
7.b HIV testing (rapid) (7.4.1)	Yes	No	
7.c Basic blood tests (haematology/biochemistry) (7.5 and 7.6)	Yes	No	
7.d Malaria blood test (7.4.14)	Yes	No	
7.e TB sputum smears (ZN stain) + (STI screening) (7.4.13) & (7.4.4)	Yes	No	
7.f Routine testing of stool and urine (7.4.12)	Yes	No	
7.g Pregnancy Test (7.4.10)	Yes	No	
8 Pharmacy services			
8.a Secure storage space large enough for 6 months supply of ARVs (8.6.1= Yes, seen)	Yes	No	
8.b Refrigerator in pharmacy (8.6.2)	Yes	No	

18.3 ARVs & Hepatitis B&C Strength and Formulation

Regimens	Strength	Formulations
ADULT FIRST LINE ARV REGIMENS		
Tenofovir Disoproxil Fumarate / Lamivudine /Dolutegravir -FDC(TLD)	TDF 300mg/3TC 300mg/ DTG 50mg	Tablets
Dolutegravir/ Lamivudine/ Tenofovir Alafenamide -FDC (TAF-LD)	TAF 25mg/3TC 300mg/ DTG 50mg	Tablets
Tenofovir /Lamivudine /Efavirenz Tablets -FDC (TLE)	TDF 300mg/3TC 300mg/ EFV 400mg	Tablets
Abacavir Sulphate / Lamivudine + Dolutegravir	ABC 600mg/3TC 300mg +DTG 50mg	Tablets
ADULT SECOND LINE ARV REGIMENS		
Tenofovir Disoproxil Fumarate / Emtricitabine + Atazanavir / Ritonavir	TDF 300mg/FTC 200mg+ATV/r 300/100mg	Tablets
Abacavir Sulphate / Lamivudine + Atazanavir / Ritonavir	ABC600mg/3TC 300mg+ATV/r 300/100mg	Tablets
Zidovudine /Lamivudine + Atazanavir / Ritonavir Tablets	AZT300mg/3TC 150mg +ATV/r 300/100mg	Tablets

Regimens	Strength	Formulations
Tenofovir Disoproxil Fumarate/ Emtricitabine + Lopinavir/ Ritonavir Tablets	TDF 300mg/FTC 200mg+LPV/r 200/50mg	Tablets
Abacavir Sulphate / Lamivudine + Lopinavir/ Ritonavir Tablets	ABC 600mg/3TC 300mg + LPV/r 200/50mg	Tablets
Zidovudine /Lamivudine + Lopinavir/ Ritonavir Tablets	AZT 300/3TC 150mg +LPV/r 200/50mg	Tablets
Zidovudine /Lamivudine + Dolutegravir tablets	AZT 300mg/3TC 150mg +DTG 50mg	Tablets
PEDIATRIC 1ST & 2ND LINE ARV REGIMENS		
Abacavir / Lamivudine + Dolutegravir Dispersible tablets	ABC 120MG/3TC 60MG + pDTG 10mg	Dispersible tablets
Abacavir / Lamivudine/Dolutegravir Dispersible tablets-FDC-(pALD)	ABC 60MG/3TC 30mg/ pDTG 5mg	Dispersible tablets
Lamivudine /Zidovudine +Dolutegravir Dispersible tablets	AZT 60mg / 3TC 30mg + pDTG 10mg	Dispersible tablets
ADULT THIRD LINE ARV REGIMEN		
Darunavir/ritonavir/Lamivudine / Zidovudine/ Dolutegravir tablets	DRV/r 400/50mg+ DTG50mg + 3TC/AZT 150/300mg	Tablets
Darunavir/ritonavir/Lamivudine / Dolutegravir tablets	DRV/r 400/50mg+ DTG /3TC 50/300mg(Dovato)	Tablets
PEDIATRIC THIRD LINE ARV REGIMEN		
Darunavir/ritonavir/Lamivudine / Zidovudine/ Dolutegravir dispersible tablets	pDTG 10mg+DRV/r 400/50mg+AZT/3TC 60/30mg (<20kg)	Tablets
Darunavir/ritonavir/Lamivudine / Zidovudine/ Dolutegravir tablets	DTG 50mg+DRV/r 400/50mg+AZT/3TC 300/150mg (≥20kg)	Tablets
PMTCT PROPHYLAXIS		
Nevirapine Oral Suspension with Syringe 10mg/ml	NVP 10mg/MI	Oral Suspension
Lamivudine 30mg /Zidovudine 60mg + Nevirapine Oral Suspension with Syringe 10mg/ml	AZT 60mg / 3TC 30mg + NVP 10mg/MI	Dispersible tablets + Oral Suspension
PrEP		
Tenofovir Disoprol Fumarate/ Emtricitabine Tablets (For PrEP)	TDF 300mg/FTC 200mg For PrEP	Tablets
HEPATITIS B&C		
Tenofovir disoproxil fumarate 300mg tablet, bottle of 30 tablet	TDF 300mg	Tablets
Entecavir 0.5mg tablet, pack of 30 tablets	Entecavir 0.5mg	Tablets
Daclatasvir dihydrochloride/Sofosbuvir 60/400mg tablet, bottle of 28 tablets	Sofos/Dacla 400/60mg	Tablets

18.4 Pediatric Dosing Chart 1 - 3

DRUG NAME	ABC/ 3TC Adult	ABC/ 3TC Baby	AZT/ 3TC Adult	AZT/ 3TC Baby	TDF/ 3TC/ EFV(TLE) Tabs	TDF/ FTC Tabs	TDF/3TC/ DTG (TLD) Tabs	Pediatric Dolutegravir (pDTG) Tabs	Dolutegravir(DTG) Tabs
DOSE	ONCE daily	ONCE daily	TWICE daily	TWICE daily	ONCE daily	ONCE daily	ONCE daily	ONCE Daily	ONCE Daily
STRENGTH	600 mg ABC/ 300 mg 3TC tablet	120 mg ABC/ 60 mg 3TC tablet	300 mg AZT/ 150 mg 3TC tablet	60 mg AZT/ 30 mg 3TC tablet	300mg TDF/ 300mg 3TC/ 400mg or 600mg EFV	300 mg TDF/ 200 mg FTC	300mg TDF / 300mg3TC / 50mg DTG	pDTG 10mg	50mg DTG
Weight range (kg)									
3-4.9		1 tab OD		1 tab BD				0.5 tab OD	
5-5.9		1 tab OD		1 tab BD				0.5 tab OD	
6-6.9		1.5 tab OD		1.5 tab BD				1.5 tab OD	
7-7.9		1.5 tab OD		1.5 tab BD				1.5 tab OD	
8-8.9		1.5 tab OD		1.5 tab BD				1.5 tab OD	
9-9.9		1.5 tab OD		1.5 tab BD				1.5 tab OD	
10-10.9		2 tab OD		2 tab BD				2 tab OD	
11-11.9		2 tab OD		2 tab BD				2 tab OD	
12-13.9		2 tab OD		2 tab BD				2 tab OD	
14-16.9		2.5 tab OD		2.5 tab BD				2.5 tab OD	
17-19.9		2.5 tab OD		2.5 tab BD				2.5 tab OD	
20-24.9	0.5 tab OD	3 tab OD	0.5 tab BD	3 tab BD					1 tab OD
25-29.9	1 tab OD		1 tab BD						1 tab OD
30-34.9	1 tab OD		1 tab BD		1 tab OD	1 tab OD	1tab OD		1tab OD
≥35	1 tab OD		1 tab BD		1 tab OD	1 tab OD	1tab OD		1tab OD

NOTE

DTG 10mg is not recommended for children under one month of age regardless of weight.

TDF is only approved for children above 12 years with weight ≥ 30 kg because of TDF-associated reduction in bone density.

DTG 50mg is recommended for children with weight ≥ 20 kg

Chart 2.

DRUG NAME	Lopinavir/ritonavir (LPV/r) Tabs					Atazanavir/ ritonavir (ATV/r)
DOSE	TWICE daily					ONCE daily
STRENGTH	SYRUP 80 mg LPV/20 mg ritonavir/ml		GRANULES sachet 40 mg LPV/ 10 mg ritonavir/ sachet	PEDIATRIC 100 mg LPV/ 25 mg ritonavir tabs	ADULT TABLETS 200 mg LPV/ 50 mg ritonavir tabs	ADULT Tablets 300mg ATV/ 100mg ritonavir
Weight range (kg)	< 4 Weeks	≥ 4 Weeks				
2-2.9	0.6ml BD					
3-3.9	0.8ml BD	1 ml BD	2 Sachets BD			
4-4.9	1ml BD	1 ml BD	2 Sachets BD			
5-5.9		1 ml BD	2 Sachets BD			
6-6.9		1.5 ml BD	3 Sachets BD			
7-7.9		1.5 ml BD	3 Sachets BD			
8-8.9		1.5 ml BD	3 Sachets BD			
9-9.9		1.5 ml BD	3 Sachets BD			
10-10.9		2 ml BD	4 Sachets BD	2 tabs AM, 1tabs PM		
11-11.9		2 ml BD	4 Sachets BD	2 tabs AM, 1tabs PM		
12-13.9		2 ml BD	4 Sachets BD	2 tabs AM ,1tabs PM		
14-16.9		2.5 ml BD	5 Sachets BD	2 tabs BD	1 tab BD	
17-19.9		2.5 ml BD	5 Sachets BD	2 tabs BD	1 tab BD	
20-24.9		3 ml BD	6 Sachets BD	2 tabs BD	1 tab BD	
25-29.9				3 tabs BD	2 tabs AM 1 tab PM	1tab OD
30-34.9				3 tabs BD	2 tabs AM 1 tab PM	1 tab OD
≥35					2 tabs BD	1 tab OD

NOTE

Do not use LPV/r solution in infants aged < 2 weeks of age.
Consider giving LPV/r tablets for children weighing $\geq 10\text{kg}$ to reduce giving large numbers of granules.

LPV/r tablet must be swallowed whole and should not be split, chewed, dissolved, or crushed as it loses effectiveness.

ATV/r 300mg tabs can be used in children with body weight $\geq 25\text{kgs}$.
LPV / r Oral Granules 40 mg / 10 mg must be taken after meals twice daily. LPV / r Oral Granules 40 mg / 10 mg should be sprinkled/mixed with soft food such as porridge or mixed with a liquid such as water. LPV / r Oral Granules 40 mg / 10 mg should not be chewed or crushed.

Chart 3.

DRUG NAME	Darunavir + ritonavir (DRV+ r)	
DOSE		
STRENGTH	Adult Tablets Darunavir 600mg	Adult ritonavir 100mg
Weight range (kg)		
14-14.9	300mg BD	50mg BD
15-29.9	375mg BD	50mg BD
30-39.9	450mg BD	100mg BD
≥ 40	600mg BD	100mg BD

18.5 ART Readiness Assessment Form

ART Readiness Assessment Form	YES	No
A. Psychosocial/Knowledge Criteria (applies to RoC and caregivers)		
1. Understands the nature of HIV infection and benefits of ART?		
2. Has screened negative for alcohol or other drug use disorder, (see Chapter XIII)		
3. Has screened negative for depression or other psychiatric illness, (see Chapter XII)		
4. Is willing to disclose/has disclosed HIV status, ideally to a family member or close friend?		
5. A demonstration of how to take/administer ART and other prescribed medications has been conducted for the RoC?		
6. Has received information on probable side effects of ART and understands what steps to take in case of these side effects?		
7. For RoC recipient of care dependent on a caregiver: caregiver is committed to long-term support of the recipient of care, daily administration of ART, and meets the criteria above?		
8. Other likely barriers to adherence have been identified and there is a plan in place to address them (e.g. frequent travel for work, plan to deal with unexpected travel, distance from clinic, etc.)?		

ART Readiness Assessment Form		YES	No
9.	RoC Recipient of care/caregiver has provided accurate locator information and contact details?		
10.	RoC Recipient of care/caregiver feels ready to start ART today?		
B. Support Systems Criteria (applies to recipients of care and caregivers)			
1.	Has identified convenient time/s of day for taking ART, and/or associated dose/s with daily event/s?		
2.	Treatment supporter has been identified and engaged in HIV education, or will attend the next counselling session?		
3.	RoC is aware of support group meeting time/s?		
4.	If the facility has an SMS reminder system: Has enrolled into the SMS reminder system?		
5.	Other support systems are in place or planned (e.g. setting phone alarm, pill box)?		
C. Medical Criteria (applies to recipients of care)			
1.	Newly diagnosed with TB: delay ART initiation until RoC can tolerate anti-TB medication; initiate ART as soon as possible preferably within 2 weeks; for TB meningitis consider delaying ART for up to 8 weeks); monitor closely for IRIS		
2.	Newly diagnosed cryptococcal meningitis (CM), or symptoms consistent with CM (progressive headache, fever, malaise, neck pain, confusion): delay ART until completed 5 weeks of CM treatment and symptoms resolved, or until CM has been ruled out as the cause of symptoms; monitor closely for IRIS		
3.	TB screen conducted to rule out TB, If eligible initiate TPT		
*If the response to any of the psychosocial criteria or support systems criteria is “No”: develop a strategy to address the issue as quickly as possible and consider assigning a case manager. ART may be initiated with adequate adherence support while the criteria is being addressed, on a case-by-case basis			

18.6 Check list for assessment of elevated viral Load –(PROBE assessment)

Probe for adherence issues	Probe for factors that may be barriers to adherence; • Cost of clinic visits to RoC, e.g. transport, loss of income, cost of social responsibilities • Medication side-effects • Unpalatable medications • Depression or other mental health conditions • Alcohol or substance abuse • Poor social support and/or GBV • Non-disclosure Pregnant women may experience nausea/ vomiting, heartburn, and constipation. Assess the need for symptomatic treatment with an anti-emetic, anti-diarrhoea agent, or fibre supplement. Adherence difficulties in young children are often linked to poor tolerability of unpalatable formulations. It is important to ask the caregiver about how the child tolerates the medication e.g. does the child refuse to swallow, does the child vomit the medication	Tips Ask open ended questions “What makes it difficult for you to take your treatment?”, and “How many doses have you missed this week?” Statements like “we all miss a dose now and then” can encourage a client to be more open. Create a safe and non-judgmental space for your client to discuss challenges.
Rule out infection	Check for symptoms and signs of infection. Do a TB and STI screen.	Remember that immune compromised, malnourished, and pregnant recipients of care may not exhibit overt symptoms of TB. If in doubt, do a TB GXP
On correct dose	Is the client on the correct dose for their weight? This is especially applicable to growing children, or RoC with deteriorating renal function or previous renal impairment	
Be aware of Drug-Drug interaction	Review • Prescribed treatment e.g. rifampicin, anti-epilepsy drugs and pregnancy supplements (iron, calcium) • Over the counter treatment e.g., antacids, multivitamins • Other supplements and traditional medications	See also “Drug Interactions with DTG and Rifampicin-containing TB Treatment

Evaluate Resistance	After ruling out other causes of virological failure consider HIV drug resistance if the RoC adherence is objectively good	Refer to the algorithm Management of Virological Failure
Barrier of adherence	Interventions	EAC indicated?
Difficulty getting to facility to collect treatment	Reduce unnecessary visits through enrolling client to multi-month dispensing (MMD)	No need for EAC
Drug side effects or unpalatability impacting adherence?	Change to more palatable regimen	No need for EAC
Challenges with taking/ remembering to take treatment	Provide EAC	

Reference, Adopted and modified from South African guideline

18.7 EAC Form for Children, Adolescents and Adults with Low and High level Viremia (For Enhanced Adherence Counselling and Second/Third line Consideration)

RoC Information	
RoC Name : _____	RoC ID: _____
Date of Birth: _____	Sex: M/F Facility Name : _____
ARV Information	
ARV Regimen: _____	Date of Initiation: _____/_____/_____
Viral Load before EAC : _____ cp/ml Date: _____/_____/_____	
Previous HVL (if any): _____ cp/ml Date: _____/_____/_____	
Reason for ARV regimen change (Substitution/Switching) ☐ Treatment failure ☐ Side effects ☐ Change of guidelines ☐ Drug interaction ☐ Any other specify: _____	
Enhanced Adherence Counselling (To be filled by the counsellor)	
Your Impression about RoC adherence before EAC ☐ Likely to be good ☐ Likely to be NOT good (relevant barriers identified) ☐ Clearly poor (defaulter) For each session, assess major barriers for possible poor adherence Date of 1 st Session: _____/_____/_____ ☐ Behavioural: _____ Cognitive: _____ ☐ Emotional: _____ Socio economic: _____ ARV-Intake demonstration by RoC/caretaker done? Y/N Pill Count done? Y/N Pill Intake: _____ % Planned Interventions: _____ _____ _____ Summary: _____ Date of Next Session: _____ Date of 2 nd Session: _____/_____/_____ ☐ Behavioural: _____ ☐ Cognitive: _____	

☐	Emotional:										
☐	Socio economic:										
	ARV-Intake demonstration by RoC /caretaker done?	Y/N	Pill Count done?	Y/N	Pill Intake: _____ %						
	Planned Interventions:	_____									
	Summary:	_____									
	Date of Next Session:	_____									
	Date of 3 rd Session:	_____/_____/_____									
☐	Behavioural:	_____									
☐	Cognitive:	_____									
☐	Emotional:	_____									
☐	Socio economic:	_____									
	ARV-Intake demonstration by RoC /caretaker done?	Y/N	Pill Count done?	Y/N	Pill Intake: _____ %						
	Planned Interventions:	_____									

	Summary:	_____									
	Date of Next Session:	_____									
	Date of Extra Session:	_____/_____/_____									
☐	Behavioural:	_____									
☐	Cognitive:	_____									
☐	Emotional:	_____									
☐	Socio economic:	_____									
	ARV-Intake demonstration by RoC /caretaker done?	Y/N	Pill Count done?	Y/N	Pill Intake: _____ %						
	Planned Interventions:	_____									

	Summary:	_____									
Did	the	RoC	attend	all	the	appointments?	Y/N	If	No,	any	reason?

ASSESSMENT SECTION Your Impression about RoC adherence during and after EAC ☐ Likely to be good ☐ Likely to be NOT good (relevant barriers identified and not cleared) Clearly poor (defaulter)* <i>If RoC is defaulting repeat viral load should be deferred and EAC extended Share decision with the team</i> Major remaining barriers identified after EAC sessions: ☐ Behavioural Y/N if Yes: _____ ☐ Cognitive Y/N if Yes: _____ ☐ Emotional Y/N if Yes: _____ ☐ Socio economic Y/N if Yes: _____ Date of collection of repeat Viral load _____ / _____ / _____ Counsellor name: _____ Date of assessment _____ / _____ / _____ Repeat Viral load result: _____ cp/ml Date _____ / _____ / _____ Was it a significant drop in the viral load (good response to EAC?) Y/N (Log drop _____) Is RoC presenting with OIs' or signs of immunosuppression? Y/N if yes, describe _____	
OUTCOME	
Regarding the ARV Regimen, what is the plan? ☐ Continue current regimen ☐ Refer to Facility MDT for discussion Name of Assessor : _____ Date of Assessment _____ / _____ / _____	
Outcome for ROCs with persistent high viral load : To be filled by the treatment failure Review Team (FMDT)	
What is the plan for this RoC? ☐ Remain on current regimen ☐ Switch to 2nd Line ☐ Refer RoC for GART ☐ State any other action Comment _____ Review Team Lead Name: _____ Date : _____ / _____ / _____	

18.8 MORSIKY Adherence Form

ASSESSMENT OF MOTIVATION AND KNOWLEDGE TOWARD MEDICATION-TAKING BEHAVIOUR

CLIENT NAME.....

SNO	QUESTION	PATIENT RESPONSE	
		MOTIVATION	KNOWLEDGE
1	Do you sometimes forget to take your medications? IF YES How often do you have difficulty remembering to take all your Medications? 1. never/rarely-4 2. once in a while -3 3. sometime- 2 4. usually-1 5. all the time-0	NO(1)	
2	People sometimes miss taking their medications for reasons other than forgetting. Thinking over the past two weeks, were there any days when you did not take your medications?	NO(1)	
3	Have you ever cut back or stopped taking your medications without telling your doctor, because you felt worse when you took it?		NO (1)
4	When you travel or leave home, do you sometimes forget to bring along your medications?		NO(1)
5	When you feel like your health condition is under control, do you Sometimes stop taking you medications?		NO(1)
6	Taking medications every day is a real inconvenience for some People. Do you ever feel hassled about sticking to your treatment Plan?		NO(1)
	TOTAL SCORE	0-1=LOW MOTIVATION >1=HIGH MOTIVATION	0-1=LOW KNOWLEDGE 2-4HIGH KNOWLEDGE

EXAMPLE 4 POINTS HIGH KNOWLEDGE AND 2 MOTIVATION POINTS HAS GOOD ADHERENCE

18.9 Weight and Age Based Dosing Charts

- i) 3HR based regimen recommended dosing chart for the treatment of tuberculosis infection in children and adolescents under 15 years.

Strength: 3HR 75mg/50mg Dispersible FDC		Strength: 3HR 150mg/75mg FDC	
Weight Bands	3HR Tablets	Weight bands	3HR Tablets
4-7 kgs	1	25-37 kgs	2
8-11 kgs	2		
12-15 kgs	3	38-54 kgs	3
16-24 kgs	4		
≥25 kgs	Use Adult Formulations	≥55 kgs	4

- ii) 3HP dosing Chart for adults above 15years

Medicine	Formulation	Weight Bands				
		30–35 kg	36–45 kg	46–55 kg	56–70 kg	>70 kg
Isoniazid	300 mg	3	3	3	3	3
Rifapentine	150 mg	6	6	6	6	6
Isoniazid + Rifapentine (FDC)	300m/300 mg	3	3	3	3	3

- iii) INH Dosing Chart

Weight Ranges (Kgs)	Dose Given (mgs)	Tablets of INH (of 100mg) per Dose
<5 kgs	50 mg	½ tab
5.1 - 9.9 Kg	100 mg	1 tab
10 - 13.9 Kg	150 mg	1 ½ tab (or ½ adult tab)
14 - 19.9 Kg	200 mg	2 tabs
20 - 24.9 Kg	250 mg	2 ½ tabs
≥25 Kg	300 mg	1 adult tablet

18.10 Drug-Drug Interactions between Rifamycin Based TPT and Available Art Regimens in Tanzania

ARV Regimen	Type of interaction	Recommended action
1st Line regimen (Paediatric)		
ABC+3TC +pDTG/ DTG AZT + 3TC + DTG	Reduced concentration of DTG	If <20Kg, double* the dose of pDTG If above 20Kg double* the dose of DTG 50mg *One dose in the morning and one dose in the evening
TDF+3TC+DTG	Reduced concentration of DTG	Double the dose to 50mg bid
ABC/3TC+LPV/r AZT/3TC+LPV/r	Reduced concentration of LPV/r	Double the dose of LPV/r
2nd Line regimen (Paediatric)		
AZT/3TC+LPV/r ABC+3TC+LPV/r	Reduced concentration of LPV/r	Double the dose of LPV/r
TDF+3TC+DTG AZT+3TC+DTG	Reduced concentration of DTG	If <20Kg, double* the dose of pDTG if above 20Kg double* the dose of DTG 50mg *One dose in the morning and one dose in the evening
3rd Line regimen (Paediatric)		
pDTG + DRV + RTV + AZT + 3TC DTG + DRV + RTV + AZT + 3TC	No studies	Don't use
1st Line regimen (Adolescents and Adults)		
TDF +3TC +DTG (TLD) ABC + 3TC+ DTG AZT + 3TC + DTG	Reduced concentration of DTG	Increase the dose of DTG by 50mg
TDF + 3TC +EFV (TLE400)	Reduced concentration	No dose adjustment
2nd Line regimen (Adolescents and Adults)		
AZT/3TC+ATV/r TDF/FTC+ATV/r	Reduced concentration of ATV/r	Don't use
AZT/3TC+LPV/r	Reduced concentration of LPV/r	Double the dose to 800mg/200mg

ARV Regimen	Type of interaction	Recommended action
3 rd Line regimen (Adolescents and Adults)		
DTG + DRV/r + AZT/3TC	No studies	Don't use
DTG (BD) + LPV/r + (AZT/3TC or TDF/FTC)	Reduced concentration of DTG Reduced concentration of LPV/r	Double the dose to 800mg/200mg
DTG+ATV/r+ AZT/3TC	Reduced concentration of ATV/r	Don't use

18.11 Co-trimoxazole formulations and dosage for infants and children living with HIV or exposed to HIV

Recommended daily dosage	Suspension (5 ml of syrup 200 mg/ 40 mg)	Child tablet (100 mg/ 20 mg)	Single strength adult tablet (400 mg/ 80 mg)	Double strength Adult tablet (800 mg/ 160 mg)
<6 months (<5 kg) 100 mg Sulfamethoxazole/ 20 mg trimethoprim	2.5 ml	One tablet	¼ tablet, Possibly Mixed with feeding	
6 months- 5 years (5-15 kg)200 mg Sulfamethoxazole/ 40 mg trimethoprim	5mlb	Two tablets	½ tablet	-
6 - 14 years(15- 30 kg) 400 mg Sulfamethoxazole/ 80 mg trimethoprim	10 mlb	Four tablets	One tablet	½ tablet
>14 years (>30 kg) 800 mg Sulfamethoxazole/ 160 mg trimethoprim			Two tablets	One tablet
Frequency- once a day				

- a Splitting tablets into quarters is not considered best practice. This should be done only if syrup is not available.
- b Children of these ages (6 months–14 years) may swallow crushed tablets

18.12 Fomu ya Kusaidia Utambuzi wa Uwezekano wa Kuwepo kwa Magonjwa ya Ngono na via vya Uzazi

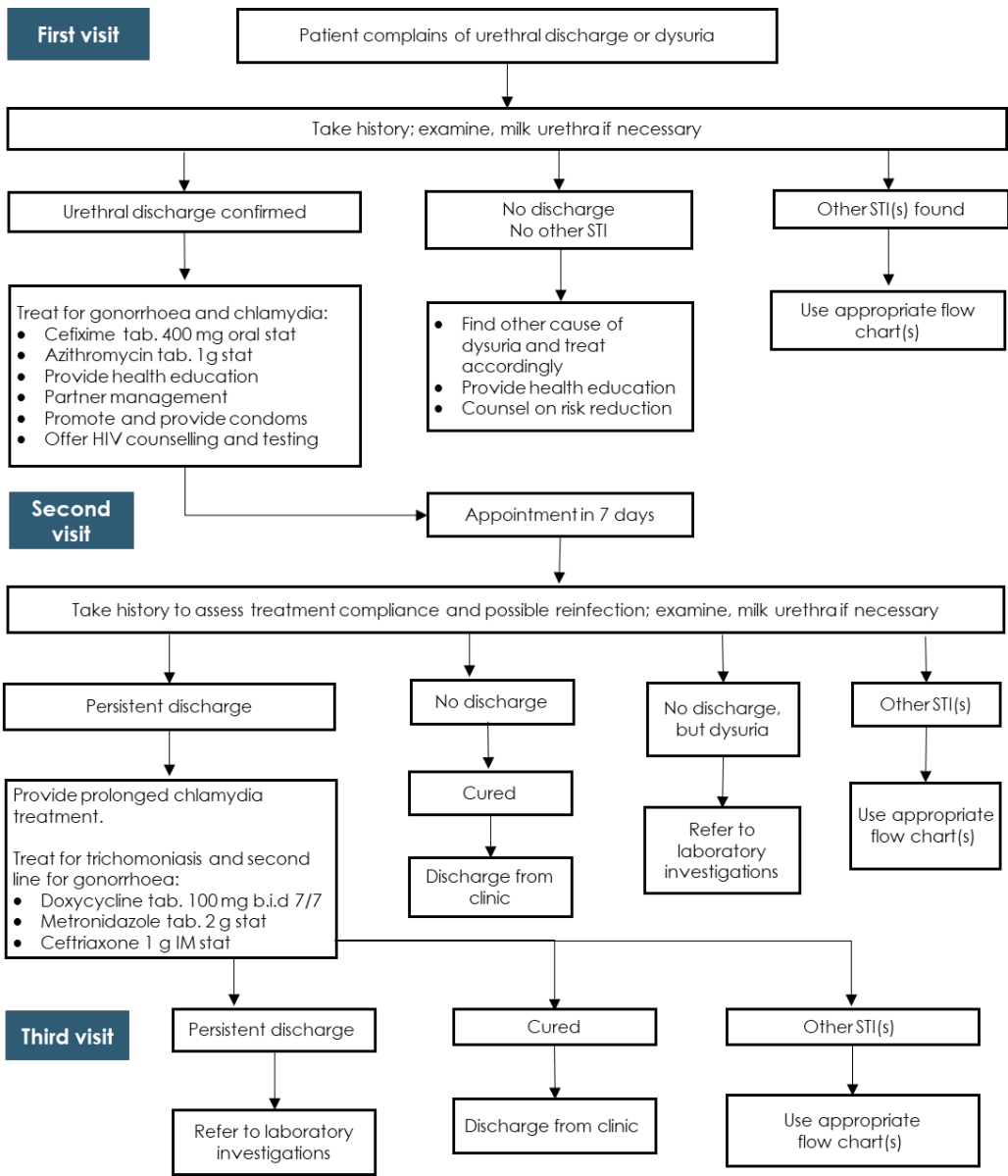
Jina la mteja:.....

Namba ya usajili: Umri: Jinsi: Me/Ke

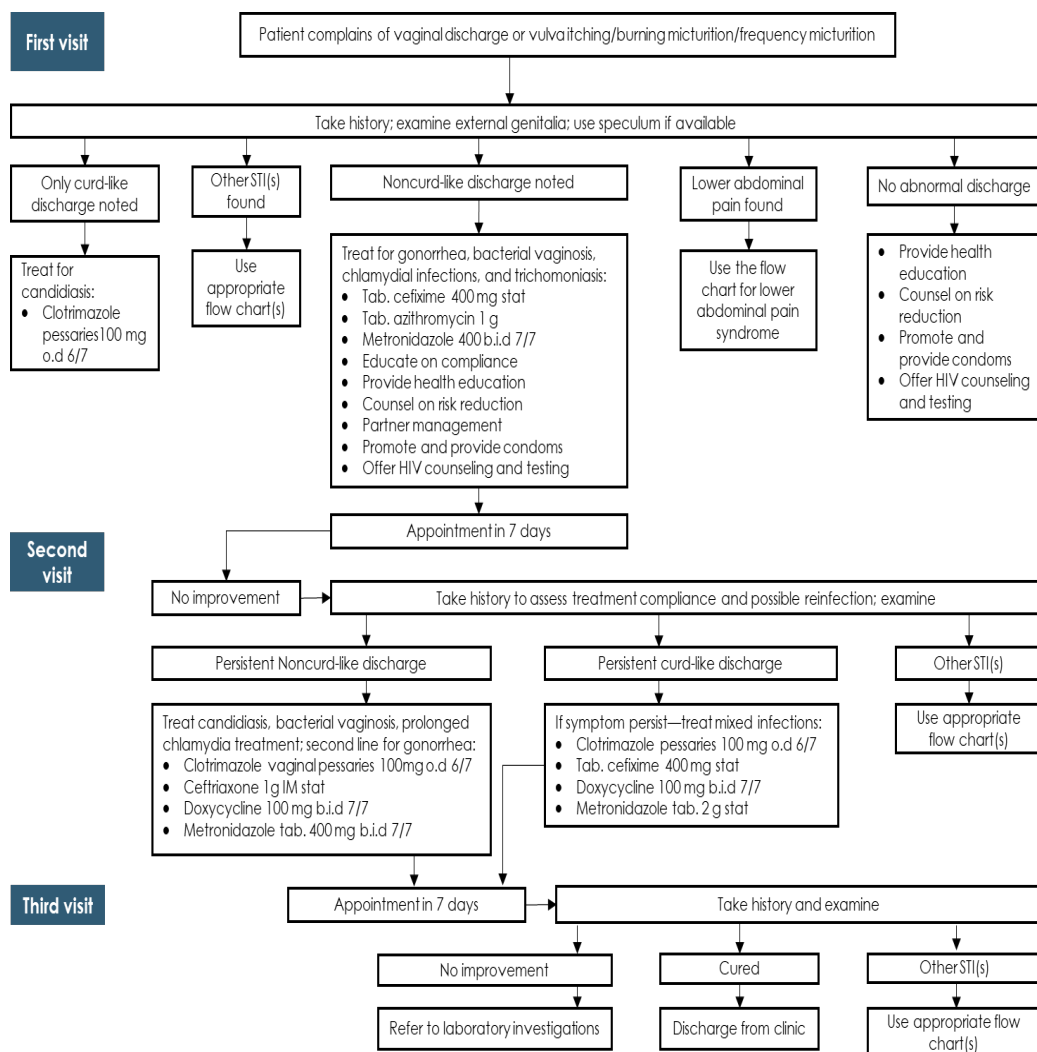
Tarehe ya mahudhurio.....

Weka alama ya vema panapohusika	N	H	N	H
Je, Unatokwa na usaha/majimaji yasiyo ya kawaida ukeni /Uumeni?				
Je, Unapata maumivu wakati wa kujisaidia haja ndogo?				
Je, Una kidonda/kipele kwenye uume au uke?				
Je, Unapata maumivu chini ya kitovu au wakati wa kujamiiana? (Kwa wanawake)				
Je, Una viotea au kuwashwa sehemu za ukeni/ uumeni?				
Je, Unapata maumivu au kutokwa na majimaji katika njia ya haja kubwa ?				
Je, Unahisi maumivu wakati wa kumeza au koo kukauka?				

18.13 Flow Chart 1: Syndromic Management of STI's/RTI's a) Management of Urethral Discharge Syndrome (UDS)

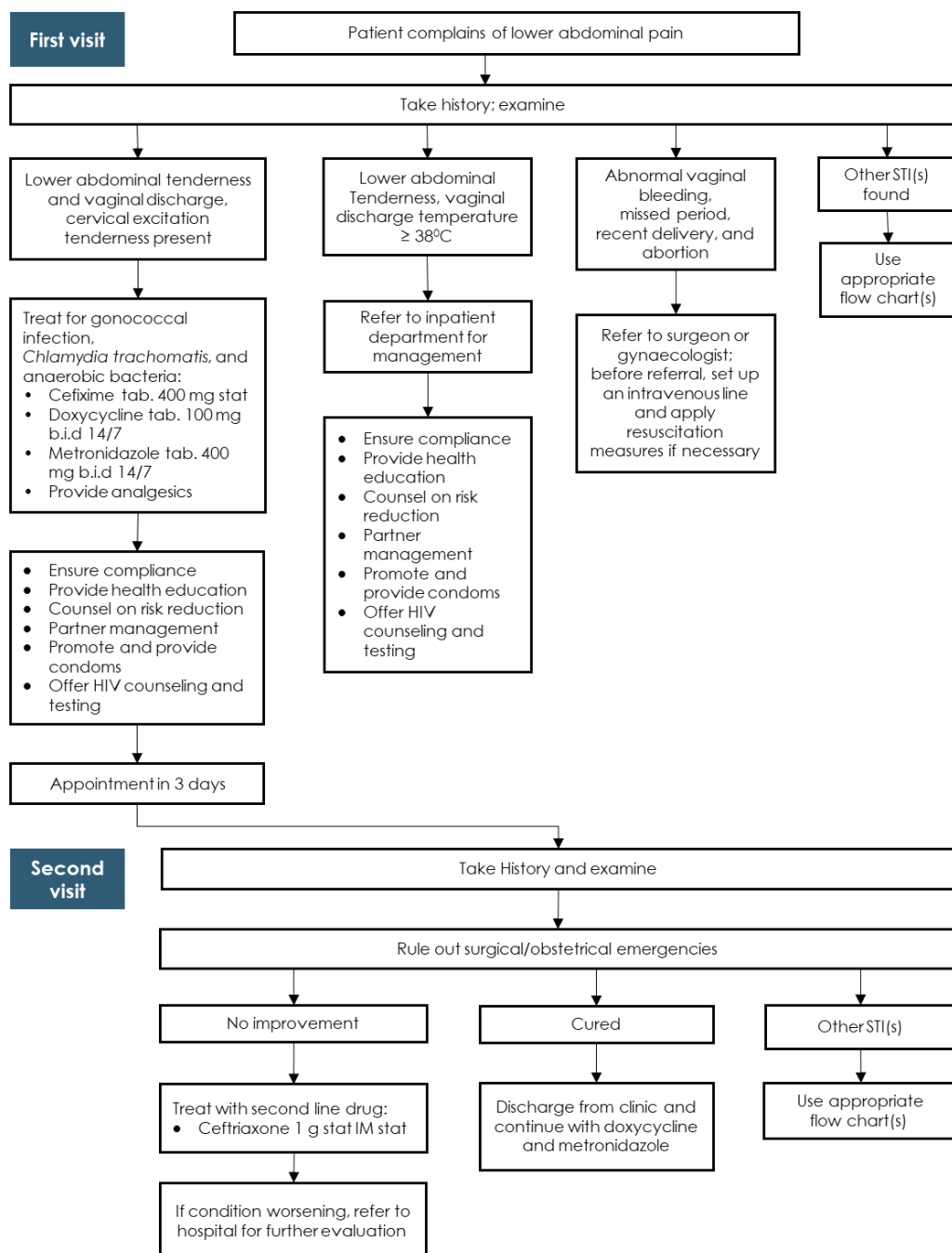


b) Management of Vaginal Discharge Syndrome (VDS)

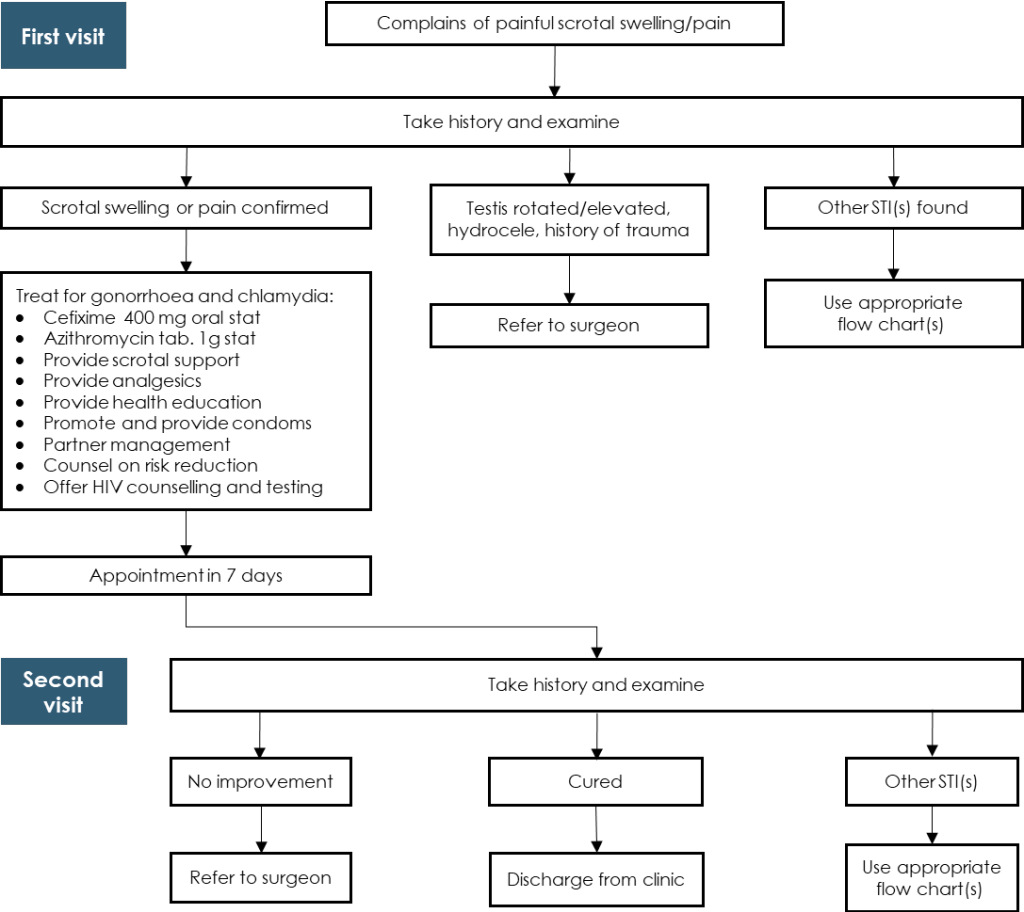


- Do not give Metronidazole in 1st trimester of pregnancy:
- Do not give Doxycycline in pregnancy or to lactating mother: substitute with Erythromycin 500 mg t.i.d 7/7 and Ceftriaxone 250 mg i.m. stat.

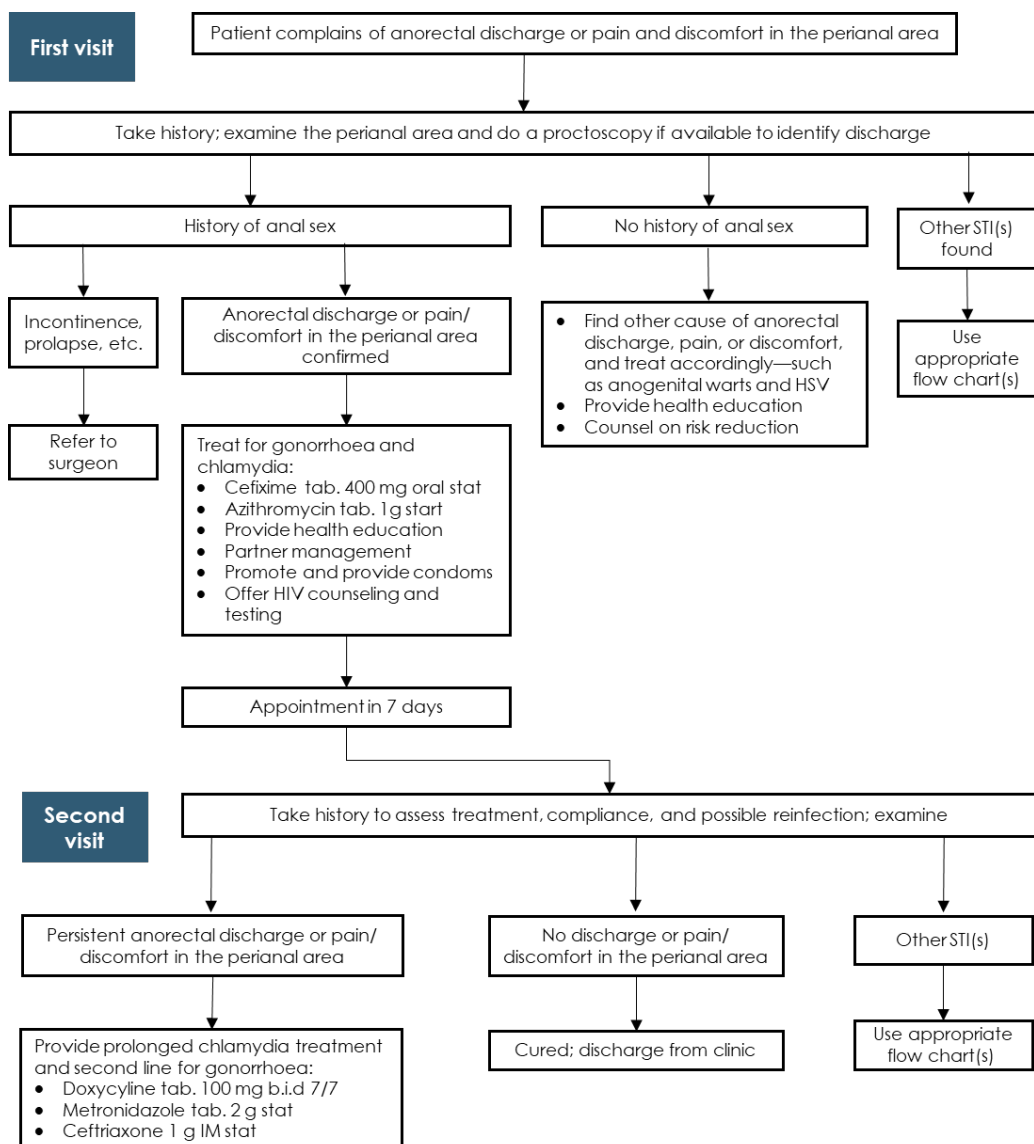
c) Management of Lower Abdominal Pain Syndrome (PID)



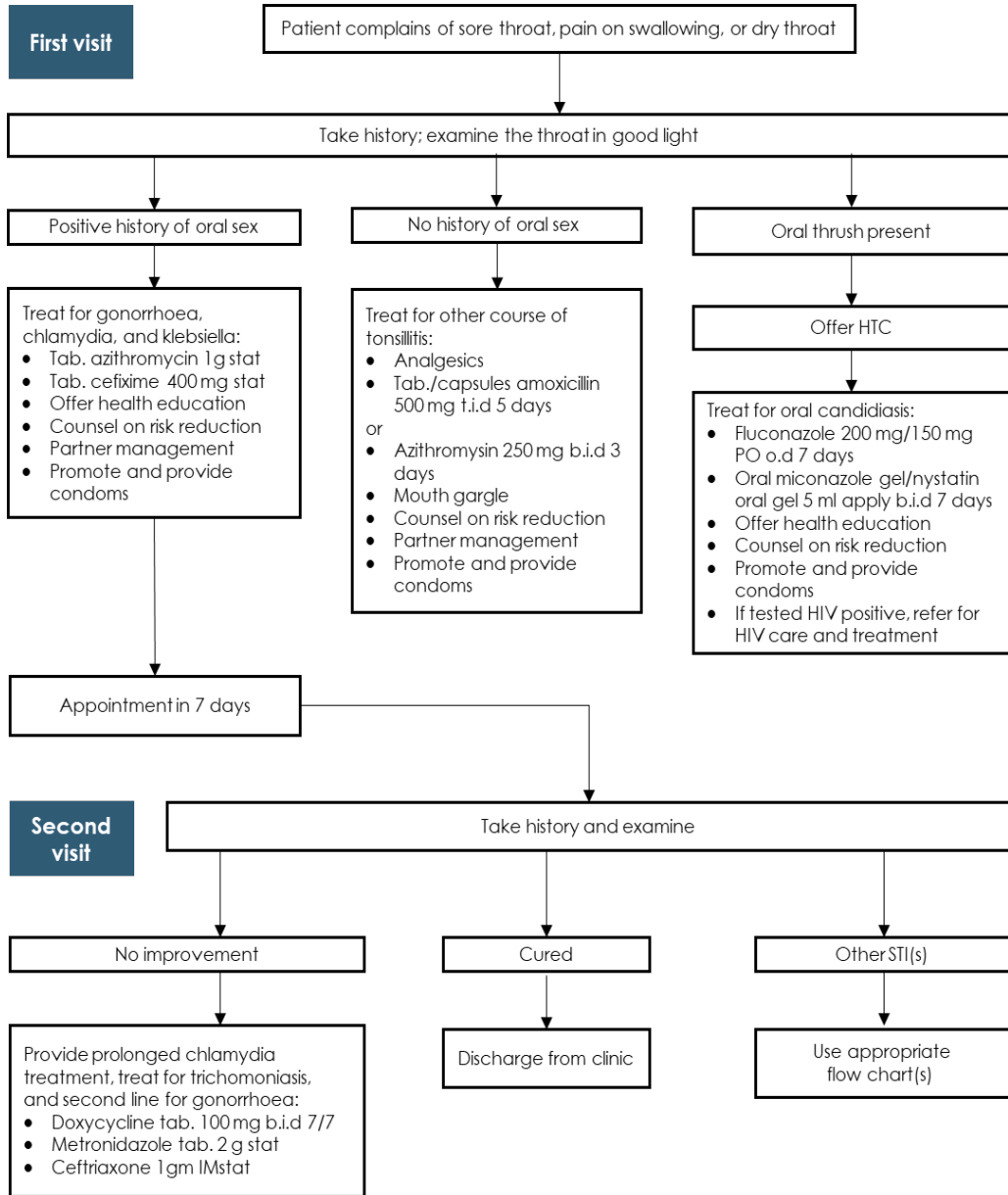
Flow Chart 2: Management of Painful Scrotal Swelling (PSS)



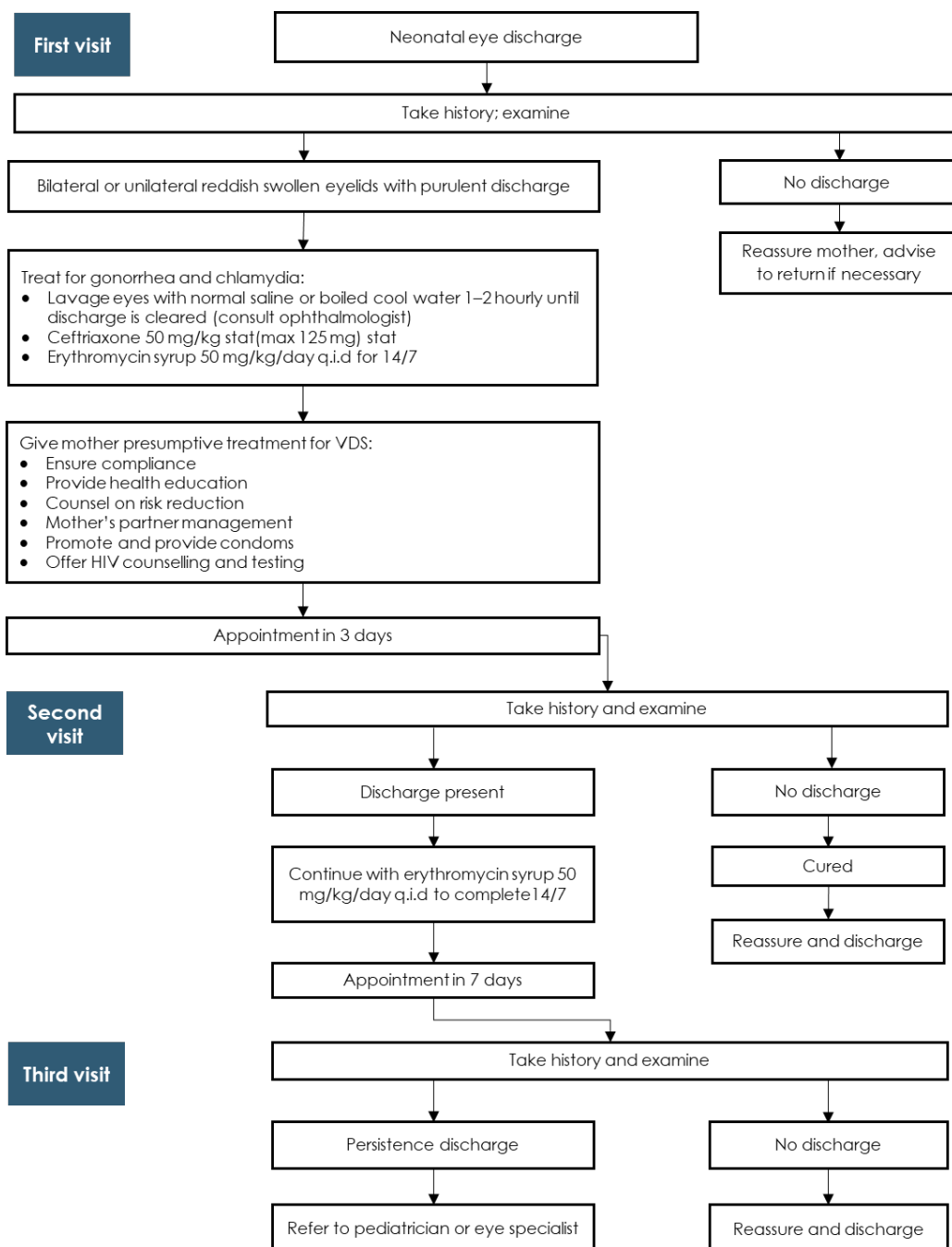
d) Management of Anorectal Syndrome (ARS)



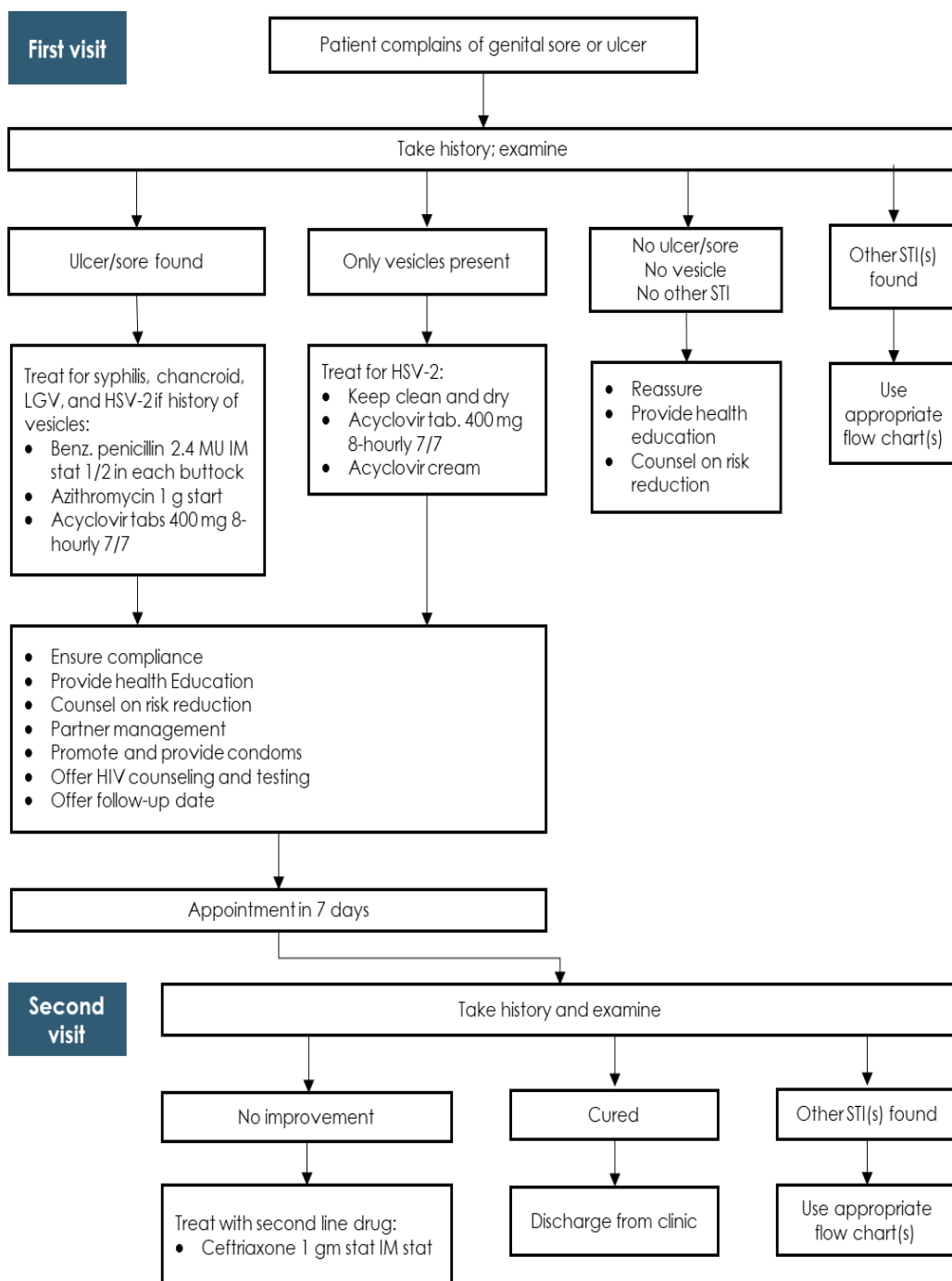
e) Management of Oropharyngeal Syndrome



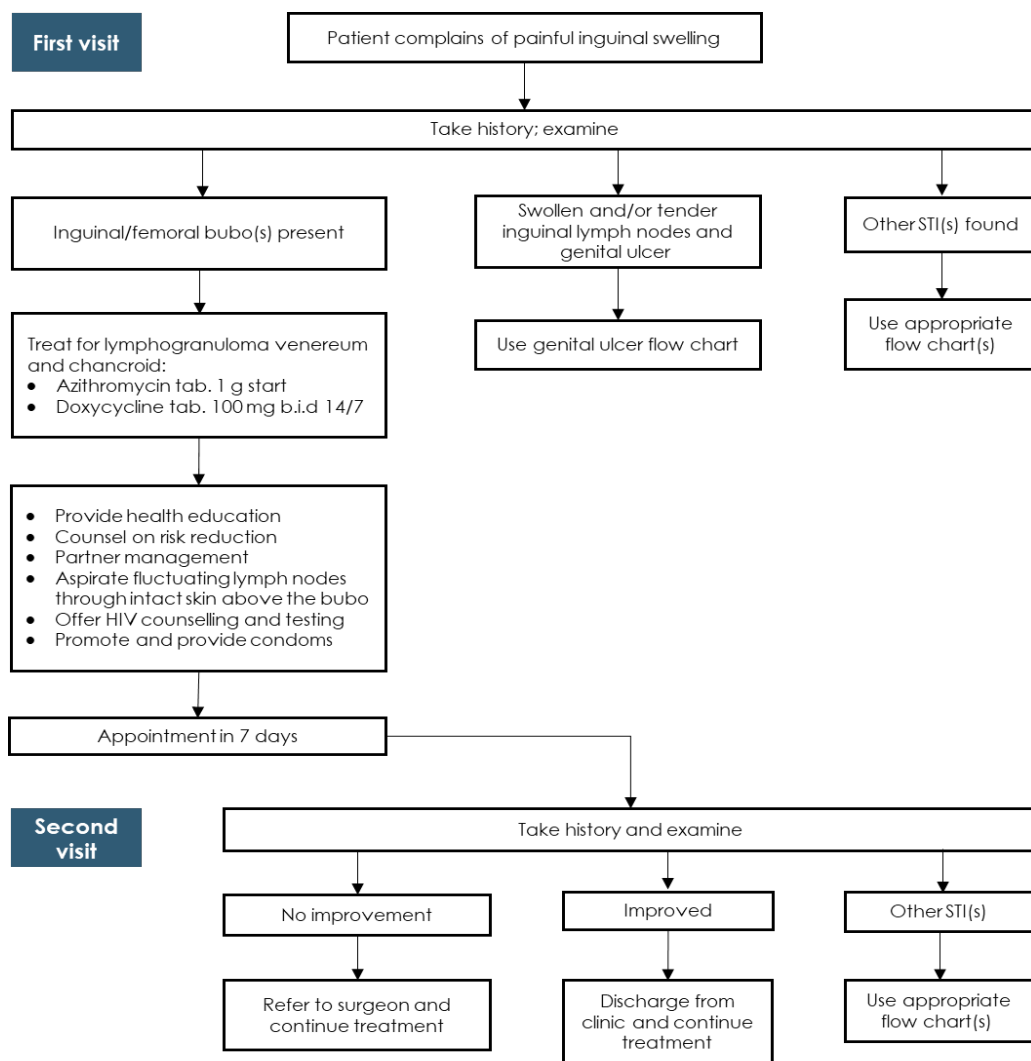
f) Management of Neonatal Conjunctivitis



g) Management of Genital Ulcer Syndrome



h) Management of Inguinal Bubos (IB)



18.14 Fomu ya Uchunguzi wa Afya ya Akili - CTC

Namba ya CTC..... Umri..... Jinsia.....
Kazi..... ART anazotumia sasa.....

A. Generalized Anxiety Disorder (GAD – 2) Ugonjwa wa Wasiwasi uliopitiliza.

	Katika wiki 2 zilizopita, ni mara ngapi umesumbuliwa na matatizo yafuatayo?	0 - Sivyo kabisa 1 - Siku kadhaa 2 - Zaidi ya nusu ya siku 3 - Karibu kila siku	Date	Date	Date	Date	Date	Date
1	Kuhisi wasiwasi au hofu	0 1 2 3						
2	Kutokuwa na uwezo wa kuacha kuhisi wasiwasi	0 1 2 3						
	Jumla ya Alama							

GAD – 2; Alama inapatikana kwa kujumlisha pointi kwa kila swali (jumla ya pointi). Alama ya pointi 3 na zaidi ndiyo inaashiria ugonjwa.

	Katika wiki 2 zilizopita, ni mara ngapi umesumbuliwa na matatizo yafuatayo?	0 - Sivyo kabisa 1 - Siku kadhaa 2 - Zaidi ya nusu ya siku 3 - Karibu kila siku	Date	Date	Date	Date	Date	Date
1	Nia ndogo au kukosa raha katika kufanya mambo. Kutojisikia raha kama vile wakati wa kula, kugusa, au ngono	0 1 2 3						
2	Hisia ya chini, huzuni au kutokuwa na tumaini. Kujisikia ya huzuni	0 1 2 3						
	Jumla ya Alama							

PHQ – 2; Alama inapatikana kwa kujumlisha pointi kwa kila swali (jumla ya pointi).

Alama ya pointi 3 na zaidi ndiyo inaashiria ugonjwa.

C. Post-Traumatic Stress Disorders (PC PTSD – 5) - Mfadhaiko wa Baada ya Kiwewe.

Wakati mwingine matukio hutokea kwa watu ambayo si ya kawaida au ya kiwewe kwa mfano ajali mbaya au moto.								
Je, umewahi kushuhudia matukio haya katika kipindi cha mwezi uliopita? (ndio endelea na swali namba 1, Hapana funga dodoso)								
	Andika N – Ndiyo, au H - Hapana	N au H	N au H	N au H	N au H	N au H	N au H	N au H
1	Je ulishawahi kufikiria kuhusu matukio haya yaliyotokea wakati hukutaka kuyafikiria?							
2	Je ulishawahi kujizuia kufikiria matukio haya?							
3	Je unahisi umekuwa mwangalifu sana siku hizi?							
4	Je ulishawahi kujitenga na wenzako?							
5	Je ulishawahi kujilaumu kuhusu matukio yaliyotokea?							
Jumla ya NDIYO								

PC PTSD – 5; Ikiwa mgonjwa atajibu NDIYO kwa maswali yoyote 3 au zaidi inachukuliwa kuwa matokeo “chanya

D. Drug Use AUDIT-3 - Matumizi ya Pombe.

			Date	Date	Date	Date	Date	Date
1	Ni mara ngapi unakunywa kinywaji kilicho na pombe?	0. Sivyo kabisa 1. kila mwezi au chini 2. Mara 2-4 kwa mwezi 3. Mara 2-3 kwa wiki 4. 4 au zaidi kwa wiki						
2	Unakunywa vinywaji vingapi vyenye pombe kwa siku?	0. 1 au 2 1. 3 hadi 4 2. 5 hadi 6 3. 7 hadi 9 4. 0 au zaidi						
3	Ni mara ngapi unakunywa vinywaji sita au Zaidi kwa tukio moja	0. Kila siku au karibu kila siku 1. Kila wiki, 2. Kila mwezi, 3. Chini ya kila mwezi, 4. Kamwe						

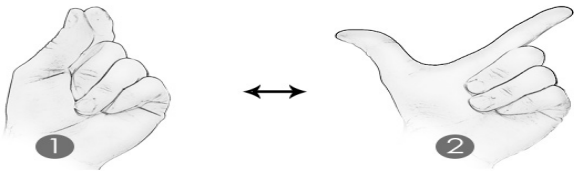
AUDIT-C imewekwa kwa alama 0-12 (alama za 0 zinaonyesha kutotumia pombe). Kwa wanaume, alama ya 4 au zaidi inachukuliwa kuwa chanya; kwa wanawake, alama 3 au zaidi inachukuliwa kuwa chanya. Kwa ujumla, kadiri alama ya AUDIT-C inavyokuwa juu, ndivyo uwezekano wa unywaji wa mgonjwa kuathiri afya na usalama wake.

**E. Modified Psychosis Screening Questionnaire (PSQ) -
Ugonjwa wa kuchangantikiwa**

		Date	Date	Date	Date	Date	Date
	Swali la uchunguzi (Ndiyo au Hapana)	N au H	N au H	N au H	N au H	N au H	N au H
1	Katika mwaka uliopita, kumekuwa na nyakati ambapo ulijisikia furaha sana bila mapumziko kwa siku kumi mfululizo?						
	a) Je, jamaa/marafiki zako walifikiri ni jambo geni au kulalamika?						
2	Katika mwaka uliopita umeshawahi kuwa na hisia kwamba mawazo fulani ambayo unayo si mawazo yako mwenyewe? Kama Ndiyo,						
	a) Ulifikiri ni mawazo ya nani?						
3	Katika mwaka uliopita, kumekuwa na nyakati ambapo ulihisi kwamba uko karibu kudhuriwa au kudhulumiwa na wengine kwa njia fulani?						
4	Katika mwaka uliopita, kumekuwa na nyakati ambapo ulihisi kwamba jambo lisilo la kawaida lilikuwa likitendeka? Kama Ndiyo,						
	a) Je, ni jambo gani lisilo la kawaida uliwahi kuhisi?						
5	Katika mwaka uliopita, kuna nyakati ambapo ulisikia au kuona kitu ambacho hakipo? Mambo ambayo watu wengine hawakuweza kuyasikia au kuona?						

Kumbuka; Kama mteja ana alama ya NDIYO katika mojawapo ya dalili zilizo hapo juu rejelea kwa uchunguzi zaidi.

F. International HIV Dementia Scale (IHDS) - Dementia itokanayo na VVU.

1	Usajili wa kumbukumbu	1. Toa maneno manne ya kukumbuka (mbwa, kofia, maharagwe, nyekundu) – sekunde 1 kusema kila moja. 2. Kisha muulize mgonjwa maneno yote manne baada ya kusema. 3. Rudia maneno ikiwa mgonjwa hatayakumbuka yote mara moja. 4. Mwambie mgonjwa baadaye utauliza maneno hayo tena. Alama
2	Kasi ya neva. 0-4 pointi	 1. Mwambie mgonjwa aguse vidole viwili vya kwanza vya mkono usio na nguvu kwa upana na haraka iwezekanavyo. Angalia picha hapo juu 2. Hesabu idadi ya miguso iliyotekelezwa kwa sekunde 5: 3. Rekodi idadi ya migusano ndani ya sekunde 5 Scoring: ≥ 15 taps = 4 points 11-14 taps = 3 points 7-10 taps = 2 points 3-6 taps = 1 points 0-2 taps = 0 points Score/Alama
3.	Kasi ya uelewa na utendaji.	Mruhusu mgonjwa kufanya harakati kwa mkono usio na nguvu haraka iwezekanavyo;  1. Nyosha mkono kwa ngumi kwenye uso tambarare. 2. Weka mkono juu ya meza na kiganja kikitazama chini. 3. Weka mkono sambamba na meza Score/Alama..... Onyesha na umruhusu mgonjwa afanye mazoezi mara 2 kwa vitendo. Hesabu idadi ya mifuatano iliyofanywa kwa sekunde 10

		Scoring: ≥ 4 mifuatano = 4 points 3 mifuatano = 3 points 2 mifuatano = 2 points 1 mifuatano = 1 point Score/ Alama.....
4.	Upimaji wa Kumbukumbu.	Uliza mgonjwa kukumbuka maneno manne kwenye usajili wa kumbukumbu. Kwa maneno ambayo hayajakumbukwa, uliza kwa kidokezo cha kisemantiki kama ifuatavyo. Mnyama (Mbwa), kipande cha nguo (Kofia), mboga (Maharagwe), Rangi (Nyekundu). Toa alama 1 kwa kila neno lililokumbukwa kwa hiari, toa alama 0.5 kwa jibu sahihi baada ya kidokezo. Jumla ya alama 4.

Alama ya juu = 12. Alama <10 inapaswa kutathminiwa zaidi kwa uwezekano wa shida ya Dementia.

Maoni:.....

.....

Rufaa kwenda.....

Psychiatric Unit.....

Aliyetoa rufaa..... Tarehe.....

18.15 Management of Malnutrition for PLHIV using Therapeutic Food Products

Age Group	Entry criteria	Product	Transition/exit criteria
Children 6 - 59 month	<u>SAM</u> MUAC: <11.5cm W/H < -3 SD	0– less than 6 Inpatient Stabilisation: - If no oedema, 130 ml of F-100-Diluted/kg of body weight/day. - If oedema, F-75 according to SAM protocol Transition and rehabilitation: - F-100-Diluted according to SAM protocol (F-100-Diluted if < 6 months of age) 6 months–14 years Inpatient Stabilisation: 130 ml of F-75/kg of body weight/day (100 ml if severe oedema) Transition: - Days 1 and 2: Same amount of F-100; Day 3: Increase each feed by 10 ml until child reaches rehabilitation phase	6–59 months: No bilateral pitting oedema for two consecutive visits, MUAC ≥ 11.5 cm, WHZ ≥ – 2 OR 15% weight gain on two consecutive visits AND appetite AND medical problems stabilised or subsiding AND continued weight gain of more than 5 g/kg of body weight/day
	<u>MAM</u> - MUAC: 11.5cm - <12.5cm - W/H -3 SD to <-2 SD W/H -3 Z-scores to <-2	- If the child was treated for SAM, 1 packet of RUTF per day PLUS 100 g of FBF/day - for children 6 months to 9 years of age 200 g of FBF/day - for children 10–14 years of age for 1 month - If the child was NOT treated for SAM, only FBF as above (no RUTF)	6–59 months: MUAC ≥ 12.5 cm OR WHZ OR BMI-for-age ≥ –2 for two consecutive visits 5–9 years: MUAC ≥ 14.5 cm 10–14 years: MUAC ≥ 18.5 cm

Age Group	Entry criteria	Product	Transition/exit criteria
Children aged 6 to 14 years	<u>SAM</u> - Bilateral pitting oedema OR - severe visible wasting OR - MUAC 5–9 years: < 13.5 cm 10–14 years: < 16.0 cm OR WHZ OR - BMI-for-age < -3	Inpatient Stabilization: - If no oedema, 130 ml of F-100-Diluted/kg of body weight/ day. - If oedema, F-75 according to SAM protocol Transition and rehabilitation: - F- 100-Diluted according to SAM protocol (F-100-Diluted if < 6 months of age) 10–14 years Inpatient Stabilization: 130 ml of F-75/kg of body weight/day (100 ml if severe oedema) Transition: - Days 1 and 2: Same amount of F-100; - Day 3: Increase each feed by 10 ml until child reaches rehabilitation phase	15% weight gain on two consecutive visits AND appetite AND medical problems stabilized or subsiding AND continued weight gain of more than 5 g/kg of body weight/day (Need to be confirmed if correct
	<u>(MAM)</u> - Confirmed weight loss of more than 5% since last visit OR - MUAC 5–9 years: ≥ 13.5 to < 14.5 cm 10–14 years: ≥ 16.0 to < 18.5 cm OR WHZ OR - BMI-for-age -3 to <-2	Children 6 months to 9 years of age - If the child was treated for SAM, 1 packet of RUTF per day PLUS 100 g of FBF/day for 200 g of FBF/day for children 10–14 years of age - for 1 month If the child was NOT treated for SAM, only FBF as above (no RUTF)	5–9 years: MUAC ≥ 14.5 cm 10–14 years: MUAC ≥ 18.5 cm

Age Group	Entry criteria	Product	Transition/exit criteria
Adolescents 15–17 year	<u>SAM</u> - Confirmed unintentional weight loss of more than 10% since the last visit - Bilateral pitting oedema OR - MUAC less than 18.5 cm OR BMI less than 16.0	Inpatient Stabilization: 50 kcal of F- 75/ kg of body weight/day Transition: 50 kcal of F100/kg of body weight/day Rehabilitation: 3 packets of RUTF/day PLUS 300 g of FBF/day Outpatient 3 packets of RUTF/day PLUS 300 g of FBF/day	MUAC $\geq$ 18.5 cm
	<u>MAM</u> - Confirmed unintentional weight loss of more than 5% since last visit - MUAC 18.5 to less than 22.0 cm OR BMI 16.0 to less than 18.5	300 g of FBF/day	MUAC $\geq$ 22.0 cm
Adults (non-pregnant/ $\leq$ 6 months post-partum)	<u>SAM</u> - Bilateral pitting oedema OR - MUAC less than 18.5 cm OR BMI less than 16.0	Inpatient Stabilization: 53 ml of F- 75 OR 40 kcal of F-75/ kg of body weight/day Transition: 50 ml of F-100 OR 40 kcal of F-75/kg of body weight/day - Rehabilitation: 3 packets of RUTF/day PLUS 300 g of FBF/day Outpatient 3 packets of RUTF/day PLUS 300 g of FBF/day	MUAC $\geq$ 19.0 cm OR BMI $\geq$ 16.0 and $<$ 18.5 OR sustained weight gain
	<u>MAM</u> - MUAC 18.5 to less than 22.0 cm OR - BMI 16.0 to less than 18.5	300 g of FBF/day	MUAC 22.0 cm or greater OR BMI 18.5 or greater for two consecutive visits

Age Group	Entry criteria	Product	Transition/exit criteria
Pregnant women and women ≤ 6 months postpartum	<u>SAM</u> - Bilateral pitting oedema OR - MUAC less than 19.0 cm	Inpatient Stabilization: 53 ml of F- 75 OR 50 kcal of F-75/kg body weight/day Transition: 40 kcal of F- 100/kg body weight/day Rehabilitation: - packets of RUTF/day PLUS 300 g of FBF/day Outpatient 3 packets of RUTF/day PLUS 300 g of FBF/day	MUAC 19.0 cm or greater
	<u>MAM</u> - Poor weight gain OR - MUAC 19.0 to less than 23.0 cm	300 g of FBF/day	- MUAC 23.0 cm or greater OR - over 6 months post-partum

18.16 Nutrition guidance for multiple medications including ARV

Antiretroviral medications (ARVs)		
Nucleoside and nucleotide reverse transcriptase inhibitors (NRTIs)		
Medication	Possible side effects	Nutrition guidance
Abacavir (ABC)	Nausea, vomiting, fever, allergic reaction, anorexia, abdominal pain, diarrhoea, anaemia, rash, hypotension, pancreatitis, dyspnea, weakness and insomnia, cough, headache	Can be taken with or without food, though taking with food reduces side effects. Avoid alcohol.
Emtricitabine (FTC)		Can be taken with or without food Avoid alcohol
Lamivudine (3TC)	Nausea, vomiting, headache, dizziness, diarrhoea, abdominal pain, nasal symptoms, cough, fatigue, pancreatitis, anaemia, insomnia, muscle pain, and rash	Can be taken with or without food. Avoid alcohol.
Tenofovir (TDF)	Headache, diarrhoea, nausea, vomiting, abdominal pain, rash, headache, flatulence, anorexia, dizziness, insomnia, depression, sweating, renal function impairment	Can be taken with or without food. Avoid alcohol
Zidovudine (ZDV, AZT)	Anorexia, anaemia, nausea, vomiting, bone marrow suppression, headache, fatigue, constipation, fever, dizziness, dyspnea, insomnia, muscle pain, rash	Can be taken with or without food. But avoid a meal with high-fat. Avoid alcohol.
Non-nucleoside reverse transcriptase inhibitors (NNRTIs)		
Medication	Possible side effects	Nutrition guidance
Efavirenz (EFZ)	Elevated blood cholesterol levels, elevated triglycerides, rash, dizziness, anorexia, nausea, vomiting, diarrhoea, dyspepsia, abdominal pain, flatulence	Can be taken with or without food, but avoid a meal with high-fat. Should be taken just before bedtime. Avoid alcohol.
Nevirapine (NVP)	Nausea, vomiting, rash, fever, headache, skin reactions, fatigue, stomatitis, abdominal pain, drowsiness, paresthesia, high hepatotoxicity	Take with or without food. Avoid using St. John's wort plant.
Protease inhibitors (PIs)		
Medication	Nutrition guidance	Possible side effects
Atazanavir (IDV)	Gastrointestinal complaints, renal toxicity (especially when renal function is already reduced)	Should be taken only with food.

Medication	Nutrition guidance	Possible side effects
Lopinavir/ Ritonavir (LPV/r)	Nausea, vomiting, weakness, diarrhoea, headache, dizziness, abdominal pain, fever, diabetes, anorexia, hepatitis, jaundice	Take with or without food. Avoid the use of St. John's wort plant.
Ritonavir (RTV)	Nausea, vomiting, diarrhoea, hepatitis, jaundice, weakness, anorexia, abdominal pain, fever, diabetes, headache, dizziness, possible increased risk of lipodystrophy	Should be taken only with food. Avoid St. John's wort.
Once daily single tablet regimen		
TLE Fixed Dose Combination (Efavirenz, Lamivudine, Tenofovir)	See above for Efavirenz, Lamivudine and Tenofovir.	Take with or without food.
Atazanavir/ Ritonavir /Tenofovir (ATV/r/FTC/ TDF)	See above for Atazanavir, Ritonavir and Tenofovir.	Take with food.
	Food reduces absorption of isoniazid	Do not take with food. Take 1 hour before or 2 hours after meals
	May affect vitamin B6 metabolism	Daily consumption of food sources of vitamin B6 such as white beans, maize avocado, meat, and fish, or vitamin B6 (25 to 50mg daily) supplementation is recommended
	Increased risk of hepatitis when combined with alcohol	Avoid alcohol
	Anorexia i.e loss of appetite	Eat small and frequent meals.
	Diarrhea	Drink plenty of fluids and eat energy- and nutrient-rich foods.
Rifampicin	Gastrointestinal irritation, anaemia, jaundice, pancreatitis, altered taste, anorexia	Take on an empty stomach, 1 hour before or 2 hours after a meal. Supplement with 10 mg vitamin B6 daily. Do not take with alcohol.

Medication	Nutrition guidance	Possible side effects
Rifapentine	Stomach upset, nausea, vomiting, loss of appetite, or headache may occur.	Take with food. Taking rifapentine with food increases its bioavailability and reduces gastrointestinal upset. Rifapentine tablets can be crushed and mixed with semisolid food if necessary
Co-trimoxazole	Rash, itching, sore throat, fever or chills, severe diarrhea (watery or bloody stools) that may occur with or without fever and stomach cramps (may occur up to 2 months or more after your treatment) shortness of breath, cough	Take it with food, although you can still take it on an empty stomach. Drink plenty of fluid such as water while you are taking Co-Trimoxazole
Rifamycin	Temporary discoloration (yellow, reddish-orange, or brown color) of your skin, teeth, saliva, urine, stool, sweat, and tears), itching, Flushing, headache, drowsiness, dizziness. lack of coordination.	Take 1 hour before or 2 hours after meal. May take with small snack if needed. Take 1 hour before antacids. Avoid alcohol

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